Class 7
Mathematics

Table of Content

1.	Number System	1 – 33
2.	Comparing Quantities	34 – 83
3.	Linear Equations in One Variable	84 – 91
4.	Triangles and Congruence of Triangles	92 – 130
5.	Data Handling	131 – 143
6.	Mensuration	144 – 195
7.	Algebra and Exponents & Powers	196 – 250
8.	Square Roots and Cube Roots	251 – 273

1. NUMBER SYSTEM

Solutions

LEVEL -I

1. Correct option: (B)

The greatest two-digit number = 99
The smallest two-digit number = 10
∴The required difference = 99 − 10 = 89.

2. Correct option: (A)

The greatest two-digit number = 99
The smallest two-digit number = 10
∴The required product = 99 × 10 = 990.

3. Correct option: (C)

A prime number is a whole number greater than 1 whose only factors are 1 and itself.
A factor is a whole number that can be divided evenly into another number.
The first few prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29.
∴The required two-digit prime number is primer is 11.

4. Correct option: (B)

The total numbers between 1000 and 2000 (including 1000 and 2000) = 1001.
Of these, 501 are even numbers and the remaining 500 are odd numbers.
∴The required Number of even numbers between 1000 and 2000 including both is 501.

5. Correct option: (A)

Since 5321 is the greatest and 1235 is the least number then 5321 − 1235 = 4086.
∴The required number is 4086.

6. Correct option: (C)

By trial, we find that the smallest number consists entirely of seven and is exactly divisible by 13 is 777777.
On dividing it by thirteen, we get 59829 as the quotient which makes it the smallest number.
The given Divisor = 13 and Dividend = 777777

```
          5 9 8 2 9
      ┌─────────────
1 3 | 7 7 7 7 7 7
      6 5
      ---
      1 2 7
      1 1 7
      -----
        1 0 7
        1 0 4
        -----
            3 7
            2 6
            ---
            1 1 7
            1 1 7
            -----
                0
```

The Quotient is 59829 and the Remainder is 0.
∴The required number is 59829.

7. Correct option: (B)

Composite numbers are only those numbers that have more than two factors.

50, 51, 52, 54, 55, 56, 57, 58, 60, 62, 63, 64, 65, 66, 68, 69, 70, 72, 74, 75, 76, 77, 78, 80, 81,82, 84, 85, 86, 87, 88, 90, 91, 92, 93, 94, 95, 96, 98, 99, 100 are the composite numbers between 50 and 100.

Thus, there are total 41 composite numbers between 50 and 100.

8. Correct option: (B)

Given that (a+b) and (a-b) are twin primes then

$\Rightarrow |a - b - a - b| = 2$

$\Rightarrow |-2b| = 2$

$\Rightarrow 2b = \pm 2$

$\Rightarrow \frac{2b}{2} = \pm\frac{2}{2}$

$\Rightarrow b = \pm 1 \cdot$

Since b must be equal to 1 then b is always 1, there can be only one common factor between a and b.

Hence a and b are always co-prime.

9. Correct option: (A)

Converting the given numbers to like decimals:

0.0005, 0.005, 0.15, 0.175, 0.5, 3.5 = $\frac{5}{1000}, \frac{50}{1000}, \frac{150}{1000}, \frac{1750}{1000}, \frac{3500}{1000}$

Treating them as whole numbers, the numbers are 5, 50, 150, 1750,3500.

Finding the prime factorization for the numbers are 5, 50, 150, 1750,3500.

$5 = 1 \times 5.$

$50 = 2 \times 5 \times 5$

$150 = 2 \times 3 \times 5 \times 5$

$1750 = 2 \times 5 \times 5 \times 5 \times 7.$

$3500 = 2 \times 2 \times 5 \times 5 \times 5 \times 7.$

The common factor for the numbers 5, 50, 150, 1750,3500 is 5.

The highest common factor for the numbers 5, 50, 150, 1750,3500 is 5.

The HCF for the numbers is $\frac{5}{1000}, \frac{50}{1000}, \frac{150}{1000}, \frac{1750}{1000}, \frac{3500}{1000} = \frac{5}{1000}$.

Therefore, the HCF of 0.0005, 0.005, 0.15, 0.175, 0.5, 3.5 = $\frac{5}{1000} = 0.0005$.

10. Correct option: (C)

Let the numbers be x and x + 2.

Then, $(x + 2)^2 - x^2 = 84$

$\Rightarrow 4x + 4 = 84$

$\Rightarrow 4x = 80$

$\Rightarrow x = 20.$

$\therefore$ The required sum = $x + (x + 2) = 2x + 2 = 42.$

11. Correct option: (C)

L.C.M of a group of numbers is the smallest number that is the multiple of all the numbers.

H.C.F of a group of numbers is the greatest factor of the numbers.

H.C.F of fractions = $\frac{\text{H.C.F of numerators}}{\text{L.C.M of denominators}}$.

H.C.F of (2, 4, 8) = 2

L.C.M of (3, 12, 15) = $2 \times 2 \times 3 \times 5 = 60.$

Required H.C.F = $\frac{2}{60} = \frac{1}{30}$.

$\therefore$ The required H.C.F is $\frac{1}{30}$.

12. Correct option: (C)

List out all of the prime factors for each number:

- Prime Factors for 182: 2, 7, and 13
- Prime Factors for 273: 3, 7, and 13
- Prime Factors for 637: 7 and 13

Here the common prime factors are 7 and 13.

On Multiplying 7 and 13 to get the HCF of the given numbers.

H.C.F of (182, 273, 637) = $7 \times 1 = 91$.

∴ The required H.C.F is 91.

The LCM of following numbers

13. Correct option: (C)

List out all of the prime factors for each number:

- The prime factors of 18 are 2, 3 and 3.
- The prime factors of 16 are 2, 2, 2 and 2.
- The prime factors of 22 are 2 and 11.

The least common multiple is the product of all factors in the greatest number of their occurrence.

Here the common and uncommon prime factors are 2, 2, 2, 2, 2, 3, 3, and 11.

On Multiplying 2, 2, 2, 2, 2, 3, 3, and 11 to get the LCM of the given numbers.

L.C.M of (18, 16, 22) = $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 11 = 1{,}584$.

∴ The required L.C.M least common multiple of 18, 16 and 22 is 1,584.

14. Correct option: (A)

Since, Prime numbers between 0 and 50 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43 and 47. So there are total of 15 prime numbers between 0 and 50.

15. Correct option: (D)

Let the digit in the tens place be x and one's place be y

Hence the number = $(10 \times x) + y = 10x + y$

Number obtained by reversing the digits = $10y + x$

Given that $10y + x = \frac{5}{6} \times (10x + y)$

$\Rightarrow 60y + 6x = 50x + 5y$

$\Rightarrow 44x - 55y = 0$

$\Rightarrow 4x - 5y = 0 \ldots (1)$

It is also given that difference of digits is 1.

$\Rightarrow x - y = 1 \ldots (2)$

Multiplying equation (2) with 4, we get

$\Rightarrow 4x - 4y = 4 \ldots (3)$

On subtracting (1) and (3)

$\Rightarrow 4x - 5y - 4x + 4y = 0 - 4$

$\Rightarrow -y = -4$

$\therefore y = 4$

Putting $y = 4$ in (1) we get $\Rightarrow 4x - 5y = 0$

$\Rightarrow 4x = 5(4)$

$\therefore x = 5$

∴The required number is 54.

16. Correct option: (B)

Given that the number 6p545 divisible by 3.

Suppose p = 0 then $\frac{60545}{3} = 20181.66666.$

p = 1 then $\frac{61545}{3} = 20515.$

Therefore, the least value of p, so that the number 6p545 divisible by 3 is 1.

17. Correct option: (C)

The Cyclic table for 4 is as follows:

	Unit digit
$4^1 = 4$	4
$4^2 = 16$	6
$4^3 = 64$	4
$4^4 = 256$	6
$4^5 = 1024$	4

The unit digit pattern is 6,4,6, 4, ...

Hence, the power cycle of 4 contains only 2 numbers 4 & 6, which appear in case of odd and even powers respectively.

$4^{(\text{odd number})} = 4.$

$4^{(\text{even number})} = 6$

Since 1970 is even number.

$\therefore$ The number in units place 4^{1970} is 6.

18. Correct option: (B)

The Cyclic table for 5 is as follows:

	Unit digit
$5^1 = 5$	5
$5^2 = 25$	5
$5^3 = 125$	5
$5^4 = 625$	5
$5^5 = 3125$	5

The unit digit pattern is 5.

Hence, the power cycle of 5 contains only 1 number 5, which appear in case of odd and even powers respectively.

$5^{(\text{oddnumber})} = 5.$

$5^{(\text{even number})} = 5.$

Since 5555 is odd number.

$\therefore$ The number in units place 5^{5555} is 5.

19. Correct option: (D)

The Cyclic table for 9 is as follows:

	Unit digit
$9^1 = 9$	9
$9^2 = 81$	1
$9^3 = 729$	9
$9^4 = 6561$	1
$9^5 = 59049$	9

The unit digit pattern is 9.

Hence, the power cycle of 9 contains only 2 numbers 9 & 1, which appear in case of odd and even powers respectively.

$9^{(\text{odd number})} = 9.$

$9^{(\text{even number})} = 1.$

Since 1999 is odd number.

$\therefore$ The number in units place 9^{1999} is 9.

20. Correct option: (A)

An exponent of a number, represents the number of times the number is multiplied to itself.

If 8 is multiplied by itself for n times, then, it is represented as:

$8 \times 8 \times 8 \times 8 \times \ldots\ldots \text{n times} = 8^n$

The above expression, 8^n, is said as 8 raised to the power n. Therefore, exponents are also called power or sometimes indices.

$7^6 = 7 \times 7 \times 7 \times 7 \times 7 \times 7 = 117649$

$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64.$

On Subtracting we get,

$7^6 - 2^6 = 117649 - 64 = 117585$

By trial and error,

$\frac{117585}{5} = 23517.$

$\frac{117585}{6} = 19597.5.$

$\frac{117585}{7} = 16797.8571.$

$\frac{117585}{8} = 14698.125.$

$\therefore$ $7^6 - 2^6$ is divisible by 5.

LEVEL -II

1. Correct option: (A)

A number that is multiple of 125 must be divisible by 125.

The number formed by the last three digits must be divisible by 125 hence it can be 000, 125, 250, 375, 500, 675,750,875.

The number which is divisible by 125 must be 9999875 since we have to subtract the minimum possible number.

$\Longrightarrow 9999999 - 9999875 = 124.$

Therefore, the least number is 124.

2. Correct option: (A)

First, find the LCM of 31, 36 and 40.

L.C.M of (31, 36, 40) = $2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 31 = 11160.$

Smallest 5 - digit no.=10000

Divide 10000 by 11160.

Remainder = 10000

Required number $= 10000 + (11160 - 10000)$

$= 10000 + 1160$

$= 11160$

Therefore, 11160 is the smallest 5-digit number exactly divisible by 31, 36, 40.

3. Correct option: (C)

On dividing 123123123 by 123 equals 1001001 with a remainder of 0.

Therefore, 123123123 ÷123 = 1001001.

4. Correct option: (D)

Here, $\mathbf{n = 63}$

We know, the sum of the first n natural numbers $= \frac{\mathbf{n(n+1)}}{\mathbf{2}}$

$\Rightarrow$ Sum of first 63 natural numbers $= \frac{\mathbf{63(63+1)}}{\mathbf{2}}$

$\Rightarrow$ Sum of first 63 natural numbers $= \frac{\mathbf{63(64)}}{\mathbf{2}}$

$\Rightarrow$ Sum of first 63 natural numbers $= \frac{\mathbf{4032}}{\mathbf{2}}$

$\therefore$ Sum of first 63 natural numbers = **2016.**

5. Correct option: (B)

L.C.M of a group of numbers is the smallest number that is the multiple of all the numbers.

H.C.F of a group of numbers is the greatest factor of the numbers.

H.C.F of fractions = $\frac{\text{H.C.F of numerators}}{\text{L.C.M of denominators}}$.

H.C.F of (22, 66, 77, 99) = 11 .

L.C.M of (81, 36, 72, 18) = $2 \times 2 \times 2 \times 3 \times 3 \times 9 = 648$

Required H.C.F = $\frac{11}{648}$.

∴ The required H.C.F is $\frac{11}{648}$.

6. Correct option: (B)

i. The prime factorization of 157

$157 = 1 \times 157$

ii. The prime factorization of 225

$225 = 1 \times 3 \times 3 \times 5 \times 5$

iii. To find the HCF, multiply all the prime factors common to both numbers that is 1.

Therefore, HCF (225, 157) = 1.

7. Correct option: (D)

i. Find the prime factorization of 247

$247 = 13 \times 19$

ii. Find the prime factorization of 91

$91 = 7 \times 13$

iii. To find the GCF, multiply all the prime factors common to both numbers that is 13.

Therefore, HCF (247, 91) is 13.

8. Correct option: (D)

Let the required numbers be $3x, 2x$ and x.

Then, their $H.C.F. = x$.

So, x = 12.

- $3x = 3 \times 12 = 36$
- $2x = 2 \times 12 = 24$
- $x = 12$.

∴The numbers are $36, 24$ and 12.

9. Correct option: (C)

List out all of the prime factors for each number:

- The prime factors of 18 are 2, 3 and 3.
- The prime factors of 24 are 2, 2, 2 and 3.
- The prime factors of 36 are 2, 2, 3 and 3.
- The prime factors of 45 are 3, 3 and 5.

The least common multiple is the product of all factors in the greatest number of their occurrence.

Here the common and uncommon prime factors are 2, 2, 3, 3, 2, 5.

On Multiplying 2, 2, 3, 3, 2, 5 to get the LCM of the given numbers.

L.C.M of (18, 24, 36, 45) = $2 \times 2 \times 2 \times 3 \times 3 \times 5 = 360$.

∴ The required L.C.M least common multiple of 18,24,36,45 is 360.

10. Correct option: (C)

Given that the L.C.M. of A and B is B.

So, we have B is a multiple of A.

It is also given that the L.C.M. of B and C is C.

So, we have C is a multiple of B.

Now, we will find the L.C.M. of A, B, and C.

Since B is a multiple of A and C is a multiple of B, then by transitive property, we have C as a multiple of A.

Thus the L.C.M. of A, B and C is C which is the multiple of both A and B.

11. Correct option: (A, B, C)

The numbers 61, 52, 63, 94, 46 and 18. Since these numbers on being reversed given out 16, 25, 36, 49, 64, 81 the two-digit perfect square numbers.

Therefore, 61, 52, 94 are the two-digit numbers which on being reversed gives out perfect square numbers.

12. Correct option: (A, D)

For a number (represented as x^y) to be both a perfect cube as well as perfect square, y needs to be a multiple of 6.

Check for the given options: $1 = 1^2 = 1^3$.

$4 = 2^2 \neq 2^3$.

$16 = 4^2 \neq 3^3$.

$64 = 8^2 = 4^3$.

From the given options, only 1 and 64 satisfies this criterion.

Therefore, the numbers 1 and 64 are perfect square and perfect cube numbers.

13. Correct option: (A, C)

We know that the LCM of a prime number and a composite number is equal to their product.

So, $mn = 4199$

Now, $m + n = 240$

$\Rightarrow n = 240 - m$

Substituting the value of n in $mn = 4199$, we will get $m(240 - m) = 4199$

$\Rightarrow 240m - m^2 = 4199$

$\Rightarrow m^2 - 240m + 4199 = 0$

$\Rightarrow m^2 - 19m - 221m + 4199 = 0$

$\Rightarrow m(m - 19) - 221(m - 19) = 0$

$\Rightarrow (m - 19)(m - 221) = 0$

$\Rightarrow (m - 19) = 0$ or $(m - 221) = 0$

$\Rightarrow m = 19, 221$

$\therefore m = 19$ and $n = 221$ or

$m = 221$ and $n = 19$

Therefore, the values of $m = 19$ and $n = 221$.

14. Correct option: (B)

Given that LCM of 248 & 868 is 1736.

Since The product of two numbers 'a' and 'b' is equal to the product of their L.C.M. and H.C.F.

$a \times b = H.C.F. \times L.C.M.$

$\Rightarrow LCM \times HCF = 248 \times 868.$

$\Rightarrow HCF = \frac{248 \times 868}{L.C.M}$

$\Rightarrow HCF = \frac{248 \times 868}{1736}$.

$\Rightarrow HCF = \frac{248}{2}$

$\Rightarrow HCF = 124.$

Therefore, HCF is 124.

15. Correct option: (D)

Product of two numbers = HCF of two numbers × LCM of two numbers

$\Rightarrow 86480 = 23 \times$ LCM of two numbers

$\Rightarrow$ LCM of two numbers $= \frac{86480}{23} = 3760.$

Therefore, the LCM of two numbers is 3760.

16. Correct option: (B)

Given, HCF=31

LCM=1488.

Let the two numbers be a and b.

Let a=161. We have to find b.

We know that $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$

$\Rightarrow 31 \times 1488 = 24 \times b$

$\Rightarrow \frac{31 \times 1488}{24} = b$

$\Rightarrow b = \frac{46128}{24}$.

$\therefore b = 1922$.

Therefore, the other number is 1922.

17. Correct Matching: a-S, b-P, c-Q, d-R

a) HCF of 222,216 and 606

HCF (222, 216, 606) = 6.

b) HCF of 50,150 and 450

HCF (50, 150, 450) =50.

c) LCM of 21, 10 and 18

LCM (21, 10, 18) = 630

d) LCM of 25,15 and 45

LCM (25, 15, 45) =225

18. Correct Answer: 4

- A prime number would be a whole number greater than one with only the number one and itself as factors.
- Composite numbers are those that have more than two factors.

Each of the numbers 302, 303, 304, 305, 306, 308, 309, 310, 312, 314, 315, 316, 318 and 319 is clearly a composite number.

Out of 307, 311, 313 and 317 clearly everyone is prime.

Hence, there are 4 prime numbers between 301 and 320.

19. Correct Answer: 3

Given numbers are 0.5,0.6,0.7.

The highest number of digits after the decimal point in the given number is 2

Thus, in order to get rid of the decimal point we need to multiply them with 10 they are as follows

- 0.5 x 10 = 5
- 0.6 x 10 = 6
- 0.7 x 10 = 7

On finding the LCM of 5,6, 7.

L.C.M of (5, 6, 7) = $5 \times 6 \times 7 = 210$.

The decimal point in the given number is after 1 place from the rightmost.

Therefore, the LCM (0.5, 0.6, 0.7) =21.0.

Now, $\frac{\text{LCM of } 0.5,0.6,0.7}{7} = = \frac{21}{7}$ =3.

20. Correct Answer: 1

Finding the prime factorization for the numbers are 2001 and 2002.

- $2001 = 1 \times 3 \times 23 \times 29$.
- $2002 = 1 \times 2 \times 7 \times 11 \times 13$.

The common factor for the numbers 2001, 2002 is 1.

Therefore, the HCF (2001, 2002) = 1.

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

The total numbers between 200000 and 600000 (including 200000 and 600000) = 400001.
Of these, 400001 are even numbers and the remaining 400000 are odd numbers.
∴The required odd numbers between 200000 and 600000 including both is 4,00,000.

2. Correct option: (B)

Converting them to like decimals we have 0.90, 0.36 and 1.08.
Treating them as whole numbers the numbers are 90, 36 and 108.
Finding the prime factorization for the numbers 90, 36, 108.
$90 = 2 \times 3^2 \times 5$
$36 = 2^2 \times 3^2$
$108 = 2^2 \times 3^3$,
The common factor for the numbers 90, 36, 108 is $2 \times 3^2 = 2 \times 9 = 18$.
The highest common factor for the numbers 90, 36, 108 is 18.
Therefore, the HCF of 0.90, 0.36 and 1.08 = 0.18. (∴ All of them have two places of decimal).

3. Correct option: (A)

L.C.M of a group of numbers is the smallest number that is the multiple of all the numbers.
H.C.F of a group of numbers is the greatest factor of the numbers.
H.C.F of fractions = $\frac{\text{H.C.F of numerators}}{\text{L.C.M of denominators}}$.
H.C.F of (16, 32, 64) = $2 \times 2 \times 2 \times 2 = 16$
L.C.M of (24, 72, 48) = $2 \times 2 \times 2 \times 2 \times 3 \times 3 = 144$.
Required H.C.F = $\frac{16}{144} = \frac{1}{9}$.
∴ The required H.C.F is $\frac{1}{9}$.

4. Correct option: (C)

L.C.M of a group of numbers is the smallest number that is the multiple of all the numbers.
H.C.F of a group of numbers is the greatest factor of the numbers.
H.C.F of fractions = $\frac{\text{H.C.F of numerators}}{\text{L.C.M of denominators}}$.
H.C.F of (8, 16, 4, 4) = $2 \times 2 = 4$.
L.C.M of (3, 9, 12, 3) = $2 \times 2 \times 3 \times 3 = 36$
Required H.C.F = $\frac{4}{36} = \frac{1}{9}$.
∴ The required H.C.F is $\frac{1}{9}$.

5. Correct option: (C)

L.C.M of a group of numbers is the smallest number that is the multiple of all the numbers.
H.C.F of a group of numbers is the greatest factor of the numbers.
$\text{L.C.M} = \frac{\text{L.C.M of numerators}}{\text{H.C.F of denominators}}$.
L.C.M of (10, 60, 70, 80) = $2 \times 2 \times 2 \times 2 \times 3 \times 5 \times 7 = 1{,}680$
H.C.F of (5, 35, 25, 65) = 5
Required L.C.M = $\frac{1680}{5} = 336$.
∴ The required L.C.M is 336.

6. Correct option: (C)

Given numbers are 0.39,0.3,0.63.
The highest number of digits after the decimal point in the given number is 2
Thus, in order to get rid of the decimal point we need to multiply them with 100 they are as follows
0.39 x 100 = 39
0.3 x 100 = 30

0.63 x 100 = 63
On finding the LCM of 39,30,63.
L.C.M of (39,30,63) = $3 \times 3 \times 7 \times 13 = 8{,}190$.
The decimal point in the given number is after 2 places from the rightmost.
Therefore, the LCM (0.39, 0.3, 0.63) =81.9.

7. Correct option: (B)

Given numbers are 0.21,0.36,0.42.
The highest number of digits after the decimal point in the given case is 2
Thus, in order to get rid of the decimal point we need to multiply them with 100 they are as follows:
0.21 x 100 = 21
0.36 x 100 = 36
0.42 x 100 = 42
On finding the LCM of 21,36,42.
L.C.M of (21, 36, 42) = $2 \times 2 \times 3 \times 3 \times 7 = 252$.
The decimal point in the given number is after 2 places from the rightmost.
Therefore, the LCM (0.21,0.36,0.42) = 2.52.

8. Correct option: (B)

A composite number is a natural number that has more than two factors.
The composite numbers up to 100 are listed below.
4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39, 40, 42, 44, 45, 46, 48, 49, 50, 51, 52, 54, 55, 56, 57, 58, 60, 62, 63, 64, 65, 66, 68, 69, 70, 72, 74, 75, 76, 77, 78, 80, 81, 82, 84, 85, 86, 87, 88, 90, 91, 92, 93, 94, 95, 96, 98, 99, 100.
Therefore, there are 74 numbers between the 1 to 100 composite numbers.

9. Correct option: (B)

The Cyclic table for 4 is as follows:

	Unit digit
$4^1 = 4$	4
$4^2 = 16$	6
$4^3 = 64$	4
$4^4 = 256$	6
$4^5 = 1024$	4

The unit digit pattern is 6,4,6, 4, ...
Hence, the power cycle of 4 contains only 2 numbers 4 & 6, which appear in case of odd and even powers respectively.
$4^{(\text{odd number})} = 4$.
$4^{(\text{even number})} = 6$
Since 1971 is odd number.
$\therefore$ The number in units place 4^{1971} is 4.

10. Correct option: (B)

The Cyclic table for 9 is as follows:

	Unit digit
$9^1 = 9$	9
$9^2 = 81$	1
$9^3 = 729$	9
$9^4 = 6561$	1
$9^5 = 59049$	9

The unit digit pattern is 9.

Hence, the power cycle of 9 contains only 2 numbers 9 & 1, which appear in case of odd and even powers respectively.

$9^{(\text{odd number})} = 9.$

$9^{(\text{even number})} = 1.$

Since 2008 is even number.

∴ The number in units place 9^{2008} is 1.

11. Correct option: (A)

The factors of $2310 = 2 \times 3 \times 5 \times 7 \times 11.$

Therefore, the required factors of 2310 are 2,3,5,7,11.

12. Correct option: (A, C, D)

Let three prime numbers are $X, X + 36$ and Y.

According to questions,

$\Rightarrow X + X + 36 + Y = 100.$

$\Rightarrow 2X + Y = 64 \quad ---------- (1)$

All three are prime number so 2X should be an even number. Also, to get sum as 100 (even number) Y should be an even prime number i.e 2.

Putting $Y = 2$ in equation 1.

$\Rightarrow 2X + 2 = 64$

$\Rightarrow X = 31$

Thus, prime numbers are 31, 67 and 2.

Therefore, the three prime numbers are 31, 67 and 2.

13. Correct option: (B)

Even numbers are those numbers that are completely divisible by 2.

A simple trick to identify even numbers is to check the last digit which should be 0, 2, 4, 6 or 8.

For example: 50, 22, 68, 86, …….

There is a total of 50 even numbers that come between 1 to 100.

∴ From 1 to 100 the number of even numbers is 50.

14. Correct option: (B)

On Substituting the values for $n = 3, 4, 5, 9$ in the expression 2n+1.

- For $n = 3, 2n + 1 = 2(3) + 1 = 6 + 1 = 7.$
- For $n = 4, 2n + 1 = 2(4) + 1 = 8 + 1 = 9.$
- For $n = 5, 2n + 1 = 2(5) + 1 = 10 + 1 = 11.$
- For $n = 9, 2n + 1 = 2(9) + 1 = 18 + 1 = 19.$

Since 9 is not prime when $n = 4$ in the given expression $2n + 1$.

Therefore, the smallest value of n for which $2n + 1$ is not a prime number is 4.

15. Correct Matching: a-T, b-P, c-S, d-Q

a) The unit digit in $55^{725} + 73^{5810} + 22^{853}$.

- Units place digit in 55^{725} is 5 ($\because 5^n$units place is 5)
- Units place digit in73^{5810} is $3^{5810} = 3^{4 \times 1452 + 2} = 3^2 = 9.$ ($\because 3^{4n+x} = 3^x$)
- Units place digit in 22^{853} is $2^{853} = 2^{4 \times 213 + 1} = 2^1$

∴ Units place digit in $(55^{725} + 73^{5810} + 22^{853})$ is $= 5 + 9 + 2 = 16.$

In terms of finding the unit's digit of the expression, $(55^{725} + 73^{5810} + 22^{853})$ is same as $55^1 + 73^2 + 22^1$).

The respective units digit for the three parts of the expression would be: 5+9+2=16→ required answer is 6.

b) The unit digit is $264^{102} + 264^{103}$.

unit digit

$4^1 \rightarrow 4$

$4^2 \rightarrow 16 \rightarrow 6$

$4^3 \rightarrow 64 \rightarrow 4$

Rule:

* When 4 has odd power, then unit digit is: 4
* When 4 has even power, then unit digit is: 6

Since 264^{102} has even power, then unit digit is 6.

264^{103} has odd power, then unit digit is 4.

$\therefore$ unit digit = 6 + 4 = 10 → 0.

c) the unit digit in 259^{59}

First let us find the powers for 259 in order and its unit place.

$259^1 = 259 = 9$ (unit place)

$(259^2 = 67081 = 1$

$(259^3 = 17373979 = 9$

$(259)^4 = 4499860561 = 1$

So the unit place comes in the order 1,9,1,9.....

So it is in a form of cycle with 2 alternatives.

So when n is a multiple of 2 , $(259)^n = 1$

Since 59 is not a multiple of 2 259^{59} the unit place is not equal to 1 and it is equal to 9.

d) Unit digit in $223 \times 329 \times 524 \times 674$ will be same as unit digit $3 \times 9 \times 4 \times 4$.

$3 \times 9 \times 4 \times 4 = 27 \times 16 = 432$.

Since the unit digit of 432 is 2.

Therefore, Unit digit in $223 \times 329 \times 524 \times 674$ is 2.

16. Correct Answer: 8 pairs

A pair of prime numbers whose difference is 2, is called twin prime.

The twin primes between 1 and 100 are; (3, 5), (5, 7), (11, 13), (17, 19), (29, 31), (41, 43), (59, 61), (71, 73).

Therefore, there are 8 pairs of twin prime numbers between 1 to 100.

17. Correct Answer: 0

The unit digit of 10 is 0 because any number that ends in a 0 is divisible by 10.

Now, the first 50 natural numbers.

Group them into pairs of factors that multiply to 10 (e.g., 2 and 5, 4 and 5, 6 and 5, etc.).

There are 5 factors that are equal to 10 (namely 10, 20, 30, 40, and 50). Therefore, the product of the first 50 natural numbers is divisible by 10^5.

Since the unit digit of any number that is divisible by 10^n (where n is a positive integer) is 0, the unit digit of the product of the first 50 natural numbers is 0.

18. Correct Answer: 1

Step 1) Find all the factors of the first 100 natural numbers:

1 = 1

2 = 1, 2

3 = 1, 3

.

.

98 = 1, 2, 7, 14, 49

99 = 1, 3, 11, 33, 99

100 = 1, 2, 4, 5, 8, 10, 20, 25, 50, 100

Step 2) Find the highest common factor (HCF) of all the factors above which is 1.

Therefore, the HCF of the first 100 natural numbers is 1.

19. Correct Answer: 5

Step 1) Find all the prime factors of the given numbers:

125 = 5×5 ×5

75 = 3×5×5

5 = 5

Step 2) Find the prime factors that are common between all the numbers: Here, the only common prime factor between all 3 numbers is 5

Step 3) Since 5 is the only common prime factor, the HCF of 125, 75 and 5 is 5

Therefore, the HCF of 125, 75 and 5 is 5.

20. Correct Answer: 6

L.C.M of a group of numbers is the smallest number that is the multiple of all the numbers.

H.C.F of a group of numbers is the greatest factor of the numbers.

$L.C.M = \frac{L.C.M\ of\ numerators}{H.C.F\ of\ denominators}$.

L.C.M of (4,4,2,36) = $2 \times 2 \times 3 \times 3 = 36$

H.C.F of (3, 9, 15, 21) = 3

Required L.C.M = $\frac{36}{3} = 12$.

∴ The required L.C.M is 12.

Now, $\frac{LCM\ of\ \frac{4}{3},\frac{4}{9},\frac{2}{15}\ and\ \frac{36}{21}}{2} = \frac{12}{2} = 6$.

Therefore, $\frac{LCM\ of\ \frac{4}{3},\frac{4}{9},\frac{2}{15}\ and\ \frac{36}{21}}{2} = 6$.

Rational Numbers

LEVEL – I

1. Correct option: (A)

The standard form of a fraction involves simplifying the numerator and denominator to their lowest terms. To simplify $-\frac{102}{202}$, we can find the greatest common divisor (GCD) of the numerator and denominator, and then divide both by that GCD to reduce the fraction.

GCD of 102 and 202 is 2.

Dividing both the numerator and denominator by 2:

$$-102 \div 2 = -51$$
$$202 \div 2 = 101$$

So, the simplified standard form of $-\frac{102}{202}$ is $-\frac{51}{101}$.

Then the correct option is A)

2. Correct option: (B)

To determine the value of 'a' for which the rational numbers $\frac{2}{8}$ and $\frac{a}{48}$ are equivalent, we need to find a value of 'a' that makes the fractions equal when simplified to their lowest terms.

Simplify the fractions:

$$\frac{2}{8} = \frac{1}{4}\ (by\ dividing\ both\ numerator\ and\ denominator\ by\ 2)$$
$$\frac{a}{48} = \frac{a}{48}\ (no\ simplification\ possible)$$

For the fractions to be equivalent, they need to be equal after simplification:

$$\frac{1}{4} = \frac{a}{48}$$
$$1 \times 48 = a \times 4$$
$$48 = 4a$$

$$a = \frac{48}{4}$$
$$a = 12$$

So, the value of 'a' that makes the two rational numbers equivalent is 12.
Then the correct option is B)

3. Correct option: (C)

The property illustrated by the given statement is the Associative Property.
The Associative Property states that for any three numbers a, b, and c, the way we group them when adding or multiplying does not affect the result.
In this case, the statement shows that we can group the fractions differently within the addition operation, but the final result remains the same.
Then the correct option is C)

4. Correct option: (A)

Let the number to be subtracted from $\frac{25}{13}$ be x.
The equation can be set up as follows: $\frac{25}{13} - x = -\frac{3}{5}$
To solve for x, isolate it on one side of the equation: $x = \frac{25}{13} + \frac{3}{5}$

$$x = \frac{25 \times 5 + 3 \times 13}{13 \times 5}$$
$$x = \frac{125 + 39}{65}$$
$$x = \frac{164}{65}$$

Then the correct option is A)

5. Correct option: (D)

To determine the value of 'b' for which the rational numbers $\frac{1}{4}$ and $\frac{b}{8}$ are equivalent, we need to find a value of 'b' that makes the fractions equal when simplified to their lowest terms.
Simplify the fractions: $\frac{1}{4} = \frac{1}{4}$ $(no\ further\ simplification)$

$$\frac{b}{8} = \frac{b}{8}\ (no\ simplification\ possible)$$

For the fractions to be equivalent, they need to be equal as they are: $\frac{1}{4} = \frac{b}{8}$

$$1 \times 8 = b \times 4$$
$$8 = 4b$$
$$b = \frac{8}{4}$$
$$b = 2$$

So, the value of b that makes the two rational numbers equivalent is 2.
Then the correct option is D)

6. Correct option: (A)

Let's substitute the given values of a and b into the expression and solve step by step:
Given: $a = 2, b = 3$

$$\left(\frac{a+b}{b}\right)^2 = \left(\frac{2+3}{3}\right)^2 = \left(\frac{5}{3}\right)^2$$

Now, square the fraction:

$$\left(\frac{5}{3}\right)^2 = \left(\frac{5}{3}\right) \times \left(\frac{5}{3}\right) = \frac{25}{9}$$

So, the value of $\left(\frac{a+b}{b}\right)^2$ when $a = 2$ and $b = 3$ is $\frac{25}{9}$.
Then the correct option is A)

7. Correct option: (A)

Let's simplify the given expression step by step:

Given expression: $(-\frac{a}{b} \times \frac{c}{a}) - (1 \times \frac{1}{c}) + (\frac{1}{a} \times \frac{1}{b})$

Step 1: Simplify each term within parentheses.

$$(-\frac{a}{b} \times \frac{c}{a}) = -\frac{c}{b}$$

$$\left(1 \times \frac{1}{c}\right) = \frac{1}{c}$$

$$\left(\frac{1}{a} \times \frac{1}{b}\right) = \frac{1}{ab}$$

Step 2: Combine the terms.

$$-\frac{c}{b} - \frac{1}{c} + \frac{1}{ab}$$

Step 3: Find a common denominator for the fractions.

The common denominator for $a, b, c \Rightarrow (abc)$

Step 4: Rewrite each fraction with the common denominator.

$$-\frac{abc \times c}{abc \times b} - \frac{abc \times 1}{abc \times c} + \frac{abc}{ab \times abc}$$

Step 5: Simplify each fraction and combine the fraction with same denominator.

$$-\frac{ac \times c}{abc} - \frac{ab}{abc} + \frac{c}{abc} \Rightarrow \frac{-ac^2 - ab + c}{abc}$$

Step 6: Common out the common multiplier $\frac{-a(c^2+b)+c}{abc}$

So,

$$\left(-\frac{a}{b} \times \frac{c}{a}\right) - \left(1 \times \frac{1}{c}\right) + \left(\frac{1}{a} \times \frac{1}{b}\right) \Rightarrow \frac{-a(c^2 + b) + c}{abc}$$

Then the correct option is A)

8. Correct option: (A)

The additive inverse of a number "a" is the number that, when added to "a," gives zero. The additive inverse of 2 is -2.

The multiplicative inverse (reciprocal) of a number "b" is the number that, when multiplied by "b," gives 1. The multiplicative inverse of 3 is $\frac{1}{3}$.

So, the question is asking for the sum of the additive inverse of 2 =>(-2) and the multiplicative inverse of 3 $(\frac{1}{3})$

$$Sum = (-2) + \left(\frac{1}{3}\right) = -\frac{6}{3} + \frac{1}{3} = -\frac{5}{3}$$

Then the correct option is A)

9. Correct option: (B)

To determine the value of $'x'$ for which the rational numbers $\frac{10}{9}$ and $\frac{x}{3}$ are equivalent, we need to find a value of $'x'$ that makes the fractions equal when simplified to their lowest terms.

Simplify the fractions:

$$\frac{10}{9} = \frac{10}{9} \ (no\ further\ simplification)$$

$$\frac{x}{3} = \frac{x}{3} \ (no\ simplification\ possible)$$

For the fractions to be equivalent, they need to be equal as they are:

$$\frac{10}{9} = \frac{x}{3}$$

Cross multiplication

$$10 \times 3 = x \times 9$$

$$30 = 9x$$
$$x = \frac{30}{9}$$
$$x = \frac{10}{3}$$

So, the value of $'x'$ that makes the two rational numbers equivalent is $\frac{10}{3}$.

Then the correct option is B)

10. Correct option: (A)

Let's substitute the value of x into the expression and solve:

Given: $x = 5$

Expression: $-\frac{x+1}{x-1}$

Substitute $x = 5$:

$$= -\left(\frac{5+1}{5-1}\right)$$
$$= -\left(\frac{6}{4}\right)$$
$$= -\frac{3}{2}$$

So, $$=> x = 5, -\frac{x+1}{x-1} => -\frac{3}{2}$$

Then the correct option is A)

11. Correct option: (A)

The multiplicative inverse (also known as the reciprocal) of a number "a" is the number that, when multiplied by "$a,$" gives 1.

For the number 2, its multiplicative inverse is $= \frac{1}{2}$

Then the correct option is A)

12. Correct option: (D)

The additive inverse of a number "a" is the number that, when added to "$a,$" gives zero. The additive inverse of 2 is -2.

The multiplicative inverse (reciprocal) of a number "b" is the number that, when multiplied by "$b,$" gives 1. The multiplicative inverse of 2 is $\frac{1}{2}$.

So, the expression we have is:

$$=> \left(\frac{(-2)}{\left(\frac{1}{2}\right)}\right)$$

To divide by a fraction, we can multiply by its reciprocal.

The reciprocal of $\frac{1}{2}$ is $\frac{2}{1}$.

$$\left(\frac{(-2)}{\left(\frac{1}{2}\right)}\right) = (-2) \times \left(\frac{2}{1}\right) = -4$$

Then the correct option is D)

13. Correct option: (A)

Given: $a = 1, b = 0$

Expression:

$$\frac{a^2 + 3ab + b^2}{a + b}$$

Substitute the values of a and b:

$$= \frac{1^2 + 3(1)(0) + 0^2}{1 + 0}$$

$$= \frac{1 + 0 + 0}{1}$$
$$= 1$$
$$=> a = 1, b = 0, => \frac{a^2 + 3ab + b^2}{a + b} = 1$$

Then the correct option is A)

14. Correct option: (A)

Let's break down the problem step by step:

Given: $a = \frac{2}{3}, b = \frac{3}{2}$

Expression: $(a + b) \times (a - b)$

Step 1: Find the sum and difference of a and b.

$$Sum\ of\ a\ and\ b: a + b = (\frac{2}{3}) + (\frac{3}{2})$$
$$Difference\ of\ a\ and\ b: a - b = (\frac{2}{3}) - (\frac{3}{2})$$

Step 2: Simplify the sum and difference.

$$Sum: \left(\frac{2}{3}\right) + \left(\frac{3}{2}\right) = \left(\frac{4}{6}\right) + \left(\frac{9}{6}\right) = \frac{13}{6}$$
$$Difference: \left(\frac{2}{3}\right) - \left(\frac{3}{2}\right) = \left(\frac{4}{6}\right) - \left(\frac{9}{6}\right) = -\frac{5}{6}$$

Step 3: Multiply the sum by the difference.

$$\left(\frac{13}{6}\right) \times \left(-\frac{5}{6}\right) = \left(-\frac{65}{36}\right)$$

So, the result of multiplying the sum of $\frac{2}{3}$ and $\frac{3}{2}$ by their difference is $-\frac{65}{36}$.

Then the correct option is A)

15. Correct option: (C)

The additive inverse of a fraction $\frac{a}{b}$ is the fraction that, when added to $\frac{a}{b}$, gives zero.

For the fraction $\frac{a}{b}$, its additive inverse is:

$$-\left(\frac{a}{b}\right)$$

Then the correct option is C)

16. Correct option: (A)

Let's denote the unknown rational number as "x".We are given that the product of the two rational numbers is 20, and one of the numbers is 12. So, we can set up an equation:

$$12 \times x = 20$$

To solve for x, divide both sides of the equation by 12:

$$x = \frac{20}{12}$$
$$x = \frac{5}{3}$$

Then the correct option is A)

17. Correct option: (C)

A) The product of two positive numbers is always positive, not negative. When we multiply two positive numbers, the result is positive.

B) Zero does not have a reciprocal. The reciprocal of a number "a" is $\frac{1}{a}$. Since division by zero is undefined, the reciprocal of zero does not exist.

C) The reciprocal of a positive number is positive. If we take the reciprocal of a positive number, the result will also be positive. For example, the reciprocal of 2 is $\frac{1}{2}$, which is positive.

D) The product of two negative rational numbers is indeed negative. When we multiply two negative numbers, the result is positive.
So, the only statement that is true is C) The reciprocal of a positive number is positive.
Then the correct option is C)

18. Correct option: (A)

Let's analyze each option and determine whether it represents a terminating decimal number or not.

A) $\frac{1}{5} = 0.2$

B) $\frac{1}{7} \approx 0.142857\ldots$ (repeating decimal, not terminating)

C) $\frac{1}{3} = 0.333\ldots$ (repeating decimal, not terminating)

D) $\frac{1}{21} = 0.047619\ldots$ (repeating decimal, not terminating)

Explanation:

A) $\frac{1}{5}$ is equivalent to 0.2, which is a terminating decimal. It has a finite number of digits after the decimal point.

B) $\frac{1}{7}$ is approximately 0.142857... This is a repeating decimal, where the sequence 142857 repeats infinitely.

C) $\frac{1}{3}$ is equivalent to 0.333..., where the digit 3 repeats infinitely.

D) $\frac{1}{21}$ is approximately 0.047619..., with the repeating sequence 047619.

Among the options, only option A) $\frac{1}{5}$ is a terminating decimal. It has a finite number of digits after the decimal point, and it does not repeat.

We can use another method of checking prime factorization of denominator.

The power of 2 and 5 rule is a method used to determine whether a fraction can be simplified to a terminating decimal or a repeating decimal. It involves checking the prime factorization of the denominator of the fraction.

Here's how the rule works:

A fraction can be expressed as a terminating decimal if its denominator, when expressed in prime factorization, consists only of powers of 2 and/or powers of 5.

Only option A) satisfies the rule.

Then the correct option is A)

19. Correct option: (B)

The additive inverse of a fraction $\frac{a}{b}$ is the fraction that, when added to $\frac{a}{b}$ gives zero.

The additive inverse of $\frac{11}{14}$ is the fraction that, when added to $\frac{11}{14}$, results in 0. Mathematically, it can be represented as:

$$\frac{11}{14} + \left(-\frac{11}{14}\right) = 0$$

So, the additive inverse of $\frac{11}{14}$ is $-\frac{11}{14}$.

Then the correct option is B)

20. Correct option: (A)

Let's denote the unknown rational number as "x". We are given that the sum of two rational numbers is $-\frac{4}{3}$, and one of the numbers is $-\frac{15}{7}$. So, we can set up an equation:

$$\left(-\frac{15}{7}\right) + x = -\frac{4}{3}$$

To solve for x, subtract $(-\frac{15}{7})$ from both sides of the equation:

$$x = -\frac{4}{3} + \frac{15}{7}$$

To add these fractions, we need a common denominator, which is the least common multiple (LCM) of 3 and 7, which is 21:

$$x = \left(-\frac{28}{21}\right) + \left(\frac{45}{21}\right)$$
$$x = \frac{45 - 28}{21}$$
$$x = \frac{17}{21}$$

So, the unknown rational number is $\frac{17}{21}$.
Then the correct option is A)

LEVEL – II

1. Correct option: (B)
Let's denote the unknown rational number as "x". We are given that subtracting this unknown number from $\frac{3}{7}$ should result in $\frac{5}{4}$. So, we can set up an equation:

$$\frac{3}{7} - x = \frac{5}{4}$$

To solve for x, subtract $\frac{3}{7}$ from both sides of the equation: $-x = \frac{5}{4} - \frac{3}{7}$
To subtract these fractions, we need a common denominator, which is the least common multiple (LCM) of 4 and 7, which is 28:

$$-x = \left(\frac{35}{28}\right) - \left(\frac{12}{28}\right)$$
$$-x = \frac{35 - 12}{28}$$
$$-x = \frac{23}{28}$$

Now, multiply both sides of the equation by -1 to solve for x: $x = -\frac{23}{28}$
So, the unknown rational number is $-\frac{23}{28}$.
Then the correct option is B)

2. Correct option: (C)
The multiplicative inverse of a number"a" is the number that, when multiplied by"a," gives1.
First, let's calculate the result of the multiplication:

$$\left(-\frac{4}{3}\right) \times \left(\frac{5}{2}\right) = \frac{(-4 \times 5)}{3 \times 2} = -\frac{20}{6} = -\frac{10}{3}$$

Now, to find the multiplicative inverse of $-\frac{10}{3}$, we need to find the number that, when multiplied by $-\frac{10}{3}$, gives 1. This number is the reciprocal of -$\frac{10}{3}$:
Reciprocal of $-\frac{10}{3} = -\frac{3}{10}$
So, the multiplicative inverse of $\left(-\frac{4}{3} \times \frac{5}{2}\right)$ is $-\frac{3}{10}$.
Then the correct option is C)

3. Correct option: (B)
To simplify the fraction $\frac{70}{-84}$, we can find the greatest common divisor (GCD) of the numerator and the denominator and then divide both by the GCD to get the simplest form.

$$GCD(70, -84) = 14$$

Dividing both the numerator and denominator by the GCD:

$$70 \div 14 = 5$$
$$-84 \div 14 = -6$$

So, the simplest form of $\frac{70}{-84}$ is $\frac{5}{-6}$, which is the same as $-\frac{5}{6}$.
Then the correct option is B)

4. Correct option: (A)

Let's simplify the expression step by step:

$$\left(-5 \times \frac{2}{15}\right) - \left(-6 \times \frac{2}{9}\right)$$

First, let's deal with the negative signs:

$$-\left(-6 \times \frac{2}{9}\right) = \frac{12}{9} = \frac{4}{3}$$

So, the expression becomes:

$$\left(-5 \times \frac{2}{15}\right) + \left(\frac{4}{3}\right)$$

Now, we need a common denominator to add these fractions. The least common multiple (LCM) of 15 and 3 is 15:

$$\left(-\frac{10}{15}\right) + \left(\frac{20}{15}\right)$$

Now, add the fractions:

$$\frac{-10 + 20}{15} = \frac{10}{15} = \frac{2}{3}$$

So, the simplified result of the expression is $\frac{2}{3}$.

Then the correct option is A)

5. Correct option: (D)

To solve for x, we'll isolate x on one side of the equation:

$$\frac{x}{2} + \frac{1}{3} = 1$$

First, subtract $\frac{1}{3}$ from both sides:

$$\frac{x}{2} = 1 - \frac{1}{3}$$

$$\frac{x}{2} = \frac{2}{3}$$

Now, multiply both sides by 2 to solve for x:

$$x = \left(\frac{2}{3}\right) \times 2$$

$$x = \frac{4}{3}$$

So, the value of x is $\frac{4}{3}$.

Then the correct option is D)

6. Correct option: (D)

Let's break down the steps to solve this expression:

1. Find the sum of $-\frac{13}{5}$ and $\frac{12}{7}$:

$$\left(-\frac{13}{5}\right) + \left(\frac{12}{7}\right) = \left(-\frac{91}{35}\right) + \left(\frac{60}{35}\right) = \frac{-91 + 60}{35} = -\frac{31}{35}$$

2. Find the product of $-\frac{31}{7}$ and $-\frac{1}{2}$:

$$\left(-\frac{31}{7}\right) \times \left(-\frac{1}{2}\right) = \left(\frac{31}{14}\right)$$

3. Divide the sum by the product:

$$\left(-\frac{31}{35}\right) \div \left(\frac{31}{14}\right) = \left(-\frac{31}{35}\right) \times \left(\frac{14}{31}\right) = -\frac{2}{5}$$

So, the result of dividing the sum of $-\frac{13}{5}$ and $\frac{12}{7}$ by the product of $-\frac{31}{7}$ and $-\frac{1}{2}$ is $-\frac{2}{5}$.

Then the correct option is D)

7. Correct option: (C)

Let's go through each option:

A) Terminating decimal: A terminating decimal is a decimal representation of a rational number that ends after a finite number of digits. For example, the fraction $\frac{1}{4}$ can be expressed as the terminating decimal 0.25. Not all rational numbers are terminating decimals. For instance, the fraction $\frac{1}{3}$ cannot be expressed as a terminating decimal; it becomes a repeating decimal (0.333...).

B) Repeating decimal: A repeating decimal is a decimal representation of a rational number that has a repeating pattern of digits after a certain point. For example, the fraction $\frac{1}{3}$ is a repeating decimal (0.333...), and so is the fraction $\frac{2}{7}$ (0.285714...). However, not all rational numbers are repeating decimals. Rational numbers that can be expressed as terminating decimals are not repeating decimals.

C) Either terminating decimal or repeating decimal: This option is correct. Every rational number can be expressed as either a terminating decimal or a repeating decimal. It depends on the factors of the denominator. If the denominator has only 2s and/or 5s as prime factors, the decimal representation will be terminating. If the denominator has prime factors other than 2 and 5, the decimal representation will be repeating.

D) Neither terminating decimal nor repeating: This option is incorrect. Every rational number falls into one of the categories: terminating decimal or repeating decimal. There are no rational numbers that do not fit into these categories.

In summary, option C is correct because it correctly states that every rational number can be expressed as either a terminating decimal or a repeating decimal.

Then the correct option is C)

8. Correct option: (B)

To find the other rational number, we can set up the following equation:

$$\left(\frac{14}{27}\right) \times x = -\frac{28}{81}$$

Now, solve for x by dividing both sides by $\left(\frac{14}{27}\right)$:

$$x = \frac{\left(-\frac{28}{81}\right)}{\frac{14}{27}}$$

Dividing fractions is the same as multiplying by the reciprocal of the divisor:

$$x = \left(-\frac{28}{81}\right) \times \left(\frac{27}{14}\right)$$

Now, calculate the product:

$$x = -\frac{756}{1134}$$

To simplify this fraction, divide both the numerator and denominator by their greatest common divisor, which is 126:

$$x = \frac{\left(-\frac{756}{126}\right)}{\left(\frac{1134}{126}\right)}$$

$$x = -\frac{6}{9}$$

$$x = -\frac{2}{3}$$

So, the other rational number is $-\frac{2}{3}$.

Then the correct option is B)

9. Correct option: (B)

To convert the decimal 3.59 into $\frac{p}{q}$ form, we can write it as a fraction with the appropriate denominator.

The given decimal is 3.59. Since there are two digits after the decimal point, we'll use 100 as the denominator to convert it into fraction form:

$$3.59 = \frac{359}{100}$$

So, the decimal 3.59 can be expressed as the fraction $\frac{359}{100}$.

Then the correct option is B)

10. Correct option: (A)

To find the cost of cloth per meter, we need to divide the total cost by the total length of cloth.

Given: Cost of $2\frac{1}{3}$ m of cloth = Rs. $75\frac{1}{4}$

First, let's convert the mixed fraction to an improper fraction for easier calculations:

$$2\frac{1}{3} = \frac{2 \times 3 + 1}{3} = \frac{7}{3}$$

$$75\frac{1}{4} = \frac{75 \times 4 + 1}{4} = \frac{301}{4}$$

Now, let's find the cost per meter:

$$Cost\ per\ meter = \frac{Total\ cost}{Total\ length\ of\ cloth}$$

$$Cost\ per\ meter = \frac{\frac{301}{4}}{\frac{7}{3}}$$

To divide by a fraction, we multiply by its reciprocal:

$$Cost\ per\ meter = \left(\frac{301}{4}\right) \times \left(\frac{3}{7}\right)$$

Now, multiply the numerators and denominators:

$$Cost\ per\ meter = \frac{301 \times 3}{4 \times 7} = \frac{903}{28} = \frac{129 \times 7}{4 \times 7} = \frac{129}{4}$$

So, the cost of cloth per meter is $\frac{129}{4}$.

Then the correct option is A)

11. Correct option: (A)

Let's break down the steps to solve this expression:

1. Find the sum of $\frac{65}{12}$ and $\frac{12}{7}$:

$$\left(\frac{65}{12}\right) + \left(\frac{12}{7}\right)$$

To add these fractions, we need a common denominator, which is the least common multiple of 12 and 7, which is 84.

Rewrite the fractions with the common denominator:

$$\left(\frac{65}{12}\right) + \left(\frac{12}{7}\right) = \frac{65 \times 7}{12 \times 7} + \frac{12 \times 12}{7 \times 12} = \frac{455}{84} + \frac{144}{84} = \frac{599}{84}$$

2. Find the difference between $\frac{65}{12}$ and $\frac{12}{7}$:

$$\left(\frac{65}{12}\right) - \left(\frac{12}{7}\right)$$

Again, use the common denominator of 84:

$$\left(\frac{65}{12}\right) - \left(\frac{12}{7}\right) = \frac{65 \times 7}{12 \times 7} - \frac{12 \times 12}{7 \times 12} = \frac{455}{84} - \frac{144}{84} = \frac{311}{84}$$

3. Divide the sum by the difference:

$$\left(\frac{599}{84}\right) \div \left(\frac{311}{84}\right)$$

To divide fractions, multiply by the reciprocal of the divisor:

$$\left(\frac{599}{84}\right) \times \left(\frac{84}{311}\right)$$

Now, multiply the fractions:

$$\frac{599 \times 84}{84 \times 311} = \frac{599}{311}$$

So, the result of dividing the sum of $\frac{65}{12}$ and $\frac{12}{7}$ by their difference is $\frac{599}{311}$.

Then the correct option is A)

12. Correct option: (A)

Let's simplify the given expression step by step:

$$6 \times \left(\frac{12}{60}\right) \times \left(\frac{24}{120}\right)$$

First, simplify the fractions:

$$\frac{12}{60} = \frac{1}{5}$$
$$\frac{24}{120} = \frac{1}{5}$$

Now, substitute these simplified fractions back into the expression:

$$6 \times \left(\frac{1}{5}\right) \times \left(\frac{1}{5}\right)$$

Now, multiply the fractions and the whole number:

$$6 \times \frac{1}{5} \times \frac{1}{5} = \frac{6}{25}$$

So, the value of the expression is $\frac{6}{25}$.

Then the correct option is A)

13. Correct option: (B) (see question 8)

To find the number by which $-\frac{33}{16}$ should be divided to get $-\frac{11}{4}$, we need to perform the division:

Number needed = $\left(-\frac{33}{16}\right) \div \left(-\frac{11}{4}\right)$

When dividing by a fraction, it's the same as multiplying by its reciprocal:

Number needed = $\left(-\frac{33}{16}\right) \times \left(-\frac{4}{11}\right)$

Now, multiply the numerators and denominators:

Number needed = $\frac{33 \times 4}{16 \times 11} = \frac{132}{176}$

To simplify this fraction, divide both the numerator and denominator by their greatest common divisor, which is 44:

Number needed = $\frac{\frac{132}{44}}{\frac{176}{44}} = \frac{3}{4}$

So, by dividing $-\frac{33}{16}$ by $\frac{3}{4}$, we get $-\frac{11}{4}$.

Then the correct option is B)

14. Correct option: (A) (see question 15)

Let's break down the steps to solve this expression:

1. Find the sum of $\frac{65}{12}$ and $\frac{12}{7}$:

$$\left(\frac{65}{12}\right) + \left(\frac{12}{7}\right)$$

To add these fractions, we need a common denominator, which is the least common multiple of 12 and 7, which is 84.

Rewrite the fractions with the common denominator:

$$\left(\frac{65}{12}\right) + \left(\frac{12}{7}\right) = \frac{65 \times 7}{12 \times 7} + \frac{12 \times 12}{7 \times 12} = \frac{455}{84} + \frac{144}{84}$$

Now, add the fractions:

$$\frac{455 + 144}{84} = \frac{599}{84}$$

2. Find the difference between $\frac{65}{12}$ and $\frac{12}{7}$:

$$\left(\frac{65}{12}\right) - \left(\frac{12}{7}\right)$$

Again, use the common denominator of 84:

$$\left(\frac{65}{12}\right) - \left(\frac{12}{7}\right) = \frac{65 \times 7}{12 \times 7} - \frac{12 \times 12}{7 \times 12} = \frac{455}{84} - \frac{144}{84} = \frac{311}{84}$$

3. Divide the sum by the difference:

$$\left(\frac{599}{84}\right) \div \left(\frac{311}{84}\right)$$

To divide fractions, multiply by the reciprocal of the divisor:

$$\left(\frac{599}{84}\right) \times \left(\frac{84}{311}\right)$$

Now, multiply the fractions:

$$\frac{599 \times 84}{84 \times 311} = \frac{599}{311}$$

Then the correct option is A)

15. Correct option: (B)

To express $-\frac{5}{6}$ as a rational number with the numerator as -15, we need to find the corresponding denominator that results in a fraction equal to $-\frac{15}{d}$.

Let's solve for the denominator:

$$-\frac{5}{6} = -\frac{15}{d}$$

Cross-multiply to solve for d:

$$6 \times (-15) = -5 \times d$$
$$-90 = -5d$$
$$d = 18$$

So, the equivalent fraction of $-\frac{5}{6}$ with a numerator of -15 is $-\frac{15}{18}$.

Then the correct option is B)

16. Correct option: (C)

The notation $0.\overline{4}$ indicates a repeating decimal, which means that the digit 4 repeats infinitely after the decimal point. To convert this repeating decimal to a fraction, we'll set up an equation.

Let $x = 0.\overline{4}$

Then, we can subtract x from $10x$ (since $0.\overline{4}$ is equivalent to $4.\overline{4}$) to eliminate the repeating decimal:

$$10x - x = 4.\overline{4} - 0.\overline{4}$$
$$9x = 4$$

Now, solve for x:

$$x = \frac{4}{9}$$

So, the repeating decimal $0.\overline{4}$ is equivalent to the fraction $\frac{4}{9}$.

Then the correct option is C)

17. Correct option: (A)

To add repeating decimals, we can use the concept of infinite geometric series. Let's first consider the repeating decimals as fractions and then add them.

$0.\overline{23}$ can be represented as the fraction x:

$$x = 0.\overline{23}$$
$$100x = 23.\overline{23}$$

Subtract the equations:

$$100x - x = 23.\overline{23} - 0.\overline{23}$$
$$99x = 23 => x = \frac{23}{9}$$

Similarly, $0.\overline{22}$ can be represented as the fraction y:

$$y = 0.\overline{22}$$
$$100y = 22.\overline{22}$$

Subtract the equations:

$$100y - y = 22.\overline{22} - 0.\overline{22}$$
$$99y = 22 => y = \frac{22}{9}$$

Now, add the fractions x and y:

$$x + y = \frac{23}{99} + \frac{22}{99}$$
$$x + y = \frac{23 + 22}{99}$$
$$x + y = \frac{45}{99}$$

Since 45 and 99 have a common factor of 9, we can simplify the fraction further:

$$x + y = \frac{5}{11}$$

So, the sum of $0.\overline{23}$ and $0.\overline{22}$ is $\frac{5}{11}$.

$$\frac{5}{11} => 0.454545 \ldots => 0.\overline{45}$$

Then the correct option is A)

18. Correct option: (A)

To solve this problem using repeating decimals, let's convert the repeating decimals to fractions first.

1. Convert $4.\overline{72}$ to a fraction:

Let $x = 4.\overline{72}$

Then $100x = 472.\overline{72}$

Subtracting the two equations:

$$100x - x = 472.\overline{72} - 4.\overline{72}$$
$$99x = 468$$
$$x = \frac{468}{99} = \frac{52}{11}$$

2. Convert $3.\overline{27}$ to a fraction:

Let $y = 3.\overline{27}$

Then $100y = 327.\overline{27}$

Subtracting the two equations: $100y - y = 327.\overline{27} - 3.\overline{27}$

$$99y = 324$$
$$y = \frac{324}{99} = \frac{36}{11}$$

Now, let's add the two fractions:

$$x + y = \frac{52}{11} + \frac{36}{11} = \frac{88}{11} = 8 \approx 7.\overline{99}$$

Or we can find the value of $4.\overline{72} + 3.\overline{27}$ directly$7.\overline{99}$.

Then the correct option is A)

19. Correct option: (D)

To convert the repeating decimal $0.\overline{67}$ to a fraction in terms of $\frac{p}{q}$, we can follow a similar approach as before:

Let $x = 0.\overline{67}$

Multiplying both sides of the equation by 100 to shift the decimal point two places to the right:

$$100x = 67.\overline{67}$$

Subtracting the original equation from the second equation to eliminate the repeating part:

$$100x - x = 67.\overline{67} - 0.\overline{67}$$

$$99x = 67 => x = \frac{67}{99}$$

So, the value of $0.\overline{67}$ as a fraction is $\frac{67}{99}$.

Then the correct option is D)

20. Correct option: (A)

To find the value of $0.\overline{89} \div 100$, we can follow these steps:

Let $x = 0.\overline{89}$

Write x in repeating number term as

$$x = 0.89898989\ldots$$

$$\frac{x}{100} = \frac{0.89898989\ldots}{100} = 0.008989898989\ldots = 0.00\overline{89}$$

Or we can use our old method decimal to fraction and fraction to back decimal conversion.

Then the correct option is A).

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

To find the number that needs to be added to $\frac{9}{5}$ to get $\frac{13}{5}$, we can set up an equation:

$$\frac{9}{5} + x = \frac{13}{5}$$

Here, x represents the number we're trying to find. To solve for x, isolate it on one side of the equation:

$$x = \frac{13}{5} - \frac{9}{5}$$

$$x = \frac{4}{5}$$

So, the number that needs to be added to $\frac{9}{5}$ to get $\frac{13}{5}$ is $\frac{4}{5}$.

Then the correct option is B)

2. Correct option: (D)

Let's compare the given fractions with their decimal equivalents: $\frac{1}{5} = 0.2$

$$\frac{1}{4} = 0.25$$

Now, let's evaluate the options: $A) \frac{12}{50} = 0.24$

$B) 0.23$

$C) 0.21$

Comparing the decimal equivalents:

$$0.24\ is\ between\ 0.2\ and\ 0.25$$

$$0.23\ is\ between\ 0.2\ and\ 0.25.$$
$$0.21\ is\ between\ 0.2\ and\ 0.25.$$

All of the provided values (A, B, and C) are lies between $\frac{1}{5}\ and \frac{1}{4}$

Then the correct option is D)

3. Correct option: (C)

To determine which of the given options is not lying between 0 and 1, let's compare their decimal representations:

A) $\frac{1}{3} = 0.33$

B) 0.67

C) 2

D) $\frac{1}{4} = 0.25$

Among these options, 2 is not lying between 0 and 1.

Then the correct option is C)

4. Correct option: (D)

Let's simplify the given expression step by step:

Expression: $\frac{4}{9} + \frac{5}{3} - \frac{7}{2} + \frac{2}{9} - \frac{1}{3} + \frac{9}{2}$

Step 1: Find a common denominator for all the fractions. The common denominator for 9, 3, and 2 is 18.

Step 2: Rewrite each fraction with the common denominator.

$$\left(\frac{4}{9}\right) + \left(\frac{5}{3}\right) - \left(\frac{7}{2}\right) + \left(\frac{2}{9}\right) - \left(\frac{1}{3}\right) + \left(\frac{9}{2}\right)$$
$$= \left(\frac{8}{18}\right) + \left(\frac{30}{18}\right) - \left(\frac{63}{18}\right) + \left(\frac{4}{18}\right) - \left(\frac{6}{18}\right) + \left(\frac{81}{18}\right)$$

Step 3: Add and subtract the fractions.

$$= \left(\frac{38}{18}\right) - \left(\frac{63}{18}\right) + \left(\frac{4}{18}\right) - \left(\frac{6}{18}\right) + \left(\frac{81}{18}\right)$$
$$= \frac{38 - 63 + 4 - 6 + 81}{18}$$
$$= \frac{123 - 69}{81}$$

Step 4: Simplify the numerator.

$$= \frac{54}{18} = 3$$

Then the correct option is D)

5. Correct option: (A)

Let's compare the given options with their decimal equivalents:

A) $\frac{1}{3} = 0.333\ldots$

B) $\frac{10}{11} = 0.909..$

C) $\frac{5}{18} = 0.277..$

Comparing the decimal equivalents:

$$-\ 0.333\ldots\ is\ between\ 0.3\ and\ 0.4.$$
$$-\ 0.909\ldots\ is\ not\ between\ 0.3\ and\ 0.4.$$
$$-\ 0.277\ldots\ is\ not\ between\ 0.3\ and\ 0.4.$$

Out of the given options, only option A) $\frac{1}{3}$ (approximated as 0.333...) is a rational number that lies between 0.3 and 0.4.

Then the correct option is A)

6. Correct option: (B)

Let's denote the unknown rational number as "x". We are given that the multiplication of the two rational numbers is $-\frac{3}{4}$ and that one of the numbers is $\frac{4}{5}$. So, we can set up an equation:

$$\left(\frac{4}{5}\right) \times x = -\frac{3}{4}$$

To solve for x, multiply both sides of the equation by the reciprocal of $\frac{4}{5}$, which is $\frac{5}{4}$:

$$x = \left(-\frac{3}{4}\right) \times \left(\frac{5}{4}\right)$$

$$x = -\frac{15}{16}$$

So, the unknown rational number is $-\frac{15}{16}$.

Then the correct option is B)

7. Correct option: (A)

Given: $a = 3$

Expression: $-\frac{a+1}{a}$

Substitute the value of a:

$$= -\left(\frac{3 + 1}{3}\right)$$

$$= -\left(\frac{4}{3}\right)$$

$$= -\frac{4}{3}$$

So, when $a = 3$, the value of $-\frac{a+1}{a}$ is $-\frac{4}{3}$.

Then the correct option is A)

8. Correct option: (B)

To find the cost per meter, we need to divide the total cost by the total length of the rope.

Given: $Cost\ of\ 6\frac{2}{3}\ meters\ of\ rope = Rs.11$

Step 1: Convert the mixed fraction to an improper fraction.

$$6\frac{2}{3} = \frac{6 \times 3 + 2}{3} = \frac{20}{3}$$

Step 2: Divide the total cost by the total length of the rope.

$$Cost\ per\ meter = \frac{Total\ Cost}{Total\ Length}$$

$$Cost\ per\ meter = \frac{11}{\frac{20}{3}}$$

To divide by $\frac{20}{3}$ we can multiply by it's multiplicative inverse.

$$Cost\ per\ meter = 11 \times \left(\frac{3}{20}\right)$$

$$Cost\ per\ meter = \frac{33}{20}$$

So, the cost per meter of the rope is $\frac{33}{20}$.

Then the correct option is B)

9. Correct option: (B)

To add fractions, they must have the same denominator. In this case, both fractions have the same denominator, which is 11. Therefore, we can simply add the numerators and keep the common denominator:

$$\left(-\frac{4}{11}\right) + \left(\frac{7}{11}\right) = \frac{-4 + 7}{11} = \frac{3}{11}$$

Then the correct option is B)

10. Correct option: (B)

To multiply fractions, simply multiply the numerators together and the denominators together. Let's multiply $-\frac{8}{7}$ by $\frac{14}{5}$:

$$\left(-\frac{8}{7}\right)\times\left(\frac{14}{5}\right)=\frac{-8\times 14}{7\times 5}=-\frac{112}{35}$$

Now, let's simplify the result:

$-\frac{112}{35}$ can be simplified by dividing both the numerator and denominator by their greatest common divisor, which is 7:

$$-112\div 7=-16$$
$$35\div 7=5$$

So, $-\frac{112}{35}$ simplifies to $-\frac{16}{5}$.

Then the correct option is B)

11. Correct option: (A)

Let's simplify the given expression step by step:

1. Simplify each multiplication within parentheses:

$$\left(-\frac{9}{4}\times\frac{5}{3}\right)=-\frac{45}{12}$$
$$\left(\frac{13}{2}\times\frac{5}{6}\right)=\frac{65}{12}$$

2. Now, add the two results together:

$$\left(-\frac{45}{12}\right)+\left(\frac{65}{12}\right)=\frac{65-45}{12}=\frac{20}{12}$$

3. Finally, simplify the fraction:

$\frac{20}{12}$ can be simplified by dividing both the numerator and denominator by their greatest common divisor, which is 4:

$$20\div 4=5$$
$$12\div 4=3$$

So, $\left(-\frac{9}{4}\times\frac{5}{3}\right)+\left(\frac{13}{2}\times\frac{5}{6}\right)$ simplifies to $\frac{5}{3}$.

Then the correct option is A)

12. Correct option: (B)

To find the number that should divide $-\frac{33}{16}$ to get $-\frac{11}{4}$, we can set up the following equation:

$$\left(-\frac{33}{16}\right)\div x=-\frac{11}{4}$$

Now, let's solve for x by cross-multiplying:

$$-\frac{33}{16}=\left(-\frac{11}{4}\right)\times x$$

Multiply both sides by -4 to solve for x:

$$x=\left(-\frac{33}{16}\right)\times\left(-\frac{4}{11}\right)$$

Now, multiply the numerators and denominators:

$$x=\frac{33\times 4}{16\times 11}$$
$$x=\frac{132}{176}$$
$$x=\frac{3}{4}$$

So, the number that should divide $-\frac{33}{16}$ to get $-\frac{11}{4}$ is $\frac{3}{4}$.

Then the correct option is B)

13. Correct option: (C)

Let's calculate the time Shiva spends in the library and in school first:

Time in library = $\frac{1}{12}\ day$

Time in school = $\frac{1}{3}\ day$

Now, we can find the total time Shiva has spent:

$$Total\ time\ spent\ =\ Time\ in\ library\ +\ Time\ in\ school$$

$$Total\ time\ spent\ = \frac{1}{12}\ + \frac{1}{3}\ = \left(\frac{1}{12}\right) + \left(\frac{4}{12}\right) = \frac{5}{12}\ day$$

The time left in the day is the difference between the total day and the time spent:

$$Time\ left\ =\ Total\ day\ -\ Total\ time\ spent$$

$$Time\ left\ =\ 1\ day\ - \frac{5}{12}\ day\ = \left(\frac{12}{12}\right)\ day\ - \left(\frac{5}{12}\right)\ day\ = \frac{7}{12}\ day$$

To convert this fraction of a day into hours, we know that there are 24 hours in a day:

$$Time\ left\ in\ hours\ = \left(\frac{7}{12}\right)\ day\ \times 24\ hours/day\ =\ 14\ hours$$

So, the number of hours left in the day is 14 hours.

Then the correct option is C)

14. Correct option: (C)

To find the fraction of the garden that was planted in one day, we need to add the fractions of what Ashok planted in the morning and afternoon.

Given:

Fraction planted in the morning = $\frac{1}{5}$

Fraction planted in the afternoon = $\frac{2}{7}$

Total fraction planted = Fraction in the morning + Fraction in the afternoon

Total fraction planted = $\frac{1}{5}\ + \frac{2}{7}$

To add these fractions, we need a common denominator, which is the least common multiple of 5 and 7, which is 35.

Rewrite the fractions with the common denominator:

Total fraction planted = $\frac{1 \times 7}{5 \times 7}\ + \frac{2 \times 5}{7 \times 5}\ = \frac{7}{35}\ + \frac{10}{35}$

Now, add the fractions:

Total fraction planted = $\frac{7 + 10}{35}\ = \frac{17}{35}$

So, the fraction of Ashok's garden that was planted in one day is $\frac{17}{35}$.

Then the correct option is C)

15. Correct option: (D)

The repeating decimal $0.\overline{37}$ can be represented as a fraction by using the concept of repeating decimals.

Let $x\ =\ 0.\overline{37}$

Multiply both sides of x by 100 to shift the repeating part to the left of the decimal point:

$$100x\ =\ 37.\overline{37}$$

Subtract x from $100x$:

$$100x\ -\ x\ =\ 37.\overline{37}\ - 0.\overline{37}$$

$$99x\ =\ 37$$

$$x\ = \frac{37}{99}$$

So, the value of the repeating decimal $0.\overline{37}$ is $\frac{37}{99}$.

Then the correct option is D)

16. Correct option: (B)

To express the repeating decimal $0.\overline{621}$as a fraction, we can use the following steps:

Let $x = 0.\overline{621}$

Multiply both sides of the equation by 1000 to remove the repeating decimal:

$$1000x = 621.\overline{621}$$

Subtract the original equation from the multiplied equation:

$$1000x - x = 621.\overline{621} - 0.\overline{621}$$

$$999x = 621$$

Now, solve for x:

$$x = \frac{621}{999}$$

Simplify the fraction by dividing both the numerator and denominator by their greatest common divisor, which is 27:

$$x = \frac{23 \times 27}{37 \times 27} = \frac{23}{37}$$

So, the repeating decimal $0.\overline{621}$ is equivalent to the fraction $\frac{23}{37}$.

Then the correct option is B)

17. Correct option: (A)

To express the repeating decimal $125.\bar{3}$ as a fraction, we can follow these steps:

Let $x = 125.\bar{3}$

Step 1: Multiply both sides of the equation by 10 to move the repeating decimal to the left of the decimal point: $10x = 1253.\bar{3}$

Step 2: Subtract the original equation from the equation obtained in step 1:

$$10x - x = 1253.\bar{3} - 125.\bar{3}$$

$$9x = 1128$$

Step 3: Solve for x:

$$x = \frac{1128}{9} = \frac{376}{3}$$

So, the fraction equivalent to the repeating decimal $125.\bar{3}$is $\frac{376}{3}$.

Then the correct option is A)

18. Correct option: (C)

To add the repeating decimals $0.\overline{32}$ and $0.\bar{4}$ we'll first convert them into fractions and then perform the addition:

Let $x = 0.\overline{32}$

Then, $100x = 32.\overline{32}$

Subtracting the first equation from the second equation:

$$100x - x = 32.\overline{32} - 0.\overline{32}$$

$$99x = 32$$

$$x = \frac{32}{99}$$

Let $y = 0.\bar{4}$

Then, $10y = 4.\bar{4}$

Subtracting the first equation from the second equation:

$$10y - y = 4.\bar{4} - 0.\bar{4}$$

$$9y = 4$$

$$y = \frac{4}{9}$$

Now, let's add the fractions:

$$x + y = \frac{32}{99} + \frac{4}{9}$$

To add these fractions, we need a common denominator, which is the least common multiple of 99 and 9, which is 891.
Rewrite the fractions with the common denominator:

$$x + y = \frac{32 \times 9}{99 \times 9} + \frac{4 \times 99}{9 \times 99} = \frac{288}{891} + \frac{396}{891} => \frac{684}{891}$$

Simplify the fraction by dividing both the numerator and denominator by their greatest common divisor, which is 9:

$$\mathrm{x} + y = \frac{\left(\frac{684}{9}\right)}{\frac{891}{9}} = \frac{76}{99}$$

Convert the fraction back to a repeating decimal:

$$\frac{76}{99} = 0.767676 \ldots\ldots = 0.\overline{76}$$

So, the result of adding $0.\overline{32}$ and $0.\overline{4}$ is $0.\overline{76}$.
Then the correct option is C)

19. Correct option: (A)

To convert repeating decimals to fractions, we can use algebraic manipulation. Let's first consider the decimal $5.\overline{73}$:
Let $x = 5.\overline{73}$
Multiplying both sides of the equation by 100 to shift the decimal point two places to the right:

$$100x = 573.\overline{73}$$

Subtracting the original equation from the second equation to eliminate the repeating part:

$$100x - x = 573.\overline{73} - 5.\overline{73}$$

$$99x = 568 => \frac{568}{99}$$

Now, let's consider the decimal $8.\overline{61}$:
Let $y = 8.\overline{61}$
Multiplying both sides of the equation by 100 to shift the decimal point two places to the right:

$$100y = 861.\overline{61}$$

Subtracting the original equation from the second equation to eliminate the repeating part:

$$100y - y = 861.\overline{61} - 8.\overline{61}$$

$$99y = 853 => \frac{853}{99}$$

Now, let's add the two fractions together:

$$x + y = \frac{568}{99} + \frac{853}{99}$$

$$x + y = \frac{568 + 853}{99}$$

$$x + y = \frac{1421}{99}$$

$$=> \frac{1421}{99} => 14.35353535 \ldots\ldots = 14.\overline{35}$$

Then the correct option is A)

20. Correct option: (B)

The value of the repeating decimal $0.\overline{09}$ can be calculated as follows:
Let $x = 0.\overline{09}$
Multiplying both sides of the equation by 100 to shift the decimal point two places to the right:

$$100x = 9.\overline{09}$$

Subtracting the original equation from the second equation to eliminate the repeating part:

$$100x - x = 9.\overline{09} - 0.\overline{09}$$

$$99x = 9 \Rightarrow x = \frac{9}{99} = \frac{1}{11}$$

So, the value of the repeating decimal $0.\overline{09}$ is $\frac{1}{11}$.

Then the correct option is B)

21. Correct option: (C)

To convert the repeating decimal $15.\bar{6}$ to a fraction, let's follow a similar process as before:

Let $x = 15.\bar{6}$

Multiplying both sides of the equation by 10 to shift the decimal point one place to the right:

$$10x = 156.\bar{6}$$

Subtracting the original equation from the second equation to eliminate the repeating part:

$$10x - x = 156.\bar{6} - 15.\bar{6}$$

$$9x = 141 \Rightarrow x = \frac{141}{9} = \frac{47}{3}$$

Therefore, the value of the repeating decimal $15.\bar{6}$ in terms of $\frac{p}{q}$ is:

$$15.\bar{6} = \frac{47}{3}$$

Then the correct option is C)

22. Correct option: (C)

The repeating decimal 0.77777… can be expressed as a fraction by considering it as an infinite geometric series. Let's call this decimal x:

$$x = 0.77777\ldots$$

Multiplying both sides by 10 to shift the decimal point one place to the right:

$$10x = 7.77777\ldots$$

Now, subtracting the original equation (x) from the second equation (10x):

$$10x - x = 7.77777\ldots - 0.77777\ldots$$

$$9x = 7 \Rightarrow x = \frac{7}{9}$$

Therefore, the value of the repeating decimal 0.77777… as a fraction is $\frac{7}{9}$.

Then the correct option is C)

23. Correct option: (D)

To find the product of the repeating decimal $97.2\overline{81}$ and 100, we can follow these steps:

Let $x = 97.2\overline{81}$

Write x in repeating number term as

$$x = 97.2818181\ldots$$

$$100x = 97.2818181\ldots \times 100 = 9728.181818 = 9728.\overline{18}$$

Or we can use our old method decimal to fraction and fraction to back decimal conversion.

Then the correct option is D)

2. Comparing Quantities

Solutions

RATIO, PROPORTION AND ITS APPLICATIONS

LEVEL – I

1. Correct option: (A)

The number of terms in a ratio is 2.

Then the correct option is A) 2

2. Correct option: (A)

In the ratio, the first term is called the "Antecedent."

Then the correct option is A) Antecedent.

3. Correct option: (A)

Given:

$$A:B = 3:4$$
$$B:C = 2:3$$

Find A in terms of B:

$$A/B = 3/4$$
$$A = \left(\frac{3}{4}\right) \times B$$

Find C in terms of B:

$$C/B = 3/2$$
$$C = \left(\frac{3}{2}\right) \times B$$

Now, express the ratio A:B:C:

$$A:B:C = \left[\left(\frac{3}{4}\right) \times B\right] : B : \left[\left(\frac{3}{2}\right) \times B\right]$$

Now, simplify the ratio by canceling out B from all terms:

To eliminate fractions, multiply all terms by 4 (to make the denominators integers):

$$A:B:C = 3 : 4 : 6$$

So, the ratio $A:B:C$ is $3:4:6$.

Therefore, the correct option is A) 3:4:6.

4. Correct option: (B)

To find the ratio of the two numbers given that 0.7 of one number is equal to 0.075 of another number, we can set up the equation: $0.7x = 0.075y$

Now, solve for the ratio of x to y:

$$\frac{x}{y} = \left(\frac{0.075}{0.7}\right) = \frac{75}{1000} \times \frac{10}{7} = \frac{75}{700} = \frac{3}{28}$$

Ratio of the number is $x:y = 3:28$

Therefore, the correct option is B) 3:28.

5. Correct option: (C)

The ratio of income-tax to income can be calculated using the given information:

Income = Rs. 56000

Income-tax = Rs. 9600

$$\text{Ratio} = \frac{\text{Income tax}}{\text{Income}}$$
$$\text{Ratio} = \frac{9600}{56000}$$
$$\text{Ratio} = \frac{24}{140}$$

$$\text{Ratio} = \frac{\frac{24}{4}}{\frac{140}{4}}$$

$$\text{Ratio} = \frac{6}{35}$$

So, the ratio of the income-tax to the income is 6:35.
Therefore, the correct option is C) 6:35.

6. Correct option: (A)

To determine which ratio is greater between 5:8 and 7:10, we can cross-multiply and compare the results:
For 5:8,

$$\frac{5}{8} = \frac{5 \times 10}{8 \times 10} = \frac{50}{80}$$

For 7:10,

$$\frac{7}{10} = \frac{7 \times 8}{10 \times 8} = \frac{56}{80}$$

Now, compare $\frac{50}{80}$ and $\frac{56}{80}$:

$$\frac{50}{80} < \frac{56}{80}$$

So, 7:10 is greater than 5:8.
The correct option is A) 7:10.

7. Correct option: (A)

To divide the sum of Rs 1216 between A and B in the ratio of $\frac{1}{3}$ to $\frac{1}{5}$, we can first simplify the ratio:

$$\frac{1}{3}:\frac{1}{5} = \left(\frac{1}{3}\right) \times \left(\frac{5}{5}\right) : \left(\frac{1}{5}\right) \times \left(\frac{3}{3}\right) = \frac{5}{15} : \frac{3}{15}$$

Now, the ratio is 5:3.
Next, we can find B's share by calculating:

$$\text{B's share} = \left(\frac{3}{5 + 3}\right) \times \text{Rs } 1216$$

$$\text{B's share} = \left(\frac{3}{8}\right) \times \text{Rs } 1216$$

Now, calculate B's share:

$$\text{B's share} = \left(\frac{3}{8}\right) \times 1216 = 3 \times 152 = \text{Rs } 456$$

So, B's share is Rs 456.
Therefore, the correct answer is A) 456.

8. Correct option: (D)

If the ratio of two quantities is 3:4, and the antecedent (the first term) is 15, we can find the consequent (the second term) using the given ratio.
The ratio is 3:4, and the antecedent is 15, so we can set up the following proportion:

$$\frac{3}{4} = \frac{15}{x}$$

To solve for x (the consequent), cross-multiply:

$$3 \times x = 4 \times 15$$

$$3x = 60$$

$$x = \frac{60}{3}$$

$$x = 20$$

So, the consequent is 20.
Therefore, the correct option is D) 20.

9. Correct option: (B)

To find the ratio $x^2y + xy^2 : x^3 + y^3$, we can first substitute the given values for x and y from the ratio x : y = 3 : 4.

Given:

x : y = 3 : 4

Let $x = 3k, y = 4k$

Now, let's calculate $x^2y + xy^2$ and $x^3 + y^3$:

$$x^2y + xy^2 = (3k)^2(4k) + (3k)(4k)^2 = 36k^3 + 48k^3 = 84k^3$$

$$x^3 + y^3 = (3k)^3 + (4k)^3 = 27k^3 + 64k^3 = 91k^3$$

Now, we can find the ratio:

$$(x^2y + xy^2) : (x^3 + y^3) = 84k^3 : 91k^3$$

Now, cancel the common factor of k^3:

$$\left(\frac{84\,k^3}{k^3}\right) : \left(\frac{91k^3}{k^3}\right) = 84 : 91 = 12 : 13$$

So, the ratio $x^2y + xy^2 : x^3 + y^3$ is 12 : 13.

Therefore, the correct option is B) 12:13.

10. Correct option: (A)

The fourth proportional to 2a, a^3, and c can be found by setting up a proportion:

$$2a : a^3 :: c : X$$

To find X, we can cross-multiply:

$$2a \times X = a^3 \times c$$

Now, solve for X by dividing both sides by 2a:

$$X = \frac{a^3 \times c}{2a}$$

Now, simplify the expression by canceling common factors:

$$X = \frac{a^2 \times c}{2}$$

So, the fourth proportional is:

$$X = \frac{a^2c}{2}$$

Therefore, the correct option is A) $\frac{a^2c}{2}$.

11. Correct option: (C)

To find the ratio x : y from the given proportion:

$$(5x - 3y) : (5y - 3x) = 3 : 4$$

First, cross-multiply:

$$4(5x - 3y) = 3(5y - 3x)$$

Now, distribute:

$$20x - 12y = 15y - 9x$$

$$20x + 9x = 15y + 12y$$

Combine like terms and solve for ratio

$$29x = 27y$$

$$\frac{x}{y} = \frac{27}{29}$$

So, the ratio x : y is 27 : 29.

Therefore, the correct option is C) 27 : 29.

12. Correct option: (B)

Given:

$$A : B = 3 : 4$$

$$B : C = 5 : 2$$

Find A in terms of B:

$$A/B = 3/4$$

$$A = \left(\frac{3}{4}\right) \times B$$

Find C in terms of B:

$$\frac{C}{B} = \frac{2}{5}$$

$$C = \left(\frac{2}{5}\right) \times B$$

Now, express the ratio A:B:C:

$$A:B:C = \left[\left(\frac{3}{4}\right) \times B\right] : B : \left[\left(\frac{2}{5}\right) \times B\right]$$

Now, simplify the ratio by canceling out B from all terms:

To eliminate fractions, multiply all terms by 4×5 (to make the denominators integers):

$$A:B:C = 15 : 20 : 8$$

So, the ratio $A:B:C$ is $15:20:8$.

Therefore, the correct option is B) 15:20:8.

13. Correct option: (A)

If the numbers 81, x, 256, and x are in proportion, it means that the ratios of consecutive terms are equal.

So, we can set up two ratios:

$$81 : x$$

$$x : 256$$

Now, equate the ratios:

$$\frac{81}{x} = \frac{x}{256}$$

To solve for x, cross-multiply:

$$81 \times 256 = x \times x$$

$$20736 = x^2$$

Now, take the square root of both sides:

$$x = \sqrt{20736}$$

$$x = 144$$

So, x is equal to 144.

Therefore, the correct option is A) 144.

14. Correct option: (C)

To find the number of shawls 6 persons can weave in 15 days, we can use the concept of work and time being inversely proportional.

Given that 5 persons weave 180 shawls in 12 days, we can find their combined work rate:

$$\text{Work rate} = \frac{\text{Total work}}{\text{Total time}} = \frac{180 \text{ shawls}}{12 \text{ days}} = 15 \text{ shawls per day}$$

Now, to find how many shawls 6 persons can weave in 15 days, we can calculate their combined work rate:

$$\text{Work rate for 6 persons} = \left(\frac{6}{5}\right) \times \text{Work rate for 5 persons}$$

$$\text{Work rate for 6 persons} = \left(\frac{6}{5}\right) \times 15 \text{ shawls per day} = 18 \text{ shawls per day}$$

Now, calculate the number of shawls 6 persons can weave in 15 days:

$$\text{Number of shawls} = \text{Work rate for 6 persons} \times \text{Time (15 days)}$$

$$\text{Number of shawls} = 18 \text{ shawls per day} \times 15 \text{ days} = 270 \text{ shawls}$$

So, 6 persons can weave 270 shawls in 15 days.

Therefore, the correct option is C) 270.

15. Correct option: (B)

Given: $\frac{a}{3} = \frac{b}{4} = \frac{c}{5}$

Let's denote the common ratio as k. So, a = 3k, b = 4k, and c = 5k.

Now, we need to find $\frac{a+b+c}{b}$:

$$\frac{a + b + c}{b} = \frac{3k + 4k + 5k}{4k}$$

$$= \frac{12k}{4k} = \frac{12}{4} = 3$$

So, $\frac{a+b+c}{b} = 3$.

Therefore, the correct option is B) 3.

16. Correct option: (A)

The number of terms in a proportion is 4.

Therefore, the correct option is A) 4.

17. Correct option: (A)

To find the value of x in the proportion 15:8 = 60:x, we can use the concept of cross-multiplication.

$$15:8 = 60:x$$

Cross-multiply: $15x = 8 \times 60$

Now, divide both sides by 15 to isolate x:

$$x = \frac{8 \times 60}{15}$$

$$x = \frac{480}{15}$$

$$x = 32$$

$$So, x = 32.$$

Therefore, the correct option is A) 32.

18. Correct option: (A)

The fourth proportional to $\frac{1}{2}, \frac{1}{3}$, and $4\frac{1}{3}$ can be found by setting up a proportion:

$$\frac{1}{2} : \frac{1}{3} :: 4\frac{1}{3} : X$$

To find X, we can cross-multiply:

$$\left(\frac{1}{2}\right) \times X = \left(\frac{1}{3}\right) \times \left(4\frac{1}{3}\right)$$

Now, convert $4\frac{1}{3}$ to an improper fraction:

$$4\frac{1}{3} = \frac{4 \times 3 + 1}{3} = \frac{13}{3}$$

Now, continue solving for X:

$$\left(\frac{X}{2}\right) = \left(\frac{1}{3}\right) \times \left(\frac{13}{3}\right)$$

Multiply both sides by 2 to isolate X:

$$X = \left(\frac{2}{3}\right) \times \left(\frac{13}{3}\right)$$

$$X = \frac{2 \times 13}{3 \times 3}$$

$$X = \frac{26}{9}$$

So, the fourth proportional to $\frac{1}{2}, \frac{1}{3}$, and $4\frac{1}{3}$ is $\frac{26}{9}$.

Therefore, the answer is A) $\frac{26}{9}$.

19. Correct option: (A)

To find the third proportional to 75 and 15, we can use the formula:

$$\text{Third Proportional} = \frac{\text{Second Proportional}^2}{\text{First Proportional}}$$

In this case:

First Proportional = 0.24

Second Proportional = 0.60

Now, calculate the third proportional:

$$\text{Third Proportional} = \frac{0.60^2}{0.24} = \frac{0.36}{0.24} = 1.5$$

So, the third proportional to 0.24 and 0.60 is 1.5.

Therefore, the correct option is A) 1.5.

20. Correct option: (A)

Let's denote the number to be added as "x."

The original numbers are: 5, 7, 8, and 11.

After adding "x" to each of them, you get the new numbers:

$$(5 + x), (7 + x), (8 + x), \text{and } (11 + x).$$

Now, we want these new numbers to be in proportion. This means:

$$(5 + x) : (7 + x) = (8 + x) : (11 + x)$$

Now, cross-multiply:

$$(5 + x) \times (11 + x) = (7 + x) \times (8 + x)$$

Expand and simplify both sides:

$$(55 + 5x + 11x + x^2) = (56 + 7x + 8x + x^2)$$
$$55 + 16x + x^2 = 56 + 15x + x^2$$
$$55 + 16x = 56 + 15x$$
$$16x - 15x = 56 - 55$$
$$x = 1$$

So, the number that should be added to 5, 7, 8, and 11 to maintain proportionality among the resulting numbers is 1.

Therefore, the correct option is A) 1.

LEVEL – II

1. Correct option: (A)

We want to divide Rs. 415 into three parts, denoted as A, B, and C, such that:

$$4A = 5B$$
$$4A = 7C$$
$$A + B + C = 415$$

From the first two equations, we find that $B = \left(\frac{4}{5}\right)A$ and $C = \left(\frac{4}{7}\right)A$.

Now, substitute these into the third equation:

$$A + \left(\frac{4}{5}\right)A + \left(\frac{4}{7}\right)A = 415$$

Combining fractions:

$$\left(\frac{35}{35}\right)A + \left(\frac{28}{35}\right)A + \left(\frac{20}{35}\right)A = 415$$
$$\left(\frac{83}{35}\right)A = 415$$
$$A = \frac{415 \times 35}{83}$$
$$A = 175$$

Therefore, the correct option is A) 175.

2. Correct option: (A)

Initially, 1600 men had enough food for 4 days.

So, the total amount of food is constant:

$$(1600 \text{ men}) \times (4 \text{ days}) = 6400 \text{ men days of food.}$$

Now, when 400 additional men join, the total number of men becomes

$$1600 + 400 = 2000 \text{ men.}$$

To find out how long the food will now last for 2000 men, simply divide the total amount of food (6400 men-days) by the number of men (2000 men):

$$\text{Food will last} = \frac{6400 \text{ men days}}{2000 \text{ men}} = 3.2 \text{ days}$$

So, when 400 additional men join, the food will now last for 3.2 days.

Therefore, the correct option is A) 3.2.

3. Correct option: (A)

Let's find the cost of $\frac{7}{12}$ quintals of wheat:

Given:

$\frac{2}{5}$ Quintals of wheat cost Rs. 96.

So, the cost of 1 quintal of wheat $= \text{Rs. } 96 \times \left(\frac{5}{2}\right) = \text{Rs. } 240.$

Now, we want to find the cost of $\frac{7}{12}$ quintals of wheat:

Cost of $\left(\frac{7}{12}\right)$ quintals of wheat $= \left(\frac{7}{12}\right) \times$ Cost of 1 quintal of wheat

Cost of $\left(\frac{7}{12}\right)$ quintals of wheat $= \left(\frac{7}{12}\right) \times \text{Rs. } 240$

Now, calculate:

Cost of $\left(\frac{7}{12}\right)$ quintals of wheat $= \text{Rs. } (7 \times 20) = \text{Rs. } 140.$

So, the cost of $\frac{7}{12}$ quintals of wheat is Rs. 140.

Therefore, the correct option is A) Rs. 140.

4. Correct option: (B)

To find the ratio x: z, we can first find the ratio of x: y and y: z and then multiply these ratios together.

Given:

$$x{:}\,y = \frac{1}{6} : \frac{1}{8}$$

$$y{:}\,z = \frac{1}{8} : \frac{1}{10}$$

Let's find the ratio : y , y: zfirst. Since both ratios have a common term of y, we can equate them:

$$\frac{x}{y} = \left(\frac{1}{6}\right) \times \left(\frac{8}{1}\right) = \frac{8}{6} = \frac{4}{3} \Rightarrow x = \frac{4}{3}y$$

$$\frac{y}{z} = \left(\frac{1}{8}\right) \times \left(\frac{10}{1}\right) = \frac{10}{8} = \frac{5}{4} \Rightarrow z = \frac{4}{5}y$$

$$x{:}\,z = \frac{\left(\frac{4}{3}\right)}{\left(\frac{4}{5}\right)} = \frac{5}{3}$$

So, x: z = 5: 3.

Therefore, the correct option is B) 5:3.

5. Correct option: (A)

To find the compounded ratio of 4:9 and the duplicate ratio of 3:4, we can first find the duplicate ratio of 3:4 and then multiply it with the original ratio of 4:9.

Given: $\text{Original ratio} = 4:9$

$$\text{Duplicate ratio of } 3:4 = (3^2):(4^2) = 9:16$$

Now, multiply the original ratio by the duplicate ratio:

$$\text{Compounded ratio} = (4:9) \times (9:16)$$

$$\text{Compounded ratio} = \left(\frac{4}{9}\right) \times \left(\frac{9}{16}\right) = \left(\frac{4}{16}\right)$$

Reduce this fraction to its simplest form:

$$\text{Compounded ratio} = \left(\frac{1}{4}\right)$$

So, the compounded ratio of 4:9 and the duplicate ratio of 3:4 is 1:4.

Therefore, the correct option is A) 1:4.

6. Correct option: (A)

To find an equivalent ratio to 5:3, we can scale it up or down by multiplying both the terms by the same number.

In this case, let's say we want to scale it up, so we'll multiply both terms by a common factor.

For example, if we multiply both terms by 3, we get:

$$5 \times 3 : 3 \times 3 = 15 : 9$$

So, an equivalent ratio to 5:3 is A) 15:9.

7. Correct option: (B)

To find the weight of copper in the alloy, we can use the given ratio of zinc to copper, which is 7:9, and the fact that there are 31.5 kg of zinc.

$$\frac{\text{Zinc (7 parts)}}{\text{Copper (9 parts)}} = \frac{31.5 \text{ kg}}{\text{Weight of copper (unknown)}}$$

Now, we can cross-multiply and solve for the weight of copper:

$$\frac{7}{9} = \frac{31.5 \text{ kg}}{\text{Weight of copper}}$$

To find the weight of copper, multiply both sides by 9:

$$\text{Weight of copper} = \frac{31.5 \text{ kg} \times 9}{7}$$

$$\text{Weight of copper} = 40.5 \text{ kg}$$

So, the weight of copper in the alloy is 40.5 kg.

Therefore the correct option is B)

8. Correct option: (A)

First, let's find p in terms of q and r in terms of q using the given ratios:

Given:

$$p:q = 1\frac{1}{2} : 2\frac{1}{3}$$

$$q:r = 1\frac{1}{3} : \frac{2}{5}$$

Convert the mixed numbers to improper fractions:

$$p:q = \frac{3}{2} : \frac{7}{3}$$

$$q:r = \frac{4}{3} : \frac{2}{5}$$

Now, let's find p in terms of q:

$$\frac{p}{q} = \frac{\frac{3}{2}}{\frac{7}{3}} = \left(\frac{3}{2}\right) \times \left(\frac{3}{7}\right) = \frac{9}{14} \Rightarrow p = q \times \frac{9}{14}$$

Now, let's find r in terms of q:

$$\frac{r}{q} = \frac{\frac{2}{5}}{\frac{4}{3}} = \left(\frac{2}{5}\right) \times \left(\frac{3}{4}\right) = \frac{6}{20} = \frac{3}{10} => r = q \times \frac{3}{10}$$

Then

$$p : q : r = \frac{9q}{14} : q : \frac{3q}{10} => 90 : 140 : 42 => 45 : 70 : 21$$

Simplify the fractions:

$$p : q : r = 45 : 70 : 21$$

Therefore the correct option is A)

9. Correct option: (B)

To find the ratio $x : y : z$, we can first find the missing terms and then construct the complete ratio.

Given:

$$x : y = 5 : 6$$
$$y : z = 3 : 7$$

First, let's find x in terms of y using the ratio $x : y = 5 : 6$

$$\frac{x}{y} = \frac{5}{6} => x = \frac{5y}{6}$$

Now, let's find z in terms of y using the ratio $y : z = 3 : 7$

$$\frac{y}{z} = \frac{3}{7} => z = \frac{7y}{3}$$

Then

$$x : y : z = \frac{5y}{6} : y : \frac{7y}{3}$$

Now, simplify the last term:

$$x : y : z = \left(\frac{5}{6}\right) : 1 : \left(\frac{7}{3}\right)$$

To make the ratios have whole numbers, multiply all terms by 6:

$$x : y : z = \left(\frac{5}{6} \times 18\right) : (1 \times 18) : \left(\frac{7}{3} \times 18\right)$$

Simplify:

$$x : y : z = 15 : 18 : 42$$

So, $x : y : z =$ B) $15 : 18 : 42$.

10. Correct option: (*)

We are given that the numbers are

$$x - y, x - 2, x + 2, x + 10$$

We also given that the numbers are in proportion

$$\frac{x - y}{x - 2} = \frac{x + 2}{x + 10}$$

Cross multiplying

$$(x - y)(x + 10) = (x - 2)(x + 2) = x^2 - 4$$
$$=> x^2 - xy + 10x - 10y = x^2 - 4 => 10x - 10y - xy = -4$$
$$=> x(10 - y) = 10y - 4 => x = \frac{10y - 4}{10 - y}$$

As we can see that x depends on y.

Without the value of y we cannot determine the value of x.

11. Correct option: (C)

To find out how many words Bhavana would type in 6 minutes, we can use the given rate of typing:

Bhavana types 540 words in half an hour, which is 30 minutes.

Now, we want to find out how many words she types in 6 minutes, which is $\frac{1}{5}$ of 30 minutes.

So, we can set up a proportion to find the number of words she types in 6 minutes:

$$\frac{540 \text{ words}}{30 \text{ minutes}} = \frac{x \text{ words}}{6 \text{ minutes}}$$

$$30x = 540 \times 6$$

$$30x = 3240$$

$$x = \frac{3240}{30}$$

$$x = 108$$

So, Bhavana would type 108 words in 6 minutes.

Therefore, the correct option is C) 108.

12. Correct option: (D)

To find out how many meters of cloth can be bought for Rs. 727.5, we can set up a proportion based on the given cost of 40 meters:

$$\text{Cost of 40 meters} = \text{Rs. } 1940$$

Now, let's find the cost per meter:

$$\text{Cost per meter} = \frac{\text{Cost of 40 meters}}{40 \text{ meters}} = \frac{1940}{40} = \text{Rs. } 48.50 \text{ per meter}$$

Now, we want to know how many meters can be bought for Rs. 727.5:

$$\text{Number of meters} = \frac{\text{Total cost}}{\text{Cost per meter}} = \frac{\text{Rs. } 727.5}{\text{Rs. } 48.50 \text{ per meter}}$$

Now, calculate: $\text{Number of meters} = 15 \text{ meters}$

So, we can buy 15 meters of cloth for Rs. 727.5.

Therefore, the correct option is D) 15.

13. Correct option: (B)

To find the distance Rekha would cover in 315 minutes, we can use the given information about her walking speed.

Rekha takes 125 minutes to walk 100 meters. We can find her speed as follows:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Speed} = \frac{100 \text{ meters}}{125 \text{ minutes}}$$

Now, we can use her speed to find the distance she would cover in 315 minutes:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{Distance} = \left(\frac{100}{125}\right) \text{ meters/minute} \times 315 \text{ minutes}$$

$$\text{Distance} = \left(\frac{4}{5}\right) \text{ meters/minute} \times 315 \text{ minutes}$$

$$\text{Distance} = \left(\frac{4}{5}\right) \times 315 \text{ meters}$$

$$\text{Distance} = 252 \text{ meters}$$

So, Rekha would cover 252 meters in 315 minutes.

Therefore, the correct option is B) 252.

14. Correct option: (B)

Given:

18 men can reap the field in 35 days.

Work rate is inversely proportional to the number of days required to complete the work.

So, if 18 men take 35 days, we can find their work rate:

$$\text{Work rate} = \frac{1}{\text{Number of men} \times \text{Number of days}}$$

$$\text{Work rate} = \frac{1}{18 \text{ men} \times 35 \text{ days}}$$

Now, let's find the work rate:

$$\text{Work rate} = \frac{1}{18 \times 35} = \frac{1}{630}$$

Now, we want to find out how many men are required to reap the same field in 15 days:

$$\text{Work rate} = 1 / (\text{Number of men} \times \text{Number of days})$$

We know the work rate is $\frac{1}{630}$, and we want to complete the work in 15 days:

$$\frac{1}{630} = \frac{1}{\text{Number of men} \times 15}$$

Now, solve for the number of men:

$$\text{Number of men} = \frac{1}{\frac{1}{630} \times 15} = \frac{1}{\frac{1}{42}} = 42$$

So, 42 men are required to reap the same field in 15 days.

Therefore, the correct option is B) 42.

15. Correct option: (A)

To find the value of x, we can use the given proportion:

$$(4x + 5) : (3x + 11) = 13 : 17$$

Now, cross-multiply:

$$(4x + 5) \times 17 = (3x + 11) \times 13$$
$$\Rightarrow 68x + 85 = 39x + 143$$
$$\Rightarrow 68x - 39x = 143 - 85$$
$$\Rightarrow 29x = 58$$
$$\Rightarrow x = \frac{58}{29}$$
$$x = 2$$

So, x is equal to 2.

Therefore, the correct option is A) 2.

16. Correct option: (A)

We are given that

$$\frac{a}{12} = \frac{b}{60} \Rightarrow \frac{a}{b} = \frac{1}{5}$$

We have to find the value of

$$a + b$$

Now, using the first ratio we can get the

$$a = 1, b = 5$$

Putting these values in the next expression

$$a + b = 1 + 5 \Rightarrow 6$$

Therefore, the correct option is A) 6.

17. Correct option: (A)

The sub-triplicate ratio of a ratio a: b is given by $\left(a^{\frac{1}{3}}\right) : \left(b^{\frac{1}{3}}\right)$.

In this case, the sub-triplicate ratio of 8:27 is:

$$\left(8^{\frac{1}{3}}\right) : \left(27^{\frac{1}{3}}\right)$$

Now, calculate the cube roots:

$$(2) : (3)$$

So, the sub-triplicate ratio of 8:27 is 2:3.

Now, let's find the sum of these ratios:

$$2 + 3 = 5$$

So, the sum of the sub-triplicate ratio 2:3 is 5.
Therefore, the correct option is A) 5.

18. Correct option: (B)

The mean proportional (also known as the geometric mean) of two numbers a and b is the square root of their product, i.e., $\sqrt{a \times b}$.
In this case, the mean proportional of 4 and 9 is:

$$\sqrt{4 \times 9} = \sqrt{36} = 6$$

So, the mean proportional of 4 and 9 is 6.
Therefore, the correct option is B) 6.

19. Correct option: (A)

To find the length of AX, we can use the concept of dividing a line segment into a given ratio.
Given:

$$\text{AB} = 14\,\text{cm}$$
$$\text{The ratio in which X divides AB} = 3:4$$

First, find the total number of parts into which AB is divided
$(3 + 4 = 7\,\text{parts})$.
Now, to find the length of AX (which is 3 parts out of 7 parts), we can set up a proportion:

$$\frac{\text{AX}}{\text{AB}} = \frac{3}{7}$$

Now, substitute the value of AB:

$$\frac{\text{AX}}{14} = \frac{3}{7}$$
$$3 \times 14 = 7 \times \text{AX}$$
$$42 = 7 \times \text{AX} \Rightarrow \text{AX} = 6$$

So, the length of AX is 6 cm.
Therefore, the correct option is A) 6.

20. Correct option: (D)

To find the value of x, we can use the concept of compound ratios.
Given the compound ratio of 7:5 and 8:x is 84:60, we can set up the proportion:

$$\left(\frac{7}{5}\right) \times \left(\frac{8}{\text{x}}\right) = \frac{84}{60}$$
$$\frac{7 \times 8}{5 \times \text{x}} = \frac{84}{60} \Rightarrow 60 \times 7 \times 8 = 84 \times 5 \times \text{x} \Rightarrow \frac{60 \times 7 \times 8}{84 \times 5}$$
$$\Rightarrow \text{x} = 8$$

So, x is equal to 8.
Therefore, the correct option is D) 8.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

To find the ratio between 1.2 kg and 840 grams, use the following equation:

$$\text{Ratio} = \frac{1.2\,\text{kg}}{840\,\text{grams}}$$
$$1.2\,\text{kg} \times 1000\,\text{grams/kg} = 1200\,\text{grams}$$

So, the equation becomes:

$$\text{Ratio} = \frac{1200\,\text{grams}}{840\,\text{grams}}$$

Now, simplify the ratio by dividing both terms by their greatest common divisor, which is 120:

$$\text{Ratio} = \frac{\frac{1200}{120}}{\frac{840}{120}}$$

$$\text{Ratio} = \frac{10}{7}$$

Therefore, the correct ratio is B) 10:7.

2. Correct option: (C)

To find the ratio of A to C (A:C), we can use the information provided for A:B and B:C. Given that A:B = 6:7 and B:C = 8:9, we can find the ratio A:C as follows:

$$\text{A:C} = (\text{A:B}) \times (\text{B:C})$$

Substitute the given ratios: $\text{A:C} = (6:7) \times (8:9)$

Now, multiply the corresponding terms:

$$\text{A:C} = \left(\frac{6}{7}\right) \times \left(\frac{8}{9}\right) = \left(\frac{48}{63}\right)$$

To simplify the ratio, divide both terms by their greatest common divisor, which is 3:

$$\text{A:C} = \frac{\frac{48}{3}}{\frac{63}{3}} = \frac{16}{21}$$

Therefore, the correct option is C) 16:21.

3. Correct option: (A)

Given: $\text{Ratio}_1 = 3:4$

$\text{Ratio}_2 = 7:5$

To find the compound ratio, multiply the two ratios together:

$$\text{Compound ratio} = \left(\frac{3}{4}\right) \times \left(\frac{7}{5}\right)$$

$$\text{Compound ratio} = \frac{3 \times 7}{4 \times 5}$$

$$\text{Compound ratio} = \frac{21}{20}$$

So, the compound ratio of 3:4 and 7:5 is 21:20.

Therefore, the correct option is A) 21:20.

4. Correct option: (A)

The ratio of the number of boys to the total number of students in the class can be calculated using the given information:

Number of boys = 32

Number of girls = 18

Total number of students = Number of boys + Number of girls = 50

Now, calculate the ratio:

$$\text{Ratio} = \frac{(\text{Number of boys})}{\text{Total number of students}}$$

$$\text{Ratio} = \frac{32}{32 + 18}$$

$$\text{Ratio} = \frac{32}{50}$$

$$\text{Ratio} = \frac{\frac{32}{2}}{\frac{50}{2}}$$

$$\text{Ratio} = \frac{16}{25}$$

So, the ratio of the number of boys to the total number of students in the class is 16:25.

Therefore, the correct option is A) 16:25.

5. Correct option: (A)

Given ratio: 8:5

Age of younger brother: 12 years and 6 months = 12.5 years

To find the age of the elder brother, use the ratio:

Age of elder brother = $\left(\frac{8}{5}\right) \times$ Age of younger brother

Age of elder brother = $\left(\frac{8}{5}\right) \times 12.5$ years = 20 years

So, the age of the elder brother is 20 years.

The correct option is A) 20.

6. Correct option: (A)

To find the value of $5px + 3qy : 10px + 4qy$, we can use the given ratios:

$$p : q = 2 : 3$$
$$x : y = 4 : 5$$

Let's find values for p and q:

Since $p : q = 2 : 3$, we can say that $p = 2k$ and $q = 3k$, where k is a positive constant.

Now, let's find values for x and y:

Since $x : y = 4 : 5$, we can say that $x = 4m$ and $y = 5m$, where m is a positive constant.

Now, substitute these values into the expressions:

$$5px + 3qy = 5(2k)(4m) + 3(3k)(5m) = 40km + 45km = 85km$$
$$10px + 4qy = 10(2k)(4m) + 4(3k)(5m) = 80km + 60km = 140km$$

Now, calculate the ratio:

$$(5px + 3qy) : (10px + 4qy) = (85km) : (140km)$$

Now, cancel the common factor of km:

$$\left(\frac{85km}{km}\right) : \left(\frac{140km}{km}\right) = 85 : 140$$

To simplify the ratio further, divide both terms by 5:

$$\left(\frac{85}{5}\right) : \left(\frac{140}{5}\right) = 17 : 28$$

So, the value of $5px + 3qy : 10px + 4qy$ is $17 : 28$.

Therefore, the correct answer is A) 17 : 28.

7. Correct option: (B)

Let's denote Krishna's current age as 3x and Balaram's current age as 4x, where x is a positive integer representing the common factor in their ages.

According to the given information, four years ago, the ratio of their ages was 5:7. So, four years ago, Krishna's age was (3x - 4) and Balaram's age was (4x - 4).

Given:

$$\frac{3x - 4}{4x - 4} = \frac{5}{7}$$

Now, cross-multiply:

$$7(3x - 4) = 5(4x - 4)$$
$$21x - 28 = 20x - 20$$
$$21x - 20x - 28 = -20$$
$$x - 28 = -20$$
$$x = 8$$

Now that we know x is 8, we can find Krishna and Balaram's current ages:

$$\text{Krishna's current age } (3x) = 3 \times 8 = 24 \text{ years}$$
$$\text{Balaram's current age } (4x) = 4 \times 8 = 32 \text{ years}$$

So, the present ages of Krishna and Balaram are 24 years and 32 years, respectively.

Therefore, the correct option is B) 24, 32.

8. Correct option: (C)

To find the cost of 18 articles, we can use the concept of proportionality.

If 40 articles cost Rs. 180, then the cost of 1 article can be found by dividing the total cost by the number of articles:

$$\text{Cost of 1 article} = \frac{\text{Rs. } 180}{40} = \text{Rs. } 4.5$$

Now, to find the cost of 18 articles, multiply the cost of 1 article by 18:

$$\text{Cost of 18 articles} = \text{Rs. } 4.5 \times 18 = \text{Rs. } 81$$

So, the cost of 18 articles is Rs. 81.

Therefore, the correct option is C) Rs. 81.

9. Correct option: (A)

To find the third proportional to 75 and 15, we can use the formula:

$$\text{Third Proportional} = \frac{\text{Second Proportional}^2}{\text{First Proportional}}$$

In this case:

First Proportional = 75

Second Proportional = 15

Now, calculate the third proportional:

$$\text{Third Proportional} = \frac{15^2}{75} = \frac{225}{75} = 3$$

So, the third proportional to 75 and 15 is 3.

Therefore, the correct option is A) 3.

10. Correct option: (D)

To ensure that the remainders are in proportion when subtracted from 4, 6, 10, and 16, we need to find a common difference that maintains the proportionality.

Let's denote the number to be subtracted as "x."

The original numbers are: 4, 6, 10, and 16.

After subtracting "x" from each of them, you get the new numbers: (4 - x), (6 - x), (10 - x), and (16 - x).

Now, we want these new numbers to be in proportion. This means:

$$(4 - x) : (6 - x) = (10 - x) : (16 - x)$$

Now, cross-multiply: $(4 - x) \times (16 - x) = (6 - x) \times (10 - x)$

$$64 - 4x - 16x + x^2 = 60 - 10x - 6x + x^2$$
$$64 - 20x + x^2 = 60 - 16x + x^2$$
$$64 - 20x = 60 - 16x$$
$$-20x + 16x = 60 - 64$$
$$-4x = -4$$
$$x = \frac{-4}{-4}$$
$$x = 1$$

So, the number that should be subtracted from 4, 6, 10, and 16 to maintain proportionality among the remainders is 1.

Therefore, the correct option is D) 1.

11. Correct option: (A)

To find the number of hours it takes for 45 pumps to empty the tank, we can set up a proportion based on the work done:

The number of pumps is directly proportional to the rate at which the tank is emptied, assuming all pumps work at the same rate.

So, we can set up the proportion:

$$20 \text{ pumps (12 hours)} = 45 \text{ pumps (x hours)}$$

Now, solve for x:

$$20 \times 12 = 45x$$
$$240 = 45x$$
$$x = \frac{240}{45}$$
$$x = \frac{16}{3}$$

So, it takes $\frac{16}{3}$ hours for 45 pumps to empty the tank.

Therefore, the correct option is A)

12. Correct option: (A)

Given:

$$36 \text{ men build a } 140 - \text{meter wall in 21 days.}$$

First, calculate the work rate of these 36 men:

$$\text{Work rate} = (\text{Total work})/(\text{Total time})$$
$$\text{Work rate} = \frac{140 \text{ meters}}{21 \text{ days}} = \frac{20}{3} \text{ meters per day}$$

Now, we want to find out how many men are needed to build a 50-meter wall in 18 days. Use the same work rate to calculate the number of men required:

$$\text{Number of men} = \frac{\text{Total work}}{\text{Work rate} \times \text{Time}}$$
$$\text{Number of men} = \frac{50 \text{ meters}}{\left[\left(\frac{20}{3}\right) \text{meters per day} \times 18 \text{ days}\right]}$$

Now, calculate: $\text{Number of men} = 15$

So, 15 men are needed to build the 50-meter wall in 18 days.

Therefore, the correct answer is A) 15.

13. Correct option: (B)

Let's find the ratio x : y : z using the given information.

Given:

$$x : y = 1 : 2$$
$$y : z = 2 : 3$$

First, express x in terms of y using the ratio $x : y = 1 : 2$:

$$x = \left(\frac{1}{2}\right) \times y$$

Next, express z in terms of y using the ratio $y : z = 2 : 3$:

$$z = \left(\frac{3}{2}\right) \times y$$

Now, we have:

$$x : y : z = \left(\frac{1}{2}\right)y : y : \left(\frac{3}{2}\right)y$$

Now, we can cancel the common factor "y" in all three terms:

$$x : y : z = \left(\frac{1}{2}\right) : 1 : \left(\frac{3}{2}\right)$$

To make the ratios whole numbers multiply each term by 2 (the least common multiple of the denominators):

$$x : y : z = 1 : 2 : 3$$

Therefore, the correct option is B)

14. Correct option: (C)

To find the value of x in the proportion:

$$16 : 10 = x : 5$$

We can cross-multiply:

$$16 \times 5 = 10 \times x$$
$$80 = 10x$$
$$x = 8$$

So, $x = 8$.

Therefore, the correct option is C) 8.

15. Correct option: (A)

To find an equivalent ratio to 3:5,we can scale it up or down by multiplying both terms by the same number.

In this case, let's scale it up by multiplying both terms by 2:

$$3 \times 2 : 5 \times 2 = 6 : 10$$

So, an equivalent ratio to 3:5 is A) 6:10.

16. Correct option: (A)

Let's calculate Srikanth's current distance to school:

$$\text{Speed} = 12\ \text{km/hr}$$
$$\text{Time} = 20\ \text{minutes} = \frac{20}{60}\ \text{hours (since 1 hour} = 60\ \text{minutes)}$$
$$\text{Distance} = \text{Speed} \times \text{Time}$$
$$\text{Distance} = 12\ \text{km/hr} \times \left(\frac{20}{60}\right)\ \text{hours}$$
$$\text{Distance} = 4\ \text{km}$$

Now, Srikanth wants to reach his school in 15 minutes.

Let's calculate his required speed to cover the same distance in 15 minutes:

$$\text{Distance} = \text{Speed} \times \text{Time}$$
$$4\ \text{km} = \text{Speed} \times \left(\frac{15}{60}\right)\ \text{hours}$$

Now, solve for Speed:

$$\text{Speed} = \frac{4\ \text{km}}{\frac{15}{60}\ \text{hours}}$$
$$\text{Speed} = \frac{4\ \text{km}}{\left(\frac{1}{4}\right)\ \text{hours}}$$
$$\text{Speed} = 16\ \text{km/hr}$$

So, Srikanth should have an average speed of 16 km/hr to reach his school in 15 minutes.

Therefore, the correct option is A) 16 km/hr.

17. Correct option: (B)

We are given that $x : y = 3 : 4$

We have to find the value of

$$x^3 - 7y^2$$

Now, using the first ratio

$$x = 4k, y = 3k$$

Putting these values in the next expression

$$(4k)^3 - 7(3k)^2 => k^2(64k - 63)$$

Putting the value of $k = 1$ (only way to get answer)

We get $1^2(64 \times 1 - 63) => 1$

Therefore, the correct option is B) 1.

18. Correct option: (A)

Let's calculate the compound ratio step by step:

First, simplify the given ratios:

1. The first ratio, $8:5\frac{1}{3}$, can be written as $8:\frac{16}{3} = 24:16 = 3:2$.

2. The second ratio, $3\frac{1}{5}:1\frac{1}{3}$, can be written as

$\frac{16}{5}:\frac{4}{3} = 48:20 = 12:5$

Now, let's find the inverse of the second ratio:

Inverse of $12:5 \Rightarrow 5:12$

Now, calculate the compound ratio by multiplying the simplified ratios:

Compound ratio = $\frac{3\times5}{12\times2} = \frac{5}{8}$

Now comparing the given ratio with this

$$50:x = 5:8 \Rightarrow \frac{50}{x} = \frac{5}{8} \Rightarrow x = 80$$

Then the value of $\frac{x}{10} = 8$

Therefore the correct option is A)

Percentages, Profit, Loss, and Interest

LEVEL – I

1. Correct option: (A)

18% of 1800 is: $\left(\frac{18}{100}\right) \times 1800 = (18 \times 18) = 324.$

So, 18% of 1800 is 324.

Therefore, the correct option is A) 324.

2. Correct option: (A)

To find the value of x, we can set up the equation based on the given information:

$$45\% \text{ of } x = 900$$

First, convert the percentage to a decimal by dividing by 100:

$$\left(\frac{45}{100}\right) \times x = 900$$

$$\left(\frac{45}{100}\right) \times x = 900$$

Multiply both sides by $\frac{100}{45}$ to isolate x:

$$x = (900) \times \left(\frac{100}{45}\right)$$

$$x = 2000$$

So, x is equal to 2000.

Therefore, the correct option is A) 2000.

3. Correct option: (B)

To find the number of students who are not good at math, we can use the given information that 60% of 25 students are good at math.

First, calculate the number of students who are good at math:

$$60\% \text{ of } 25 = \left(\frac{60}{100}\right) \times 25 = 0.6 \times 25 = 15$$

So, 15 students are good at math.

Now, to find the number of students who are not good at math, subtract this number from the total number of students:

Total students - Students good at math = Students not good at math

25 - 15 = 10

Therefore, the correct option is B) 10.

4. Correct option: (C)

To find the value of P, given that P is 30% of Q and Q is 20% of 500, we can follow these steps:

1. First, find the value of Q:

$$Q = \left(\frac{20}{100}\right) \times 500$$

$$Q = \left(\frac{1}{5}\right) \times 500$$

$$Q = 100$$

2. Now that we know Q is 100, find the value of P:

$$P = \left(\frac{30}{100}\right) \times Q$$

$$P = \left(\frac{3}{10}\right) \times 100$$

$$P = 30$$

So, P is equal to 30.

Therefore, the correct option is C) 30.

5. Correct option: (C)

To find the number, we can set up an equation based on the given information:

Let the number be x.

According to the information provided, the number consists of 20 plus 20% of its value, which can be represented as:

$$x = 20 + \left(\frac{20}{100}\right) \times x$$

$$x = 20 + \left(\frac{1}{5}\right) \times x$$

$$\left(\frac{4}{5}\right) \times x = 20$$

$$x = 20 \times \left(\frac{5}{4}\right)$$

$$x = 25$$

So, the number is 25.

Therefore, the correct option is C) 25.

6. Correct option: (B)

To calculate 20% of 50 and 5% of 200 and then add them together:

1. Calculate 20% of 50:

$$20\% \text{ of } 50 = \left(\frac{20}{100}\right) \times 50 = 10$$

2. Calculate 5% of 200:

$$5\% \text{ of } 200 = \left(\frac{5}{100}\right) \times 200 = 10$$

Now, add these results together:

$$10 + 10 = 20$$

So, the sum of 20% of 50 and 5% of 200 is 20.

Therefore, the correct option is B) 20.

7. Correct option: (D)

Let "x" represent the total number of games played.

According to the information provided, the team wins 7 games, which is 35% of the total games. This can be represented as:

$$\left(\frac{35}{100}\right) \times x = 7$$

$$\left(\frac{35}{100}\right) \times x = 7$$

$$x = \frac{7}{\frac{35}{100}} = 7 \times \left(\frac{100}{35}\right)$$

Now, calculate the value of x:
x = 20
So, the total number of games played is 20.
Therefore, the correct option is D) 20.

8. Correct option: (A)

Gain = Selling Price - Cost Price
In this case, the cost price of the book is ₹250, and the selling price is ₹285.
Gain = ₹285 - ₹250 = ₹35
So, the shopkeeper's gain is ₹35
Therefore, the correct option is A) ₹35.

9. Correct option: (A)

To find the cost price of 1 kg of apples, given that it is sold for ₹160 at a loss of 20%, we can use the following formula:

$$\text{Cost Price} = \frac{\text{Selling Price}}{1 - \text{Loss Percentage}}$$

Loss Percentage is 20%, which is equivalent to 0.20 as a decimal.

$$\text{Cost Price} = \frac{₹160}{1 - 0.20}$$

$$\text{Cost Price} = \frac{₹160}{0.80}$$

$$\text{Cost Price} = ₹200$$

So, the cost price of 1 kg of apples is ₹200.
Therefore, the correct option is A) ₹200.

10. Correct option: (A)

$$\text{Total Amount} = \text{Marked Price} + \text{VAT}$$

$$(\text{VAT} = \text{value added tax})$$

$$\text{VAT} = \left(\frac{\text{Percentage}}{100}\right) \times \text{Marked Price}$$

In this case, the VAT is 5% (0.05 as a decimal), and the marked price is ₹300.

$$\text{VAT} = 0.05 * ₹300 = ₹15$$

Now, add the VAT to the marked price to find the total amount:

$$\text{Total Amount} = ₹300 + ₹15 = ₹315$$

So, the total amount the buyer has to pay is ₹315.
Therefore, the correct option is A) ₹315.

11. Correct option: (B)

To calculate the cost of the bag after levying a 13% Value Added Tax (VAT), we can use the following formula: $\text{Cost with VAT} = \text{Original Cost} + \left(\frac{\text{VAT Percentage}}{100}\right) \times \text{Original Cost}$

In this case, the original cost is ₹1800, and the VAT is 13%.

$$\text{Cost with VAT} = ₹1800 + \left(\frac{13}{100}\right) \times ₹1800$$

$$\text{Cost with VAT} = ₹1800 + ₹234$$

Now, calculate the cost of the bag after adding the VAT:
Cost with VAT = ₹2034
So, the cost of the bag after levying 13% VAT is ₹2034.
Therefore, the correct option is B) ₹2034.

12. Correct option: (A)

$$\text{Discount Percentage} = \frac{\text{Discount Amount}}{\text{Marked Price}} \times 100$$

In this case, the Marked Price (M.P.) of the music system is ₹20,000, and the discount given is ₹1,500.

$$\text{Discount Percentage} = \left(\frac{₹1{,}500}{₹20{,}000}\right) \times 100$$

$$\text{Discount Percentage} = \left(\frac{3}{40}\right) \times 100$$

Now, calculate the discount percentage:
Discount Percentage = 7.5%
So, the discount percentage is 7.5%.
Therefore, the correct option is A) 7.5%.

13. Correct option: (D)

In this case, the selling price is ₹13,200, and the discount is 12%.

$$\text{Discount Amount} = \left(\frac{12}{100}\right) \times \text{Marked Price}$$

Now, we need to find the marked price (M.P.). We can rearrange the formula as:

$$\text{Marked Price} = \left(\frac{100}{100 - \text{Discount Percentage}}\right) \times \text{Selling Price}$$

$$\text{Marked Price} = \left(\frac{100}{100 - 12}\right) \times ₹13{,}200$$

$$\text{Marked Price} = \left(\frac{100}{88}\right) \times ₹13{,}200$$

$$\text{Marked Price} = \left(\frac{25}{22}\right) \times ₹13{,}200$$

Marked Price = ₹15,000
Now, we can calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{12}{100}\right) \times ₹15{,}000$$

$$\text{Discount Amount} = ₹1{,}800$$

So, the discount on the moped is ₹1,800.
Therefore, the correct option is D) ₹1,800.

14. Correct option: (A)

15. Correct option: (B)

To calculate the sale price of the LED TV after a 20% discount on the marked price, follow these steps:
1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

$$\text{Discount Amount} = \left(\frac{20}{100}\right) \times ₹1280$$

$$\text{Discount Amount} = ₹256$$

2. Calculate the sale price:

$$\text{Sale Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{Sale Price} = ₹1280 - ₹256$$

$$\text{Sale Price} = ₹1024$$

So, the sale price of the LED TV after allowing a 20% discount on its marked price is ₹1024.
Therefore, the correct option is B) ₹1024.

16. Correct option: (D)

To calculate the sale price of the dinner set after a 12% discount on its marked price, follow these steps:
1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

$$\text{Discount Amount} = \left(\frac{12}{100}\right) \times ₹560$$

$$\text{Discount Amount} = ₹67.20$$

2. Calculate the sale price:

$$\text{Sale Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{Sale Price} = ₹560 - ₹67.20$$

$$\text{Sale Price} = ₹492.80$$

So, the sale price of the dinner set after a 12% discount on its marked price is ₹492.80.
Therefore, the correct option is D) ₹492.80.

17. Correct option: (D)

1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

$$\text{Discount Amount} = \left(\frac{18}{100}\right) \times ₹279$$

$$\text{Discount Amount} = ₹50.22$$

2. Calculate the sale price:

$$\text{Sale Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{Sale Price} = ₹279 - ₹50.22$$

$$\text{Sale Price} = ₹228.78$$

So, the sale price of the laptop computer after an 18% discount on the marked price is ₹228.78.
Therefore, the correct option is D) ₹228.78.

18. Correct option: (C)

1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{List Price}$$

$$\text{Discount Amount} = \left(\frac{12}{100}\right) \times ₹1320$$

$$\text{Discount Amount} = ₹158.40$$

2. Calculate the selling price:

$$\text{Selling Price} = \text{List Price} - \text{Discount Amount}$$

$$\text{Selling Price} = ₹1320 - ₹158.40$$

$$\text{Selling Price} = ₹1161.60$$

So, the selling price of the air conditioner after a 12% off-season discount on the list price is ₹1161.60.
Therefore, the correct option is C) ₹1161.60.

19. Correct option: (C)

1. Simple interest for the first 2 years at a rate of 6%
2. Simple interest for the next 3 years at a rate of 9%
3. Simple interest for the remaining 4 years at a rate of 14%

Let P be the principal amount (the money he borrowed). We'll calculate the interest for each part and then add them up to equal the total interest paid.

1. For the first 2 years at 6%:

$$\text{Interest} = \left(\text{P} \times \frac{6}{100}\right) \times 2$$

2. For the next 3 years at 9%:

$$\text{Interest} = \left(\text{P} \times \frac{9}{100}\right) \times 3$$

3. For the remaining 4 years at 14%:

$$\text{Interest} = \left(\text{P} \times \frac{14}{100}\right) \times 4$$

The total interest paid is given as ₹11400, so we can set up the equation:

$$(\text{Interest for 2 years at 6\%}) + (\text{Interest for 3 years at 9\%}) + (\text{Interest for 4 years at 14\%}) = 11400$$

Now, calculate each part of the interest:

$$\left(2 \times \frac{6}{100} \times \text{P}\right) + \left(3 \times \frac{9}{100} \times \text{P}\right) + \left(4 \times \frac{14}{100} \times \text{P}\right) = 11400$$

Now, simplify and solve for P:

$$\left(\frac{12}{100} \times \text{P}\right) + \left(\frac{27}{100} \times \text{P}\right) + \left(\frac{56}{100} \times \text{P}\right) = 11400$$

$$\left(\frac{95}{100} \times \text{P}\right) = 11400$$

$$\text{P} = \frac{11400 \times 100}{95}$$

$$\text{P} = ₹12{,}000$$

So, Raman borrowed approximately ₹12,000.

The correct answer is C) ₹12,000.

20. Correct option: (A)

To find the time it takes to yield ₹81 as interest on ₹900 at 4.5% per annum of simple interest:

$$\text{Time (T)} = \left(\frac{\text{Interest}}{\text{Principal} \times \text{Rate}}\right) \times 100$$

$$\text{T} = \left(\frac{81}{900 \times 4.5}\right) \times 100$$

$$\text{T} = \left(\frac{81}{4050}\right) \times 100$$

$$\text{T} = 2 \text{ years}$$

So, it will take 2 years for ₹900 to yield ₹81 as interest.

The answer is A) 2 years.

LEVEL – II

1. Correct option: (A)

$$\text{Gain Percentage} = \left[\frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}}\right] \times 100$$

In this case, we have:

$$\text{Selling Price of 10 pens} = \text{Cost Price of 14 pens}$$

Let cost price is CP

The selling price of 10 pens is equal to 14CP

$$\text{Gain Percentage} = \frac{14\text{CP} - 10\text{CP}}{10\text{CP}} \times 100$$

$$\text{Gain Percentage} = \frac{4\text{CP}}{10\text{CP}} \times 100 = 40\%$$

So, the gain percentage is 40%

Therefore, the correct answer is A) 40%.

2. Correct option: (C)

To find the value of x, we can set up a proportion:

$$\left(\frac{\text{x}}{100}\right) = \left(\frac{90}{120}\right)$$

$$\left(\frac{\text{x}}{100}\right) = \left(\frac{3}{4}\right)$$

$$4\text{x} = 300$$

$$x = \frac{300}{4} => 75\ \%$$

So, x is 75%.
Therefore, the answer is C) 75%.

3. Correct option: (B)

$$\text{Gain Percentage} = [\left(\frac{\text{Profit}}{\text{Cost Price}}\right) \times 100]$$

In this case: $\text{Selling Price (SP)} = ₹100$

$$\text{Profit} = ₹20 \text{ (since he gained ₹20)}$$

$$\text{Cost Price (CP)} = \text{SP} - \text{Profit} = ₹100 - ₹20 = ₹80$$

$$\text{Gain Percentage} = [\left(\frac{20}{80}\right) \times 100]$$

$$\text{Gain Percentage} = \left(\frac{1}{4}\right) \times 100$$

$$\text{Gain Percentage} = 25\%$$

So, the man's gain percentage is 25%.
The correct answer is B) 25%.

4. Correct option: (C)

30% can be expressed as: C) 3/10

5. Correct option: (A)

0.05 can be expressed as: A) 5%

6. Correct option: (C)

Initial volume of the cube = a^3
New side length = 1.1a (10% increase)
New volume of the cube = $(1.1a)^3 = 1.331a^3$

$$\text{Increase in volume} = \text{New volume} - \text{Initial volume}$$

$$\text{Increase in volume} = 1.331a^3 - a\text{^}3 = 0.331a^3$$

To find the percentage increase, divide the increase in volume by the initial volume and multiply by 100:

$$\text{Percentage increase} = \left(\frac{0.331a^3}{a^3}\right) \times 100\% = 33.1\%$$

So, the increase in volume when the sides of the cube are increased by 10% is 33.1%.
Therefore, the correct answer is option C) 33.1%.

7. Correct option: (A)

Let's calculate both the simple and compound interest for the same principal amount, rate of interest, and time period.

$$\text{Principal amount (P)} = \text{Let's assume P}$$

$$\text{Rate of interest (R)} = 8\%$$

$$\text{Time (T)} = 3 \text{ years}$$

Now, calculate the simple interest (SI):

$$SI = \frac{P \times 8 \times 3}{100} = \frac{24P}{100} = 0.24P$$

Now, calculate the compound interest (CI):

$$CI = P \times [\left(1 + \left(\frac{8}{100}\right)\right)^3 - 1]$$

$$CI = P \times [(1.08 - 1)(1.08^2 + 1.08 + 1)]$$

$$CI = P \times [(0.08)(1.08^2 + 2.08)]$$

Now, we want to find the ratio of SI to CI:

$$\frac{SI}{CI} = \frac{0.24P}{P \times [(0.08)(1.08^2 + 2.08)]} = \frac{3}{3.2464} = \frac{30000}{32464} = \frac{1875}{2029}$$

Then the correct option is A)

8. Correct option: (B)

$$X\% \text{ of } 60 = 3$$
$$\left(\frac{X}{100}\right) \times 60 = 3$$
$$\left(\frac{X}{100}\right) \times 60 = 3$$
$$\frac{X}{100} = \frac{3}{60}$$
$$\frac{X}{100} = \frac{1}{20}$$

Now, multiply both sides by 100 to isolate X:

$$X = \left(\frac{1}{20}\right) \times 100$$
$$X = 5$$

So, the missing percentage is 5%, which corresponds to option B.

9. Correct option: (D)

1. Initially, the area of the rectangle is given by:

$$\text{Initial Area} = \text{Length} \times \text{Width}$$

2. After increasing the length by 30%, the new length is 1.3 times the initial length.

$$\text{New Length} = 1.3 \times \text{Length}$$

3. After decreasing the width by 20%, the new width is 0.8 times the initial width.

$$\text{New Width} = 0.8 * \text{Width}$$

4. The new area of the rectangle with these changes is:

$$\text{New Area} = \text{New Length} \times \text{New Width}$$
$$\text{New Area} = (1.3 \times \text{Length}) \times (0.8 \times \text{Width})$$
$$\text{Increase in Area} = (\text{Length} \times \text{Width}) \times [(1.3 \times 0.8) - 1]$$

Now, calculate the value inside the brackets:

$$(1.3 \times 0.8) - 1 = 1.04 - 1 = 0.04$$

So, Increase in Area = (Length × Width) × 0.04

This means the area is increased by 4% (0.04 × 100%).

So, the area is increased by 4%, which corresponds to option D.

10. Correct option: (C)

Since M is 20% of D, we can write it as:

$$M = 0.20D$$

We also given that sum of M and D is 60

$$M + D = 60 \Rightarrow 0.2D + D = 60 \Rightarrow D = 50 \Rightarrow M = 10$$

Now, if we divide M by 2, we get: $\frac{M}{2} = \frac{10}{2} = 05$

So, M/2 is 5.

Then the correct option is C)

11. Correct option: (A)

To find the value of x, we can set up the equation:

$$x\% \text{ of } 10 = 10\% \text{ of } 1$$
$$\left(\frac{x}{100}\right) \times 10 = 0.1 \times 1$$
$$\left(\frac{x}{10}\right) = 0.1$$
$$x = 0.1 \times 10$$
$$x = 1$$

Therefore, the correct option is A) 1.

12. Correct option: (C)

10% of 10 is equal to 1.

Therefore, the correct option is C) 1.

13. Correct option: (A)

To find the value of x, we can set up the equation:

$$25\% \text{ of } x = \frac{1}{2}$$

$$0.25 \times x = \frac{1}{2}$$

$$x = \frac{\frac{1}{2}}{0.25}$$

$$x = 2$$

Therefore, the correct option is A) 2.

14. Correct option: (D)

1. Current population: 100,000

2. Population 3 years ago:

$$\text{Population 3 years ago} = \frac{\text{Current population}}{(1 + \text{Growth rate})^3}$$

$$\text{Population 3 years ago} = \frac{100{,}000}{(1 + 0.25)^3} = \frac{100000}{(1.25)^3}$$

3. Population 2 years ago:

$$\text{Population 2 years ago} = \frac{\text{Current population}}{(1 + \text{Growth rate})^2}$$

$$\text{Population 2 years ago} = \frac{100{,}000}{(1.25)^2}$$

Now, find the difference between the two populations:

$$\text{Difference} = \text{Population 2 years ago} - \text{Population 3 years ago}$$

$$\text{Difference} = \frac{100000}{(1.25)^2} - \frac{100{,}000}{(1.25)^3} = \frac{100{,}000}{(1.25)^2}\left(1 - \frac{1}{1.25}\right)$$

$$\Rightarrow \frac{100{,}000}{(1.25)^2} \times \frac{25}{125} = \frac{100{,}000}{5 \times (1.25)^2} = 12800$$

Therefore, the difference in population between 3 years ago and 2 years ago is 12,800, which corresponds to option D) 12,800.

15. Correct option: (C)

To find the rate of interest per annum, we can use the compound interest formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$583.20 = 500\left(1 + \frac{r}{1}\right)^{1 \times 2}$$

Let's simplify further:

$$583.20 = 500(1 + r)^2$$

Now, divide both sides by 500:

$$1.1664 = (1 + r)^2$$

Take the square root of both sides:

$$\sqrt{1.1664} = \sqrt{(1 + r)^2}$$

$$1.08 = 1 + r \Rightarrow r = 0.08 \Rightarrow 8\%$$

So, the rate of interest per annum is 8%.

Then the correct option is C).

16. Correct option: (C)

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

1. The sum amounts to ₹7,000 in 2 years.
2. The same sum amounts to ₹8,000 in 3 years.
For the first case (₹7,000 in 2 years):

$$7{,}000 = P\left(1 + \frac{r}{1}\right)^2$$

For the second case (₹8,000 in 3 years):

$$8{,}000 = P\left(1 + \frac{r}{1}\right)^3$$

Divide the second equation by the first equation:

$$\left(\frac{8{,}000}{7{,}000}\right) = \frac{P(1 + r)^3}{P(1 + r)^2} = 1 + r$$

Simplify:

$$\frac{8}{7} = 1 + r \Rightarrow r = \frac{1}{7}$$

Now, we have found the annual interest rate (r). To find the principal sum (P), we can use the first equation:

$$7{,}000 = P\left(1 + \frac{1}{7}\right)^2$$

$$7{,}000 = P\left(\frac{8}{7}\right)^2$$

$$7{,}000 = P\left(\frac{64}{49}\right) \Rightarrow P = 7000 \times \frac{49}{64} = 5359.375$$

So, the principal sum is ₹5,359.37.
Therefore the correct option is C)

17. Correct option: (A)

To find the principal sum, we can use the formula for the difference between compound interest (CI) and simple interest (SI):

$$\text{Difference} = CI - SI$$

$$\text{Difference} = P \times [(1 + r)^t - 1] - \frac{P \times r \times t}{100}$$

We can set up the equation with the known values:

$$₹530 = P \times [(1 + 0.10)^2 - 1] - (P \times 0.10 \times 2)$$

$$530 = P[1.21 - 1 - 0.20] = P[0.01]$$

$$P = \frac{₹530}{0.01}$$

$$P = ₹53{,}000$$

So, the principal sum is ₹53,000.
Therefore, the correct option is A) ₹53,000.

18. Correct option: (A)

To find the amount to be paid after 2 years when ₹7,500 are borrowed at a compound interest rate of 4% per annum, we can use the formula for compound interest:

$$\text{Amount (A)} = \text{Principal (P)} \times \left(1 + \frac{\text{Rate}}{100}\right)^{\text{Time}}$$

Where:
- P = ₹7,500
- Rate = 4% per annum (0.04 as a decimal)
- Time = 2 years

Plug in these values:

$$A = ₹7{,}500 \times (1 + 0.04)^2$$

$$A = ₹7{,}500 \times (1.04)^2$$

$$A = ₹7{,}500 \times 1.0816$$
$$A = ₹8{,}112$$

So, after 2 years, the amount to be paid is ₹8,112.
Therefore, the correct option is A) ₹8,112.

19. Correct option: (A)

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$
$$₹\,9261 = ₹\,8000\left(1 + \frac{0.05}{1}\right)^{t}$$
$$₹\,9261 = ₹\,8000(1 + 0.05)^{t}$$
$$1.157625 = (1.05)^{t}$$
$$(1.05)^3 = 1.157625 = (1.05)^{t} \Rightarrow t = 3$$

So, it will take 3 years for ₹ 8000 to grow to ₹ 9261 at a rate of 5% per annum compounded annually.
The correct option is A) 3 years.

20. Correct option: (A)

To find the compound interest at the end of three years on an amount of ₹20,000 at a rate of 10% per annum, we can use the same compound interest formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$
$$A = 20000 \times \left(1 + \frac{0.10}{1}\right)^{1 \times 3}$$
$$A = 20000 \times (1 + 0.10)^3$$
$$A = 20000 \times (1.10)^3$$
$$A = 20000 \times 1.331$$
$$A = ₹26{,}620$$

Now, calculate the compound interest:

$$\text{Compound Interest (C.I.)} = A - P$$
$$\text{C.I.} = ₹26{,}620 - ₹20{,}000$$
$$\text{C.I.} = ₹6{,}620$$

So, the compound interest at the end of three years on ₹20,000 at a rate of 10% per annum is ₹6,620.
The correct option is A) ₹6,620.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

12% of 3600 is: $\left(\frac{12}{100}\right) \times 3600 = (12 \times 36) = 432.$

So, 12% of 3600 is 432.
Therefore, the correct option is B) 432.

2. Correct option: (D)

22% of 4000 is: $\left(\frac{22}{100}\right) \times 4000 = (22 \times 40) = 880.$

So, 22% of 4000 is 880.
Therefore, the correct option is D) 880.

3. Correct option: (C)

To find the value of x when it's 32% of y, and y is given as 1200, we can use the following calculation:

$$x = \left(\frac{32}{100}\right) \times y$$

Substitute the value of y, which is 1200:

$$x = \left(\frac{32}{100}\right) \times 1200$$

Now, calculate:

$$x = (0.32) \times 1200$$
$$x = 384$$

So, x is equal to 384.
Therefore, the correct option is C) 384.

4. Correct option: (B)

To find Bhavana's investment, we can use the following calculation:
Let's represent her investment as "I."
Bhavana earns 24% of her investment, which is equal to 480.
So, we can write this as an equation:

$$0.24 \times I = 480$$

Now, to find I, divide both sides by 0.24:

$$I = \frac{480}{0.24}$$
$$I = 2000$$

So, Bhavana's investment is 2000.
Therefore, the correct option is B) 2000.

5. Correct option: (B)

Here's how we can calculate the percentage improvement in Rahul's performance:
1. Calculate the initial score as a percentage of 100:
Percentage for the first internal exam = $\left(\frac{70}{100}\right) \times 100 = 70\%$
2. Calculate the second score as a percentage of 100:
Percentage for the second internal exam = $\left(\frac{90}{100}\right) \times 100 = 90\%$
3. Calculate the improvement in percentage:
Improvement percentage = (Percentage for the second internal exam - Percentage for the first internal exam)
Improvement percentage = (90% - 70%) = 20%
So, the percentage improvement in Rahul's performance is 20%.
Therefore, the correct option is B) 20%.

6. Correct option: (A)

To find the initial number of oranges the vendor had, we can set up the following equation:
Let x be the initial number of oranges.
After selling 35% of the oranges, the vendor has 260 oranges left. This can be represented as:

$$x - 0.35x = 260$$

Now, simplify the equation:

$$0.65x = 260$$
$$x = 260 / 0.65$$
$$x = 400$$

So, the initial number of oranges the vendor had was 400.
Therefore, the correct option is A) 400.

7. Correct option: (C)

To find 30% of one-fourth of the number, we can follow these steps:
1. First, find the number based on the information given:
40% of a number is 80, so let "x" be the number:

$$(40/100) \times x = 80$$

Now, solve for x:

$$\left(\frac{2}{5}\right) \times x = 80$$

$$x = 80 \times \left(\frac{5}{2}\right)$$

$$x = 200$$

So, the number is 200.

2. Next, find one-fourth of the number: $\text{One} - \text{fourth of } 200 \text{ is } \left(\frac{1}{4}\right) \times 200 = 50.$

3. Finally, find 30% of one-fourth of the number: $(30/100) \times 50 = 0.3 \times 50 = 15$

So, 30% of one-fourth of the number is 15.

Therefore, the correct option is C) 15.

8. Correct option: (A)

To find the percentage of natural numbers from 1 to 60 that have '5' or '0' in the units place, we need to count how many numbers meet this criteria and then express that count as a percentage of the total count.

Numbers with '5' or '0' in the units place are: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60. (multiple of 5)

There are 12 such numbers.

Now, calculate the percentage:

$$\left(\frac{12}{60}\right) \times 100\% = \left(\frac{1}{5}\right) \times 100\% = 20\%$$

So, the percentage of natural numbers from 1 to 60 that have '5' or '0' in the units place is 20%.

Therefore, the correct option is A) 20%.

9. Correct option: (B)

To find the maximum marks, we can set up an equation based on the given information:

Let M be the maximum marks.

The student needs to secure 40% of M to pass.

Given that he gets 150 marks and fails by 30 marks, we can represent this as:

$$40\% \text{ of } M - 30 = 150$$

$$\left(\frac{40}{100}\right) \times M - 30 = 150$$

$$\left(\frac{2}{5}\right) \times M - 30 = 150$$

Now, add 30 to both sides of the equation:

$$\left(\frac{2}{5}\right) \times M = 150 + 30$$

$$\left(\frac{2}{5}\right) \times M = 180$$

$$M = \left(\frac{5}{2}\right) \times 180$$

$$M = 450$$

So, the maximum marks are 450.

Therefore, the correct option is B) 450.

10. Correct option: (B)

Let x be the number.

According to the information provided, 26% of the number is equal to 65, which can be represented as: $\left(\frac{26}{100}\right) \times x = 65$

Now, simplify the equation: $\left(\frac{26}{100}\right) \times x = 65$

$$x = \frac{65}{\frac{26}{100}} = 65 \times \left(\frac{100}{26}\right)$$

$$x = 250$$

So, the number is 250.
Therefore, the correct option is B) 250.

11. Correct option: (C)

Loss = Cost Price - Selling Price
In this case, the cost price of the bicycle is ₹2200, and the selling price is ₹1980.

$$\text{Loss} = ₹2200 - ₹1980 = ₹220$$

So, Rohan incurred a loss of ₹220.
Therefore, the correct option is C) ₹220.

12. Correct option: (B)

Gain = Selling Price - Cost Price
In this case, the cost price of the book is ₹900, and the selling price is ₹920.

$$\text{Gain} = ₹920 - ₹900 = ₹20$$

So, Rishi's gain is ₹20.
Therefore, the correct option is B) ₹20.

13. Correct option: (C)

To calculate the loss percentage, you can use the following formula:

$$\text{Loss Percentage} = \left[\frac{\text{Cost Price} - \text{Selling Price}}{\text{Cost Price}}\right] \times 100\%$$

In this case, the cost price of the car is ₹8,00,000, and the selling price is ₹6,40,000.

$$\text{Loss Percentage} = \left[\frac{8,00,000 - 6,40,000}{8,00,000}\right] \times 100\%$$

$$\text{Loss Percentage} = \left[\frac{1,60,000}{8,00,000}\right] \times 100\%$$

$$\text{Loss Percentage} = (0.2) \times 100\%$$

$$\text{Loss Percentage} = 20\%$$

So, the loss percentage is 20%.
Therefore, the correct option is C) 20%.

14. Correct option: (B)

To calculate the gain percentage, we can use the following formula:

$$\text{Gain Percentage} = [\frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}}] \times 100\%$$

In this case, the cost price of 10 pencils is ₹50, and the selling price is ₹60.

$$\text{Gain Percentage} = \left[\frac{60 - 50}{50}\right] \times 100\%$$

$$\text{Gain Percentage} = \left(\frac{10}{50}\right) \times 100\%$$

$$\text{Gain Percentage} = \left(\frac{1}{5}\right) \times 100\%$$

$$\text{Gain Percentage} = 20\%$$

So, the gain percentage is 20%.
Therefore, the correct option is B) 20%.

15. Correct option: (D)

To calculate John's gain percentage, we can use the following formula:

$$\text{Gain Percentage} = \left[\frac{\text{Selling Price} - \text{Total Cost}}{\text{Total Cost}}\right] \times 100\%$$

In this case, the total cost includes the cost of the second-hand car and the repair expenses:

$$\text{Total Cost} = \text{Cost of Car} + \text{Repair Expenses}$$
$$\text{Total Cost} = ₹1{,}80{,}000 + ₹20{,}000$$
$$\text{Total Cost} = ₹2{,}00{,}000$$

Now, we can calculate the gain percentage:

$$\text{Gain Percentage} = \left[\frac{₹2{,}20{,}000 - ₹2{,}00{,}000}{₹2{,}00{,}000}\right] \times 100\%$$
$$\text{Gain Percentage} = \left(\frac{₹20{,}000}{₹2{,}00{,}000}\right) \times 100\%$$
$$\text{Gain Percentage} = \left(\frac{1}{10}\right) \times 100\%$$
$$\text{Gain Percentage} = 10\%$$

So, John's gain percentage is 10%.
Therefore, the correct option is D) 10%.

16. Correct option: (B)

To calculate the selling price with a 12% gain, we can use the following formula:

$$\text{Selling Price} = \text{Cost Price} + \text{Gain}$$

First, find the gain:

$$\text{Gain} = \left(\frac{12}{100}\right) \times \text{Cost Price}$$
$$\text{Gain} = \left(\frac{12}{100}\right) \times ₹20{,}000$$
$$\text{Gain} = ₹2{,}400$$

Now, add the gain to the cost price to find the selling price:

$$\text{Selling Price} = ₹20{,}000 + ₹2{,}400$$
$$\text{Selling Price} = ₹22{,}400$$

So, the selling price with a 12% gain is ₹22,400.
Therefore, the correct option is B) ₹22,400

17. Correct option: (A)

To calculate the selling price after a 15% loss, we can use the following formula:

$$\text{Selling Price} = \text{Cost Price} - \text{Loss}$$

First, find the loss:

$$\text{Loss} = \left(\frac{15}{100}\right) \times \text{Cost Price}$$
$$\text{Loss} = \left(\frac{15}{100}\right) \times ₹3{,}200$$
$$\text{Loss} = ₹480$$

Now, subtract the loss from the cost price to find the selling price:

$$\text{Selling Price} = ₹3{,}200 - ₹480$$
$$\text{Selling Price} = ₹2{,}720$$

So, Ganesh sold the watch for ₹2,720.
Therefore, the correct option is A) ₹2,720.

18. Correct option: (A)

$$\text{Gain Percentage} = \left[\frac{\text{Selling Price} - \text{Total Cost}}{\text{Total Cost}}\right] \times 100\%$$

Total Cost includes the cost of the almirah and the repair expenses:

$$\text{Total Cost} = \text{Cost of Almirah} + \text{Repair Expenses}$$
$$\text{Total Cost} = ₹6{,}250 + ₹375$$
$$\text{Total Cost} = ₹6{,}625$$

Now, we can calculate the gain percentage:

$$\text{Gain Percentage} = \left[\frac{₹6{,}890 - ₹6{,}625}{₹6{,}625}\right] \times 100\%$$

$$\text{Gain Percentage} = \left(\frac{₹265}{₹6{,}625}\right) \times 100\%$$

$$\text{Gain Percentage} = \left(\frac{1}{25}\right) \times 100\%$$

$$\text{Gain Percentage} = 4\%$$

So, Arnav's gain percentage is 4%.
Therefore, the correct option is A) 4%.

19. Correct option: (C)

1. Convert the loss percentage to a decimal:

$$\text{Loss Percentage} = 6\frac{2}{3}\% = \frac{20}{3}\%$$

$$\text{Loss Percentage} = \frac{20}{3}\%$$

2. Calculate the loss amount:

$$\text{Loss Amount} = \text{Loss Percentage} \times \text{Cost Price}$$

$$\text{Loss Amount} = \frac{20}{3} \times \frac{₹720}{100}$$

$$\text{Loss Amount} = ₹48$$

3. Subtract the loss amount from the cost price to find the selling price:

$$\text{Selling Price} = \text{Cost Price} - \text{Loss Amount}$$

$$\text{Selling Price} = ₹720 - ₹48$$

$$\text{Selling Price} = ₹672$$

So, Rashmi sold the calculator for ₹672.
Therefore, the correct option is C) ₹672.

20. Correct option: (D)

$$\text{Cost Price} = \frac{\text{Selling Price}}{1 - \text{Loss Percentage}}$$

In this case, the selling price is ₹987, and the loss percentage is 6% (0.06 as a decimal).

$$\text{Cost Price} = \frac{₹987}{1 - 0.06}$$

$$\text{Cost Price} = \frac{₹987}{0.94}$$

Now, calculate the cost price:

$$\text{Cost Price} = \frac{₹987}{0.94} = ₹1{,}050$$

So, the cost price is ₹1,050.
Therefore, the correct option is D) ₹1,050.

21. Correct option: (B)

If the selling price (SP) of an item is equal to $1\frac{1}{3}$ (or$\frac{4}{3}$) times its cost price (CP), then the gain can be calculated as follows:

$$\text{Gain Percentage} = \left[\frac{\text{SP} - \text{CP}}{\text{CP}}\right] \times 100\%$$

In this case, SP is $\frac{4}{3}$ times CP, so $\text{SP} = (4/3) \times \text{CP}$.
Now, plug this value into the formula:

$$\text{Gain Percentage} = \left[\frac{\left(\frac{4}{3}\right) \times \text{CP} - \text{CP}}{\text{CP}}\right] \times 100\%$$

$$\text{Gain Percentage} = \left[\frac{\left(\frac{4}{3} - 1\right) \times \text{CP}}{\text{CP}}\right] \times 100\%$$

$$\text{Gain Percentage} = \left[\frac{\left(\frac{1}{3}\right) \times \text{CP}}{\text{CP}}\right] \times 100\%$$

$$\text{Gain Percentage} = \left(\frac{1}{3}\right) * 100\%$$

$$\text{Gain Percentage} = 33.33\%$$

So, the gain percentage is 33.33%.
Therefore, the correct option is B) 33.33%.

22. Correct option: (D)

To calculate the total amount the buyer has to pay with a 15% value-added tax (VAT) on a marked price of ₹500, we can use the following formula:

$$\text{Total Amount} = \text{Marked Price} + \text{VAT}$$

$$\text{VAT} = \left(\frac{\text{Percentage}}{100}\right) \times \text{Marked Price}$$

In this case, the VAT is 15% (0.15 as a decimal), and the marked price is ₹500.

$$\text{VAT} = 0.15 \times ₹500 = ₹75$$

Now, add the VAT to the marked price to find the total amount:
Total Amount = ₹500 + ₹75 = ₹575
So, the total amount the buyer has to pay is ₹575.
Therefore, the correct option is D) ₹575.

23. Correct option: (A)

To calculate the selling price (SP) when a buyer gets a 20% discount on a marked price (MP) of ₹150, we can use the following formula:

$$\text{SP} = \text{MP} - \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{MP}$$

In this case, the discount is 20%, which means the discount percentage is $\frac{20}{100}$ or 0.20 as a decimal. The marked price is ₹150.

$$\text{SP} = ₹150 - (0.20 \times ₹150)$$

$$\text{SP} = ₹150 - ₹30$$

Now, subtract the discount amount from the marked price to find the selling price:
SP = ₹120
So, the selling price of the good is ₹120.
Therefore, the correct option is A) ₹120.

24. Correct option: (B)

To find the original price of the designer pair of shoes before the 6% Goods and Services Tax (GST) was added, we can use the following formula:

$$\text{Original Price} = \frac{\text{Selling Price}}{1 + \text{GST Percentage}}$$

In this case, the selling price (inclusive of GST) is ₹210, and the GST is 6%.

$$\text{Original Price} = \frac{₹210}{1 + 0.06}$$

$$\text{Original Price} = \frac{₹210}{1.06}$$

Now, calculate the original price:
Original Price ≈ ₹198.11
So, the original price of the designer pair of shoes is approximately ₹198.11.
Therefore, the closest option to this value is B) ₹198.

25. Correct option: (C)

1. Calculate the total sales made by the 30 customers:

$$\text{Total Sales} = \text{Number of Customers} \times \text{Cost per Customer}$$

$$\text{Total Sales} = 30 \times ₹120 = ₹3600$$

2. Calculate the commission received by the sales manager (5% of total sales):

$$\text{Commission} = \left(\frac{\text{Commission Percentage}}{100}\right) \times \text{Total Sales}$$

$$\text{Commission} = \left(\frac{5}{100}\right) \times ₹3600 = ₹180$$

So, the combined commission received by the sales manager is ₹180.

Therefore, the correct option is C) ₹180.

26. Correct option: (B)

To calculate the sale price of the box with a 10% discount, we can use the following formula:

$$\text{Sale Price} = \text{M.P.} - \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{M.P.}$$

In this case, the M.P. (Marked Price) of the box is ₹10, and the discount is 10%.

$$\text{Sale Price} = ₹10 - \left(\frac{10}{100}\right) \times ₹10$$

$$\text{Sale Price} = ₹10 - ₹1$$

Now, calculate the sale price: Sale Price = ₹9

So, the sale price of the box with a 10% discount is ₹9.

Therefore, the correct option is B) ₹9.

27. Correct option: (B)

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

In this case, the Marked Price (M.P.) of the pen is ₹20, and the discount is 15%.

$$\text{Discount Amount} = \left(\frac{15}{100}\right) \times ₹20$$

$$\text{Discount Amount} = \left(\frac{3}{20}\right) \times ₹20$$

$$\text{Discount Amount} = ₹3$$

So, the discount allowed on the pen is ₹3.

Therefore, the correct option is B) ₹3.

28. Correct option: (A)

In this case, the selling price is ₹680, and the discount is 15%.

$$\text{Discount Amount} = \left(\frac{15}{100}\right) \times \text{Marked Price}$$

Now, we can rearrange the formula to calculate the M.P.:

$$\text{Marked Price} = \frac{\text{Selling Price}}{1 - \text{Discount Percentage}}$$

$$\text{Marked Price} = \frac{₹680}{1 - 0.15}$$

$$\text{Marked Price} = \frac{₹680}{0.85}$$

$$\text{Marked Price} = ₹800$$

So, the marked price of the jacket is approximately ₹800.

Therefore, the correct option is A) ₹800.

29. Correct option: (A)

$$\text{Original Price} = \frac{\text{Total Price}}{1 + \left(\frac{\text{Sales Tax Rate}}{100}\right)}$$

In this case, the total price of the cycle with the sales tax is ₹660, and the sales tax rate is 10%.

$$\text{Original Price} = \frac{₹660}{1 + \left(\frac{10}{100}\right)}$$

$$\text{Original Price} = \frac{₹660}{1 + 0.10}$$

$$\text{Original Price} = ₹660 / 1.10$$

$$\text{Original Price} = ₹600$$

So, the original price of the cycle before the sales tax was added is ₹600.
Therefore, the correct option is A) ₹600.

30. Correct option: (C)

1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

$$\text{Discount Amount} = \left(\frac{5}{100}\right) \times ₹46,000$$

$$\text{Discount Amount} = ₹2,300$$

2. Calculate the discounted price (the price after the 5% discount):

$$\text{Discounted Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{Discounted Price} = ₹46,000 - ₹2,300$$

$$\text{Discounted Price} = ₹43,700$$

3. Calculate the sales tax amount:

$$\text{Sales Tax Amount} = \left(\frac{\text{Sales Tax Rate}}{100}\right) \times \text{Discounted Price}$$

$$\text{Sales Tax Amount} = \left(\frac{10}{100}\right) \times ₹43,700$$

$$\text{Sales Tax Amount} = ₹4,370$$

4. Calculate the final amount Nazim has to pay:

$$\text{Final Amount} = \text{Discounted Price} + \text{Sales Tax Amount}$$

$$\text{Final Amount} = ₹43,700 + ₹4,370$$

$$\text{Final Amount} = ₹48,070$$

So, the amount Nazim has to pay to purchase the motorcycle, including the discount and sales tax, is ₹48,070.
Therefore, the correct option is C) ₹48,070.

31. Correct option: (A)

1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

$$\text{Discount Amount} = \left(\frac{15}{100}\right) \times ₹475$$

$$\text{Discount Amount} = ₹71.25$$

2. Calculate the sale price:

$$\text{Sale Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{Sale Price} = ₹475 - ₹71.25$$

$$\text{Sale Price} = ₹403.75$$

So, the sale price of the cell phone after a 15% discount on its marked price is ₹403.75.
Therefore, the correct option is A) ₹403.75.

32. Correct option: (B)

1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

$$\text{Discount Amount} = \left(\frac{9}{100}\right) \times ₹260$$

$$\text{Discount Amount} = ₹23.40$$

2. Calculate the sale price:

$$\text{Sale Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{Sale Price} = ₹260 - ₹23.40$$

$$\text{Sale Price} = ₹236.60$$

So, the sale price of the toy after a 9% discount on its marked price is ₹236.60.
Therefore, the correct option is B) ₹236.60.

33. Correct option: (B)

1. Calculate the discount amount:

$$\text{Discount Amount} = \left(\frac{\text{Discount Percentage}}{100}\right) \times \text{Marked Price}$$

$$\text{Discount Amount} = \left(\frac{20}{100}\right) \times ₹1675$$

$$\text{Discount Amount} = ₹335$$

2. Calculate the sale price:

$$\text{Sale Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{Sale Price} = ₹1675 - ₹335$$

$$\text{Sale Price} = ₹1340$$

So, the sale price of the refrigerator after a 20% discount on the marked price is ₹1340.
Therefore, the correct option is B) ₹1340.

34. Correct option: (C)

To calculate the time it took for Sam to return the money, you can use the formula for simple interest:

$$\text{Simple Interest (SI)} = \frac{\text{Principal Amount} \times \text{Rate} \times \text{Time}}{100}$$

Given:

$$\text{Principal Amount (P)} = ₹12{,}000$$

$$\text{Simple Interest (SI)} = ₹15{,}600 - ₹12{,}000 = ₹3{,}600$$

$$\text{Rate (R)} = 6\% \text{ per annum}$$

We need to find the time (T). Rearrange the formula to solve for time:

$$\text{Time (T)} = (\text{SI} * 100) / (\text{P} * \text{R})$$

$$\text{Time (T)} = (₹3{,}600 * 100) / (₹12{,}000 * 6)$$

$$\text{Time (T)} = (360{,}000) / (72{,}000)$$

$$\text{Time (T)} = 5 \text{ years}$$

So, it took Sam 5 years to return the borrowed ₹12,000.
The correct answer is C) 5 years.

35. Correct option: (B)

Given:

$$\text{Simple Interest (SI)} = ₹9{,}600$$

$$\text{Rate (R)} = 16\% \text{ per annum}$$

$$\text{Time (T)} = 5 \text{ years}$$

We need to find the Principal Amount (P). Rearrange the formula to solve for P:

$$\text{Principal Amount (P)} = \frac{\text{SI} \times 100}{\text{R} \times \text{T}}$$

$$\text{Principal Amount (P)} = \frac{₹9{,}600 \times 100}{16 \times 5}$$

$$\text{Principal Amount (P)} = \frac{₹960{,}000}{80}$$

$$\text{Principal Amount (P)} = ₹12{,}000$$

So, Kabir took a loan of ₹12,000.
The correct answer is B) ₹12,000.

36. Correct option: (A)

Let's assume the rate of interest for SBI is 'R' percent and the rate of interest for ICICI is also 'R' percent.

Given:

$$\text{Principal amount (P1)} = ₹500 \text{ (for SBI)}$$
$$\text{Principal amount (P2)} = ₹300 \text{ (for ICICI)}$$
$$\text{Time (T1)} = 2 \text{ years (for SBI)}$$
$$\text{Time (T2)} = 4 \text{ years (for ICICI)}$$
$$\text{Total interest received from both banks} = ₹220$$

We can use the formula for simple interest to set up two equations:

1. For SBI:

$$SI1 = \frac{P1 \times R \times T1}{100}$$
$$SI1 = \frac{500 \times R \times 2}{100}$$

2. For ICICI:

$$SI2 = \frac{P2 \times R \times T2}{100}$$
$$SI2 = \frac{300 \times R \times 4}{100}$$

According to the given information, the total interest received is ₹220:

$$SI1 + SI2 = 220$$

Substitute the expressions for SI1 and SI2:

$$\frac{500 \times R \times 2}{100} + \frac{300 \times R \times 4}{100} = 220$$

Now, solve for 'R':

$$(10R) + (12R) = 220$$
$$22R = 220$$
$$R = \frac{220}{22}$$
$$R = 10$$

So, the rate of interest for both SBI and ICICI is 10%.

The correct answer is A) 10%.

37. Correct option: (₹900)

Let's find the rate of interest for the first investment using the formula for simple interest:

$$\text{Simple Interest (SI)} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

We know the following:

$$\text{Simple Interest (SI)} = ₹2600 - ₹2000 = ₹600 \text{ (the interest earned)}$$
$$\text{Principal (P)} = ₹2000$$
$$\text{Time (T)} = 5 \text{ years}$$

We need to find the Rate (R). Rearrange the formula to solve for R:

$$\text{Rate (R)} = \frac{SI \times 100}{P \times T}$$
$$\text{Rate (R)} = \frac{600 \times 100}{2000 \times 5} = 6\%$$

Now that we know the rate for the first investment is 6%, we can calculate the interest for the second investment with a rate 3% more, which is 9%.

$$\text{Principal (P)} = ₹2000$$
$$\text{Time (T)} = 5 \text{ years}$$
$$\text{Rate (R)} = 6\% + 3\% = 9\%$$
$$SI = (2000 \times 9 \times 5)/100 = ₹900$$

So, if Ram had invested ₹2000 at a rate of 9% for 5 years, he would have earned ₹900 in the same time.

38. Correct option: (B)

Let's denote the principal amount as "P" and the rate of interest as "R%."

Given:

1. Simple Interest (SI) for 2 years = ₹600
2. Compound Interest (CI) for 2 years = ₹645

Compound interest: $CI = P\left[\left(1 + \frac{R}{100}\right)^{T} - 1\right]$

Let's first find the rate of interest (R) using the given simple interest:

Simple interest

$$600 = \frac{P \times R \times 2}{100} \Rightarrow 300 = \frac{P \times R}{100}$$

Take $\frac{R}{100} = x \Rightarrow 300 = \frac{P \times R}{100} = Px \Rightarrow P = \frac{300}{x}$

Now, let's find the compound interest formula for 2 years and equate it to ₹645.

$$645 = P\left[\left(1 + \frac{R}{100}\right)^{2} - 1\right]$$

Putting the value of P

$$\Rightarrow 645 = \frac{300}{x}[(1 + x)^2 - 1]$$

Solving the equation

$$645 = 300\left[\frac{1 + x^2 + 2x - 1}{x}\right] = 300(x + 2)$$

$$\frac{645}{300} = x + 2 \Rightarrow x = \frac{645 - 600}{300} = \frac{45}{300} = \frac{15}{100}$$

Putting back the value of $x = \frac{R}{100}$

$$\frac{15}{100} = \frac{R}{100} \Rightarrow R = 15\%$$

Therefore the correct option is B)

39. Correct option: (C)

Let's denote the principal amount as "P" and the rate of interest as "R%."

Given: $SI = \left(\frac{9}{25}\right) \times P, \text{and } T = 4 \text{ years.}$

So, we can write the equation as:

$$\left(\frac{9}{25}\right) \times P = \frac{P \times R \times 4}{100}$$

Now, let's solve for R:

$$\left(\frac{9}{25}\right) \times 100 = 4 \times R \Rightarrow R = \frac{9}{25} \times \frac{1}{4} = \frac{9}{100} = 9\%$$

So, the annual rate of interest is 9%.

Therefore, option C) 9% is the correct answer.

40. Correct option: (A)

Let principal amount be P then,

Total Amount = 6P

Interest = Total Amount - Principal = $6P - P = 5P$

$$5P = \frac{P \times R \times 40}{100}$$

$$500P = P \times R \times 40$$

$$R = \frac{500}{40} = 12.5\ \%$$

So, the rate of interest is 12.5%.

Therefore, option A) 12.5% is the correct answer.

41. Correct option: (B)

1. Calculate the total cost of the T.V. when paid in cash:
   Total Cost = ₹20,000
2. Calculate the down payment:
   Down Payment = ₹4,000
3. Calculate the remaining amount to be paid in installments:
   Remaining Amount = ₹20,000 - ₹4,000 = ₹16,000
4. Calculate the annual interest on the remaining amount:

$$\text{Annual Interest} = \frac{\text{Rate of Interest} \times \text{Remaining Amount}}{100}$$

$$\text{Annual Interest} = \frac{8\% \times ₹16{,}000}{100} = ₹1{,}280$$

Monthly Interest = Annual Interest / 12
Monthly Interest = ₹1,280 / 12 = ₹106.67 (approximately)
Monthly Installment = (₹16,000 / 4) + ₹106.67
Monthly Installment = ₹4,000 + ₹106.67
Monthly Installment = ₹4,106.67

So, the monthly installment for the T.V. is approximately ₹4,106.67. Therefore, option B) ₹4,106.67 is the correct answer.

42. Correct option: (B)

$$\text{Loss Percentage} = \left[\frac{\text{Cost Price} - \text{Selling Price}}{\text{Cost Price}}\right] \times 100$$

In this case, we have:

$$\text{Selling Price of 18 oranges} = \text{Cost Price of 16 oranges}$$

Let cost price is CP
The selling price of 18 oranges is equal to 16CP

$$\text{Loss Percentage} = \frac{18CP - 16CP}{18CP} \times 100$$

$$\text{Loss Percentage} = \frac{2CP}{18CP} \times 100 = \frac{100}{9}\% = 11\frac{1}{9}\%$$

So, the Loss percentage is $11\frac{1}{9}\%$

Therefore, the correct answer is B) $11\frac{1}{9}\%$.

43. Correct option: (D)

Let's denote Saritha's monthly salary as "S."
According to the information provided, she spends 65% of her salary and saves the rest, which is 100% - 65% = 35% of her salary.

$$35\% \text{ of } S = ₹2940$$

$$0.35S = ₹2940 \Rightarrow S = \frac{2940}{0.35} = 8400$$

So, Saritha's monthly salary is ₹8400.
The correct answer is D) 8400.

44. Correct option: (B)

Let's denote the total number of students in the class as "T." According to the information given:

$$\text{Girls} = 60\% \text{ of } T$$

$$\text{Boys} = 18$$

$$0.60T = \text{Girls}$$

We also know that there are 18 boys.
So, the total number of students in the class is: $T = \text{Girls} + \text{Boys}$

$$T = 0.60T + 18$$

Now, subtract 0.60T from both sides:

$$0.40T = 18$$

$$T = \frac{18}{0.40}$$

$$T = 45$$

So, the total number of students in the class is 45.

The correct answer is B) 45.

45. Correct option: (B)

$$\text{Gain Percentage} = \left[\frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}}\right] \times 100$$

In this case:

$$\text{Cost Price (CP)} = ₹80$$

$$\text{Selling Price (SP)} = ₹100$$

$$\text{Gain Percentage} = \left[\frac{100 - 80}{80}\right] \times 100$$

$$\text{Gain Percentage} = \left(\frac{20}{80}\right) \times 100$$

$$\text{Gain Percentage} = \left(\frac{1}{4}\right) \times 100$$

$$\text{Gain Percentage} = 25\%$$

So, the man's gain percentage is 25%.

The correct answer is B) 25%.

46. Correct option: (A)

$$\text{Loss Percentage} = \left[\frac{\text{Cost Price} - \text{Selling Price}}{\text{Cost Price}}\right] \times 100$$

In this case:

$$\text{Cost Price (CP)} = ₹120$$

$$\text{Selling Price (SP)} = ₹105$$

Now, plug these values into the formula:

$$\text{Loss Percentage} = \left[\frac{120 - 105}{120}\right] \times 100$$

$$\text{Loss Percentage} = \left(\frac{15}{120}\right) \times 100$$

$$\text{Loss Percentage} = \left(\frac{1}{8}\right) \times 100$$

$$\text{Loss Percentage} = 12.5\%$$

So, the loss percentage is 12.5%.

Therefore the correct option is A) $12\frac{1}{2}\%$

47. Correct option: (A)

Let's denote the cost price (CP) of the racket as "x."

Given that the shopkeeper gains 10% when selling it for ₹198, we can use the formula for calculating the selling price (SP) when there is a profit:

$$SP = CP + \text{Profit}$$

Then

$$SP = x + 0.10x$$

$$SP = 1.10x$$

$$1.10x = ₹198$$

$$x = \frac{₹198}{1.10}$$

$$x = ₹180$$

So, the cost price of the racket is ₹180.

The correct answer is A) ₹180.

48. Correct option: (A)

Let's denote the man's outlay (cost price) for the jug as "x."

From the information we can say $SP1 = \left(\frac{6}{7}\right) \times x$

Now, we are given that the jug is sold for ₹189. We can calculate the gain by finding the difference between the selling price (SP2) and the cost price (x):

$$Gain = SP2 - x$$

$$Gain = ₹189 - x$$

From above equation $x = \left(\frac{7}{6}\right) \times SP1$

$$x = \left(\frac{7}{6}\right) \times ₹144$$

$$x = ₹168$$

$$\text{Gain Percentage} = \left[\left(\frac{₹189}{₹168}\right) - 1\right) \times 100]$$

$$\text{Gain Percentage} = [(1.125 - 1) \times 100] => 12.5\%$$

So, the gain percentage is 12.5%.

The correct answer is A) 12.5%.

49. Correct option: (A)

If the sides of a square are increased by 20%, the new side length will be:

$$\text{New Side Length} = S + 20\% \text{ of } S$$

$$\text{New Side Length} = 1.20 \times \text{Original Side Length}$$

$$\text{Area} = (\text{Side Length})^2$$

$$\text{Original Area} = S^2$$

$$\text{New Area} = (1.20 \times S)^2 = 1.44 \times S^2$$

$$\text{Percentage Increase} = \left[\frac{1.44 \times S^2 - S^2}{S^2}\right] \times 100$$

$$\text{Percentage Increase} = \left[\frac{0.44 \times S^2}{S^2}\right] \times 100$$

$$\text{Percentage Increase} = 44\%$$

So, if the sides of a square are increased by 20%, the area of the square will be increased by 44%.

The correct answer is A) 44%.

50. Correct option: (D)

Let's denote the cost price (CP) of the article as "x."

When selling at a gain of 8%, the selling price (SP) is given by:

$$SP = CP + 8\% \text{ of } CP$$

$$SP = x + 0.08x$$

$$SP = 1.08x$$

When selling at a loss of 8%, the selling price (SP) is given by:

$$LSP = CP - 8\% \text{ of } CP$$

$$LSP = x - 0.08x$$

$$LSP = 0.92x$$

According to the given information, SP1 is ₹12 more than SP2:

$$SP = LSP + ₹12$$

$$1.08x = 0.92x + ₹12$$

$$0.16x = ₹12$$

$$x = ₹12 / 0.16$$

$$x = ₹75$$

So, the cost price of the article is ₹75.

The correct answer is D) ₹75.

51. Correct option: (D)

Let's denote the cost price (CP) of the article as "x."

When the article is sold for a 20% gain, the selling price (SP_1) is given by:

$$SP_1 = CP + 20\% \text{ of } CP$$
$$SP_1 = x + 0.20x$$
$$SP_1 = 1.20x$$

Now, if it is sold for twice the first selling price $2SP_1$, the selling price SP_2 is:

$$SP_2 = 2 \times SP_1$$
$$SP_2 = 2 \times 1.20x$$
$$SP_2 = 2.40x$$
$$\text{Gain Percentage} = \left[\frac{SP2 - CP}{CP}\right] \times 100$$
$$\text{Gain Percentage} = \left[\frac{2.40x - x}{x}\right] \times 100$$
$$\text{Gain Percentage} = \left[\frac{1.40x}{x}\right] \times 100$$
$$\text{Gain Percentage} = 140\%$$

So, the gain percentage when it is sold for twice the first selling price is 140%.

The correct answer is D) 140%.

52. Correct option: (B)

If the side of the cube is increased by 10%, the new side length becomes 1.1a (10% increase means adding 10% of a to a).

The new surface area (A') of the cube is given by:

$$A' = 6(1.1a)^2 = 6(1.21a^2) = 1.21(6a^2)$$

Now, let's find the increase in surface area (ΔA):

$$\Delta A = A' - A = 1.21(6a^2) - 6a^2 = 0.21(6a^2)$$

To find the percentage increase, divide ΔA by the original surface area (A) and multiply by 100%:

$$\text{Percentage Increase} = \left(\frac{\Delta A}{A}\right) \times 100\% = \left(\frac{0.21(6a^2)}{6a^2}\right) \times 100\% = 0.21 \times 100\% = 21\%$$

So, the increase in surface area is 21%, which corresponds to option B.

53. Correct option: (C)

If the sides of the cube are decreased by 10%, the new side length becomes 0.9a (10% decrease means subtracting 10% of a from a).

The new volume (V') of the cube is given by:

$$V' = (0.9a)^3 = 0.729a^3$$

Now, let's find the decrease in volume (ΔV):

$$\Delta V = V - V' = a^3 - 0.729a^3 = 0.271a^3$$

To find the percentage decrease, divide ΔV by the original volume (V) and multiply by 100%:

$$\text{Percentage Decrease} = \left(\frac{\Delta V}{V}\right) \times 100\% = \left(\frac{0.271a^3}{a^3}\right) \times 100\% = 27.1\%$$

So, the decrease in volume is 27.1%, which corresponds to option C.

54. Correct option: (B)

To find the gain percentage, we can use the formula you provided:

$$\text{Gain Percentage} = \left(\frac{\text{Gain}}{\text{Cost Price}}\right) \times 100\%$$

In this case, the cost price (CP) of the tricycle is ₹1120, and it was sold for ₹1260.

$$\text{Gain} = \text{Selling Price (SP)} - \text{Cost Price (CP)}$$
$$\text{Gain} = ₹1260 - ₹1120 = ₹140$$

Now, use the formula to calculate the gain percentage:

$$\text{Gain Percentage} = \left(\frac{₹140}{₹1120}\right) \times 100\% = (0.125) \times 100\% = 12.5\%$$

So, the gain percentage is 12.5%.

Therefore the correct option is B)

55. Correct option: (A)

To calculate the selling price (SP) at which Ramu should sell the rickshaw to gain 12%, we can use the formula:

$$SP = CP + Gain$$

$$\text{Loss}\,\% = 12\%$$

$$\text{Loss}\,\% = \left(\frac{\text{Loss}}{CP}\right) \times 100$$

$$12\% = \left(\frac{\text{Loss}}{CP}\right) \times 100$$

$$\text{Loss} = \left(\frac{12}{100}\right) \times CP$$

$$\text{Loss} = \left(\frac{3}{25}\right) \times CP$$

$$CP = SP + \text{Loss}$$

$$CP = ₹9240 + \left(\frac{3}{25}\right) \times CP$$

Now, solve for CP:

$$\left(\frac{22}{25}\right) \times CP = ₹9240$$

$$CP = \frac{₹9240 \times 25}{22}$$

$$CP = ₹10{,}500$$

Now that we have found CP, we can calculate the selling price (SP) at which Ramu should sell to gain 12%:

$$SP = CP + Gain$$

$$SP = ₹10{,}500 + 12\%\ \text{of}\ ₹10{,}500$$

$$SP = ₹10{,}500 + \left(\frac{12}{100}\right) \times ₹10{,}500$$

$$SP = ₹10{,}500 + ₹1{,}260$$

$$SP = ₹11{,}760$$

So, Ramu should sell the rickshaw for ₹11,760 to gain 12%, which corresponds to option A.

56. Correct option: (B)

To calculate the compound interest for two years with different interest rates, we can break it down into two steps:

1. Calculate the amount after the first year at a 10% interest rate.
2. Calculate the amount after the second year at a 12% interest rate (10% + 2%).

Amount after the first year

$$(A_1) = P + \left(P \times \frac{R_1}{100}\right)$$

$$A_1 = ₹77{,}500 + \left(₹77{,}500 \times \frac{10}{100}\right)$$

$$A_1 = ₹77{,}500 + ₹7{,}750$$

$$A_1 = ₹85{,}250$$

Now our principle amount is ₹85,250

Amount after the second year

$$A_2 = P + \left(P \times \frac{R_1}{100}\right)$$

$$A2 = ₹85{,}250 + \left(₹85{,}250 \times \frac{12}{100}\right)$$
$$A2 = ₹85{,}250 + ₹10{,}230$$
$$A2 = ₹95{,}480$$

Now, let's find the compound interest (CI) for the entire two-year period:

$$CI = A2 - P$$
$$CI = ₹95{,}480 - ₹77{,}500$$
$$CI = ₹17{,}980$$

Then the correct option is B)

57. Correct option: (A)

Let's calculate each part of the expression:

$$20\% \text{ of } 5: \left(\frac{20}{100}\right) \times 5 = 1$$
$$5\% \text{ of } 20: \left(\frac{5}{100}\right) \times 20 = 1$$

Now, add these two results together: $1 + 1 = 2$

So, 20% of 5 + 5% of 20 equals 2, which corresponds to option A.

58. Correct option: (A)

To find the value of x, we can set up the equation:

$$x\% \text{ of } 20 = 1$$
$$\left(\frac{x}{100}\right) \times 20 = 1$$
$$\left(\frac{x}{5}\right) = 1$$
$$x = 1 \times 5$$
$$x = 5$$

Therefore, the correct option is A) 5.

59. Correct option: (B)

Let's assume that A initially gave P an amount of "x" rupees, and Q received the remaining amount of "25500 - x" rupees.

According the given information

$$\text{Amount} = \text{Principal} \times \left(1 + \left(\frac{\text{Rate}}{100}\right)\right)^{\text{Time}}$$

For P:

$$P_{25} = x \times \left(1 + \frac{4}{100}\right)^{12}$$

(since P is 13 years old and will receive the money after 12 years)

For Q:

$$Q_{25} = (25500 - x) \times \left(1 + \frac{4}{100}\right)^{11}$$

(since Q is 14 years old and will receive the money after 11 years)

According to the given information, P and Q will receive the same amount when they turn 25:

$$P_{25} = Q_{25}$$
$$x \times \left(1 + \frac{4}{100}\right)^{12} = (25500 - x) \times \left(1 + \frac{4}{100}\right)^{11}$$
$$x \times \left(1 + \frac{4}{100}\right) = (25500 - x)$$
$$x \times (1.04) = (25500 - x)$$
$$1.04x = 25500 - x$$
$$1.04x + x = 25500$$

$$2.04x = 25500$$

$$x = \frac{25500}{2.04} = 12500$$

So, A initially gave Q $25500 - x = 25500 - 12500 = 13000$.
Therefore, the correct option is B) ₹13,000.

60. Correct option: (D)

Let's assume that Raj gave Prakash an amount of "x" rupees. Then, he gave Prashant the remaining amount of "1301 - x" rupees.
For Prakash:

$$\text{Prakash's Amount} = x \times \left(1 + \frac{4}{100}\right)^{17}$$

For Prashant:

$$\text{Prashant's Amount} = (1301 - x) \times \left(1 + \frac{4}{100}\right)^{19}$$

According to the given information, Prakash and Prashant receive the same amount:

$$x \times \left(1 + \frac{4}{100}\right)^{17} = (1301 - x) \times \left(1 + \frac{4}{100}\right)^{19}$$

Now, we can solve this equation for x:

$$x = (1301 - x) \times (1.04)^2 = 1301 \times 1.0816 - x \times 1.0816$$

$$x(1 + 1.0816) = 1407.1616 => x = \frac{1407.1616}{2.0816} = 676$$

So, Raj initially gave Prashant $1301 - 676 = 625$.
Therefore, the correct option is D) ₹625.

61. Correct option: (D)

Let's assume the principal amount Shamik invested in both simple interest (SI) and compound interest (CI) is "P" rupees.
Given that the difference in both interests is ₹50, we can set up the equation:

$$CI - SI = 50$$

Now, let's plug in the values and solve:

$$P \times \left[\left(1 + \left(\frac{4}{100}\right)\right)^2 - 1\right] - \left(P \times 4 \times \frac{2}{100}\right) = 50$$

Simplify:

$$P \times [(1.04^2 - 1) - 0.08] = 50$$

$$P \times [(1.0816 - 1) - 0.08] = 50$$

$$P \times [0.0816 - 0.08] = 50$$

$$P \times 0.0016 = 50 => P = \frac{50}{0.0016} = 31250$$

So, Shamik initially invested ₹31,250 in both simple interest and compound interest.
Therefore, the correct option is D) ₹31,250.

62. Correct option: (B)

To find the compound interest for 9 months at a 6% per annum rate, compounded quarterly, we can use the formula:

$$A = P\left(1 + \left(\frac{r}{n}\right)\right)^{nt}$$

Where:
A = the final amount
P = the principal amount (₹5000)
r = the annual interest rate (6% or 0.06 as a decimal)
n = the number of times interest is compounded per year (quarterly, so n = 4)
t = the time period in years (9 months or 9/12 years)

1. Convert the time to years: 9 months = 9/12 years.

$$A = 5000 \times \left[1 + \left(\frac{0.06}{4}\right)\right]^{4 \times \frac{9}{12}}$$

$$A = 5000 \times [1 + 0.015]^3$$

$$A = 5000 \times (1.015)^3 = 5000 \times 1.045678375 \approx 5228.39$$

$$\text{Compound Interest} = A - P$$

$$\text{Compound Interest} \approx ₹5228.39 - ₹5000$$

$$\text{Compound Interest} \approx ₹228.39$$

So, the compound interest on ₹5000 for 9 months at a 6% per annum rate, compounded quarterly, is approximately ₹228.39.

Therefore the correct option is B)

63. Correct option: (A)

Using the formula: $A = P\left(1 + \left(\frac{r}{n}\right)\right)^{nt}$

Where:

A = the final amount

P = the principal amount (₹20,000)

r = the annual interest rate (4% or 0.04 as a decimal)

n = the number of times interest is compounded per year (half-yearly, so n = 2)

t = the time period in years (2 years)

1. Plug in the values:

$$A = 20000 * \left[1 + \left(\frac{0.04}{2}\right)\right]^{2 \times 2}$$

$$A = 20000 \times [1 + 0.02]^4$$

$$A = 20000 \times (1.02)^4$$

$$A \approx 20000 \times 1.082432$$

$$A \approx ₹21648.64$$

$$\text{Compound Interest} = A - P$$

$$\text{Compound Interest} \approx ₹21648.64 - ₹20000$$

$$\text{Compound Interest} \approx ₹1648.64$$

So, the compound interest on ₹20,000 for 2 years at a 4% per annum rate, compounded half-yearly, is approximately ₹1648.64.

Therefore, the correct option is A) ₹1648.64.

64. Correct option: (B)

To find the compound interest on ₹8,500 at 4% per annum for 2 years, compounded annually, we can use the formula:

$$A = P\left(1 + \left(\frac{r}{100}\right)\right)^t$$

$$A = 8500 \times (1 + 0.04)^2$$

$$A = 8500 \times (1.04)^2$$

$$A = 8500 \times 1.0816$$

$$A = ₹9193.6$$

$$\text{Compound Interest} = A - P$$

$$\text{Compound Interest} = ₹9193.6 - ₹8500$$

$$\text{Compound Interest} = ₹693.6$$

So, the compound interest on ₹8,500 at 4% per annum for 2 years, compounded annually, is ₹693.6.

Therefore, the correct option is B).

65. Correct option: (A)

$$\text{Amount} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^{\text{Time}}$$

In this case, the initial amount doubles in 10 years:

$$2 \times \text{Principal} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^{10}$$

We can find the value of Rate

$$2 = \left(1 + \frac{\text{Rate}}{100}\right)^{10} \Rightarrow \left(2^{\frac{1}{10}} - 1\right) \times 100 = \text{Rate}$$

Now, we want to find out when the amount becomes 8 times the initial amount:

$$8 \times \text{Principal} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^{x}$$

Putting the value of Rat e

$$8 = \left(1 + \frac{\left(\left(2^{\frac{1}{10}} - 1\right)100\right)}{100}\right)^{x}$$

$$8 = \left(1 + 2^{\frac{1}{10}} - 1\right)^{x} = 2^{\frac{x}{10}}$$

$$8 = 2^3 = 2^{\frac{x}{10}}$$

Since base is same, we can compare the exponent

$$3 = \frac{x}{10} \Rightarrow x = 30$$

So, it will take 30 years for the money to become eight times the initial amount
Therefore the correct option is A)

66. Correct option: (C)

To find the principal sum, we can use the formula for the difference between compound interest (CI) and simple interest (SI):

$$\text{Difference} = \text{CI} - \text{SI}$$

$$\text{Difference} = P \times [(1 + r)^t - 1] - \frac{P \times r \times t}{100}$$

We can set up the equation with the known values:

$$₹392 = P \times [(1 + r)^2 - 1] - (P \times r \times 2)$$

$$392 = 20000 \times [1 + r^2 + 2r - 1 - 2r] = P[r^2]$$

$$r^2 = \frac{392}{20000} = 196 \times \frac{1}{10000} = 14^2 \times \left(\frac{1}{100}\right)^2 = \left(\frac{14}{100}\right)^2$$

$$r = \frac{14}{100} \Rightarrow r = 14\%$$

So, the rate is 14%.
Therefore, the correct option is C).

67. Correct option: (D)

To find the simple interest (SI) for the same sum, period, and rate, we can use the information given about the compound interest (CI) and the formula for CI:

$$\text{CI} = P\left[\left(1 + \frac{r}{n}\right)^{nt} - 1\right]$$

$$127 = P\left[\left(1 + \frac{1}{6}\right)^3 - 1\right]$$

$$127 = P\left[\left(\frac{7}{6}\right)^3 - 1\right]$$

$$127 = P\left[\left(\frac{343}{216}\right) - 1\right]$$

$$127 = P\left[\frac{343 - 216}{216}\right]$$

$$127 = P\left[\left(\frac{127}{216}\right)\right]$$

$$P = 216$$

$$SI = P \times r \times t$$

$$SI = 216 \times \frac{50}{300} \times 3$$

$$SI = \frac{216}{2} = 108$$

So, the simple interest on the same sum for the same period and rate is ₹108.
Therefore the correct option is D)

68. Correct option: (A)

1. Amount (A) after 3 years with compound interest: $A = P \times (1 + r)^t$
2. Compound Interest (CI): $CI = A - P$

1. Calculate the amount (A) after 3 years:

$$A = ₹8{,}000 \times (1 + 0.05)^3$$

$$A = ₹8{,}000 \times (1.05)^3 =$$

$$A = ₹8{,}000 \times 1.157625 = 9261$$

$$CI = A - P$$

$$CI = ₹9{,}261 - ₹8{,}000$$

$$CI = ₹1{,}261$$

So, the compound interest on ₹8,000 at 5% per annum for 3 years when CI is compounded yearly is ₹1,261.
Therefore, the correct option is A) ₹1,261.

69. Correct option: (A)

$$A = P\left(1 + \left(\frac{r}{n}\right)\right)^{nt}$$

$$A = 5000 \times \left(1 + \left(\frac{0.04}{2}\right)\right)^{2 \times \left(\frac{3}{2}\right)}$$

$$A = 5000 \times (1 + 0.02)^3$$

$$A = 5000 \times (1.02)^3$$

$$A = 5000 \times 1.061208$$

$$A = ₹\,5306.04$$

Now, calculate the compound interest:

$$\text{Compound Interest (C.I.)} = A - P$$

$$C.I. = ₹\,5306.04 - ₹\,5000$$

$$C.I. = ₹\,306.04$$

So, the compound interest on ₹ 5000 at 4% per annum compounded half-yearly for $\frac{3}{2}$ years is ₹ 306.04.
The correct option is A) ₹ 306.04.

70. Correct option: (A)

$$\text{Amount} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^{\text{Time}}$$

In this case, the initial amount doubles in 10 years:

$$3 \times \text{Principal} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^3$$

We can find the value of Rate

$$3 = \left(1 + \frac{\text{Rate}}{100}\right)^3 \Rightarrow \left(3^{\frac{1}{3}} - 1\right) \times 100 = \text{Rate}$$

Now, we want to find out when the amount becomes 8 times the initial amount:

$$9 \times \text{Principal} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^x$$

Putting the value of Rat e

$$9 = \left(1 + \frac{\left(\left(3^{\frac{1}{3}} - 1\right)100\right)}{100}\right)^x$$

$$9 = \left(1 + 3^{\frac{1}{3}} - 1\right)^x = 3^{\frac{x}{3}}$$

$$9 = 3^2 = 3^{\frac{x}{3}}$$

Since base is same, we can compare the exponent

$$2 = \frac{x}{3} \Rightarrow x = 6$$

So, it will take 6 years for the money to become 9 times the initial amount

Therefore the correct option is A)

71. Correct option: (A)

$$\text{Amount} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^{\text{Time}}$$

In this case, the initial amount doubles in 10 years:

$$2 \times \text{Principal} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^4$$

We can find the value of Rate

$$2 = \left(1 + \frac{\text{Rate}}{100}\right)^4 \Rightarrow \left(2^{\frac{1}{4}} - 1\right) \times 100 = \text{Rate}$$

Now, we want to find out when the amount becomes 8 times the initial amount:

$$8 \times \text{Principal} = \text{Principal} \times \left(1 + \frac{\text{Rate}}{100}\right)^x$$

Putting the value of Rat e

$$8 = \left(1 + \frac{\left(\left(2^{\frac{1}{4}} - 1\right)100\right)}{100}\right)^x$$

$$8 = \left(1 + 2^{\frac{1}{4}} - 1\right)^x = 2^{\frac{x}{4}}$$

$$8 = 2^3 = 2^{\frac{x}{4}}$$

Since base is same, we can compare the exponent

$$3 = \frac{x}{4} \Rightarrow x = 12$$

So, it will take 12 years for the money to become 8 times the initial amount

Therefore the correct option is A)

3. Linear Equation in One Variable

Solutions

LEVEL – I

1. Correct option: (C) 5

Given $3P + 1 = 16$

Then
$$3P = 15$$
$$P = 5$$

2. Correct option: (A) 15

Given $\frac{x}{3} - 5 = 0$

Then
$$\frac{x}{3} = 5$$
$$x = 15$$

3. Correct option: (D) 1

Given $6(2a - 1) + 8 = 14$

Then
$$6(2a - 1) = 14 - 8$$
$$12a - 6 = 6$$
$$12a = 12$$
$$a = \frac{12}{12}$$
$$a = 1$$

4. Correct option: (A) 2

Given $16(x + 7) = 144$

Then
$$16x + 112 = 144$$
$$16x = 32$$
$$x = 2$$

5. Correct option: (D) 8

Let the number be x

Given that six times of a number is 48
$$6x = 48$$
$$x = 8$$

6. Correct option: (C) 6

The number to be divided on equation $6a = 30$, is 6 to get value of a

7. Correct option: (B) 8

The number to be added on both side of equation $y - 8 = 6$ is 8, to get value of y

8. Correct option: (A) b + x

Given a = b

Then
$$a + x = b + x$$

9. Correct option: (C) 21/16

Given $8t + \frac{5}{2} = 13$

Then
$$8t = 13 - \frac{5}{2}$$
$$8t = \frac{26 - 5}{2}$$
$$t = \frac{21}{16}$$

10. Correct option: (C) 0

Given $11n + 1 = 1$

Then

$$11n = 1 - 1$$
$$11n = 0$$
$$n = 0$$

11. Correct option: (A) 1/5

Given $2x + \frac{4}{5} = x + 1$

Then

$$2x - x = 1 - \frac{4}{5}$$
$$x = \frac{5 - 4}{5}$$
$$x = \frac{1}{5}$$

12. Correct option: (D) 1

Given $15(L - 1) + 4(L + 3) = 2(7 + L)$

Then

$$15L - 15 + 4L + 12 = 14 + 2L$$
$$19L - 3 - 2L = 14$$
$$17L = 14 + 3$$
$$17L = 17$$
$$L = 1$$

13. Correct option: (A) $\frac{-5}{6}$

Given $5(I + 2) - 2(3 - 4I) = 3(I + 5) - 4(4 - I)$

Then

$$5I + 10 - 6 + 8I = 3I + 15 - 16 + 4I$$
$$13I + 4 = 7I - 1$$
$$13I - 7I = -1 - 4$$
$$6I = -5$$
$$I = \frac{-5}{6}$$

14. Correct option: (B) 2

Given $\frac{5x+2}{2x+3} = \frac{12}{7}$

Then

$$7(5x + 2) = 12(2x + 3)$$
$$35x + 14 = 24x + 36$$
$$35x - 24x = 36 - 14$$
$$11x = 22$$
$$x = 2$$

15. Correct option: (C) 9/23

Given $\frac{8p-5}{7p+1} = \frac{-2}{4}$

Then

$$4(8p - 5) = -2(7p + 1)$$
$$32p - 20 = -14p - 2$$
$$32p + 14p = 20 - 2$$
$$46p = 18$$
$$p = \frac{18}{46}$$
$$p = \frac{9}{23}$$

16. Correct option: (C) 12

Given $7x - 81 = 3$

Then

$$7x = 81 + 3$$
$$7x = 84$$
$$x = 12$$

17. Correct option: (D) 40

The number 40 when divided by 8 gives 5

18. Correct option: (A) 1

Given $3(m + 6) = 21$

Then

$$3m + 18 = 21$$
$$3m = 3$$
$$m = 1$$

19. Correct option: (B) 2.5

Given q = 2

Then

$$pq - 3 = 2$$
$$p(2) = 2 + 3$$
$$2p = 5$$
$$p = \frac{5}{2}$$
$$p = 2.5$$

20. Correct option: (A) 1

Given $0.5x + 3.5 = 3.5x + 0.5$

Then

$$3.5x - 0.5x = 3.5 - 0.5$$
$$3x = 3$$
$$x = 1$$

LEVEL – II

1. Correct option: (B) 3

Given $\frac{x}{3} - \frac{1}{4} = \frac{3}{2} - \frac{x}{4}$

Then

$$\frac{x}{3} + \frac{x}{4} = \frac{3}{2} + \frac{1}{4}$$
$$\frac{4x + 3x}{12} = \frac{6 + 1}{4}$$
$$\frac{7x}{12} = \frac{7}{4}$$
$$4x = 12$$
$$x = 3$$

2. Correct option: (A) 4

Given $(x + 2)^2 = x^2 + ax + 4$

Then

$$x^2 + 4x + 4 = x^2 + ax + 4$$
$$4x = ax$$
$$a = 4$$

3. Correct option: (B) 20/3

Given equation $(5x-1)(9x+1)$ and $3a(3x-1)$ are equal at x = 1

So, put x = 1 and equate equations

$$[5(1)-1][9(1)+1] = 3a[3(1)-1]$$
$$(5-1)(9+1) = 3a(3-1)$$
$$(4)(10) = 3a(2)$$
$$a = \frac{40}{6}$$
$$a = \frac{20}{3}$$

4. Correct option: (C) 3

Given $\frac{3x+5}{3x-5} = \frac{7}{2}$

Then
$$2(3x+5) = 7(3x-5)$$
$$6x+10 = 21x-35$$
$$21x-6x = 35+10$$
$$15x = 45$$
$$x = 3$$

5. Correct option: (D) 0

Given $x + \frac{1}{x} = 2$

Squaring both sides

$$\left(x+\frac{1}{x}\right)^2 = (2)^2$$
$$x^2 + \frac{1}{x^2} + 2 = 4$$
$$x^2 + \frac{1}{x^2} = 2$$
$$x^2 + \frac{1}{x^2} - 2 = 0$$

6. Correct option: (C) 2

Given $\frac{10z+8}{2z} = 7$

Then
$$10z+8 = 14z$$
$$4z = 8$$
$$z = 2$$

7. Correct option: (A) 4

Given $3y-3 = 4y-7$

Then
$$4y-3y = 7-3$$
$$y = 4$$

8. Correct option: (A)

Given $2x-1 = \frac{3x-1}{2}$

Then
$$2(2x-1) = 3x-1$$
$$4x-2 = 3x-1$$
$$x = 1$$

9. Correct option: (B) 1/2

Given $8(2p-5)-6(3p-7) = 1$

Then
$$16p-40-18p+42 = 1$$
$$2-2p = 1$$
$$2p = 1$$
$$p = \frac{1}{2}$$

10. Correct option: (B) 32

Given r = 6

Then value of

$$\frac{9r}{2} + 5$$
$$= \frac{9(6)}{2} + 5$$
$$= 9(3) + 5$$
$$= 27 + 5$$
$$= 32$$

11. Correct option: (B) 8

Given

$$3(n-4) + 2(4n-5) = 5(n+2) + 16$$

Then

$$3n - 12 + 8n - 10 = 5n + 10 + 16$$
$$11n - 22 = 5n + 26$$
$$11n - 5n = 26 + 22$$
$$6n = 48$$
$$n = 8$$

12. Correct option: (B) -5/6

Given

$$5(x+2) - 2(3-4x) = 3(x+5) - 4(4-x)$$

Then

$$5x + 10 - 6 + 8x = 3x + 15 - 16 + 4x$$
$$13x + 4 = 7x - 1$$
$$13x - 7x = -1 - 4$$
$$6x = -5$$
$$x = \frac{-5}{6}$$

13. Correct option: (B) 6

Let the number be x

Four times of a number reduced by 5 is equal to 19

So

$$4x - 5 = 19$$
$$4x = 19 + 5$$
$$4x = 24$$
$$x = 6$$

14. Correct option: (C) 9/2

Given $\frac{x}{2} - \frac{1}{4} = \frac{x}{3} + \frac{1}{2}$

Then

$$\frac{x}{2} - \frac{x}{3} = \frac{1}{2} + \frac{1}{4}$$
$$\frac{3x - 2x}{6} = \frac{2+1}{4}$$
$$\frac{x}{6} = \frac{3}{4}$$
$$x = \frac{18}{4}$$
$$x = \frac{9}{2}$$

15. Correct option: (C) 145/21

Given $\frac{n}{5}-\frac{5}{7}=\frac{2}{3}$

Then

$$\frac{n}{5}=\frac{2}{3}+\frac{5}{7}$$
$$\frac{n}{5}=\frac{14+15}{21}$$
$$\frac{n}{5}=\frac{29}{21}$$
$$n=\frac{145}{21}$$

16. Correct option: (C) -4/7

Given $3(t-3)=5(2t-1)$

Then

$$3t-9=10t-5$$
$$5-9=10t-3t$$
$$7t=-4$$
$$t=\frac{-4}{7}$$

17. Correct option: (D) -1

Given $8m+9=7m+8$

Then

$$8m-7m=8-9$$
$$m=-1$$

18. Correct option: (B) 13/6

Given $x+\frac{4}{3}=3\frac{1}{2}$

then

$$x=\frac{7}{2}-\frac{4}{3}$$
$$x=\frac{21-8}{6}$$
$$x=\frac{13}{6}$$

19. Correct option: (B)

20. Correct option: (A)

STUDENT PRACTICE QUESTIONS

1. Correct option: (A) 3

Given $7z-5=16$

Then

$$7z-5=15$$
$$7z=21$$
$$z=3$$

2. Correct option: (C) 9

Given $x+3=12$

Then

$$x=12-3$$
$$x=9$$

3. Correct option: (A) 3

Given $4x - 1 = 2x + 5$

Then

$$4x - 2x = 5 + 1$$
$$2x = 6$$
$$x = 3$$

4. Correct option: (A) 1.2

Given $0.8 + \frac{x}{4} = 0.5x + \frac{1}{2}$

Then

$$\frac{x}{4} - 0.5x = \frac{1}{2} - 0.8$$
$$\frac{x - 2x}{4} = \frac{1 - 1.6}{2}$$
$$\frac{-x}{4} = \frac{-0.6}{2}$$
$$\frac{x}{4} = \frac{0.6}{2}$$
$$x = \frac{2.4}{2}$$
$$x = 1.2$$

5. Correct option: (C) 3

Given $8(C - 3) - (6 - 2C) = 2(C + 2) - 5(5 - C)$

Then

$$8C - 24 - 6 + 2C = 2C + 4 - 25 + 5C$$
$$10C - 30 = 7C - 21$$
$$10C - 7C = 30 - 21$$
$$3C = 9$$
$$C = 3$$

6. Correct option: (C) -3

Given $\frac{x+7}{3x+16} = \frac{4}{7}$

Then

$$7(x + 7) = 4(3x + 16)$$
$$7x + 49 = 12x + 64$$
$$12x - 7x = 49 - 64$$
$$5x = -15$$
$$x = -3$$

7. Correct option: (A) -2

Given $7.5x + 4 = 6.5x + 2$

Then

$$7.5x - 6.5x = 2 - 4$$
$$x = -2$$

8. Correct option: (D) 88/77

Given $0.5x + 0.3 = 0.23x + 1.18$

Then

$$0.5x - 0.23x = 1.18 - 0.3$$
$$0.27x = 0.88$$
$$x = \frac{0.88}{0.27}$$
$$x = \frac{88}{27}$$

9. Correct option: (D) -2

Given $\frac{2x-1}{2x-3} = \frac{3x+1}{3x-1}$

Then

$$(2x-1)(3x-1) = (2x-3)(3x+1)$$
$$6x^2 - 2x - 3x + 1 = 6x^2 + 2x - 9x - 3$$
$$1 - 5x = -7x - 3$$
$$7x - 5x = -1 - 3$$
$$2x = -4$$
$$x = -2$$

10. Correct option: (B) 6

Given $x - \left(2x - \frac{3x-4}{7}\right) = \frac{4x-27}{3} - 3$

Then

$$x - \frac{14x - 3x + 4}{7} = \frac{4x - 27 - 9}{3}$$
$$x - \frac{11x + 4}{7} = \frac{4x - 36}{3}$$
$$\frac{7x - 11x - 4}{7} = \frac{4x - 36}{3}$$
$$\frac{-4x - 4}{7} = \frac{4x - 36}{3}$$
$$-12x - 12 = 28x - 252$$
$$28x + 12x = 252 - 12$$
$$40x = 240$$
$$x = 6$$

11. Correct option: (B)

Given $\frac{x+7}{3x+16} = \frac{4}{7}$

Then

$$7x + 49 = 12x + 64$$
$$64 - 49 = 7x - 12x$$
$$5x = -15$$
$$x = -3$$

12. Correct option: (A) 5

Let present age of Marry be x

After 15 year, Marry's age will be four times the age of present age

$$x + 15 = 4x$$
$$3x = 15$$
$$x = 5$$

13. Correct option: B

4. Triangles and Congruence of Triangles

Solutions

LEVEL – I

1. Correct option: (C)

Let's denote the three angles of the triangle:

$$\angle 1: 3x°$$
$$\angle 2: 4x°$$
$$\angle 3: 5x°$$

According to the given ratio, the sum of the angles in a triangle is 180°.

So, we can write the equation: $\angle 1 + \angle 2 + \angle 3 = 180°$

$$3x° + 4x° + 5x° = 180°$$
$$12x° = 180°$$
$$x° = \frac{180°}{12}$$
$$x° = 15°$$

Now that we have the value of x, we can find the measures of the angles:

$$\angle 1: 3x° = 3 \times 15° = 45°$$
$$\angle 2: 4x° = 4 \times 15° = 60°$$
$$\angle 3: 5x° = 5 \times 15° = 75°$$

Therefore, the angles of the triangle are 45°, 60°, and 75°.

Then the correct option is C)

2. Correct option: (C)

Given the relationships in triangle ABC: $\frac{2}{3}\angle A = \frac{3}{4}\angle B = \frac{5}{6}\angle C$

Let's denote the angles as follows:

$$\angle A = x°$$
$$\angle B = y°$$
$$\angle C = z°$$

According to the given relationships:

$$\left(\frac{2}{3}\right)x = \left(\frac{3}{4}\right)y = \left(\frac{5}{6}\right)z$$

Solve for x, y, and z:

$$x = \left(\frac{3}{4}\right) \times \left(\frac{3}{2}\right)y => \frac{9}{8}y$$
$$z = \left(\frac{6}{5}\right) \times \left(\frac{3}{4}\right)y => \frac{9}{10}y$$

So our equation become

$$x° + y° + z° = \frac{9}{8}y° + y° + \frac{9}{10}y° => \frac{45+36}{40}y° = \frac{81}{40}y° = 180°$$
$$y° = 180° \times \frac{40}{81} = \left(\frac{80}{9}\right)°$$

Then x and z are

$$x° = \left(\frac{9}{8} \times \frac{80}{9}\right)° = 10°$$
$$z° = \left(\frac{9}{10} \times \frac{80}{9}\right)° = 8°$$

So, the ratios of the angles are:

$$\angle A : \angle B : \angle C = x° : y° : z° = 10° : \left(\frac{80}{9}\right)° : 8° => 45 : 40 : 36$$

3. Correct option: (A)

In triangle ⊿ABC, if AB = AC and BC is produced to point D such that ∠ACD=100°, we can find the measure of angle A as follows:

Since AB = AC, it means that triangle ⊿ABC is an isosceles triangle, and angles B and C are congruent.

Given that ∠ACD = 100°, we know that ∠ACD and ∠ACB are supplementary angles because they form a straight line.

Therefore, ∠ACB = 180° - ∠ACD = 80°.

Since triangle ⊿ABC is isosceles, ∠B and ∠C are congruent, so ∠B = ∠C = 80°.

To find the value of ∠A we use triangle internal angle sum property.

$$\angle A + \angle B + \angle C = 180°$$

Putting the values $\angle A = 180° - 160° = 20°$

Hence, the measure of ∠A in triangle ABC is 20° (∠A = 20°).

4. Correct option: (A)

In a triangle, the sum of the exterior angles is always 360°. Therefore, we can set up an equation using the given exterior angles:

$$(2x + 11) + (3x - 6) + (2x + 40) = 360$$

Combine like terms:

$$7x + 45 = 360$$

$$7x = 315$$

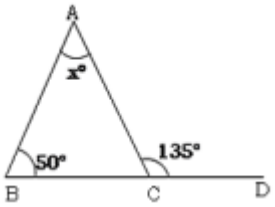
$$x = \frac{315}{7}$$

$$x = 45$$

Therefore, the value of x is: A) 45

5. Correct option: (B)

Given triangle is



Angle BOC can be determined using the linear pair of lines as

$$\angle BCA + 135° = 180° => \angle BOC = 45°$$

To find the value of x we use the triangle internal angles sum property as

$$\angle BCA + \angle BAC + \angle ABC = 180°$$

Putting the values of angles our equation become

$$x + 50 + 45 = 180 => x = 180 - 95$$

$$x = 85$$

Then the value of ∠BAC = 85° option B)

6. Correct option: (A)

Let's denote the three interior angles of the triangle as A, B, and C. The exterior angle is 135° and is equal to the sum of the two non-adjacent interior angles.

So, ∠A + ∠B = 135°.

Given that the interior opposite angles are in the ratio of 4:5, we can write:

$$\angle A = 4x$$

$$\angle B = 5x$$

Now we substitute these values into the equation A + B = 135°:

$$4x + 5x = 135$$

$$9x = 135$$

$$x = 15$$

Now we can find the values of angles A and B:

$$\angle A = 4x = 4 \times 15 = 60°$$

$$\angle B = 5x = 5 \times 15 = 75°$$

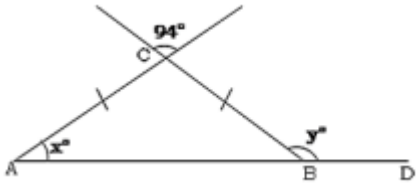
The third angle, angle C, can be found using the fact that the sum of the interior angles of a triangle is always 180°:

$$\angle C = 180 - (\angle A + \angle B) = 180 - (60 + 75) = 180 - 135 = 45$$

So, the angles of the triangle are A) 60°, 75°, and 45°.

7. Correct option: (A)

Given triangle is



We have to find the value of the y°.

Using the property of the triangle we can easily find the value of y°.

For angle ACB we can use opposite angle property as

$$\angle ACB = 94°$$

From the given figure we can easily find that triangle is isosceles as AC = BC.

Then the

$$\angle CAB = \angle CBA = x°$$

Using the triangle sum law

$$x° + x° + 94° = 180°$$

$$x = \frac{86}{2} = 43$$

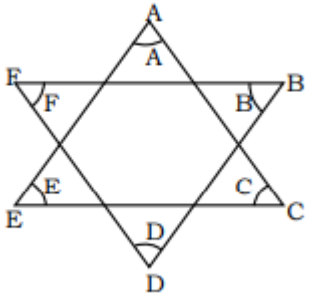
Value of x = 43

We can easily find the value of y using x as $\angle CBA + y° = 180°$

Then the value of the $y° = 180° - 43° = 137°$ as option A)

8. Correct option: (B)

Given figure is



We have to find the value of the $\angle A + \angle B + \angle C + \angle D + \angle E + \angle F$.

For triangle AEC

Using the triangle sum law

$$\angle A + \angle C + \angle E = 180° \ldots 1$$

For triangle FBD

$$\angle F + \angle D + \angle B = 180° \ldots .2$$

Adding both of the equations

$$\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 180° + 180° = 360°$$

Then the correct option is B)

9. Correct option: (C)

The sum of the angles in a triangle is always 180°. So, we can write an equation using the given angles:

$$(3x - 5) + (2x + 15) + (5x - 50) = 180$$

Now, let's solve for x:

$$3x + 2x + 5x - 5 + 15 - 50 = 180$$
$$10x - 40 = 180$$
$$10x = 220$$
$$x = 22$$

Now that we have the value of x, we can find the measures of the angles:

1. First angle: $3x - 5 = 3 \times 22 - 5 = 66 - 5 = 61°$
2. Second angle: $2x + 15 = 2 \times 22 + 15 = 44 + 15 = 59°$
3. Third angle: $5x - 50 = 5 \times 22 - 50 = 110 - 50 = 60°$

Out of these three angles, the greatest is 60°, and the least is 59°.

So, the correct answer is:

C) 61°, 59°

10. Correct option: (B)

In a rhombus, all sides are equal in length and opposite angles are equal. If the diagonals of a rhombus are equal, it implies that the angles between the diagonals are right angles (90°).

A rhombus with right angles between its diagonals is a specific type of rhombus known as a "square." A square is a rhombus where all angles are right angles.

So, the correct answer is: B) A square

11. Correct option: (D)

Let's analyze the steps and arrange them in sequential order:

d) Given $\angle P : \angle Q : \angle R : \angle S = 1 : 2 : 3 : 4$

b) Let the angles be $\angle P = x, \angle Q = 2x, \angle R = 3x, \text{and } \angle S = 4x$

e) $\angle P + \angle Q + \angle R + \angle S = 360° \Rightarrow x + 2x + 3x + 4x = 360°$

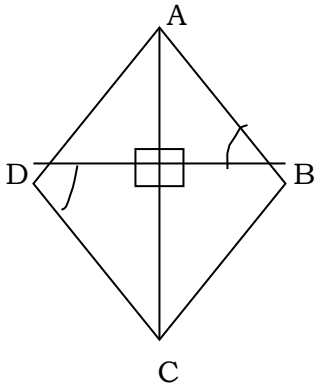
a) $10x = 360° \Rightarrow x = 36°$

c) The largest angle $4(36°) = 144°$

Then the correct option is D) D) dbeac

12. Correct option: (B)

In a kite, the two pairs of adjacent sides are equal in length, and the diagonals are perpendicular to each other.



Given that AB = AD and CB = CD in kite ABCD, and the angles ∠ABD = 25° and ∠BDC = 35°, we can find angle ∠ACD using the property that the diagonals of a kite are perpendicular bisectors of each other:

Angle∠ ACD = 90° - angle ∠BDC = 90° - 35° = 55°.

Look at the triangle BCD (isosceles triangle)

So $\angle ACD = \angle ACB = 55°$

$$\angle C = \angle ACD + \angle ACB = 110°$$

For $\angle A$

Angle$\angle$ BAC = 90° - angle $\angle ABD = 90° - 25° = 65°$.

Look at the triangle BAD (isosceles triangle)

So $\angle BAC = \angle DAC = 65°$

$$\angle A = \angle BAC + \angle DAC = 130°$$

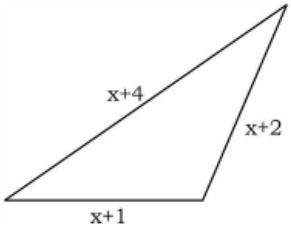
Now, to find $\angle A - \angle C$:

$\angle A - \angle C = 130° - 110° = 20°$

So, the correct answer is: B) 20°

13. Correct option: (A)

Given triangle



The perimeter of a triangle if sides are given as $x, y, z \Rightarrow x + y + z$

Putting the value into the formula

$$P = x + 4 + x + 1 + x + 2 = 3x + 7$$

As given perimeter is 19 cm

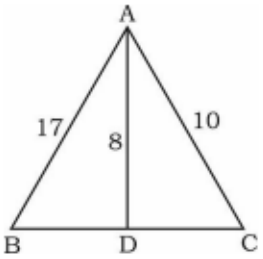
Putting the values in equation

$$19 = 3x + 7 \Rightarrow 3x = 12 \Rightarrow x = 4$$

Then the correct option is A)

14. Correct option: (A)

Given triangle



The perimeter of a triangle if sides are given as $x, y, z \Rightarrow x + y + z$

We are given the values of AC=10, AB=17

To find the perimeter of the triangle ABC we need value of BC.

Finding the value of BC.

As looking at the figure the $\angle ADC = \angle ADB = 90°$

Now we can say that the triangles ADC and ADB are right angle triangle.

The Pythagorean Theorem is a math rule for right triangles. It says that in a right triangle, the square of the longest side (the hypotenuse) is equal to the sum of the squares of the other two sides. This rule is written as

($c^2 = a^2 + b^2$), where c is the hypotenuse and a and b are the other sides.

So using Pythagorean Theorem we can find BC, as(BC = BD+DC).

$$(DC)^2 = 100 - 64 = 36 \Rightarrow DC = 6$$

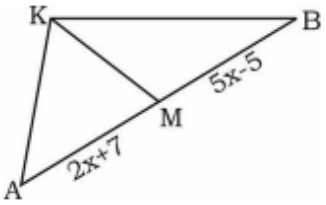
$$(BD)^2 = 289 - 64 = 225 \Rightarrow BD = 15$$

Then BC = 15 + 6 = 21

At last the parimeter of the triangle ABC will be $21 + 10 + 17 \Rightarrow 38$ option A)

15. Correct option: (B)

Our triangle is



Given that KM is median and we have to find the value of AB

Using the property of median of a triangle that it cut the base into two equal half

In our figure we can say that AM = MB (median property)

Putting the values to determine the value of x

$$2x + 7 = 5x - 5$$

$$3x = 12 \Rightarrow x = 4$$

From the above figure AB = AM + MB

$$AB = 2x + 7 + 5x - 5 = 7x + 2 \Rightarrow 28 + 2 = 30$$

Then the correct option is B)

16. Correct option: (A)

Pythagorean triplets are sets of three positive integers that satisfy the Pythagorean theorem, which states that in a right triangle, the square of the length of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the lengths of the other two sides.

The Pythagorean theorem can be written as: $(a^2 + b^2 = c^2)$, where a, b and c are the lengths of the sides of the triangle.

Let's check the options to see which one is a Pythagorean triplet:

A) $(12^2 + 35^2 = 144 + 1225 = 1369)$, but $(37^2 = 1369)$, so this is a Pythagorean triplet.

B) $(5^2 + 12^2 = 25 + 144 = 169)$, but $(14^2 = 196)$, so this is not a Pythagorean triplet.

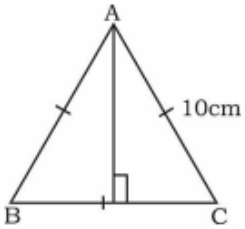
C) $(7^2 + 8^2 = 49 + 64 = 113)$, and $(9^2 = 81)$, so this is not a Pythagorean triplet.

D) $(10^2 + 14^2 = 100 + 196 = 296)$, and $(15^2 = 225)$, so this is not a Pythagorean triplet.

Based on the above analysis, the Pythagorean triplet among the options is A) 12, 35, 37.

17. Correct option: (D)

Our triangle is



In an equilateral triangle, all sides are of equal length, and all angles are 60°. To find the length of the altitude of an equilateral triangle with a given side length, we can use the following formula:

$$\text{Altitude length} = \frac{\text{Side length} \times \sqrt{3}}{2}$$

Given that the side length is 10 cm, we can plug this into the formula:

$$\text{Altitude length} = \frac{10 \text{ cm} \times \sqrt{3}}{2} = 5\sqrt{3}$$

So the correct option is D)

18. Correct option: (B)

To determine if a set of side lengths forms a valid triangle, we need to apply the triangle inequality theorem. According to this theorem, the sum of the lengths of any two sides of a triangle must be greater than the length of the third side.

Let's go through the given options: A) $8\text{ cm}, 4\text{ cm}, 20\text{ cm}$

Sum of the two smaller sides: $8\text{ cm} + 4\text{ cm} = 12\text{ cm}$

The sum is not greater than 20 cm (the longest side).

So, this set does not form a triangle.

B) $9\text{ cm}, 17\text{ cm}, 25\text{ cm}$

Sum of the two smaller sides: $9\text{ cm} + 17\text{ cm} = 26\text{ cm}$

The sum is greater than 25 cm (the longest side). This set can form a triangle.

C) $11\text{ cm}, 16\text{ cm}, 28\text{ cm}$

Sum of the two smaller sides: $11\text{ cm} + 16\text{ cm} = 27\text{ cm}$

The sum is less than 28 cm (the longest side). So, this set does not form a triangle.

Based on the triangle inequality theorem, option B) with side lengths 9 cm, 17 cm, and 25 cm is the set of measures that forms a valid triangle.

Therefore, the correct answer is B) 9 cm, 17 cm, 25 cm.

19. Correct option: (C)

In a quadrilateral ABCD where AC is perpendicular to BD and AB is equal to AD, the quadrilateral ABCD is a rectangle.

This can be explained through the following points:

1. Perpendicular Diagonals: The fact that AC is perpendicular to BD indicates that the diagonals of the quadrilateral intersect at a 90° angle. This is a characteristic feature of rectangles.

2. Equal Opposite Sides: The given information that AB = AD means that opposite sides of the quadrilateral are equal in length. In a rectangle, opposite sides are always equal, and this property is fulfilled here.

3. Right Angles: Since AC and BD are perpendicular, the angles formed between the sides of the quadrilateral and these diagonals are all right angles. This alignment of right angles is a defining property of rectangles.

In conclusion, with the given conditions, the quadrilateral ABCD satisfies the criteria of having perpendicular diagonals, equal opposite sides, and right angles, making it a rectangle.

Therefore, the correct answer is C) Rectangle.

20. Correct option: (B)

The properties of a square include having all sides equal in length and all angles equal to 90°.

Let's analyze the given options:

A) A rhombus whose adjacent angles are equal: While a rhombus does have all sides equal in length, it does not necessarily have all angles equal to 90°.

A rhombus can have equal adjacent angles, but this alone does not guarantee that all angles are 90 degrees. Therefore, option A is not necessarily a square.

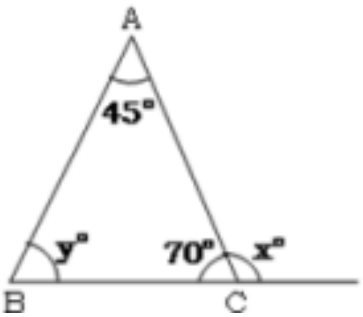
B) A rectangle whose adjacent sides are equal: A rectangle has all angles equal to 90° and opposite sides that are equal in length. If adjacent sides are equal in a rectangle, then it has all the properties of a square. Therefore, option B can indeed represent a square.

Then the correct answer is B) A rectangle whose adjacent sides are equal. ‘

LEVEL – II

1. Correct option: (D)

Given figure



From the figure $\angle A = 45°, \angle B = y°, \angle C = 70°$

We have to find the value $x° + y°$.

To find the value of $x°$

$\angle ACB = \angle C$ and $\angle x$ are adjacent angles and we know that adjacent angles on a straight line add up to 180°.

$$\angle ACB + \angle x = 180° \Rightarrow \angle x = 180° - 70° = 110°$$

To find the value of $y°$

As point A, B and C make a triangle and using the property of triangle

We can say that $\angle A + \angle B + \angle C = 180°$

Putting the values we get $y°$ as

$$\angle A + \angle B + \angle C = 180° \Rightarrow 45° + y° + 70° = 180°$$

$$y° = 65°$$

Then the value of $x° + y° = 110° + 65° = 175°$ option D)

2. Correct option: (D)

In a right-angled triangle, one angle is always 90° (the right angle), and the other two angles are acute angles whose sum is 90°.

Let's denote the acute angles as 2x and 3x, where x is a positive constant.

According to the given ratio, we have: $2x + 3x = 90$

Simplifying this equation: $5x = 90$

$$x = 18$$

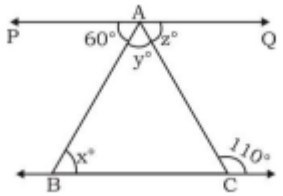
Now that we have the value of x, we can find the measures of the acute angles:

$$\text{Angle } 1 = 2x = 2 \times 18 = 36°, \text{Angle } 2 = 3x = 3 \times 18 = 54°$$

So, the correct answer is D) 36°, 54°.

3. Correct option: (C)

Given figure



From the figure $\angle CAB = 60° + y°, \angle ABC = x°$

We have to find the value $y°$.

To find the value of $y°$

We can use the traversal line property (alternative interior angle will be same)

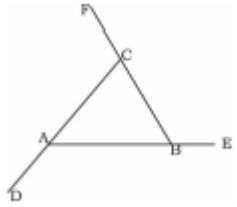
Now our equation become $\angle CAB = 110° \Rightarrow 60° + y° = 110°$

$$y° = 50°$$

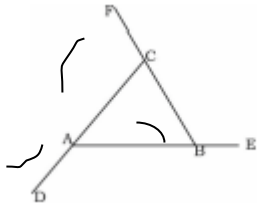
Then the correct option is C)

4. Correct option: (D)

Given figure



From the figure



We have to find the sum of exterior angle as specify $\angle DAB + \angle EBC + \angle FCA$

Let $\angle DAB = x°, \angle EBC = y°, \angle FCA = z°$

Their adjacent angles are interior angles of the triangle ABC and values are $\angle CAB = 180° - x°, \angle ABC = 180° - y°, \angle BAC = 180° - z°$

According triangle property

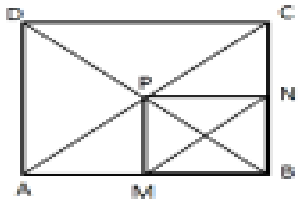
$$\angle CAB + \angle ABC + \angle BAC = 180°$$

$$180° - x° + 180° - y° + 180° - z° = 180 => x° + y° + z° = 240°$$

Then the value of $\angle DAB + \angle EBC + \angle FCA = x° + y° + z° = 240°$ option D)

5. Correct option: (D)

Given figure



Given that ABCD is square of side length 18cm and $\overline{PN} \perp \overline{BC}, \overline{PM} \perp \overline{AB}$

We have to find the value of MN.

In a square, when a perpendicular line is drawn from a side to the point of intersection of the diagonals, it serves as a perpendicular bisector for the side it intersects.

We know that PN bisects CB, so $BN = \frac{BC}{2} = 9$ cm.

Similarly, PM bisects AB, so $BM = \frac{AB}{2} = 9$ cm.

In triangle MBN, we have an angle ABC = 90° (given) and it's also a right triangle because one angle is 90°.

Therefore, we can use the Pythagorean Theorem to find MN:

$$MN^2 = BM^2 + BN^2$$

$$MN^2 = 9^2 + 9^2$$

$$MN^2 = 162$$

$$MN = \sqrt{162}$$

$$MN = 9\sqrt{2}\text{ cm}$$

So, the value of MN is $9\sqrt{2}$ cm option D)

6. Correct option: (C)

If the medians of a triangle are equal, then the triangle is equilateral.

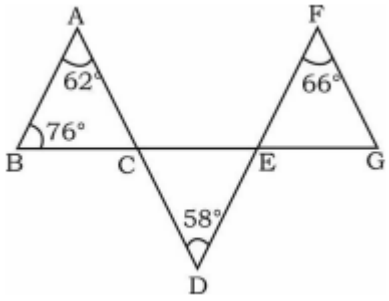
In a triangle, the medians are the line segments drawn from each vertex to the midpoint of the opposite side. The point where the medians intersect is called the centroid.

For an equilateral triangle, all three sides are of equal length, and all three angles are also equal (each measuring 60°). When you draw the medians in an equilateral triangle, they all intersect at the centroid, and they are also equal in length.

Then the correct option is C)

7. Correct option: (B)

Given figure



Given

$$\angle CAB = 62°, \angle CBA = 76°, \angle ADE = 58°, \angle DFG = 66°$$

We have to find the value of $\angle FGE$.

Looking at the triangle ABC

$$\angle ACB = 180° - 62° - 76° = 42°$$

The angle ACB and angle ECD are equal because they are opposite angle for pair of line AD and BG

$$\angle ACB = \angle ECD = 42°$$

Now looking at the triangle CDE

$$\angle DEC = 180° - \angle CDE - \angle DCE = 80°$$

The angle FEG and angle DEC are equal because they are opposite angle for pair of line FD and BG

$$\angle DEC = \angle FEG = 80°$$

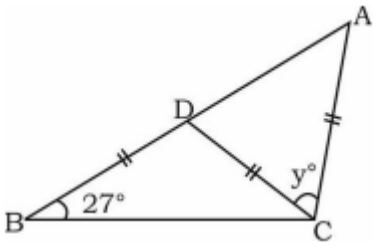
Consider the triangle EFG we can calculate the required angle FGE as

$$\angle FGE = 180° - 80° - 66° = 34°$$

Then the correct option is B)

8. Correct option: (C)

Given figure



Given

$$BD = CD = AC, \angle ABC = 27°, \angle ACD = y°$$

We have to find the value of y

Looking at the isosceles triangle BDC

$$\text{as } BD = CD \Rightarrow \angle DBC = \angle DCB \Rightarrow \angle DCB = 27°$$

For $\angle BDC \Rightarrow 180° - 27° - 27° = 126°$ (Triangle BDC)
As $\angle BDC, \angle CDA$ is adjacent angle pair so we can conclude

$$\angle CDA = 180° - \angle BDC = 54°$$

Consider the isosceles triangle ADC we can calculate the required angle ACD as

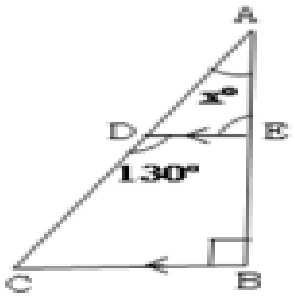
$$\angle CDA = \angle CAD = 54°$$

$$\angle ACD = 180° - \angle CDA - \angle CAD = 72°$$

Value of $y = 72$.
Then the correct option is C)

9. Correct option: (A)

Given figure



Consider the ACB the angle ACB can be calculated using parallel line and traversal line property as CB and DE is parallel and AC is traversal line.
So the angle ACB is equal to ADE
For the value of angle ADE we use adjacent angle property as

$$\angle ADE = 180° - \angle CDE = 50°$$

Then the value of $\angle ACB = 50°$.
To find the value of angle CAB we use triangle ABC (angle sum property).

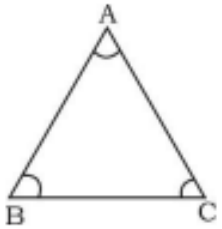
$$\angle CAB + \angle ABC + \angle ACB = 180°$$

$$\angle ABC = 90°, \angle ACB = 50° \Rightarrow \angle CAB = 180° - 90° - 50° = 40°$$

Then the value of $x = 40$ correct option A)

10. Correct option: (A)

Given figure



In a triangle ABC, $\angle BAC > 90°$ then $\angle ABC$ and $\angle ACB$ must both be acute angles.
An obtuse triangle is defined as a triangle with one angle greater than 90°. Given that angle BAC is obtuse ($\angle BAC > 90°$), the sum of angles ABC and ACB, being the other two angles in the triangle, must be less than 90° combined due to the fact that the angles in a triangle sum up to 180°.
To summarize:

$$\angle BAC > 90° \text{ (obtuse angle)}$$

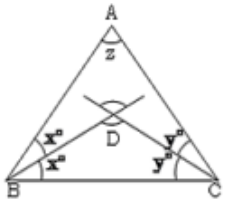
$$\angle ABC < 90° \text{ (acute angle)}$$

$$\angle ACB < 90° \text{ (acute angle)}$$

Both angles ABC and ACB must be acute angles because of the constraint posed by the obtuse angle $\angle BAC$ and the total sum of angles in a triangle being 180°.
Then the correct option is A)

11. Correct option: (B)

Given figure



The sum of the interior angles in a triangle is always 180°.

This property holds true for all types of triangles, whether they are equilateral, isosceles, scalene, acute, obtuse, or right-angled. Regardless of the triangle's shape or angles, the sum of the measures of the three interior angles will always add up to 180°.

To find the value of ∠D we can apply above property to triangle BCD.

$$x° + y° + \angle D = 180° \Rightarrow \angle D = 180° - x° - y°$$

Then the correct option is B)

12. Correct option: (C, D)

Pythagorean triplets are sets of three positive integers that satisfy the Pythagorean Theorem, which states that in a right-angled triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides.

Let's check the options:

A) 3, 4, 6: Not a Pythagorean triplet. $3^2 + 4^2 \neq 6^2$

B) 5, 12, 14: Not a Pythagorean triplet. $5^2 + 12^2 \neq 14^2$

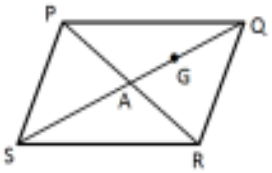
C) 8, 15, 17: Pythagorean triplet. $8^2 + 15^2 = 17^2$

D) 7, 24, 25: Pythagorean triplet. $7^2 + 24^2 = 25^2$

So, the correct answers are C) 8, 15, 17 and D) 7, 24, 25 since they are Pythagorean triplets.

13. Correct option: (A)

Figure according the question



In a triangle, the centroid divides each median in a ratio of 2:1, where the longer segment is closer to the centroid. Given that AG = 3 cm, this means that the longer segment, GQ, is $\frac{2}{3}$AQ cm, the shorter segment, GA, is $\frac{1}{3}$AQ = 3cm.

$$AQ = 9cm$$

Then value of $GQ = 9 - 3 = 6cm$

Further values of $SQ = 2 \times AQ = 18cm$, $SG = AS + AG = 9 + 3 = 12cm$.

Then the correct option is A)

14. Correct option: (C)

For ⊿AEF

$$\angle FAE + \angle AFE + \angle FEA = 180° \Rightarrow \angle AFE = 60°$$

Using the pair of line AD, BE we can say that

$$x° = \angle AFE = 60°$$

Then the correct option is C)

15. Correct option: (D)

For $\measuredangle BEC$

$$\angle EBC + \angle ECB + \angle BEC = 180° => z° = 50°$$

Then the correct option is D)

16. Correct option: (B)

For $\measuredangle ABD$

$$\angle ABD + \angle ADB + \angle BAD = 180° => w° + y° + 55° = 180°$$

$$w° + y° = 125°$$

Then the correct option is B)

17. Correct Matching is: a-R; b-S; c-T; d-P

Solving for the values of $x°, y°, z°, w°$

Using the pair of parallel line AE and BC with traversal line AC

We can conclude that $x° = 56°$ (alternative interior angles pair)

Using the $\measuredangle BFC$

$$\angle BFC = 180° - \angle AFB \text{ (Pair of lines) } \angle BFC = 95°$$

We can write $y° + x° + \angle BFC = 180° => y° = 29°$

To find the value of $w°$ concentrate on vertex C (sum of all angle on a vertex on a line is always 180°)

$$x° + 2w° = 180° => w° = \frac{124°}{2} = 62°$$

To find the value of $z°$ concentrate on vertex B

$$z° + y° + \angle ABF = 180°$$

Value of $\angle ABF$ can be determined by the $\measuredangle ABF$

$$\angle ABF = 180° - 85° - \frac{3}{2}x° => 11°$$

Then $z° + 29° + 11° = 180° => z° = 140°$

Let's match the given angle values with above conclusions:

$$\textbf{a}) \; x° = 56° \longrightarrow (\textbf{R})$$

$$\textbf{b}) \; y° = 29° \longrightarrow (\textbf{S})$$

$$\textbf{c}) \; \frac{z°}{2} = 70° \longrightarrow (\textbf{T})$$

$$\textbf{d}) \; w° = 62° \longrightarrow (\textbf{P})$$

18. Correct Answer is: (5)

Let's consider the situation as described:

Two vertical poles of heights 7 m and 11 m stand on a plane ground, and the distance between their feet is 3 m.

To find the distance between their tops, we can treat this as forming a right triangle with the line connecting the tops of the poles being the hypotenuse.

Let A be the top of the shorter pole (7 m) and B be the top of the taller pole (11 m). The distance between their feet is the base of the triangle (3 m).

Using the Pythagorean Theorem, we can find the distance between their tops:

$$AB^2 = (\text{Height difference})^2 + (\text{Distance between feet})^2$$

$$AB^2 = (11 - 7)^2 + 3^2$$

$$AB^2 = 4^2 + 3^2$$

$$AB^2 = 16 + 9$$

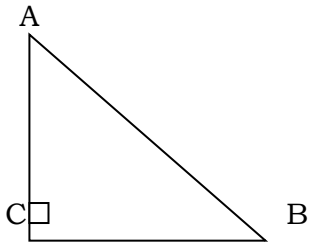
$$AB^2 = 25$$

$$AB = 5 \text{ m}$$

So, the distance between the tops of the two poles is 5 meters.

19. Correct Answer is: (6)

Figure according the question



We can write according the question $CB = AC$ and $AC = 6cm$

We can use the Pythagorean Theorem to find the value of AB.

Let's denote the legs of the triangle as AC = BC = a and the hypotenuse as AB = c.

The Pythagorean Theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the legs:

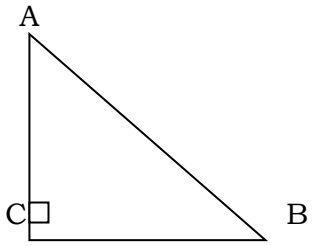
$$c^2 = a^2 + a^2$$
$$c^2 = 2a^2$$
$$c^2 = 2 \times AC^2$$

Therefore, the value of AB^2 (c^2) in terms of AC^2 (a^2) is:

$$AB^2 = 2 \times AC^2 => \frac{AB^2}{2} = AC^2 => \frac{AB}{\sqrt{2}} = AC = 6cm$$

20. Correct Answer is: (9)

Figure according the question



We can write according the question $CB = 12$ and $AB = 15cm$

We can use the Pythagorean Theorem to find the value of AB.

Let's denote the unknown leg of the triangle as $BC = a$.

The Pythagorean Theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the legs:

$$AB^2 = BC^2 + AC^2 => a^2 = 225 - 144 = 81 = 9^2 => a = 9$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

Given the relationships $\angle A = 4\angle B$ and $\angle C = 5\angle B$ in triangle ABC:

Let's denote the angles as follows:

$$\angle A = 4B°$$
$$\angle B = B°$$
$$\angle C = 5B°$$

The sum of the angles in a triangle is 180°. So, we can write the equation:

$$\angle A + \angle B + \angle C = 180°$$

Substitute the angle values:

$$4\angle B° + \angle B° + 5\angle B° = 180°$$

Combine like terms:

$$10\angle B° = 180°$$

Now, solve for B:

$$\angle B° = \frac{180°}{10}$$

$$\angle B° = 18°$$

Now that we have the value of $\angle B$, we can find the measures of the angles:

$$\angle A = 4B° = 4 \times 18° = 72°$$

$$\angle B = B° = 18°$$

$$\angle C = 5B° = 5 \times 18° = 90°$$

Therefore, the angles of the triangle ABC are 72°, 18°, and 90°, respectively.

The correct option is A)

2. Correct option: (C)

In triangle ABC, given that $\angle A + \angle B = 145°$ and $\angle B + \angle C = 90°$, we can solve for the measure of angle B as follows:

Using the first equation:

$$\angle A + \angle B = 145°$$

$$\angle A = 145° - \angle B$$

Using the second equation:

$$\angle B + \angle C = 90°$$

$$\angle C = 90° - \angle B$$

Since the angles in a triangle sum up to 180°:

$$\angle A + \angle B + \angle C = 180°$$

Substitute the expressions for $\angle A$ and $\angle C$:

$$(145° - \angle B) + \angle B + (90° - \angle B) = 180°$$

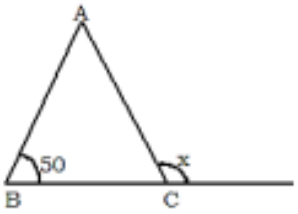
$$-\angle B = 180° - 145° - 90° => -55°$$

$$\angle B = 55°$$

Then the correct option is C)

3. Correct option: (A)

Given triangle is



Value of x can be determined using the property of isosceles triangle and linear pair of angles of lines.

$$\angle ACB + x° = 180° => \angle ACB = 180° - x°$$

As the triangle is isosceles then $\angle BAC = \angle BCA = 180° - x°$

We know that the sum of all internal angles is 180°

Putting the values in equation

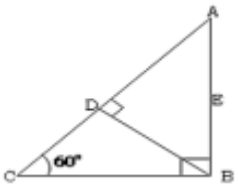
$$50° + 180° - x° + 180° - x° = 180°$$

$$-2x° = -230° => x° = \frac{230}{2} = 115°$$

Then the correct option A)

4. Correct option: (C)

Given triangle is



We have to find the value of the ∠ABD

From the figure we can say that the given triangle is an isosceles triangle.

$$\angle ACB = \angle CAB$$

Then the value of the $\angle CAB = 60°$

We know that the value of $\angle ADB = 90°$.

Looking in the triangle ADB the angles are $\angle ADB + \angle DAB + \angle ABD = 180°$

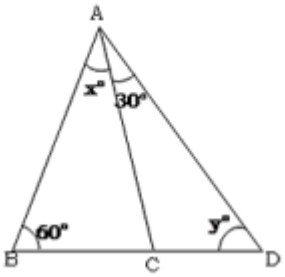
Putting the values to find the unknown angle $\angle ABD = 180° - 90° - 60°$

$$\angle ABD = 30°$$

Then the correct option C)

5. Correct option: (A)

Given triangle is



We have to find the value of the $\frac{x+y}{2}$

For triangle ABD

Using the triangle sum law $\angle A + \angle B + \angle D = 180°$

Putting the values in the equation as

$$60 + 30 + x + y = 180$$

$$90 + x + y = 180 => x + y = 90$$

Then the value of $\frac{x+y}{2} = \frac{90}{2} = 45$

Then the correct option is A)

6. Correct option: (D)

A linear pair of angles is formed when two angles are adjacent and their non-common sides form a straight line. In a rhombus, opposite angles are equal. If the opposite angles of a rhombus are a linear pair (meaning they are adjacent and their non-common sides are collinear), then their measures add up to 180°.

Let's denote the angles of the rhombus as A, B, C, and D. If A and C are opposite angles forming a linear pair, then: $\angle A + \angle C = 180°$

Since opposite angles in a rhombus are equal:

$$\angle A = \angle C$$

So, we have: $\angle A + \angle A = 180°$

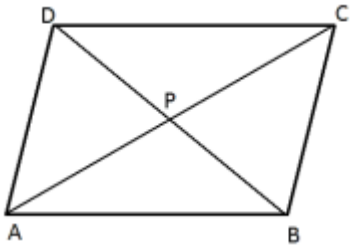
$$2\angle A = 180°$$

$$\angle A = 90°$$

Each angle of the rhombus is 90°.

Therefore, the correct answer is: D) 90°

7. Correct option: (D)



In a parallelogram, the diagonals bisect each other, meaning they divide each other into two equal segments. This property is known as the diagonal bisecting theorem.
Given that AC = 14 cm and BD = 10 cm, and considering the diagonal bisecting theorem at point P, we can write:

$$AP + BP = \frac{AC}{2} + \frac{BD}{2}$$

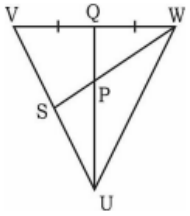
Substitute the given values:

$$AP + BP = \frac{14}{2} + \frac{10}{2}$$
$$AP + BP = 7 + 5$$
$$AP + BP = 12\,\text{cm}$$

So, the value of AP + BP is 12 cm.
Then the correct option is D)

8. Correct option: (B)

Given triangle



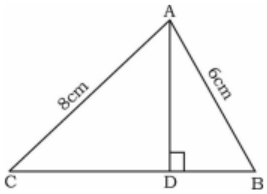
Given
QU and WS are medians of ⊿UVW
QV = $7x - 1$ and QW = $4x + 5$
Then we have to find the value of QV.
From the above figure we can conclude that the QV = QW
Now we can construct our equation

$$7x - 1 = 4x + 5 \Rightarrow 7x - 4x = 5 + 1 \Rightarrow 3x = 6$$
$$x = 2$$

Then the value of QV is $7x - 1 = 14 - 1 = 13$ option B)

9. Correct option: (B)

Our triangle is



Given that value of side AB = 6cm and value of side AC = 8cm also the triangle ABC is right angle triangle.
Using the right angle triangle property we can write as $AC^2 + AB^2 = CB^2$

Solving the equation we get CB is 10cm.
Let BD = x then CD = 10 – x and AD = y.
Then we can write the equations from the $\triangle ABD$ and $\triangle ADC$

$$AD^2 + CD^2 = AC^2 \Rightarrow y^2 + (10 - x)^2 = 64$$
$$y^2 + x^2 + 100 - 20x = 64 \ldots 1$$
$$AD^2 + BD^2 = AB^2 \Rightarrow y^2 + x^2 = 36 \ldots 2$$

Putting the value of $y^2 + x^2$ into equation 1 from the equation 2.

$$36 + 100 - 20x = 64 \Rightarrow -20x = -72 \Rightarrow x = \frac{72}{20} = 3.6$$

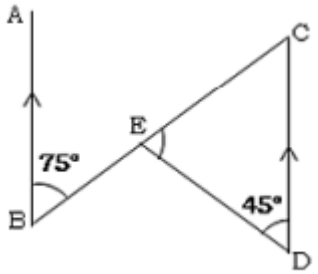
Then the value of y

$$y^2 + 12.96 = 36 \Rightarrow y^2 = 23.04 \Rightarrow y = 4.8$$

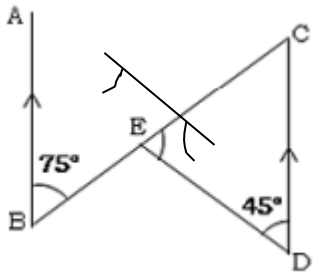
Value of AD = 4.8 cm option B)

10. Correct option: (A)

Our figure is



Given that the line AB is parallel to line CD.
We have to find the value of $\angle CED$
To provide a clearer visualization, let's extend the lines.



Using the property of traversal line the $\angle ADC = \angle DAB$ (alternative interior angle property)
$\angle AEB = \angle CED$ (Opposite angle property)
Looking at the triangle $\triangle AEB$

$$\angle BAE + \angle AEB + \angle EBA = 180°$$

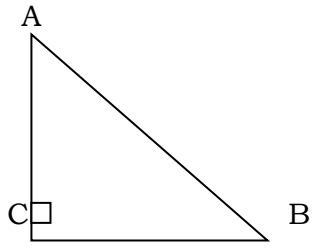
Putting the values to find the $\angle CED$.

$$45° + E + 75° = 180° \Rightarrow E = 60°$$

Then the value of E is 60° option A).

11. Correct option: (A)

Figure according the question



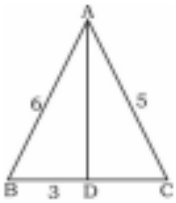
We can write according the question $CB = AC$
We can use the Pythagorean Theorem to find the value of AB^2.
Let's denote the legs of the triangle as AC = BC = a and the hypotenuse as AB = c.
The Pythagorean Theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the legs:

$$c^2 = a^2 + a^2$$
$$c^2 = 2a^2$$
$$c^2 = 2 * AC^2$$

Therefore, the value of AB^2 (c^2) in terms of AC^2 (a^2) is: $AB^2 = 2 * AC^2$
Then the correct option is A)

12. Correct option: (A)

Figure according the question



Given that AD is the angle bisector of angle BAC in triangle ABC; we can apply the angle bisector theorem to find the value of DC.
The angle bisector theorem states that in a triangle, if a line bisects an angle and intersects the opposite side, it divides that side into segments that are proportional to the other two sides of the triangle.
In triangle ABC:

$$AB = 6\text{ cm}$$
$$AC = 5\text{ cm}$$
$$BD = 3\text{ cm (Since BD is given, it's a part of angle bisector theorem)}$$

According to the angle bisector theorem:

$$\frac{BD}{CD} = \frac{AB}{AC}$$

Substitute the given values:

$$\frac{3}{CD} = \frac{6}{5}$$
$$CD = \frac{15}{6}$$
$$CD = 2.5\text{ cm}$$

So, DC = 2.5 cm. correct option is A)

13. Correct option: (C)

For $\triangle ABC$

$$\angle BAC + \angle ABC + \angle ACB = 180° => \angle BAC = x° = 70°$$

Then the correct option is C)

14. Correct option: (D)

For $\triangle DAB$

$$AB = DB => \angle ADB = \angle DAB = y°$$

Using the pair of line AB and DE

$$\angle ABD = 180° - \angle ABC = 120°$$

In $\triangle DAB$

$$\angle ADB + \angle DAB + \angle ABD = 180° => 2y° = 60 => y° = 30$$

In $\triangle DAC$

$$\angle ADC + \angle DAC + \angle DCA = 180° => \angle DAC = \angle DAB + \angle BAC = y° + x° = 180° - 30° - 50° = 100°$$

Then the correct option is A)

15. Correct Matching is: a-S; b-T; c-P; d-R

Solving for the values of $x°, y°, z°, w°$

Extending the line CB to E

As line DE and FG is parallel then using the property of line we can say $\angle BEA = \angle BCF = 23°$

For line EC at vertex B

$$\angle EBA = 180° - \angle ABC = 98°$$

Using the $\triangle AEB$

$$z° = 180° - \angle BEA - \angle EBA = 59°$$

Using the pair of line AC and ED at vertex A the vertical angle pair

$$Z° + y° = 125° \Rightarrow y° = 125° - 59° = 66°$$

Using the $\triangle ABC$ $\quad x° = 180° - 82° - y° \Rightarrow x° = 32°$

Using the AC and FG at vertex C using the vertical angle pair

$$x° + 23° = w° \Rightarrow w° = 55°$$

Let's match the given angle values with above conclusions:

a) $2x° = 64° \longrightarrow$ (**S**)

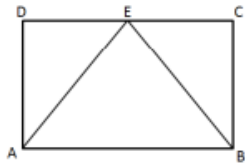
b) $y° = 66° \longrightarrow$ (**T**)

c) $z° = 59° \longrightarrow$ (**P**)

d) $w° = 55° \longrightarrow$ (**R**)

16. Correct Answer is: (2)

Figure according the question



A rectangle has all of its interior angles measuring 90°.

Given that E is midpoint of DC

Then the value of DE and EC will be $\frac{DC}{2} = \frac{24}{2} = 12$

Applying the Pythagorean Theorem in triangle ECB and triangle EDA

$$DE^2 + AD^2 = AE^2, CB^2 + CE^2 = BE^2,$$

Putting the values in above equation to find the value of AE and BE for perimeter.

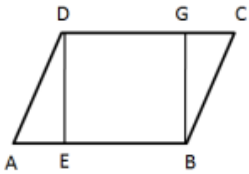
$$AE = 13, BE = 13$$

Perimeter of the triangle AEB $= 13 + 13 + 24 = 50$

$$50 = 2 \times 5^2 = 2 \times 5^x \Rightarrow x = 2$$

17. Correct Answer is: (4)

Figure according the question



Given ABCD is a parallelogram , $DE \perp AB$, $BG \perp CD$ and EBGD square side length 12cm and $BC = 13cm$

We have to find the value of AB

Looking the triangle BGC ($BG \perp DC$) a right angle triangle

$$BC^2 = AG^2 + CG^2 \Rightarrow CG^2 = 169 - 144 \Rightarrow CG = 5$$

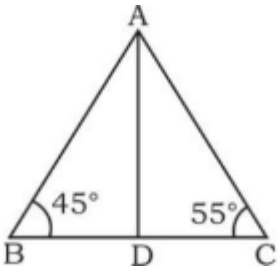
Value of $DC = DG + GC = 12 + 5 = 17$

$$DC = AB \Rightarrow \text{using parallelogram property}$$

$$AB = 17 = 1 + 4^2 \Rightarrow AB - 1 = 4^2 = x^2 \Rightarrow x = 4$$

18. Correct Answer is: (4)

Figure according the question



Given AD is angle bisector of $\angle A, \angle B = 45°, \angle C = 55°$

Consider the $\triangle ABC$

Sum of interior angle of a triangle always $180°$

$$\angle A + \angle B + \angle C = 180°$$

$$\angle A = 180° - 45° - 55° = 80°$$

As AD is angle bisector of $\angle A \Rightarrow \angle BAD = \angle CAD = \frac{80°}{2} = 40°$

We have to find the value of $\angle \frac{BAD}{10}$

$$\Rightarrow \frac{40}{10} = 4$$

19. Correct Answer is: (5)

Let A be the smallest angle.

1. The sum of 3 distinct angles is equal to the sum of 2 right angles, which is

$$90° + 90° = 180°$$

$$A + B + C = 180°$$

2. The difference between two pairs of the angles is $10°$.

$$B - A = 10°$$

$$C - B = 10°$$

Now, we can express B and C in terms of A using these equations:

$$B = A + 10°$$

$$C = B + 10°$$

Now, substitute these expressions for B and C back into the sum equation:

$$A + (A + 10°) + (A + 10° + 10°) = 180°$$

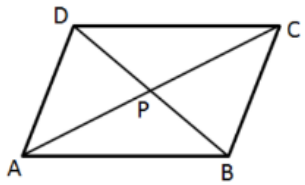
Now, simplify and solve for A:

$$3A + 30° = 180° \Rightarrow 3A = 150° \Rightarrow A = 50°$$

Now that we've found A, the smallest angle, we can find $\frac{1}{10}$th of it: $\frac{1}{10} \times 50° = 5°$

20. Correct Answer is: (4)

Figure according the question



Given that ABCD is a parallelogram where BD and AC are diagonals

From the property of parallelogram we conclude that the diagonals bisect each other as

$$DP = \frac{1}{2}BD = \frac{20}{2} = 10, CP = \frac{1}{2}AC = \frac{30}{2} = 15$$

We have to find the value of

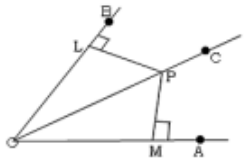
$$\frac{CP + DP}{5} = \frac{10 + 15}{5} = 5$$

Congruency of Triangles

LEVEL – I

1. Correct option: (A)

Given figure



Given conditions $PL \perp OB, PM \perp OA, PL = PM$

Since PL = PM and both PL and PM are perpendicular to their respective sides (OB and OA), we have a case of the hypotenuse-leg congruence (HL congruence) for right triangles.

In more detail: PL = PM (given)

$\angle LPO = \angle MPO$ (both are right angles, as PL ⊥ OB and PM ⊥ OA)

Hypotenuse OP = Hypotenuse OP (both are equal to OP)

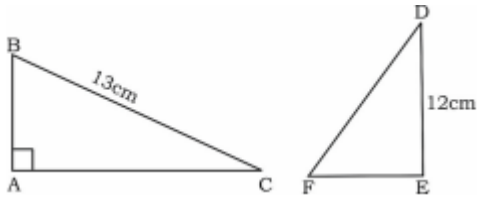
By the hypotenuse-leg congruence (HL congruence (RHS)) theorem, if the hypotenuse and one leg of one right triangle are equal to the hypotenuse and corresponding leg of another right triangle, then the two triangles are congruent.

Therefore, we can say that $\triangle PLO \cong \triangle PMO$ due to the HL congruence theorem, given that PL = PM and same hypotenuse.

Then the correct option is A)

2. Correct option: (B)

Given figure



Given conditions $\triangle CAB \cong \triangle DEF$

We have to find the value of EF.

According to the given condition we can write that

$$AB = EF, DE = AC, BC = DF, \angle BAC = \angle DEF$$

Looking into $\triangle DEF$ and applying Pythagorean Theorem.

$$FE^2 = 13^2 - 12^2 = 25 \Rightarrow FE = 5$$

Then the correct option is B)

3. Correct option: (A)

If $\triangle PQR \cong \triangle P'Q'R'$, their corresponding angles are equal in measure.

Given that $\angle P$ is twice $\angle Q'$, and $\angle Q'$is 30°, we can find the measure of $\angle P$:

$$\angle P = 2 \times \angle Q'$$
$$\angle P = 2 \times 30°$$
$$\angle P = 60°$$

Since $\triangle PQR \cong \triangle P'Q'R'$, $\angle R$ in $\triangle PQR$ is equal to the corresponding $\angle R'$ in $\triangle P'Q'R'$,.

The sum of the angles in a triangle is always 180°, so:

$$\angle R = 180° - (\angle P + \angle Q)$$
$$\angle R = 180° - (60° + 30°)$$
$$\angle R = 180° - 90°$$
$$\angle R = 90°$$

Therefore, the correct answer is A) 90°.

4. Correct option: (D)

If $\triangle ABC \cong \triangle PQR$, their corresponding angles are equal in measure.

Given that

$$\angle A = 2x, \angle Q = \frac{3}{2}x, \angle C = 40°$$

Since $\triangle ABC \cong \triangle PQR$, $\angle Q = \angle B = \frac{3}{2}x$

The sum of the angles in a triangle is always 180°, so:

$$\angle A + \angle B + \angle C = 180°$$

$$2x + \frac{3}{2}x + 40° = 180°$$

$$\frac{7}{2}x = 140 => x = 40°$$

$$\angle P = \angle A = 2x = 80$$

Therefore, the correct answer is D) 80°.

5. Correct option: (A)

Given conditions $\triangle ABC \cong \triangle A'B'C', AC = 3x + 4, B'C' = 2x + 1, P(\triangle A'B'C') = 30$

We have to find the value of A'B' in terms of x.

According to the given condition we can write that

$$AC = A'C' => A'C' + B'C' + A'B' = P(\triangle A'B'C') = 30$$

Putting the values in equation $3x + 4 + 2x + 1 + A'B' = 30 => A'B' = 25 - 5x$

Then the correct option is A)

6. Correct option: (B)

The correct answer is B) A.A.A.

Explanation:

S.S.S. (Side-Side-Side): This is a valid method to prove triangles congruent. If all three sides of one triangle are congruent to the corresponding three sides of another triangle, the triangles are congruent.

A.A.A. (Angle-Angle-Angle): This method cannot be used to prove that two triangles are congruent because having the same angles does not necessarily mean the triangles have the same shape and size. This is because two different triangles can have the same set of angles, but their side lengths could be different.

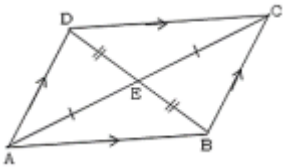
S.A.S. (Side-Angle-Side): This is a valid method to prove triangles congruent. If two sides and the included angle of one triangle are congruent to the corresponding two sides and included angle of another triangle, the triangles are congruent.

A.A.S. (Angle-Angle-Side): This is a valid method to prove triangles congruent. If two angles and a non-included side of one triangle are congruent to the corresponding two angles and non-included side of another triangle, the triangles are congruent.

So, the method that cannot be used to prove that two triangles are congruent is B) A.A.A.

7. Correct option: (D)

Given figure



In ABCD, let E be the intersection point of the diagonals AC and BD. When diagonals intersect, they create multiple congruent triangles. Here are the congruent triangles that can be identified:

1. $\triangle DEC \cong \triangle BEA$:

By the (S.A.S.) congruence criterion:

$$\angle AEB = \angle CED \text{ (both are opposite vertical angles)}$$

$EB = ED$ (given)

$EA = EC$ (given)

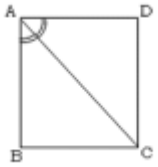
Therefore, $\triangle DEC \cong \triangle BEA$ (by S.A.S.).

Same as we can also prove that $\triangle BEC \cong \triangle DEA$.

Then the correct option is D)

8. Correct option: (A, B, C, D)

Given figure



Given figure can be a square or rectangle. (Without assuming we cannot prove $\triangle ABC \cong \triangle ADC$)

If ABCD is a rectangle or a square and AC is a diagonal, the postulate that can be used to prove that $\triangle ABC \cong \triangle ADC$ is (A.S.A, S.A.S, A.A.S, R.H.S and S.S.S).

Explanation:

- In a rectangle or a square, all angles are right angles (90°). So,

$$\angle ABC = \angle ADC = 90°$$

- Since ABCD is a rectangle or a square, opposite sides are parallel and congruent.

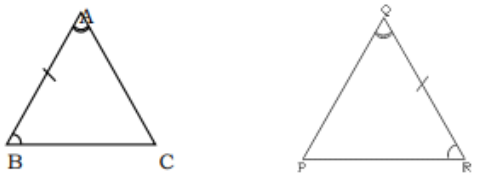
$$AB = DC, AD = BC, \angle BAC = \angle DCA$$

Therefore taking any three combinations we can prove that $\triangle ABC \cong \triangle ADC$.

All of the provided options are correct.

9. Correct option: (A)

Given figure



From the above figure we can conclude that $AB = QR, \angle A = \angle Q, \angle B = \angle R$

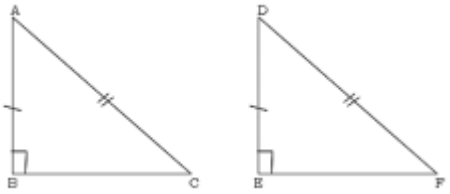
So using the A.S.A we can say that

$$\triangle ABC \cong \triangle QRP$$

Then the correct option is A)

10. Correct option: (C)

Given figure



From the above figure we can conclude that

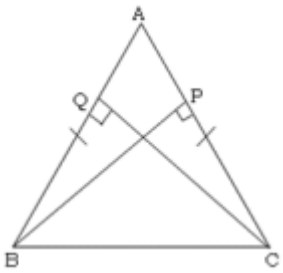
$$AB = DE, \angle B = \angle E = 90°, AB = DE$$

Given that $\triangle ABC$ and $\triangle DEF$ have right angles at B and E, AC = DF, and AB = DE, the postulate that can be used to prove that the triangles are congruent is C) RHS (Right Hypotenuse Side).

Explanation:
In the RHS (Right Hypotenuse Side) congruence criterion:
- If the hypotenuse and one leg of a right triangle are congruent to the hypotenuse and corresponding leg of another right triangle, the triangles are congruent.
In this case:
- Hypotenuse AC is congruent to hypotenuse DF.
- Leg AB is congruent to leg DE.
Therefore, using the RHS congruence criterion, we can prove that
$\triangle ABC \cong \triangle DEF$

11. Correct option: (C)

Given figure



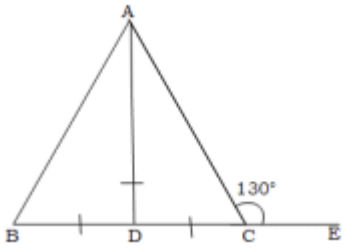
We are given that $BQ = CP, BP \perp AC, CQ \perp AB$
From the given condition and figure we can conclude that

$$\triangle BQC \cong \triangle BPC \ldots . (R.H.S)$$

As both of the triangle is right angle triangle and hypotenuse and one of the leg is equal in both of the triangle because they share the same hypotenuse (BC).
So the correct option is C)

12. Correct option: (40°)

Given figure



We are given that $\angle ACE = 130°, BD = AD = DC$
We have to find out the value of

$$\angle ABC$$

Form the adjacent angle pair we can calculate

$$\angle DCA = 180° - 130° = 50°$$

From the condition $DC = AD \Rightarrow \angle DAC = \angle DCA = 50°$
From the condition $BD = AD \Rightarrow \angle DAB = \angle ABD = x°$
Looking into triangle ABC

$$\angle A + \angle B + \angle C = 180°$$

$$\angle A = \angle DAC + \angle DAB, \angle C = \angle DCA, \angle B = \angle ABD$$

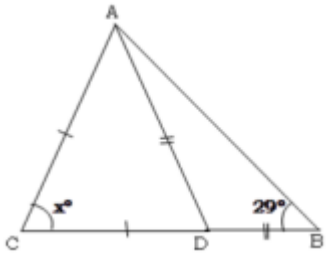
Putting the values

$$x° + 50° + x° + 50° = 180° \Rightarrow x° = 40°$$

The value of $\angle ABC = \angle ABD = 40°$

13. Correct option: (A)

Given figure



We are given that $\angle B = 29°, BD = AD, CD = AC, \angle C = x°$

We have to find out the value of x.

Form the adjacent angle pair we can calculate

$$\angle CDA = 180° - \angle BDA$$

From the condition $AD = BD \Rightarrow \angle DAB = \angle DBA = 29°$

Looking into triangle ADB, $\angle BDA = 180° - \angle DAB - \angle DBA = 122°$

Putting the values $\angle CDA = 58°$

From the condition $CD = AC \Rightarrow \angle CAD = \angle CDA$

Looking into $\triangle ACD$,

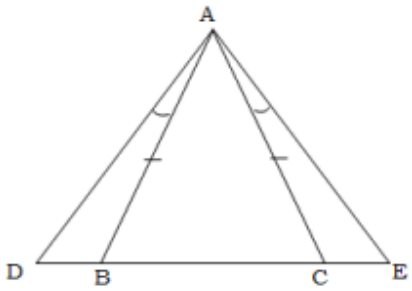
$$x° + 58° + 58° = 180° \Rightarrow x° = 64°$$

Then $x° = 64° = \angle C$

The correct option is A)

14. Correct option: (C)

Given figure



We are given that $AB = AC, \angle BAD = \angle CAE$

We have to find out the value of AD.

Let $\angle BAD = \angle CAE = x°$

From the condition $AB = AC \Rightarrow \angle ABC = \angle ACB = y°$

From the adjacent angle pair $\angle ABD = 180° - y°, \angle ACE = 180° - y°$

From triangle ABD and ACE we can conclude that

$$\angle ADB = 180° - x° - (180° - y°) = y° - x°$$

$$\angle AEC = 180° - x° - (180° - y°) = y° - x°$$

$$\angle ADB = \angle ACE$$

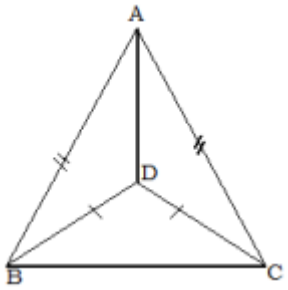
Looking in triangle ADE we have $\angle ADB = \angle ACE$ then we can say that

$$AD = AE$$

The correct option is C)

15. Correct option: (C)

Given figure



We are given that $AB = AC, BD = CD$

We have to find out the value of $\angle ABD$.

From the condition $AB = AC \Rightarrow \angle ABC = \angle ACB$

From the condition $BD = DC \Rightarrow \angle DBC = \angle DCB$

Let analyze the $\angle ABC, \angle ACB$ as $\angle ABC = \angle ABD + \angle DBC \Rightarrow \angle ABD = \angle ABC - \angle DBC$

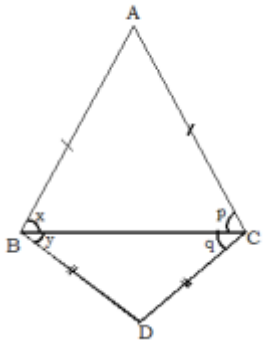
$\angle ACB = \angle ACD + \angle DCB \Rightarrow \angle ACD = \angle ACB - \angle DCB$

Putting the values from above $\angle ABD = \angle ACB - \angle DCB \Rightarrow \angle ACD$

The correct option is C)

16. Correct option: (A, D)

Given figure



We are given that $AB = AC, BD = CD$

We have to find out the value of $x - p$.

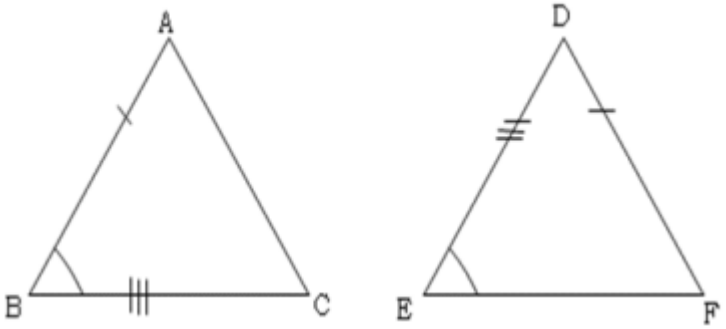
From the condition $AB = AC \Rightarrow x = p \Rightarrow x - p = 0$

From the condition $BD = DC \Rightarrow y = q \Rightarrow y - q = 0 \Rightarrow q - y = 0$

The correct option is a, D)

17. Correct option: (C)

Given figure

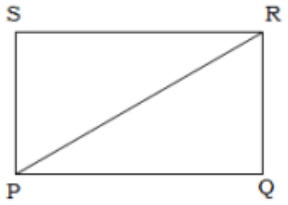


We are given that $AB = DF, BD = DE, \angle ABC = \angle FED$

Which is not the sufficient condition for the given triangle be congruent .

18. Correct option: (A)

Given figure



We are given that $PQ = QR = RS = SP$

1. Diagonal PR divides the PQRS into two congruent triangles, namely triangle PQR and triangle PSR.

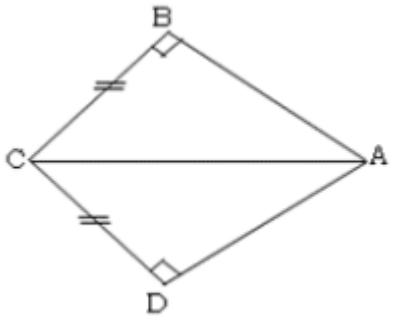
2. In each of these triangles, the sides PR (which is common) and the sides PQ, QR, PS, and RS are all equal due to the given.

By the Side-Side-Side (S.S.S.) congruence criterion, the two triangles PQR and PSR are congruent. This is because all three sides of triangle PQR are equal in length to the corresponding sides of triangle PSR.

We can also use S.A.S and A.S.A to prove the same if given quadrilateral is square or rhombus.

19. Correct option: (B)

Given figure



We are given that $BC = CD$ and $\angle B = \angle D = 90°$

Given that ABCD is a quadrilateral with right angles at B and D, and BC = DC, we can prove that $\triangle ABC \cong \triangle ADC$ using the RHS (Right Hypotenuse Side) congruence criterion.

In $\triangle ABC$ and $\triangle ADC$, we have:

$$BC = DC \text{ (Given)}$$

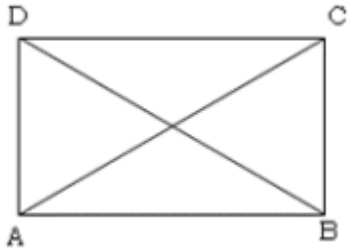
$$\angle B = \angle D \text{ (Both are right angles)}$$

Since BC = DC and both $\triangle ABC$ and $\triangle ADC$ have a right angle, we can conclude that $\triangle ABC$ and $\triangle ADC$ are right triangles with the same hypotenuse, which is side AC.

Therefore, using the RHS congruence criterion, we can prove that $\triangle ABC \cong \triangle ADC$

20. Correct option: (C)

Given figure



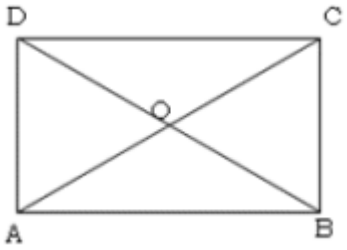
We are given that $AD \perp AB, BC \perp AB, AD = BC$
We have to find the $\triangle ABC \cong ?$ using S.A.S postulate.
To prove triangles congruent using the Side-Angle-Side (S.A.S.) postulate, we need to identify a common angle, a side between the angles, and another side opposite to one of the angles.
Looking $\triangle ABC$ and $\triangle DAB$

$$\angle DAB = \angle ABC, AD = BC$$

And we are using one common side AB so we can say that $\triangle ABC \cong \triangle BAD$

LEVEL - II

1. Correct option: (D)
Given figure

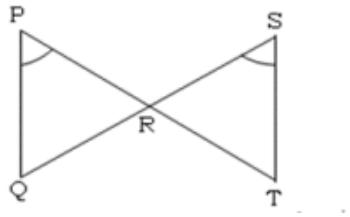


We are given that $AO = CO, BO = DO, \angle AOD = 90°$
We have to find the congruent triangle using S.A.S postulate.
To prove triangles congruent using the Side-Angle-Side (S.A.S.) postulate, we need to identify a common angle, a side between the angles, and another side opposite to one of the angles.
Let's check options
A) $\triangle AOD \cong \triangle COD \cong \triangle COB$
For $\triangle AOD \cong \triangle COD$
We are given $AO = CO$. (One of the side of triangles is equal)
$OD = OD$ (Common side)
$\angle AOD = 90° \Rightarrow \angle DOC = 180° - 90° = 90°$ (Adjacent angle for one line)
So $\triangle AOD \cong \triangle COD$ is true.
Like above we can prove that $\triangle AOD \cong \triangle COD \cong \triangle COB$
B) As well as we can prove that $\triangle BOC \cong \triangle BOA \cong \triangle DOA$
C) From the congruency of triangle given above we can also prove that

$$AB = BC = CD = AD$$

So we can say that all of the options are correct.

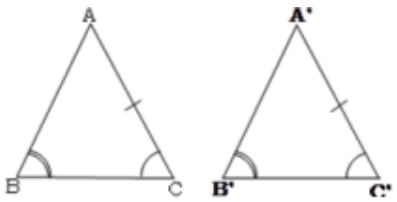
2. Correct option: (D)
Given figure



We are given that $PQ = TS, \angle P = \angle S$
We have to find the postulate which prove $\triangle PQR \cong \triangle STR$
We can conclude that $\angle PRQ = \angle SRT$ (vertical opposite angle)
Using the above known information we can conclude that using the
A.A.S postulate we can prove $\triangle PQR \cong \triangle STR$.

3. Correct option: (D)

Given figure



From the figure $AC = A'C', \angle B = \angle B', \angle C = \angle C'$

We have to find the postulate which prove $\triangle ABC \cong \triangle A'B'C'$

Here we have one side of the one of the triangle is equal to another triangle and two of its angle is also equal to two of other triangle angle so we can conclude that using A.A.S postulate we can easily prove $\triangle ABC \cong \triangle A'B'C'$.

A.A.S

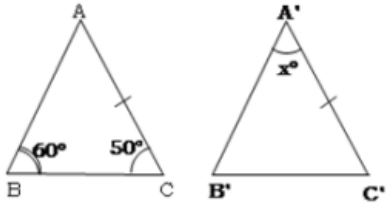
If two angles and a non-included side of one triangle are congruent to the corresponding two angles and non-included side of another triangle, the triangles are congruent.

A.S.A

If two angles and the included side of one triangle are congruent to the corresponding two angles and included side of another triangle, the triangles are congruent.

4. Correct option: (C)

Given figure



Given that $\triangle ABC \cong \triangle A'B'C'$

From the given information we can say that $\angle BAC = B'A'C' = x°$

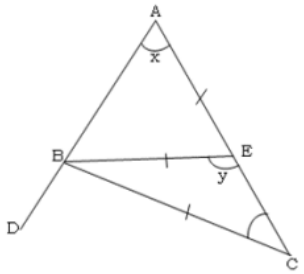
In triangle ABC $\angle BAC = 180° - 60° - 50° \Rightarrow \angle BAC = 70°$

Then the value of $x = 70$.

Correct option is C)

5. Correct option: (B)

Given figure



Given that $BC = BE = AE$

We have to find the value of $\angle y$.

From the figure we can say that $\angle AEB = 180° - y°$

Looking into $\triangle$ ABE

$$BE = AE \Rightarrow \angle EAB = \angle EBA = x°$$

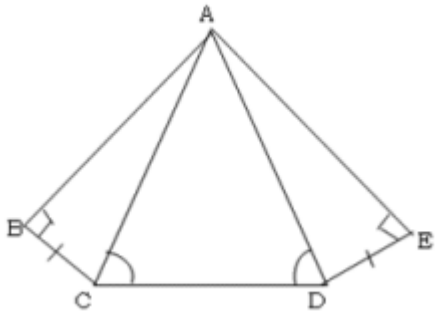
Triangle angle sum law $\angle AEB + \angle EAB + \angle EBA = 180° \Rightarrow 180° - y° + x° + x° = 180°$

$$y° = 2x°$$

Then the value of $\angle y = 2\angle x$. Option B)

6. Correct option: (A)

Given figure



Given that $\angle ACD = \angle ADC, \angle ABC = \angle AED = 90°, BC = ED$

We have to find the value of AB.

Using $\angle ACD = \angle ADC \Rightarrow AC = AD$ ($\triangle ACD$)

Comparing the triangle $\triangle ABC, \triangle AED$

From above equations we can conclude that $\triangle ABC \cong \triangle AED$ as R.H.S postulate.

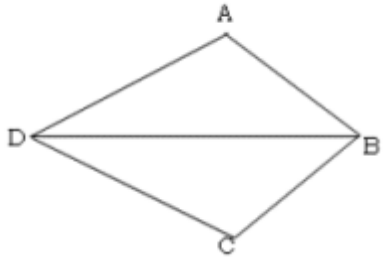
$AC = AD$(hypertonuse), $DE = BC$(one leg), $\angle ABC = \angle AED$(right angle)

As $\triangle ABC \cong \triangle AED \Rightarrow AB = AE$

Then the correct option is A)

7. Correct option: (A)

Given figure



Given that $AB = CB, AD = CD$

Let go through options

A) $\triangle DCB \cong \triangle DAB$

From the given values we can conclude that given condition is true using S.S.S postulate as

$AB = CB$(given), $AD = CD$(given), $BD = BD$(common side)

B) $\angle C = \angle B$

As we prove that $\triangle DCB \cong \triangle DAB \Rightarrow \angle C = \angle A \neq \angle B$

C) $BD = AD$

As we prove that $\triangle DCB \cong \triangle DAB \Rightarrow AD = CD \neq BD$

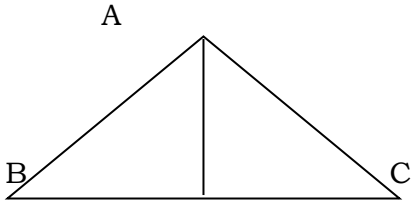
D) $\triangle DCB \cong \triangle ABD$

By above condition we can conclude that $\triangle DCB \cong \triangle DAB$, $\triangle DCB \neq \triangle ABD$

Then the correct option is A)

8. Correct option: (B)

Given figure



Given that $AB = AC$

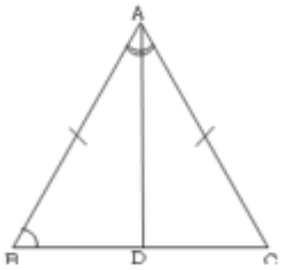
Given that $\triangle ABC$ is isosceles with AB = AC and D is the midpoint of base BC, we can determine which statement is true using congruence properties.
Since AB = AC, triangle ABC is isosceles with the equal sides being AB and AC.
Because D is the midpoint of BC, we have BD = CD.
We can consider the angles formed by the sides of the triangle and the bisector:
$\angle ABD = \angle ACD$ (Angles opposite to the equal sides in an isosceles triangle).
From the given condition we can conclude that $\triangle ABD \cong \triangle ACD$ as

$$AB = AC(\text{given}), \angle ABD = \angle ACD(\text{from above}), AD = AD(\text{common side})$$

Using the S.A.S postulate we can say that $\triangle ABD \cong \triangle ACD$.
Then the correct option is B)

9. Correct option: (B)

Given figure



Given that $AB = AC$ and AD is angle bisector of $\angle A$

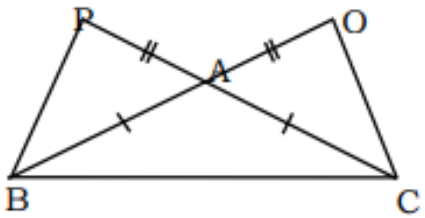
Given that triangle ABC is isosceles with AB = AC and AD is the bisector of angle A, we can determine which statement is true using congruence properties.
Since AB = AC, triangle ABC is isosceles with the equal sides being AB and AC.
So we can write $\angle ABC = \angle ACB$
Because AD is the bisector of $\angle A$, we have $\angle BAD = \angle CAD$
From the given condition we can conclude that $\triangle ABD \cong \triangle ACD$ as

$$AB = AC(\text{given}), \angle ABC = \angle ACB(\text{from above}), \angle BAD = \angle CAD$$

Using the A.S.A or S.A.S postulate we can say that $\triangle ABD \cong \triangle ACD$.
Then the correct option is B)

10. Correct option: (B)

Given figure



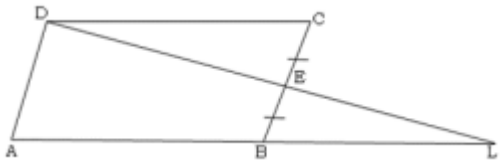
Given that $AB = AC \Rightarrow \triangle ABC(\text{isosceles}), AP = AQ$
Let's use the properties of triangle and prove the options
We have $AP = AQ(\text{given}), AB = AC(\text{given}), \angle BAP = \angle CAQ(\text{vertical angles})$
From above similarity (S.A.S postulate) we can conclude that $\triangle APB \cong \triangle AQC$.
Using $\triangle APB \cong \triangle AQC \Rightarrow \angle BPA = \angle BPC = \angle AQC = \angle BQC, PB = QC$
For $\triangle BQC, \triangle CPB$, BC is common side $\Rightarrow \triangle BQC \cong \triangle CPB \neq \triangle CBP$

$$\triangle APB \cong \triangle AQC \Rightarrow PB = QC$$
$$BQ = AB + AQ$$

Then the correct option is B)

11. Correct option: (A)

Given figure



Given that ABCD (parallelogram), E (mid point of BC)

Let's use the properties of triangle and prove the options

A) $\triangle EBL \cong \triangle ECD$

Look at the triangles given in the option.

$$EB = EC \text{ (E is midpoint od BC)}, \angle DEC = LEB \text{(vertical angle)}$$

$$\angle EDC = \angle ELB \text{ (alternative interior angle)}$$

Using the postulate A.A.S we can conclude that option A) is correct.

B) $\triangle DCE \cong \triangle BLE$

As we proved that $\triangle EBL \cong \triangle ECD \Rightarrow \triangle DCE \neq \triangle BLE$

C) AD=BL

As we proved that $\triangle EBL \cong \triangle ECD \Rightarrow BL = DC \Rightarrow BL = AB$

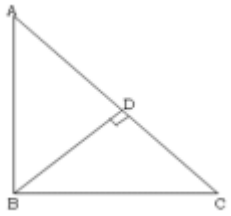
D) BC=BL

As ABCD is parallelogram $BC = AD$

Then the only correct option is A)

12. Correct option: (C)

Given figure



Given that $AB = BC \Rightarrow \triangle ABC(\text{isosceles}, \angle B = 90°), \angle BDC = 90°$

For given triangles $\triangle ABD, \triangle CBD$

We have

$$AB = BC(\text{given}), BD = BD(\text{common}), \angle BDC = \angle BDA(\text{right angles})$$

From above similarity (RHS postulate) we can conclude that $\triangle ABD \cong \triangle CBD$.

The RHS (Right Hypotenuse Side) postulate is a criterion used to prove the congruence of two right triangles. It states that if the hypotenuse and one leg of one right triangle are congruent to the hypotenuse and the corresponding leg of another right triangle, then the two triangles are congruent.

In our question $AP = BC \Rightarrow$ Hypotenuse ($\triangle ABD$) = Hypotenuse ($\triangle CBD$)

One leg is common for both of the right angle triangle.

Then the correct option is C)

13. Correct option: (C)

We have to find the value of $\angle ABC$

Looking at triangles $\triangle ABC, \triangle DCB$

$$AB = CD(\text{given}), \angle ABC = \angle DCB = 90° \text{ (given } \perp) \; BC = BC$$

From above condition we can say that $\triangle ABC, \triangle DCB$ are congruent using SAS postulate.

Then $\angle ABC = \angle BCD$

Therefore the correct option is C)

14. Correct option: (A)

Looking at triangles $\triangle ABC, \triangle DCB$

$$AB = CD(\text{given}), \angle ABC = \angle DCB = 90° \text{ (given } \perp) \; BC = BC$$

From above condition we can say that $\triangle ABC, \triangle DCB$ are congruent using SAS postulate.

Therefore the correct option is A)

15. Correct option: (A)

We have to find the value of $\angle BAC$

Looking at triangles $\triangle ABC, \triangle DCB$

$$AB = CD(\text{given}), \angle ABC = \angle DCB = 90° \text{ (given } \perp) \; BC = BC$$

From above condition we can say that $\triangle ABC, \triangle DCB$ are congruent using SAS postulate.

Then $\angle BAC = \angle BDC$

Therefore the correct option is A)

16. Correct Matching is: a-R; b-T; c-P; d-Q

Looking at the triangles $\triangle AOF, \triangle AOE$

Form the figure we can conclude that

$$\angle AFO = \angle AEO = 90°(\perp), AO = AO \text{ (common side)}$$

Consider sides OF and OE:

Since F and E are the feet of perpendiculars from O to BC and AC respectively, OF and OE are the respective heights of triangles AOF and AOE.

Since they are heights from the same base AO, they have the same length.

So, we have: OF = OE

Now, all three conditions of RHS congruence are satisfied:

1. We have right triangles AOF and AOE.
2. The hypotenuse AO is common to both triangles.
3. We have shown that the heights OF and OE are congruent.

Therefore, by the RHS congruence criterion, we can conclude that

$$\triangle AOF \cong \triangle AOE$$

Similarly we can conclude that $\triangle BFO \cong \triangle BGO$

Similarly we can conclude that $\triangle CGO \cong \triangle BGO$

Using above two congruency conditions we also conclude that

$$\triangle AOF \cong \triangle BGO \Rightarrow \triangle AOF \cong \triangle CGO$$

From the above congruency we can conclude that

$$OF = OG$$

$$GC = EC = BG$$

Let's match the given angle values with above conclusions:

$$\mathbf{a}) \triangle AOF \cong \triangle AOE \cong \triangle CGO \longrightarrow (\mathbf{R}, \mathbf{T})$$

$$\mathbf{b}) \triangle OGB \cong \triangle OGC \cong \triangle AOE \longrightarrow (\mathbf{R}, \mathbf{T})$$

$$\mathbf{c}) OF = OG \longrightarrow (\mathbf{P})$$

$$\mathbf{d}) GC = EC = BG \longrightarrow (\mathbf{Q}, \mathbf{S})$$

17. Correct Answer is: (5)

Given that triangle ABC is congruent to triangle PQR using the ASA (Angle-Side-Angle) criterion; let's use the angle information to find the value of y.

Given:

$$\angle B = x + 15$$

$$\angle Q = 70$$

$$\angle C = 5y + 10$$

$$\angle R = 35$$

Since triangles ABC and PQR are congruent by ASA, the corresponding angles must be congruent:

$$1.\angle B = \angle Q \text{ (Angle} - \text{Angle congruence)}$$
$$2.\angle C = \angle R \text{ (Angle} - \text{Angle congruence)}$$

Using the given angle values:

$$5y + 10 = 35$$
$$5y = 35 - 10 = 25$$
$$y = \frac{25}{5} = 5$$

18. Correct Answer is: (14.4)

Given that triangle ABC has ∠B = 2∠C, D is the point on BC where AD bisects ∠ BAC, and AC = BC, we need to find the value of $\frac{1}{5}\angle BAC$

Let's denote ∠C as x. Then, ∠B is 2x.

Since AC = BC, triangle ABC is isosceles, and ∠B = ∠A

So, $x + 2x + 2x = 180$

$$5x = 180$$
$$x = \frac{180}{5} = 36°$$

∠C = 36°.

∠B = 2 × 36 = 72° =∠A.

Therefore, $\frac{1}{5}\angle BAC = \frac{1}{5} \times 72° = \frac{72}{5}$.

So, the answer is $\frac{72}{5}$.

19. Correct Answer is: (1)

Given figure

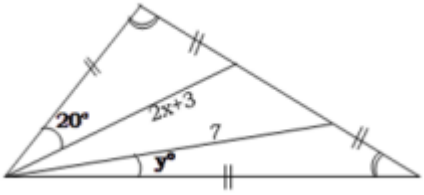
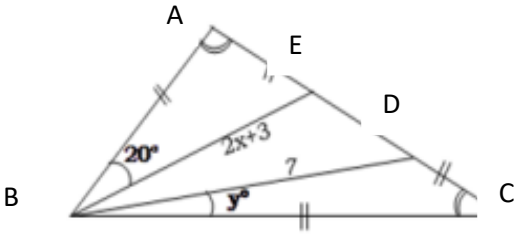


Figure with notations



According the figure $AB = BC, \angle BAC = \angle BCA, AE = DC$

For given triangles $\triangle ABE, \triangle CBD$

We have

$AB = BC(\text{given}), AE = DC(\text{given}), \angle BAC = \angle BCA(\text{given})$

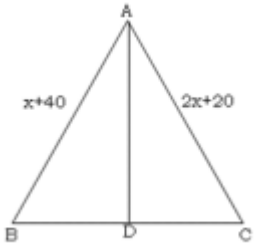
From above similarity (S.A.S postulate) we can conclude that $\triangle ABE \cong \triangle CBD$.

Using the congruent property of $\triangle ABE \cong \triangle CBD$ we can say

$$BD = BE \Rightarrow 2x + 3 = 7 \Rightarrow x = 2 \Rightarrow \frac{1}{2}x = 1$$

20. Correct Answer is: (5)

Given figure



Given that $\triangle ABD \cong \triangle ACD$

Using the congruent property of $\triangle ABD \cong \triangle ACD$ we can say

$$AB = AC \Rightarrow x + 40 = 2x + 20 \Rightarrow x = 20 \Rightarrow \frac{x}{4} = 5$$

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

Given conditions $\triangle ABC \cong \triangle PQR, \angle A = \angle P$

We have to find the value of x.

According to the given condition we can write that

$$\angle A = \angle P \Rightarrow 5x + 4 = 79 \Rightarrow 5x = 75 \Rightarrow x = 15$$

Then the correct option is B)

2. Correct option: (C)

If $\triangle ABC \cong \triangle PQR$, their corresponding sides are equal in length. Specifically, if $BC = 3x + 4$ and $QR = 2x + 7$, we can set up an equation based on their congruence:

$$BC = QR$$

Substituting the given lengths: $3x + 4 = 2x + 7$

$$3x - 2x = 7 - 4$$

$$x = 3$$

So, the value of x is 3, option C)

3. Correct option: (C)

Given conditions

$$\triangle PQR \cong \triangle P'Q'R', QR = x + 8, P'Q' = 2x - 1, PR = 2x + 3, P(\triangle P'Q'R') = 45$$

We have to find the value of Q'R'.

According to the given condition we can write that

$$QR = Q'R', PR = P'R' \Rightarrow P'Q' + Q'R' + P'R' = P(\triangle P'Q'R') = 45$$

Putting the values in equation

$$x + 8 + 2x - 1 + 2x + 3 = 45 \Rightarrow 5x = 35 \Rightarrow x = 7$$

Value of Q'R' $= x + 8 \Rightarrow 15$

Then the correct option is C)

4. Correct option: (D)

Given conditions

$$\triangle ABC \cong \triangle A'B'C', AB = A'B', BC = B'C', AC = A'C', \angle A' + \angle B' = 116°$$

We have to find the value of $\angle C$.

According to the given condition we can write that

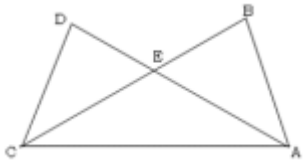
$$\angle A = \angle A', \angle B = \angle B', \angle C = \angle C'$$

$$\angle A + \angle B + \angle C = 180° \Rightarrow \angle C = 180° - (\angle A + \angle B) = 64°$$

Then the correct option is D)

5. Correct option: (A)

Given figure



We are given that $AB = CD, AD = BC$

For $\triangle ADC, \triangle CBA$ the side AC is common.

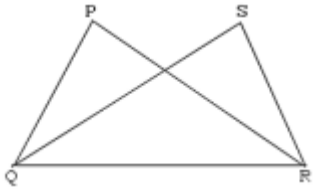
From the given condition and figure we can conclude that $\triangle ADC \cong \triangle CBA$ …. $(S.S.S)$

As one side is common and two other sides are equal as given.

So the correct option is A)

6. Correct option: (A)

Given figure



We are given that $PQ = SR, \angle PQR = \angle QRS$

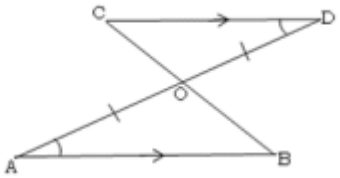
From the given condition and figure we can conclude that $\triangle PQR \cong \triangle SRQ$ …. $(S.A.S)$

As one side is common and one of the side is equal as given and one of the angle is also equal in both triangle.

So the correct option is A)

7. Correct option: (C)

Given figure



We are given that $AB \parallel CD$ and BC, AD are travesal lines

From the given condition and figure we can conclude that $\triangle OAB \cong \triangle ODC$ …. $(A.S.A)$

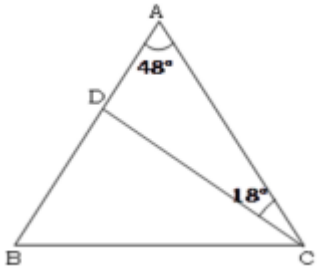
As from the parallel line and traversal line system

$$\angle AOB = \angle COD, \angle OAB = \angle ODC, AO = DO$$

Then the correct option is B)

8. Correct option: (A)

Given figure



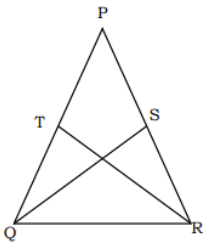
We are given that $AB = AC, \angle ACD = 18°$

From the given condition and figure we can conclude that

$\triangle ABC$ is isosceles triangle so $\angle ABC = \angle ACB = x°$
Let find the $\angle ADC$ from $\triangle ADC$, $\angle ADC = 180° - 18° - 48° = 114°$
Then the $\angle CDB = 180 - \angle ADC = 66°$ (Adjacent pair of angle)
Now look at $\triangle BDC$ $\angle BCD = \angle BCA - \angle DCA = x° - 18°, \angle ABC = \angle DBC$
Putting the values $x° + x° - 18° + 66° = 180° \Rightarrow x° = 66°$
From above information we can conclude that $\angle CBD = \angle CDB \Rightarrow BC = DC$
Then option A) is correct.

9. Correct option: (D)

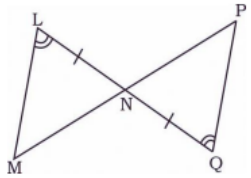
Given figure



We are given that $TR = QS$ and $\angle PQS = \angle PRT =$
Given the information that $QS = TR, \angle PQS = \angle PRT,$ and $\angle P$ is a common angle between $\triangle PQS$ and $\triangle PRT$, we can prove that $\triangle PQS$ and $\triangle PRT$ are congruent using the A.A.S congruence criterion.
Using the A.A.S. congruence criterion: $\angle PQS = \angle PRT$ (Angle)
$\angle P$ is a shared angle.
Side QS = Side TR (Side)
By the A.A.S congruence criterion, when two angles and the included side of one triangle are congruent to the corresponding two angles and included side of another triangle, the triangles are congruent.

10. Correct option: (D)

Given figure



We are given that $NP = x + 4, MN = 3x - 4, PQ = 2x + 1$
We have to find the value of LM.
Using the postulate A.S.A we can prove that $\triangle LMN \cong \triangle QPN$
$LN = NQ, \angle NLM = \angle NQP, \angle PNQ = \angle MNL \Rightarrow \triangle LMN \cong \triangle QPN$
As $\triangle LMN \cong \triangle QPN \Rightarrow NP = MN \Rightarrow x + 4 = 3x - 4 \Rightarrow x = 4$
$\triangle LMN \cong \triangle QPN \Rightarrow LM = PQ = 2 \times 4 + 1 \Rightarrow 9cm$
Then the correct option is D)

11. Correct option: (A)

We have to find the value of x
Looking at triangles $\triangle ABC, \triangle DEC$ as they are congruent triangles.
$AB = CD, ED = AB, BC = EC$
From above condition we can set the equations as $3y = 2x, x + y = 5$
Putting the value of x form equation first to equation second as

$$\frac{3}{2}y + y = 5 \Rightarrow \frac{5y}{2} = 5 \Rightarrow y = 2$$

Then $x = \frac{3}{2}y = \frac{3}{2} \times 2 = 3$
Therefore the correct option is A)

12. Correct option: (B)

We have to find the value of 2y

Looking at triangles $\triangle ABC, \triangle DEC$ as they are congruent triangles.

$$AB = CD, ED = AB, BC = EC$$

From above condition we can set the equations as $3y = 2x, x + y = 5$

Putting the value of x form equation first to equation second as

$$\frac{3}{2}y + y = 5 \Rightarrow \frac{5y}{2} = 5 \Rightarrow y = 2$$

Then $2y = 2 \times 2 = 4$

Therefore the correct option is B)

13. Correct option: (B)

We have to find the value of $\angle BAC$

Looking at triangles $\triangle ABC, \triangle DEC$ as they are congruent triangles.

$$\angle BAC = \angle CDE, \angle ABC = \angle CED, \angle ACE = \angle ECD$$

From above condition we can say that $\angle BAC = \angle CDE$

Therefore the correct option is B)

14. Correct Matching is: a-S; b-T; c-Q; d-P

Looking at the triangles $\triangle ABC$

We are given that AB=AC=>$\angle ABC = \angle ACB = x°$

We are also given that $\angle C = 2\angle A$

Using the sum of interior angle of a triangle

$$\angle C + \angle C + \angle A = 180° \Rightarrow 5\angle A = 180° \Rightarrow \angle A = 36°, \angle C = 72°, \angle B = 72°$$

So we can conclude that

$$\angle BDC = 180° - \angle\frac{B}{2} - \angle C = 72°$$

$$\angle DBC = \frac{72°}{2} = 36°, \angle BDA = 180° - \angle\frac{B}{2} - \angle A = 108° \Rightarrow \angle DBC + \angle BDA = 144°$$

$$\angle ABD = \angle\frac{B}{2} = 36°$$

Let's match the given angle values with above conclusions:

$$\mathbf{a}) \angle BDC = 72° \longrightarrow (\mathbf{S})$$
$$\mathbf{b}) \angle DBC + \angle BDA = 144° \longrightarrow (\mathbf{T})$$
$$\mathbf{c}) \angle ABD = 36° \longrightarrow (\mathbf{Q})$$
$$\mathbf{d}) \angle BDA = 108° \longrightarrow (\mathbf{P})$$

15. Correct Answer is: (2)

Given that triangle ABC is an isosceles triangle with AB = AC, and the vertex angle (angle BAC) is twice the sum of the base angles (angles ABC and ACB), we can find the value of $\frac{1}{15}$ of $\angle B$.

Let's denote angle ABC as x. Then, angle ACB is also x (since triangle ABC is isosceles). And the vertex angle BAC is 2 times the sum of the base angles, so it's $2 * (x + x) = 4x$.

Since the sum of the angles in a triangle is 180°, we can write an equation:

$$x + x + 4x = 180$$
$$6x = 180$$
$$x = 30$$

Now we know that each base angle (angle ABC and angle ACB) is 30°, and the vertex angle (angle BAC) is $4x = 4 \times 30 = 120°$.

Then the value of $\frac{1}{15}\angle B = \frac{1}{15} \times 30° = 2$

5. Data Handling
Solutions

LEVEL- I

1. Correct option: (A)

First 8 odd natural numbers $1, 3, 5, 7, 9, 11, 13, 15$

$$Mean = \frac{sum\ of\ all\ numbers}{count\ of\ numbers} = \frac{S}{N}$$

S = $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 = 64$

N = 8

Mean $= \frac{S}{N} = \frac{64}{8} = 8$

2. Correct option: (D)

In a pie chart, the sum of the angles of all its components is always **360°**.

A pie chart represents a whole, and the angles of its components (or sectors) are proportional to the relative sizes of the data they represent. Since a full circle has 360°, the sum of the angles in a pie chart must add up to 360°.

3. Correct option: (C)

In a bar graph, the height of a bar is proportional to the: C) Value of component

The height of each bar in a bar graph represents the value or quantity of the component it represents. The taller the bar, the larger the value it corresponds to. This is a fundamental characteristic of bar graphs, where the visual representation helps compare different values or categories effectively.

4. Correct option: (B)

Given number of candidates appeared for exam in consecutive 8 years as

$$12464, 14731, 13213, 14219, 15613, 11287, 14162$$

$$\boldsymbol{Range = Highest\ value - Lowest\ value}$$

Highest number of candidates appeared = 15613

Lowest number of candidates appeared = 11287

So, $Range\ of\ the\ number\ of\ candidates\ appeared = 15613 - 11287 = 4326$

5. Correct option: (C)

According to the given table,

Number of people who play between the age group 10-13 is = 17

Number of people who go to the work between the age group 25-28 is = 20

Then the Ratio of the number of people who play between the age group 10-13 to the number of people who go to the work between the age group 25-28 will be $= \frac{17}{20}$

6. Correct option: (D)

Given observation $11, 4, 12, 9, 8, 11, 12, 4, 12, 8, 8$

We have to find out the frequency of the number 12

Frequency of any number can be defined as how many times the number appears in the observation.

So the frequency of the number 12 is = 3

7. Correct option: (A)

Given the size of bat as $10, 15, 20, 30, 40, 50$

$$Range = Highest\ value - Lowest\ value$$

Greatest size of the bat = 50

Smallest size of the bat = 10

So $Range\ of\ the\ bat\ size = 50 - 10 = 40$

8. Correct option: (C)

In a bar graph, the height of the bars is proportional to the quantity they represent. If you know the relationship between the height of a bar and the units it represents, you can use that relationship to find the height for a different number of units.

Given that a bar with a height of 5 cm represents 40 units, you can set up a proportion to solve for the height of a bar representing 56 units:

$$\frac{Height1}{Units1} = \frac{Height2}{Units2}$$

Plugging in the values: $\frac{5}{40} = \frac{Height2}{56}$

Solving for Height2: $Height2 = \frac{5*56}{40}$

$$Height2 = 7$$

So, the height of a bar representing 56 units is 7 cm.

9. Correct option: (B)

The range of a set of observations is the difference between the highest and lowest values in that set.

In this case, you're given that the range is 15 and the highest score is 28. To find the least score (lowest value); you can subtract the range from the highest score:

$$Least\ score = Highest\ score - Range$$
$$Least\ score = 28 - 15$$
$$Least\ score = 13$$

Therefore, the correct answer is: B) 13

10. Correct option: (B)

In the given bar graph, the scale is $1cm = 2 * m$ units. This means that for every $1\ cm$ on the graph, it represents $2 * m\ units$

Given that the length of the bar representing $875\ units$ is $1.75\ cm$; we can use the scale to find the actual value of m.

$$1cm = 2 * m\ units$$

Multiplication both sides by 1.75

$$1.75cm = 1.75 * 2 * m => 1.75cm = 3.5 * m\ units$$

Given that $1.75cm => 875\ units$

As from earlier equations we can say that $875\ units = 3.5 * m\ units$

Solving for $m => m = \frac{875}{3.5} = \frac{8750}{35} = 225$

Therefore, the correct answer is: 225

11. Correct option: (B)

To find the value of x, we need to use the formula for the mean (average):

$$Mean = \frac{Sum\ of\ all\ values}{Number\ of\ values}$$

Given that the mean is 6, and the values are $8,\ 5,\ x,\ 4,\ x + 3, and\ 2x - 4$, we can set up the equation:

$$\frac{(8 + 5 + x + 4 + x + 3 + 2x - 4)}{6} = 6$$

Combine the terms and solve for x:

$$\frac{(4x + 16)}{6} = 6$$
$$4x + 16 = 36$$
$$4x = 20$$
$$x = 5$$

Therefore, the value of x is 5.

12. Correct option: (B)

Given the ratio of total salary: share of A: share of B as 20:5:8

Let's denote the parts of the ratio as follows:

$$Total\ salary\ =\ 20x$$
$$Share\ of\ A\ =\ 5x$$
$$Share\ of\ B\ =\ 8x$$

To find the share of C, we can use the fact that the sum of the shares of A, B, and C should equal the total salary:

$$Share\ of\ C\ =\ Total\ salary\ -\ (Share\ of\ A\ +\ Share\ of\ B)$$
$$Share\ of\ C\ =\ 20x\ -\ (5x\ +\ 8x)$$
$$Share\ of\ C\ =\ 20x\ -\ 13x$$
$$Share\ of\ C\ =\ 7x$$

Now, let's calculate the central angle for the sector that represents the share of C in a pie chart. The central angle can be calculated using the formula:

$$Central\ Angle\ =\ \left(\frac{Share\ of\ C}{Total\ Salary}\right)\times\ 360°$$

Substitute the values we calculated:

$$Central\ Angle\ =\ \left(\frac{7x}{20x}\right)\times\ 360°$$
$$Central\ Angle\ =\ \left(\frac{7}{20}\right)\times\ 360\ °$$
$$Central\ Angle\ =\ 126°$$

Therefore, the central angle of the sector representing the share of C is 126°.

13. Correct option: (B)

Given that a picture is represented by 375 units and 30 pictures represent the data, we can calculate the total number of units by multiplying the number of pictures by the number of units represented by each picture.

$$Total\ units\ =\ Number\ of\ pictures\ \times\ Units\ per\ picture$$
$$Total\ units\ =\ 30\ pictures\ \times\ 375\ units/picture$$
$$Total\ units\ =\ 11250\ units$$

Therefore, the correct answer is:11250.

14. Correct option: (C)

Tallies are usually marked in a bunch of 5. 𝍸

So, the correct answer is: C) 5

15. Correct option: (B)

According to the pictograph given in the question.

Each picture represents the 3 oranges => 3 units

According to the pictograph,

Number of oranges purchased by the Pam = 6 picture => 6*3 oranges = 18

Number of oranges purchased by the Ben = 3 picture => 3*3 oranges = 9

We have to find the difference between the number of oranges purchased by Pam and Ben. $D = 18 - 9 = 9$

16. Correct option: (111)

According to the pictograph given in the question.

Total number of motorcycles parked on all seven days = sum of motorcycle parked on each day

$$MP = s + m + t + w + th + f + sa$$
$$MP = 23 + 13 + 10 + 12 + 16 + 17 + 20$$
$$MP = 111$$

17. Correct option: (B)

An observation is a single data value in a dataset, such as a measurement or count. It contributes to the overall information used for analysis.

18. Correct option: (D)

According to the pictograph given in the question.

Each picture represents the 9 flowers.

So according to the given information

Number of rose flowers = 7*9 = 63

Number of tulip flowers = 5*9 = 45

Number of orchid flowers = 6*9 = 54

Number of lily flowers = 4*9 = 36

So total number of flowers sold by flower seller that day, $= 63 + 45 + 54 + 36 = 198$

19. Correct option: (D)

According to the given graph, revenues from different categories (in thousands) are as follows:

$$-\,Through\ oil\ =\ 100$$
$$-\,Through\ steel\ =\ 150$$
$$-\,Through\ coal\ =\ 200$$
$$-\,Through\ gold\ =\ 50$$

To achieve combined revenue of 350, the categories "steel" and "coal" can be considered. Their combined revenue is 350 (150 + 200 = 350).

20. Correct option: (D)

According to the provided bar graph, the number of trees planted in a particular city for four consecutive years is as follows:

$$-\,In\ 2003\ =\ 400$$
$$-\,In\ 2004\ =\ 550$$
$$-\,In\ 2005\ =\ 500$$
$$-\,In\ 2006\ =\ 600$$

From the given information, it is evident that the maximum number of trees was planted in the year 2006, with a count of 600 trees.

LEVEL- II

1. Correct option: (B)

Given numbers, $x, x+2, x+4, x+6, x+8$

Given mean of the total 5 numbers = 11

$$Mean = \frac{sum\ of\ numbers}{count\ of\ number}$$

$$\frac{x + x + 2 + x + 4 + x + 6 + x + 8}{5} = 11$$

Solving for the x; $5x + 20 = 55 \Rightarrow 5x = 35 \Rightarrow x = 7$

Then the numbers (putting the value of x); $7, 9, 11, 13, 19$

So, the mean of first 3 numbers will be $= \frac{7+9+11}{3} = \frac{24}{3} = 8$

2. Correct option: (C)

Given the gasoline price of different states, $\$1.79, \$1.61, \$1.96, \$1.84, \$1.75, \$\,2.09.$

$$Range = Highest\ value - Lowest\ value$$

Highest price of gasoline = \$2.09

Lowest price of gasoline = \$1.61

$Range\ of\ the\ gasoline\ price = 2.09 - 1.61 = \$0.48$

3. Correct option: (A)

Given numbers, $\frac{x}{5}, x, \frac{x}{4}, \frac{x}{2}, \frac{x}{3}$ $(x > 0)$

The median is the middle value when the numbers are arranged in order.

To find the median, let's arrange them in order: $\frac{x}{5}, \frac{x}{4}, \frac{x}{3}, \frac{x}{2}, x \; as \; x > 0$

Given that the median is 8, we need to find the value of x for which the middle value is 8.

Since there are an odd number of terms, the median is the middle value, which is $\frac{x}{3}$.

So, we have: $\frac{x}{3} = 8$

To solve for x: $$x = 8 * 3$$
$$x = 24$$

4. Correct option: (D)

The given pictograph represents the video rented in respective months.

According to the given information, 1 blue circle (key) represents 20 videos.

For October, there are 4 blue circles represented in the pictograph:

$$4 \; blue \; circles * 20 \; videos/circle = 80 \; videos$$

Therefore, the correct answer is: 80

5. Correct option: (B)

If the scale of the bar graph is 1 cm = 10 units, and the length of the bar is 5.2 cm, we can calculate the value of the quantity represented by the bar by multiplying the length of the bar by the scale factor:

$$Value = Length \; of \; bar \; (cm) * Scale \; factor \; (units/cm)$$
$$Value = 5.2 \; cm * 10 \; units/cm$$
$$Value = 52 \; units$$

Therefore, the correct answer is: B) 52 units

6. Correct option: (D)

Given a bar graph representing the favorite activities of the students in the class as follows:

$$- Sports = 7$$
$$- Dance = 5$$
$$- Painting = 6$$
$$- Drama = 8$$

To find the total number of students in the class, we add up the counts for each favorite activity: Total number of students = 7 + 5 + 6 + 8 = 26.

7. Correct option: (A)

Given a bar graph representing the favorite sports of grade 3 students as follows:

$$- Badminton = 25$$
$$- Tennis = 30$$
$$- Soccer = 20$$
$$- Baseball = 35$$

To find the difference between the numbers of students whose favorite sports are soccer and tennis: $Difference = Tennis - Soccer = 30 - 20 = 10$.

8. Correct option: (B)

In the given pie chart, the magazines account for 35%, novels account for 25%, journals account for 20%, and the remaining portion represents newsletters.

Since the total percentage must add up to 100%, we can calculate the percentage of newsletters by subtracting the percentages of magazines, novels, and journals from 100%.

$$Percentage \; of \; newsletters = 100\% - (35\% + 25\% + 20\%)$$
$$Percentage \; of \; newsletters = 100\% - 80\%$$
$$Percentage \; of \; newsletters = 20\%$$

The library has a total of 300 books. To find the number of newsletters, we need to calculate 20% of 300: $Number\ of\ newsletters\ =\ 20\%\ of\ 300$

$$Number\ of\ newsletters\ =\ 0.20\ *\ 300$$

$$Number\ of\ newsletters\ =\ 60$$

Therefore, the correct answer is: B) 60 newsletters

9. Correct option: (A)

Given numbers, $x, x+2, x+4, x+6, x+8$

$$Mean = \frac{sum\ of\ numbers}{count\ of\ number}$$

$$\frac{x+x+2+x+4+x+6+x+8}{5} = \frac{5x+20}{5}$$

Mean of the given observation in terms of x will be $x+4$

10. Correct option: (D)

Given that the scale is 1 cm = 250 population in a bar graph, and we want to find the length of the rectangle that represents 2000 population, we can use the following formula:

$$Length\ of\ rectangle\ = \frac{Population}{Scale}$$

Given: $Population\ =\ 2000$

$$Scale\ =\ 1\ cm\ =\ 250\ population$$

$$Length\ of\ rectangle\ = \frac{2000}{250}\ =\ 8\ cm$$

Therefore, the correct answer is: 8 cm

11. Correct option: (B)

To find the median of a set of values, you need to arrange the values in ascending order and then determine the middle value.

If there's an odd number of a value, the median is the middle value.

If there's an even number of values, the median is the average of the two middle values.

Given the values: $37, 31, 42, 43, 46, 25, 39, 45, 32$

First, arrange the values in ascending order: $25, 31, 32, 37, 39, 42, 43, 45, 46$

There are 9 values, which is an odd number. So, the median is the middle value, which is the fifth value in the ordered list:

Median = 39

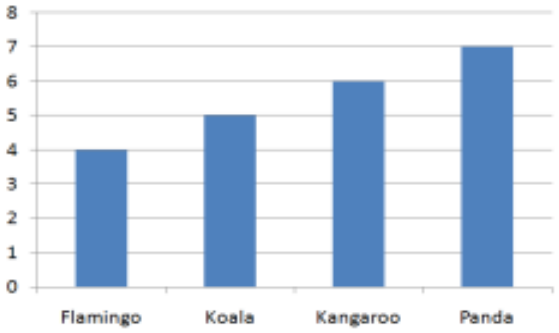
Therefore, the correct answer is: B) 39

12. Correct option: (C)

Given is a table that indicates the number of four types of animals in a particular zoo.

Animal	**Number**
Flamingo	4
Panda	7
Koala	5
Kangaroo	6

The correct bar graph representation is



13. Correct option: (B)

The range of a set of observations is the difference between the highest and lowest values in that set.

Given the observations: 7, 21, 11, 5, 9, 20, 32, 28

First, arrange the observations in ascending order: $5, 7, 9, 11, 20, 21, 28, 32$

The highest value is 32, and the lowest value is 5.

$$Range = Highest\ value - Lowest\ value$$
$$Range = 32 - 5$$
$$Range = 27$$

Therefore, the correct answer is: B) 27

14. Correct option: (A)

The table provides the enrollment of students in the school for the years 1990 to 1993:

To find the total number of students enrolled in the school from 1990 to 1993, we need to sum up the number of students for each year:

Total students = 100 + 250 + 200 + 350 = 900

Therefore, the correct answer is:A) 900

15. Correct option: (B)

Given the number of students enrolled in different years (refer the table)

To find the ratio of students enrolled in 1990 to 1991, you can divide the number of students in 1990 by the number of students in 1991:

$$Ratio = \frac{Students\ in\ 1990}{Students\ in\ 1991}$$
$$Ratio = \frac{100}{250}$$
$$Ratio = \frac{2}{5}$$

Therefore, the correct answer is: B) 2:5

16. Correct option: (C)

To calculate the decrease percentage between two values, you can use the formula:

$$Decrease\ Percentage = \left(\frac{Initial\ Value - Final\ Value}{Initial\ Value}\right) * 100$$

Given the data:

$$Initial\ Value\ (1991) = 250$$
$$Final\ Value\ (1992) = 200$$

Using the formula:

$$Decrease\ Percentage = \left(\frac{250 - 200}{250}\right) * 100 = \left(\frac{50}{250}\right) * 100 = 0.2 * 100 = 20\%$$

Therefore, the correct answer is: C) 20%

17. Correct option: (A)

To calculate the percentage increase between two values, you can use the following formula: $Percentage\ Increase = \left(\frac{New\ Value - Old\ Value}{Old} Value\right) * 100$

Given the values for 1992 and 1993:

$$Old\ Value\ (1992) = 200$$
$$New\ Value\ (1993) = 350$$
$$Percentage\ Increase = \left(\frac{350 - 200}{200}\right) * 100$$
$$Percentage\ Increase = \left(\frac{150}{200}\right) * 100$$
$$Percentage\ Increase = 0.75 * 100$$
$$Percentage\ Increase = 75\%$$

Therefore, the correct answer is: A) 75%

18. Correct option: (225)

To find the average number of students enrolled in the school from 1990 to 1993, you need to sum up the number of students for each year and then divide by the number of years.

$$Total\ number\ of\ students\ =\ 100\ +\ 250\ +\ 200\ +\ 350\ =\ 900$$

$$Number\ of\ years\ =\ 4$$

$$Average\ number\ of\ students\ =\frac{Total\ number\ of\ students}{Number\ of\ years}$$

$$Average\ number\ of\ students\ =\frac{900}{4}\ =\ 225$$

Therefore, the correct answer is not among the options provided. The correct average number of students is 225.

19. Correct Answer is: 7

To find the mode, we need to identify the number that appears most frequently in the given set: 13, 16, 12, 14, 19, 12, 14, 13, and 14.

The number 14 appears three times, which is more frequent than any other number.

Now, calculate 1/2 of the mode: $\frac{1}{2}\ of\ mode\ =\frac{1}{2}\ *\ 14\ =\ 7$

Therefore, the answer is: A) 7

20. Correct Answer is: 9

To find the median of a set of data, we need to arrange the values in ascending order and then determine the middle value.

If there's an odd number of a value, the median is the middle value.

Given the data: 9, 12, 11, 10, 8, 9, 7

First, arrange the data in ascending order: $7, 8, 9, 9, 10, 11, 12$

There are 7 values, which is an odd number. So, the median is the middle value, which is the fourth value in the ordered list: $Median\ =\ 9$

Therefore, the correct answer is: C) 9

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

Given $a, a+2, a+4, a+12,\ a+7$

Given average of the numbers = 9

$$\text{Mean} = \text{Average}\ =\frac{sum\ of\ all\ numbers}{count\ of\ numbers}=\frac{S}{N}$$

S = $a + a + 2 + a + 4 + a + 12 + a + 7 = 5a + 25$

N = 5

Mean $=\frac{S}{N}=\frac{5a+25}{5}=\boldsymbol{a+5=9}$

$$\boldsymbol{a+5=9 => a=4}$$

2. Correct option: (C)

Given Mathematics marks of 10 students as

$$25, 40, 30, 35, 21, 45, 23, 33, 10, 29$$

Given Statistics marks of 10 students as

$$30, 39, 23, 42, 20, 40, 25, 30, 18, 19$$

$$\boldsymbol{Range = Highest\ value - Lowest\ value}$$

Highest marks of Mathematics = 45, Lowest marks of Mathematics = 10

Highest marks of Statistics = 42, Lowest marks of statistics = 18

So, $Range\ of\ Mathematics\ marks = 45 - 10 = 35$

$$Range\ of\ Statistics\ marks = 42 - 18 = 24$$

3. Correct option: (B)

Given,

The central angle of a component is 72° and its value is 24, as we know that in a pie chart, the central angle of a component is proportional to its value.

We can set up a proportion to find the total value of all the components.

Let x represent the total value of all the components. The central angle of a component is directly proportional to its value, so we can set up the following proportion:

$$\frac{Central\ Angle}{Total\ Angle} = \frac{Value\ of\ Component}{Total\ Value}$$

$$\frac{72°}{360°} = \frac{24}{x}$$

$$x = \frac{360° * 24}{72°}$$

$$x = 120$$

So, the total value of all the components of the data is 120.

4. Correct option: (B)

In a pictograph, each picture represents 1000 units of a certain product. If there are 4 full pictures and $\frac{3}{4}$ of a picture in a row, we can calculate the total units produced by adding the units represented by the full pictures and the partial picture.

Number of full pictures = 4

Number of units represented by each full picture = 1000 units

Number of units represented by full pictures = Number of full pictures × Units per full picture

Number of units represented by full pictures = 4 × 1000 = 4000 units

Now, we have $\frac{3}{4}$ of a picture remaining.

Number of units represented by $\frac{3}{4}$ of a picture = $\frac{3}{4}$ × Units per full picture

Number of units represented by $\frac{3}{4}$ of a picture = $\frac{3}{4}$ × 1000 = 750 units

Now add the units from full pictures and the partial picture:

Total units produced = Units from full pictures + Units from partial picture

Total units produced = 4000 units + 750 units = 4750 units

5. Correct option: (A)

In a pie chart, the central angle of a component is directly proportional to its percentage of the total value.

To find the central angle of a component that represents 30% of the total value, you can calculate it using the following formula:

$$Central\ Angle = \left(\frac{Percentage}{100}\right) \times 360°$$

Given that the component represents 30% of the total value:

$$Central\ Angle = \left(\frac{30}{100}\right) \times 360°$$

$$Central\ Angle = 0.3 \times 360°$$

$$Central\ Angle = 108°$$

Therefore, the correct answer is:108°

6. Correct option: (D)

In a pie chart, the central angle of a component is proportional to its percentage of the total value.

To find the percentage of the value of the component represented by a central angle of 72°, you can use the following formula: $Percentage = (\frac{Central\ Angle}{360°}) \times 100$

Given that the central angle is 72°:

$$Percentage = (\frac{72°}{360°}) \times 100$$
$$Percentage = 0.2 \times 100$$
$$Percentage = 20\%$$

Therefore, the correct answer is: D) 20%

7. Correct option: (B)

According to the table given in the question.
Children wears the Lace-up shoes = 3
Children wears the Ballerina shoes = 4
Children wears the Sneakers = 3
Children wears the Keds = 5
So total number of children = (3+4+3+5) = 15
We have to find the fraction of children who wear the Lace-up shoes to the total number of children.

$$R = \frac{3}{15} = \frac{1}{5}$$

8. Correct option: (B)

According to the pictograph given in the question
We have data of people who read the different type of magazines as
Fashion magazine = 4,
Drama = 3,
Fantasy = 6,
Sports fiction = 5,
Biographies = 2,
Historical fiction = 4.
The maximum number of people preferred to read Fantasy magazine.

9. Correct option: (B)

According to the pictograph given in the question.
The monkey ate a total of 12 bananas over the given period.
Now we have to find the 4 times of the total number of bananas eaten by the monkey in the given period which will be = 4*12 =48

10. Correct option: (B)

According to the provided bar graph, the number of vehicles passing through a road in Texas during different time intervals is as follows:

$$-From\ 8-9\ AM\ =\ 50$$
$$-From\ 9-10\ AM\ =\ 400$$
$$-From\ 10-11\ AM\ =\ 250$$
$$-From\ 11-12\ AM\ =\ 250$$
$$-From\ 12-1\ PM\ =\ 150$$
$$-From\ 1-2\ PM\ =\ 100$$

To find the total number of vehicles passing through the road, we can sum up the individual counts for each time interval:

$$= 50 + 400 + 250 + 250 + 150 + 100 = 1200.$$

11. Correct option: (D)

The given information is that the total number of people who died in the earthquake was 3000.
The pie chart represents the percentages of people from villages A, B, C, and D as 30%, 35%, 20%, and 15% respectively.

To find out in which village the number of earthquake victims was less than 25%, we need to compare the percentage of each village with 25%.

$$-\ Village\ A: 30\% > 25\%$$
$$-\ Village\ B: 35\% > 25\%$$
$$-\ Village\ C: 20\% < 25\%$$
$$-\ Village\ D: 15\% < 25\%$$

From the given options, Village C and Village D have percentages less than 25%. So, the number of earthquake victims was less than 25% in these villages.
Therefore, the correct answer is: D) Village 'D'

12. **Correct option: (D)**

The meaning of a set of numbers is calculated by dividing the sum of the numbers by the count of the numbers.
Given that the mean of the 5 numbers is 27, the sum of those numbers is 5 * 27 = 135.
Let's denote the excluded number as "x". According to the information provided, when this number is excluded, the mean becomes 25.
We know that the sum of the numbers divided by the count of the numbers gives the mean. So, the sum of the remaining 4 numbers is 4 * 25 = 100.
Now, we can set up an equation based on the sum of the numbers:

$$Sum\ of\ 4\ numbers\ +\ Excluded\ number\ =\ Sum\ of\ 5\ numbers$$
$$100\ +\ x\ =\ 135$$

Solving for "x":

$$x\ =\ 135\ -\ 100$$
$$x\ =\ 35$$

Therefore, the excluded number is 35.
The correct answer is:D) 35

13. Correct option: (A)

To determine the week with the greatest increase in the number of deaths, we need to calculate the differences between consecutive weeks and identify the largest difference.
Given the data:

$$-\ 1\ July: 135\ deaths$$
$$-\ 8\ July: 170\ deaths\ (Difference: 170\ -\ 135\ =\ 35)$$
$$-\ 15\ July: 140\ deaths\ (Difference: 140\ -\ 170\ =\ -30)$$
$$-\ 22\ July: 155\ deaths\ (Difference: 155\ -\ 140\ =\ 15)$$
$$-\ 29\ July: 160\ deaths\ (Difference: 160\ -\ 155\ =\ 5)$$
$$-\ 5\ August: 150\ deaths\ (Difference: 150\ -\ 160\ =\ -10)$$
$$-\ 12\ August: 165\ deaths\ (Difference: 165\ -\ 150\ =\ 15)$$

The greatest increase occurs between 1 July and 8 July, with a rise of 35 deaths.
Therefore, the correct answer is:
A) 1 July - 8 July

14. Correct option: (C)

To find out in how many weeks more than 145 deaths were registered, you need to count the number of weeks where the number of deaths was greater than 145.
From the provided data:

$$-\ 8\ July: 170\ deaths\ (more\ than\ 145)$$
$$-\ 15\ July: 140\ deaths\ (less\ than\ 145)$$
$$-\ 22\ July: 155\ deaths\ (more\ than\ 145)$$

$$-\ 29\ July\text{: } 160\ deaths\ (more\ than\ 145)$$

$$-\ 5\ August\text{: } 150\ deaths\ (more\ than\ 145)$$

$$-\ 12\ August\text{: } 165\ deaths\ (more\ than\ 145)$$

There are a total of 4 weeks where more than 145 deaths were registered.
Therefore, the correct answer is:C) 4

15. Correct option: (C)

To find out how many weeks had less than 165 deaths, you can go through the provided data and count the number of weeks where the deaths were less than 165.
From the given data:

$$-\ 1\ July\text{: } 135\ deaths\ (less\ than\ 165)$$

$$-\ 8\ July\text{: } 170\ deaths\ (more\ than\ 165)$$

$$-\ 15\ July\text{: } 140\ deaths\ (less\ than\ 165)$$

$$-\ 22\ July\text{: } 155\ deaths\ (less\ than\ 165)$$

$$-\ 29\ July\text{: } 160\ deaths\ (less\ than\ 165)$$

$$-\ 5\ August\text{: } 150\ deaths\ (less\ than\ 165)$$

$$-\ 12\ August\text{: } 165\ deaths\ (equal\ to\ 165)$$

So, there are 5 weeks (1 July, 15 July, 22 July, 29 July and 5 Aug) where the deaths occurred less than 165.
The correct answer is:C) 5

16. Correct Answer is: 1

The mean of a set of observations is calculated by dividing the sum of the observations by the number of observations.
Given that the mean is 12 and the sum of the observations is 132, you can set up an equation: $Mean\ =\ Sum\ /\ Number\ of\ Observations$

$$12\ = \frac{132}{n}$$

Now, solve for n $$n\ = \frac{132}{12}\ =\ 11$$

So n can be written as $n = 10 + 1$
Therefore, the answer is: B) 1

17. Correct Answer is: 8

To find the value of x, we need to use the formula for the mean (average):
Mean = Sum of all values / Number of values
Given that the mean is 7 and the values are 5, 7, x, 10, 7, and 5, we can set up the equation:

$$\frac{5\ +\ 7\ +\ x\ +\ 10\ +\ 7\ +\ 5}{6}\ =\ 7$$

Combine the terms and solve for x: $\frac{34\ +\ x}{6}\ =\ 7$

Now, isolate x: $$34\ +\ x\ =\ 42$$

$$x\ =\ 42\ -\ 34\ =\ 8$$

Therefore, the value of x is: D) 8

18. Correct Answer is: 8

The mode of a set of data is the value that appears most frequently.
Given the data: 7, 8, 9, 8, 9, 10, 6, 11, 11, 8
The number 8 appears most frequently, occurring three times, more frequently than any other number.
Therefore, the correct answer is: B) 8

19. Correct Answer is: 3

To find the value of $\frac{x}{7}$ we first need to determine the median of the given data set.

Given the data in ascending order: $11, 12, 14, 18, (x+2), (x+4), 30, 32, 35, 41$

Since there are 10 values, the median is the average of the fifth and sixth values.

$$Median = \frac{x+2+x+4}{2}$$

Given that the median is 24, we can set up the equation: $24 = \frac{2x+6}{2}$

Now, solve for x;

$$(48 = 2x + 6)$$
$$2x = 42$$
$$x = 21$$

Now that we know the value of x , we can calculate $\frac{x}{7}$; $x/7 = 21/7 = 3$

Therefore, the value of $\frac{x}{7}$ is: B) 3

20. Correct Answer is: 4

The mean of a set of numbers is calculated by dividing the sum of the numbers by the count of the numbers.

The first 7 natural numbers are 1, 2, 3, 4, 5, 6, and 7.

Sum of the first 7 natural numbers = 1 + 2 + 3 + 4 + 5 + 6 + 7 = 28

Number of natural numbers = 7

Mean = Sum of numbers / Number of numbers; $Mean = \frac{28}{7} = 4$

Therefore, the correct answer is: A) 4

6. Mensuration

Solutions

Area And Perimeter

LEVEL – I

1. Correct option: (C)

Perimeter of square = 4 × side length

Perimeter = 4 × 8 cm

Perimeter = 32 cm

So, the perimeter of the square field is 32 cm.

Then the correct option is C)

2. Correct option: (C)

If we denote the side length of the square as "s" and the area of the square as $4x^2$ square units, we can set up the equation:

$$Area = s^2$$

Given that the area is $4x^2$ square units, we can now write the equation as:

$$4x^2 = s^2$$

$$\sqrt{4x^2} = \sqrt{s^2}$$

$$2x = s$$

So, the side length of the square is 2x units when the area is $4x^2$ square units.

Then the correct option is C)

3. Correct option: (B)

To find the area of a rectangle, we can use the formula:

$$Area = Length \times Breadth$$

In this case, the length of the rectangle is 20 cm, and the breadth is 14 cm.

We can calculate the area as follows:

$$Area = 20\,cm \times 14\,cm$$

$$Area = 280\,cm^2$$

So, the area of the rectangle is 280 square cm.

Then the correct option is B)

4. Correct option: (D)

$$Area = Length \times Breadth$$

Given that the length is 26 cm and the area is 286 sq.cm, we can solve for the breadth as follows:

$$286\,sq.cm = 26\,cm \times Breadth$$

$$Breadth = \frac{286\,sq.cm}{26\,cm} = 11\,cm$$

So, the breadth of the rectangular field is 11 cm.

Then the correct option is D)

5. Correct option: (A)

$$Perimeter = 2 \times (Length + Breadth)$$

Given that the breadth is 2 cm and the perimeter is 24 cm, we can solve for the length as follows:

$$24\,cm = 2 \times (Length + 2\,cm)$$

$$12\,cm = Length + 2\,cm$$

$$Length = 12\,cm - 2\,cm = 10\,cm$$

So, the length of the rectangle is 10 cm.

Then the correct option is A)

6. Correct option: (B)

Let's denote the breadth as "b" cm. According to the information provided, the length is twice the breadth, so the length is "2b" cm.

Now, we can use the formula for the area of a rectangle:

$$Area = Length \times Breadth$$

Given that the area is 512 sq. cm:

$$512\ sq.cm = (2b\ cm) \times (b\ cm)$$
$$2b^2 = 512\ sq.cm$$
$$b^2 = 256\ sq.cm$$
$$b = \sqrt{256\ sq.cm}$$
$$b = 16\ cm$$

So, the breadth of the rectangle is 16 cm, and the length is 2 times the breadth, which is 32 cm.

Now

$$Perimeter = 2 \times (Length + Breadth)$$
$$Perimeter = 2 \times (32\ cm + 16\ cm)$$
$$Perimeter = 2 \times 48\ cm$$
$$Perimeter = 96\ cm$$

The perimeter of the rectangle is 96 cm.

Then the correct option is B)

7. Correct option: (D)

The area of a square can be calculated in terms of its diagonal (d) using the following formula: $Area = \left(\frac{1}{2}\right) \times (diagonal)^2$

So, for a square with a diagonal "d," the area is:

$$Area = \left(\frac{1}{2}\right) \times d^2$$

Then the correct option is D)

8. Correct option: (C)

The area of a triangle can be calculated using the formula:

$$Area = \left(\frac{1}{2}\right) \times Base \times Height$$

In this case, the base of the triangle (b) is 8 cm, and the height (h) is 6 cm.

We can plug these values into the formula:

$$Area = \left(\frac{1}{2}\right) \times 8\ cm \times 6\ cm$$
$$Area = \left(\frac{1}{2}\right) \times 48\ cm^2$$
$$Area = 24\ cm^2$$

So, the area of triangle ABC is 24 square centimeters.

Then the correct option is C)

9. Correct option: (A)

The area of a parallelogram can be calculated using the formula:

$$Area = Base \times Altitude$$

In this case, the base of the parallelogram is 8 meters, and the altitude (height) is 4.5 meters.

$$Area = 8\ m \times 4.5\ m$$
$$Area = 36\ m^2$$

So, the area of the parallelogram is 36 square meters.

Then the correct option is A)

10. Correct option: (A)

In this case, we have a triangle ABC with a base BC of 8 cm and an altitude (height) from point A to BC of 4 cm.

$$Area = \left(\frac{1}{2}\right) \times 8\,cm \times 4\,cm$$

$$Area = 4\,cm \times 8\,cm$$

$$Area = 32\,square\,cm$$

So, the area of triangle ABC is 32 square centimeters.

Then the correct option is A)

11. Correct option: (B)

The area of a parallelogram can be calculated using the formula:

$$Area = Base \times Height$$

Provided area (80 cm²) and the height (8 cm) of a parallelogram. To find the base, we can rearrange the formula as follows:

$$Base = \frac{Area}{Height}$$

$$Base = \frac{80\,cm^2}{8\,cm}$$

$$Base = 10\,cm$$

So, the base of the parallelogram is 10 cm.

Then the correct option is B)

12. Correct option: (A)

Given that the area is 80 cm² and the height is 8 cm, we can plug these values into the formula and solve for the base:

$$80\,cm^2 = \left(\frac{1}{2}\right) \times Base \times 8\,cm$$

$$80\,cm^2 = 4\,cm \times Base$$

$$Base = \frac{80\,cm^2}{4\,cm}$$

$$Base = 20\,cm$$

So, the base of the triangle is 20 cm.

Then the correct option is A)

13. Correct option: (C)

To find the area of triangle ABC, we can use the formula for the area of a triangle, which is:

$$Area = \left(\frac{1}{2}\right) \times Base \times Height$$

In this case, triangle ABC shares its base with the rectangle, which is AB, and the height of triangle ABC is the same as the height of the rectangle, which is BC.

$$Area\ of\ triangle\ ABC = \left(\frac{1}{2}\right) \times 10\,cm \times 13\,cm$$

$$Area\ of\ triangle\ ABC = 65\,square\,cm$$

So, the area of triangle ABC is 65 square centimeters.

Then the correct option is C)

14. Correct option: (A)

The area of a rhombus can be calculated using the formula:

$$Area = \left(\frac{1}{2}\right) \times Product\ of\ Diagonals$$

In this case, the diagonals of the rhombus are given as 4 cm and 3 cm. So, we can calculate the area as follows:

$$Area = \left(\frac{1}{2}\right) \times (4\ cm \times 3\ cm)$$

$$Area = \left(\frac{1}{2}\right) \times 12\ square\ cm$$

$$Area = 6\ square\ cm$$

So, the area of the rhombus is 6 square centimeters.
Then the correct option is A)

15. Correct option: (B)

The area of an equilateral triangle can be calculated using the formula:

$$Area = \frac{\sqrt{3}}{4} \times (Side\ Length)^2$$

In this case, we are given that the height (altitude) of the equilateral triangle is "x" cm. For an equilateral triangle, the height is related to the side length as follows:

$$Height\ (h) = \frac{\sqrt{3}}{2} \times Side\ Length\ (s)$$

Now, if the height is "x" cm, we can express the side length in terms of "x":

$$x = \frac{\sqrt{3}}{2} \times s$$

$$s = \frac{x}{\frac{\sqrt{3}}{2}} = \frac{2x}{\sqrt{3}}$$

Now, we can plug this value of "s" into the formula for the area:

$$Area = \frac{\sqrt{3}}{4} \times \left[\frac{2x}{\sqrt{3}}\right]^2$$

$$Area = \frac{\sqrt{3}}{4} \times \left(\frac{4}{3}\right) \times x^2$$

$$Area = \frac{x^2}{\sqrt{3}}$$

Then the correct option is B)

16. Correct option: (A)

To find the length of the second diagonal of the rhombus, we can use the formula for the area of a rhombus:

$$Area = \left(\frac{1}{2}\right) \times (Product\ of\ Diagonals)$$

Given that the area is 96 sq. cm and one diagonal is 16 cm, we can use the formula to solve for the length of the second diagonal (let's call it "d"):

$$96\ sq.cm = \left(\frac{1}{2}\right) \times (16\ cm \times d)$$

$$96\ sq.cm = 8\ cm \times d$$

$$d = \frac{96\ sq.cm}{8\ cm}$$

$$d = 12\ cm$$

So, the length of the second diagonal of the rhombus is 12 cm.
Then the correct option is A)

17. Correct option: (D)

The area of a parallelogram can be calculated using the formula:

$$Area = Base \times Height$$

In this case, the base of the parallelogram is 12 cm, and the height is 7 cm. We can plug these values into the formula:

$$Area = 12\,cm \times 7\,cm$$
$$Area = 84\,square\,cm$$

So, the area of the parallelogram is 84 square centimeters.
Then the correct option is D)

18. Correct option: (C)

The formula for the area of an equilateral triangle is:

$$Area = \frac{\sqrt{3}}{4} \times (Side\ Length)^2 = \frac{\sqrt{3}}{4} \times (s)^2$$

Given that the area is $64\sqrt{3}$ square cm. So, we can set up the equation:

$$64\sqrt{3} = \frac{\sqrt{3}}{4} \times (s)^2$$

$$64\sqrt{3} \times \frac{4}{\sqrt{3}} = s^2 => s^2 = 256$$

$$s^2 = 256 => s = \sqrt{256} = 16$$

Then

$$Perimeter = 3 \times Side\ Length$$
$$Perimeter = 3 \times 16\,cm$$
$$Perimeter = 48\,cm$$

So, the perimeter of the equilateral triangle is 48 cm.

Then the correct option is C)

19. Correct option: (B)

From the figure we can conclude that lengths of parallel sides are 7cm and 9cm with distance between them 8cm.
The area of a trapezium can be calculated using the formula:

$$Area = \left(\frac{1}{2}\right) \times (x + y) \times (Distance\ Between\ Parallel\ Sides)$$

Where x and y are length of parallel sides.
So area of Trapezium is $A = \frac{1}{2}(7 + 9)(8) = 64\ sq.cm^2$

Then the correct option is B)

20. Correct option: (A)

To find the area of given figure we can take two triangle ABC and ADC with height 3 cm and 5cm having the same base as 6cm
So total area of the figure

$$A = area\ of\ \triangle ABC + area\ of\ \triangle ADC$$

Therefore

$$area\ of\ \triangle ABC = \frac{1}{2} \times 6 \times 3 = 9\ cm^2$$
$$area\ of\ \triangle ADC = \frac{1}{2} \times 6 \times 5 = 15\ cm^2$$

So total area

$$A = (9 + 15)cm^2 = 24\ cm^2$$

Then the correct option is A)

LEVEL – II

1. Correct option: (A)

Let the length of the rectangle be 3x cm and the breadth be 2x cm, where x is a positive constant.

Given that the area of the rectangle is 216 cm², we can write the equation for the area as:

$$Area = Length \times Breadth$$
$$216\,cm^2 = (3x\,cm) \times (2x\,cm)$$
$$216\,cm^2 = 6x^2$$
$$36\,cm^2 = x^2$$
$$x = \sqrt{36\,cm^2}$$
$$x = 6\,cm$$

Now that we know the value of x, we can find the length and breadth:

$$Length = 3x = 3 \times 6\,cm = 18\,cm$$
$$Breadth = 2x = 2 \times 6\,cm = 12\,cm$$

The perimeter of the rectangle is calculated as:

$$Perimeter = 2 \times (Length + Breadth)$$
$$Perimeter = 2 \times (18\,cm + 12\,cm)$$
$$Perimeter = 2 \times 30\,cm$$
$$Perimeter = 60\,cm$$

So, the perimeter of the rectangle is 60 cm.

Then the correct option is A)

2. Correct option: (A)

Let's denote the breadth of the rectangular field as "B" meters.

According to the given information, the length of the field is thrice its breadth, which means the length is 3B meters.

The formula for the perimeter of a rectangle is: $Perimeter = 2 \times (Length + Breadth)$

Given that the perimeter is 240 meters, we can write:

$$240 = 2 \times (3B + B)$$
$$240 = 2 \times (4B)$$
$$120 = 4B$$
$$B = \frac{120}{4}$$
$$B = 30\,meters$$

So, the breadth of the rectangular field is 30 meters. Since the length is three times the breadth, the length is: $Length = 3 \times B = 3 \times 30\,meters = 90\,meters$

Therefore, the length of the field is 90 meters.

Then the correct option is A)

3. Correct option: (B)

The area of a square can be calculated using the formula: $Area = Side^2$

Given that the area is 225 m², we can find the length of each side (S) as follows:

$$225\,m^2 = S^2$$

Take the square root of both sides to find S: $S = \sqrt{225\,m^2}$

$$S = 15\,m$$

Now that we know the side length (S), we can find the perimeter (P) of the square:

$$Perimeter = 4 \times Side$$
$$Perimeter = 4 \times 15\,m$$
$$Perimeter = 60\,m$$

So, the perimeter of the square is 60 meters.

Then the correct option is B)

4. Correct option: (C)

Let's denote the length of the rectangle as "L" cm.

The formula for the perimeter of a rectangle is:

$$Perimeter = 2 \times (Length + Breadth)$$

Given that the perimeter is 130 cm and the breadth is 30 cm, we can write:

$$130\,cm = 2 \times (L + 30\,cm)$$

Now, let's solve for L (the length):

$$130\,cm = 2L + 60\,cm$$
$$2L = 130\,cm - 60\,cm$$
$$2L = 70\,cm$$
$$L = \frac{70\,cm}{2}$$
$$L = 35\,cm$$

So, the length of the rectangle is 35 cm.

Now, let's calculate the area of the rectangle using the length and breadth:

$$Area = Length \times Breadth$$
$$Area = 35\,cm \times 30\,cm$$
$$Area = 1050\,square\,cm$$

So, the length of the rectangle is 35 cm, and its area is 1050 square cm.

Then the correct option is C)

5. Correct option: (B)

To find the length of the diagonal of a square when we know its area, we can use the following steps:

1. Start with the formula for the area of a square:

$$Area = Side^2$$

2. Given that the area is 50 cm², set up the equation:

$$50\,cm^2 = Side^2$$

3. Solve for the length of one side (Side):

$$Side = \sqrt{50\,cm^2}$$
$$Side = 5\sqrt{2}\,cm$$

4. Now, we can use the Pythagorean theorem to find the length of the diagonal (D) of the square. In a square, the diagonal divides it into two congruent right triangles. So:

$$D^2 = Side^2 + Side^2$$
$$D^2 = (5\sqrt{2}\,cm)^2 + (5\sqrt{2}\,cm)^2$$
$$D^2 = 50\,cm^2 + 50\,cm^2$$
$$D^2 = 100\,cm^2$$

5. Take the square root of both sides to find D:

$$D = \sqrt{100\,cm^2}$$
$$D = 10\,cm$$

So, the length of the diagonal of the square is 10 cm.

Then the correct option is B)

6. Correct option: (B)

To find the length of the other parallel side of the trapezium-shaped field, we can use the formula for the area of a trapezium:

$$Area = \left(\frac{1}{2}\right) \times (Sum\ of\ parallel\ sides) \times Height$$

In this case, we are given the area as 480 m², one of the parallel sides as 20 m, and the height (distance between the parallel sides) as 15 m. Let's denote the length of the other parallel side as "x" meters.

Now, plug in the given values and solve for "x":

$$480\,m^2 = \left(\frac{1}{2}\right) \times (20\,m + x) \times 15\,m$$
$$960\,m^2 = (20\,m + x) \times 15\,m$$
$$\frac{960\,m^2}{15\,m} = 20\,m + x$$
$$64\,m = 20\,m + x$$
$$64\,m - 20\,m = x$$
$$44\,m = x$$

So, the length of the other parallel side is 44 meters.
Then the correct option is B)

7. Correct option: (A)

To find the length of the other diagonal of the rhombus, we can use the formula for the area of a rhombus:

$$Area = \left(\frac{1}{2}\right) \times Product\ of\ Diagonals$$

In this case, we're given that the area is 240 cm², and one of the diagonals (let's call it D_1) is 16 cm. We can use this information to find the length of the other diagonal (D_2).

$$240\,cm^2 = \left(\frac{1}{2}\right) \times D_1 \times D_2$$

Now, solve for D_2:

$$2 \times 240\,cm^2 = D1 \times D2$$
$$480\,cm^2 = D1 \times D2$$

To find D_2, divide both sides by D_1:

$$D_2 = \frac{480\,cm^2}{D_1}$$
$$D_2 = \frac{480\,cm^2}{16\,cm}$$
$$D_2 = 30\,cm$$

So, the length of the other diagonal (D_2) is 30 cm.
Then the correct option is A)

8. Correct option: (B)

To find the area of a rhombus when we know the lengths of its diagonals, we can use the formula:

$$Area = \left(\frac{1}{2}\right) \times Product\ of\ Diagonals$$

In this case, the diagonals are 10 cm and 8.2 cm:

$$Area = \left(\frac{1}{2}\right) \times (10\,cm \times 8.2\,cm)$$
$$Area = \left(\frac{1}{2}\right) \times 82\,cm^2$$
$$Area = 41\,cm^2$$

So, the area of the rhombus is 41 square cm.
Then the correct option is B)

9. Correct option: (B)

Let's first find the area of the given triangle using the side lengths 5, 12, and 13.
This triangle is a Pythagorean triple, so it's a right triangle.
The area of a triangle can be calculated using the formula:

$$Area = \left(\frac{1}{2}\right) \times base \times height$$

In this case, the base and height of the right triangle are 5 units and 12 units, respectively.

$$Area\ of\ the\ triangle\ = \left(\frac{1}{2}\right) \times 5\ units\ \times 12\ units\ = 30\ square\ units$$

Now, let's construct a rectangle with an area equal to the area of the triangle, which is 30 square units. Since the length of the rectangle is given as 10 units, we can find the width of the rectangle (w) using the formula:

$$Area\ of\ the\ rectangle\ = length \times width$$
$$30\ square\ units\ = 10\ units\ \times w$$
$$w\ = \frac{30\ square\ units}{10\ units}\ = 3\ units$$

So, the width of the rectangle is 3 units.

Now that we know the length and width of the rectangle, we can calculate its perimeter:

$$Perimeter\ = 2 \times (length\ +\ width)$$
$$Perimeter\ = 2 \times (10\ units\ +\ 3\ units)$$
$$Perimeter\ = 2 \times 13\ units$$
$$Perimeter\ = 26\ units$$

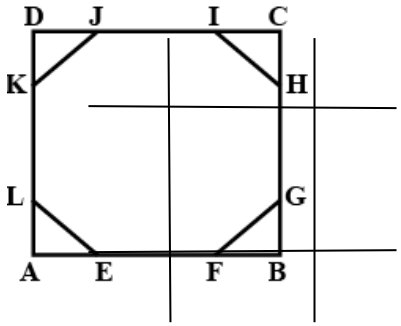
Therefore, the perimeter of the rectangle is 26 units.

Then the correct option is B)

10. Correct option: ($\frac{12}{1+\sqrt{2}}$)

Side of the

From the question we can construct the following figure as



Here value of $AB = 12$

Let the side of regular octagon is x

Then the value of

$$IC^2 = \frac{x^2}{2} => IC = \frac{x}{\sqrt{2}}$$

So the total area of the square in the form of octagon

If $JI = x = JK => DC = \left(x + \sqrt{2}x\right) = x(1 + \sqrt{2})$

We are given that the length of the side of the square is 12

$$=> x\left(1 + \sqrt{2}\right) = 12 => x = \frac{12}{1+\sqrt{2}}$$

Then the value of side when we cut the corner of a square to form an octagon

$$= \frac{12}{1 + \sqrt{2}}$$

11. Correct option: (A, C)

The formula for the perimeter (P) of a rectangle is given by:

$$P\ = 2 \times (Length\ +\ Breadth)$$

We are given the perimeter as 68 cm and the breadth as 14 cm. Let's denote the length of the rectangle as "L" cm. Now, plug in the values:

$$68\ cm\ = 2 \times (L\ cm\ +\ 14\ cm)$$

$$34\,cm = L\,cm + 14\,cm$$
$$34\,cm - 14\,cm = L\,cm$$
$$20\,cm = L\,cm$$

So, the length of the rectangle is 20 cm.
Then the correct option is A, C).

12. Correct option: (C, D)

The area of a square is given by the formula:

$$Area\ of\ Square = Side^2$$

In this case, the side of the square is 40 meters, so the area of the square is:

$$Area\ of\ Square = 40^2 = 1600\ square\ meters$$

Now, we're given that the square and the parallelogram have the same area. Therefore, the area of the parallelogram is also 1600 square meters.
The area of a parallelogram is given by the formula:

$$Area\ of\ Parallelogram = Base \times Height$$

We're given that the height of the parallelogram is 20 meters, and we need to find the base (B). So, we can rearrange the formula to solve for the base:

$$Base = \frac{Area\ of\ Parallelogram}{Height}$$
$$Base = \frac{1600\ square\ meters}{20\ meters}$$
$$Base = 80\ meters = 4 \times height$$

So, the base of the parallelogram is 80 meters.
Then the correct option is C, D)

13. Correct option: (C)

Let's denote the base of the triangle as "B" centimeters and the height as "H" centimeters.
According to the information provided, the height is double the base:

$$H = 2B$$

The formula for the area (A) of a triangle is:

$$A = \left(\frac{1}{2}\right) \times Base \times Height$$

Given that the area is 400 cm², we can substitute the values:

$$400\,cm^2 = \left(\frac{1}{2}\right) \times B \times H$$

Now, we can substitute H with its value in terms of B:

$$400\,cm^2 = \left(\frac{1}{2}\right) \times B \times (2B)$$
$$400\,cm^2 = B \times B$$
$$400\,cm^2 = B^2$$
$$\sqrt{400\,cm^2} = \sqrt{B^2}$$
$$B = 20\,cm$$

Now, we know the base (B), and since the height (H) is double the base:

$$H = 2B = 2 \times 20\,cm = 40\,cm$$

So, the height of the triangle is 40 centimeters.
Then the correct option is C)

14. Correct option: (C)

Let's denote the base of the parallelogram as "B" meters and the height as "H" meters.
We're given that the area of the parallelogram is 360 m², and the base and height are in the ratio of 5:2. So, we can express the height in terms of the base:

$$H = \left(\frac{2}{5}\right) \times B$$

Now, we can use the formula for the area of a parallelogram:

$$Area = Base \times Height$$

Substitute the expression for H:

$$360\,m^2 = B \times \left[\left(\frac{2}{5}\right) \times B\right]$$

$$360\,m^2 = \left(\frac{2}{5}\right) \times B^2$$

$$1800\,m^2 = 2 \times B^2$$

$$900\,m^2 = B^2$$

$$B = \sqrt{900\,m^2}$$

$$B = 30\,m$$

So, the base of the parallelogram is 30 meters.
Then the correct option is C)

15. Correct option: (B)

From the given information and figure we can conclude that

$$Area\ of\ Triangle = \left(\frac{1}{2}\right) \times Area\ of\ Parallelogram$$

Given that the area of the parallelogram is 360 m², we can directly calculate the area of the triangle:

$$Area\ of\ Triangle = \left(\frac{1}{2}\right) \times 360\,m^2 = 180\,m^2$$

So, the area of triangle ΔABC is 180 square meters.
Then the correct option is B)

16. Correct option: (A)

Let's denote the base of the parallelogram as "B" meters and the height as "H" meters.
According to the given information, the base and height are in the ratio of 5:2:

$$\frac{B}{H} = \frac{5}{2}$$

We're also given that the area of the parallelogram is 360 m². The formula for the area (A) of a parallelogram is:

$$A = Base \times Height$$

So, we can set up an equation using the given area:

$$360\,m^2 = B \times H$$

$$B = \left(\frac{5}{2}\right)H$$

Now, substitute this expression for B into the area equation:

$$360\,m^2 = \left(\frac{5}{2}\right)H \times H$$

$$360\,m^2 = \left(\frac{5}{2}\right)H^2$$

$$\left(\frac{2}{5}\right) \times 360\,m^2 = H^2$$

$$\left(\frac{2}{5}\right) \times 360\,m^2 = H^2$$

$$144\,m^2 = H^2$$

$$H = \sqrt{144\,m^2}$$

$$H = 12\,m$$

So, the height of the parallelogram is 12 meters.
Then the correct option is A)

17. Correct Matching is: a-Q, b-R, c-S, d-T, e-P

Here's the matching of the formulas for area (Column I) with the corresponding polygons (Column II):

a) $\frac{1}{2}\,d_1d_2\rightarrow$ Q) Rhombus (This formula calculates the area of a rhombus using its diagonals.)

b) $\frac{1}{2}h(a+b)\rightarrow$ R) Trapezium (This formula calculates the area of a Trapezium using its base and height.)

c) $\frac{1}{2}(h_1+h_2)d\rightarrow$ S) Quadrilateral (This formula calculates the area of a trapezium using its heights and one base.)

d) $bh\rightarrow$ T) Parallelogram (This formula calculates the area of a Parallelogram using its base and height, but in a square, the base and height are the same.)

e) $a^2\rightarrow$ P) Square

18. Correct Answer is: (5)

Let's denote the base of the parallelogram as "b" and the height as "h." According to the given information, the ratio of the base to the height is 1:5, which can be written as:

$$\frac{b}{h}=\frac{1}{5}$$

We're also given that the area of the parallelogram is 125 cm². The formula for the area (A) of a parallelogram is:

$$A = base \times height$$
$$125\,cm^2 = b \times h$$
$$b = \left(\frac{1}{5}\right)h$$
$$125\,cm^2 = \left[\left(\frac{1}{5}\right)h\right] \times h$$
$$125\,cm^2 = \left(\frac{1}{5}\right)h^2$$
$$h^2 = 125\,cm^2 \times 5$$
$$h^2 = 625\,cm^2$$

Now, take the square root of both sides to find h:

$$h = \sqrt{625\,cm^2}$$
$$h = 25\,cm$$

So, the height of the parallelogram is 25 cm.

Then
$$b = \left(\frac{1}{5}\right)h$$
$$b = \left(\frac{1}{5}\right) \times 25\,cm$$
$$b = 5\,cm$$

Therefore, the base of the parallelogram is 5 cm.

19. Correct Answer is: (9)

We can find the breadth (width) of the rectangle by using the formula for the area of a rectangle, which is: $Area = Length \times Breadth$

In this case, we're given the area (144 cm²) and the length (16 cm). We need to solve for the breadth (B):

$$144\,cm^2 = 16\,cm \times B$$
$$B = \frac{144\,cm^2}{16\,cm}$$
$$B = 9\,cm$$

So, the breadth of the rectangle is 9 cm.

20. Correct Answer is: (8)

To find the area of a triangle when you know its base (b) and altitude (h), we can use the following formula: $Area = \left(\frac{1}{2}\right) \times base \times altitude$

In this case, we're given the base as 12 cm and the area as 48 cm². we want to solve for the altitude (h), so the formula becomes: $48\,cm^2 = \left(\frac{1}{2}\right) \times 12\,cm \times h$

Now, solve for h: $48\,cm^2 = 6\,cm \times h$

To isolate h, divide both sides by 6 cm: $h = \frac{48\,cm^2}{6\,cm}$

$$h = 8\,cm$$

So, the altitude (height) of the triangle is 8 cm.

STUDENT PRACTICE QUESTIONS

1. Correct option: (D)

Perimeter of square = 4 × side length

Let's take the side length of the square is s then

Perimeter = 4 × s cm

$$Perimeter = 88\,cm$$

$$4s = 88 => s = \frac{88}{4} = 22$$

So, the side length of the square is 22 cm.

Then the correct option is D)

2. Correct option: (C)

To find the area of a square, we can use the formula:

$$Area = side\ length \times side\ length$$

In this case, the side length of the square is 4 cm, so we can calculate the area as follows:

$$Area = 4\,cm \times 4\,cm$$

$$Area = 16\,cm^2$$

So, the area of the square with a side length of 4 cm is 16 square cm.

Then the correct option is C)

3. Correct option: (A)

The perimeter of a rectangle can be found using the following formula:

$$Perimeter = 2 \times (Length + Breadth)$$

Given that the length is 15 cm and the breadth is 10 cm, we can plug these values into the formula:

$$Perimeter = 2 \times (15\,cm + 10\,cm)$$

$$Perimeter = 2 \times 25\,cm$$

$$Perimeter = 50\,cm$$

So, the perimeter of the rectangle is 50 cm.

Then the correct option is A)

4. Correct option: (B)

Given that the area of the rectangle is 400 square units, we can use the formula for the area of a rectangle: $Area = Length \times Breadth$

So, substituting the values: (as given ratio is 4:1)

$$400\,sq.units = (4x) \times (x)$$

$$4x^2 = 400$$

$$x^2 = 100$$

$$x = \sqrt{100}$$

$$x = 10$$

So, the breadth of the rectangle is 10 units, and the length is 4 times the breadth:

$$Length = 4x = 4 \times 10 = 40\,units$$

The length of the rectangle is 40 units.

Then the correct option is B)

5. Correct option: (D)

Given that the area is 39 cm² and the height is 13 cm, we can plug these values into the formula and solve for the base:

$$39\,cm^2 = \left(\frac{1}{2}\right) \times Base \times 13\,cm$$

$$39\,cm^2 = \left(\frac{1}{2}\right) \times 13\,cm \times Base$$

$$Base = (39\,cm^2) \times \left(\frac{2}{13}\right)$$

$$Base = 6\,cm$$

So, the base of the triangle is 6 cm.

Then the correct option is D)

6. Correct option: (C)

The area of a parallelogram can be calculated using the formula:

$$Area = Base \times Altitude$$

In this case, the base of the parallelogram is 12 cm, and the corresponding altitude (height) is 6 cm.

$$Area = 12\,cm \times 6\,cm$$

$$Area = 72\,square\,cm$$

So, the area of the parallelogram is 72 square centimeters.

Then the correct option is C)

7. Correct option: (C)

The area of a figure with a base DC and an altitude (height) to AB can be calculated using the formula for the area of a parallelogram:

$$Area = Base \times Height$$

In this case, we have:

$$Base\,DC = x$$

$$Altitude\,(height)\,to\,AB = x - 1$$

From figure we can say that $AB \parallel DC$ then area

$$Area = x \times (x - 1)$$

$$Area = x^2 - x$$

Then the correct option is C)

8. Correct option: (B)

The area of the given figure can be calculated using triangle with legs as 3cm and 4 cm

We are given 4 right angle triangles so total area of the figure can be calculated as

$$A = 4 \times (area\ of\ one\ triangle)$$

So area of one triangle is

$$area\ of\ one\ triangle = \frac{1}{2} \times 3 \times 4 = 6$$

Therefore the total area is

$$A = 4 \times 6 = 24\,sq.cm$$

Then the correct option is B)

9. Correct option: (C)

To find the area of a rhombus when we know the lengths of its diagonals, we can use the following formula:

$$Area = \left(\frac{1}{2}\right) \times (Product\ of\ Diagonals)$$

In this case, the diagonals are given as 10 cm and 24 cm. So, we can calculate the area as follows:

$$Area = \left(\frac{1}{2}\right) \times (10\ cm \times 24\ cm)$$

$$Area = \left(\frac{1}{2}\right) \times 240\ cm^2$$

$$Area = 120\ cm^2$$

So, the area of the rhombus is 120 square centimeters.

Then the correct option is C)

10. Correct option: (D)

The area of a trapezium can be calculated using the formula:

$$Area = \left(\frac{1}{2}\right) \times (x + y) \times (Distance\ Between\ Parallel\ Sides)$$

Where x and y are length of parallel sides.

In this case, 'a' and 'b' are the parallel sides, and we want to find the distance between them, denoted as 'h'. So, the formula becomes:

$$Area = \left(\frac{1}{2}\right) \times (a + b) \times h$$

We are given that area is $\frac{1}{2}(a^2 - b^2)$

So we can equate the areas

$$\left(\frac{1}{2}\right)(a^2 - b^2) = \left(\frac{1}{2}\right)(a + b) \times h$$

$$(a^2 - b^2) = (a + b) \times h$$

$$h = \frac{a^2 - b^2}{a + b} = \frac{(a-b)(a+b)}{a+b} = a - b$$

So, the distance between the parallel sides 'a' and 'b' is a-b units.

Then the correct option is D)

11. Correct option: (B, C)

To find the perimeter of a square when we know its area, we can use the following steps:

1. Start with the formula for the area of a square:
   $Area = Side^2$
2. Given that the area is 625 cm², set up the equation:
   $625\ cm^2 = Side^2$
3. Solve for the length of one side (Side):

$$Side = \sqrt{625\ cm^2}$$

$$Side = 25\ cm$$

4. Now that we know the side length (Side), we can find the perimeter (P) of the square:

$$Perimeter = 4 \times Side$$

$$Perimeter = 4 \times 25\ cm$$

$$Perimeter = 100\ cm$$

So, the perimeter of the square is 100 cm.

Then the correct option is B, C)

12. Correct option: (A, B)

Let's denote the height of the triangle as "H" centimeters.

First, let's calculate the area of the rectangle:

$$Area\ of\ Rectangle\ =\ Length\ \times\ Breadth\ =\ 20\,cm\ \times\ 15\,cm\ =\ 300\,cm^2$$

Now, we're given that the area of a rectangle is equal to the area of a triangle. The formula for the area of a triangle is:

$$Area\ of\ Triangle\ =\ \left(\frac{1}{2}\right)\ \times\ Base\ \times\ Height$$

We're given that the base of the triangle is 30 cm. So, we have:

$$Area\ of\ Triangle\ =\ \left(\frac{1}{2}\right)\ \times\ 30\,cm\ \times\ H$$

Now, equate the area of the rectangle to the area of the triangle:

$$300\,cm^2\ =\ \left(\frac{1}{2}\right)\ \times\ 30\,cm\ \times\ H$$

Now, solve for H (the height of the triangle):

$$300\,cm^2\ =\ 15\,cm\ \times\ H$$

$$H\ =\frac{300\,cm^2}{15\,cm}$$

$$H\ =\ 20\,cm$$

So, the height of the triangle, when its base measures 30 cm, is 20 cm.

Here the height of triangle is equal to length of rectangle.

Then the correct option is A, B)

13. Correct option: (B)

To find the area of a rectangle, we use the formula:

$$Area\ =\ Length\ \times\ Breadth$$

In this case, the length of the rectangle is 9 units, and the breadth of the rectangle is 5 units.

So, the area of the rectangle is:

$$Area\ =\ 9\,units\ \times\ 5\,units\ =\ 45\,square\ units$$

Therefore, the area of the rectangle is 45 square units.

Then the correct option is B)

14. Correct option: (B)

The perimeter of a rectangle can be calculated using the formula:

$$Perimeter\ =\ 2\ \times\ (Length\ +\ Breadth)$$

In this case, the length of the rectangle is 12 units, and the breadth of the rectangle is 8 units.

So, the perimeter of the rectangle is:

$$Perimeter\ =\ 2\ \times\ (12\,units\ +\ 8\,units)\ =\ 2\ \times\ 20\,units\ =\ 40\,units$$

Therefore, the perimeter of the rectangle is 40 units.

Then the correct option is B)

15. Correct Matching is: a-S, b-, c- P, d-Q, e-R

a) Side of a square is 12 cm, then its area is

- Match: S) 144 sq. cm
- Explanation: The area of a square is calculated using the formula $Area\ =\ Side^2$. In this case, Area = 12 cm × 12 cm = 144 sq. cm.

b) h = 12 cm, b = 6 cm, then the area of the triangle is

- Match: No Match
- Explanation: The area of a triangle is calculated using the formula Area = $\left(\frac{1}{2}\right)\ \times\ base\ \times\ height$. In this case, $Area\ =\ \left(\frac{1}{2}\right)\ \times\ 6\,cm\ \times\ 12\,cm\ =\ 36\,sq.cm.$

c) Perimeter of a square is 88 cm, then the side is

- Match: P) 22 cm
- Explanation: The perimeter of a square is calculated as $Perimeter = 4 \times Side$.

In this case, 88 cm = 4 × Side, so $Side = \frac{88\ cm}{4} = 22\ cm$.

d) The perimeter of a rhombus whose side is a cm is

- Match: S) 4a
- Explanation: The perimeter of a rhombus is calculated as Perimeter = 4 × Side. In this case, the side is represented as "a," so the perimeter is 4a.

e) The perimeter of a parallelogram if sides are 3 cm and 2 cm

- Match: R) 10 cm
- Explanation: $P = 2(3 + 2) = 10$

16. Correct Answer is: (4)

To find the side length of a square when we know the perimeter, we can use the formula for the perimeter of a square, which is:

$$Perimeter = 4 \times Side$$

In this case, we're given the perimeter as 16 cm, so we can set up the equation:

$$16\ cm = 4 \times Side$$
$$Side = \frac{16\ cm}{4}$$
$$Side = 4\ cm$$

So, the side of the square is 4 cm.

17. Correct Answer is: (7)

To find the length of the second diagonal of a rhombus when we know its area and one diagonal, we can use the following formula:

$$Area = \left(\frac{1}{2}\right) \times Product\ of\ Diagonals$$

In this case, we're given the area as 112 sq. cm and one diagonal as 32 cm. Let's denote the length of the second diagonal as "d_2." The formula becomes:

$$112\ sq.cm = \left(\frac{1}{2}\right) \times 32\ cm \times d_2$$
$$112\ sq.cm = 16\ cm \times d_2$$
$$d_2 = \frac{112\ sq.cm}{16\ cm}$$
$$d_2 = 7\ cm$$

So, the length of the second diagonal of the rhombus is 7 cm.

18. Correct Answer is: (8)

In an isosceles right-angled triangle, the two legs are congruent (equal) and the angles at the base are each 45°. To find the length of a side in such a triangle, we can use the following formula for the area:

$$Area = \left(\frac{1}{2}\right) \times Leg^2$$

In this case, we're given the area as 32 sq. cm. Let's denote the length of each leg as "L." The formula becomes:

$$32\ sq.cm = \left(\frac{1}{2}\right) \times L^2$$

Now, solve for L:

$$32\ sq.cm = \left(\frac{1}{2}\right) \times L^2$$
$$64\ sq.cm = L^2$$

$$L = \sqrt{64\ sq.cm}$$
$$L = 8\ cm$$

So, the length of each side of the isosceles right-angled triangle is 8 cm.

19. Correct Answer is: (8)

In an equilateral triangle, the area (A) can be calculated using the following formula:

$$A = \frac{\sqrt{3}}{4} \times Side^2$$

In this case, we're given the area as $144\sqrt{3}$. Let's denote the length of each side as "S." The formula becomes:

$$144\sqrt{3} = \frac{\sqrt{3}}{4} \times S^2$$
$$S^2 = 144 \times 4 = 576$$

Now, take the square root of both sides to find S:

$$S = \sqrt{576}$$
$$S = 24$$

So, the length of each side of the equilateral triangle is 24 units.

Now, to find $\frac{1}{3}$ of the side length:

$$\frac{1}{3} \times 24 = 8\ units$$

Therefore, $\frac{1}{3}$ of the side length is 8 units.

20. Correct Answer is: (8)

We can find the height (h) of the triangle using the formula for the area (A) of a triangle, which is: $A = \left(\frac{1}{2}\right) \times base \times height$

In this case, we're given the area as 20 cm² and the base as 5 cm. We want to solve for the height, so the formula becomes:

$$20\ cm^2 = \left(\frac{1}{2}\right) \times 5\ cm \times h$$
$$20 = \left(\frac{5}{2}\right) \times h$$
$$h = \frac{20}{\frac{5}{2}}$$
$$h = (20) \times \left(\frac{2}{5}\right)$$
$$h = \frac{40}{5}$$
$$h = 8\ cm$$

So, the height of the triangle is 8 cm.

CIRCLES AND RINGS
LEVEL – 1

1. Correct option: (C)

The circumference of a circle is given by the formula $C = 2\pi r$, where r is the radius of the circle. In this case, the wire has been bent into a circle, so the length of the wire is equal to the circumference of the circle.

Given the length of the wire is $10\ cm$, the circumference of the circle is also $10\ cm$.

Then the correct option is C)

2. Correct option: (C)

The formula for the circumference of a circle is $C = 2\pi r$, where C is the circumference and r is the radius of the circle.

Given that the circumference is 44 cm, we can rearrange the formula to solve for the radius: $r = \frac{C}{2\pi}$

Substitute the given circumference:

$$r = \frac{44\ cm}{2\pi}, \pi = \frac{22}{7} => r = 7cm$$

Then the correct option is C)

3. Correct option: (A)

The formula to calculate the circumference of a circle is $C = \pi d$, where C is the circumference and d is the diameter of the circle.

Given the diameter is 4.9 cm, we can plug it into the formula:

$$C = \pi \times 4.9\ cm$$

$$\pi = \frac{22}{7} => C = \frac{22}{7} \times 4.9\ cm = 15.4\ cm$$

Then the correct option is A)

4. Correct option: (B)

The formula for the circumference of a circle is $C = \pi d$, where C is the circumference and d is the diameter of the circle.

Given the diameter is 10.5 cm, we can plug it into the formula: $C = \pi \times 10.5\ cm$

$$\pi = \frac{22}{7} => C = \frac{22}{7} \times 10.5 = 22 \times 1.5 = 33$$

Then the correct option is B)

5. Correct option: (C)

The perimeter of a semicircle is the sum of half the circumference of the full circle and the straight side of the semicircle.

1. First, find the circumference of the full circle using the formula C = 2πr, where r is the radius:

Hence $\frac{C}{2} = \pi r$

$$\frac{C}{2} = \pi \times 2.1\ m = \frac{22}{7} \times 2.1\ m = 22 \times 0.3\ m = 6.6\ m$$

2. Finally, add the half circumference to the straight side of the semicircle (which is the diameter): $Perimeter = 6.6\ m + 2 \times 2.1\ m = 6.6 + 4.2\ m = 10.8\ m$

Then the correct option is C)

6. Correct option: (B)

To find the circumference of a semicircle with a given diameter, we can use the formula:

$$C = \pi \times r + d$$

Given that we're using π as $\frac{22}{7}$ and the diameter is 56 cm, we can calculate the circumference as follows:

1. Calculate the radius (r) by dividing the diameter by 2:

$$r = \frac{56\ cm}{2} = 28\ cm$$

2. Now, use the formula to find the circumference:

$$C = \left(\frac{22}{7}\right) \times 28\ cm + 56\ cm$$

$$C = (22 \times 4\ cm) + 56\ cm$$

$$C = 88\ cm + 56\ cm$$

$$C = 144\ cm$$

Then the correct option is B)

7. Correct option: (D)

The formula for the circumference (C) of a circle in terms of its diameter (d) is:

$$C = \pi \times d$$

Given that the circumference is 154 cm, we can solve for the diameter (d) as follows:

$$154\,cm = \pi \times d$$

$$d = \frac{154}{\pi}\,cm = \frac{154}{\frac{22}{7}} = 154 \times \frac{7}{22} = 7 \times 7$$

Then the correct option is D)

8. Correct option: (D)

To find the area of a sector, we can use the following formula:

$$Area = \left(\frac{\theta}{360}\right) \times \pi \times r^2$$

Where:

- θ is the angle of the sector in degrees (240° in this case).

Let's calculate it:

$$Area = \left(\frac{240}{360}\right) \times \left(\frac{22}{7}\right) \times (7\,cm)^2$$

$$Area = \left(\frac{2}{3}\right) \times \left(\frac{22}{7}\right) \times 49\,cm^2$$

Now, perform the calculations step by step:

$$Area = \left(\frac{2}{3}\right) \times \left(\frac{22}{7}\right) \times 49\,cm^2$$

$$Area = \left(\frac{44}{21}\right) \times 49\,cm^2$$

$$Area = \frac{2156}{21}\,cm^2$$

$$Area \approx 102.67\,cm^2$$

Then the correct option is D)

9. Correct option: (A)

To find the length of the arc of a sector, we can use the following formula:

$$Arc\ Length = \left(\frac{\theta}{360}\right) \times 2\pi r$$

Where:

- θ is the angle of the sector in degrees (72° in this case).

Let's calculate it: $Arc\ Length = \left(\frac{72}{360}\right) \times 2 \times \left(\frac{22}{7}\right) \times 14\,cm$

First, simplify the fraction: $Arc\ Length = \left(\frac{1}{5}\right) \times 2 \times \left(\frac{22}{7}\right) \times 14\,cm$

Now, perform the calculations step by step:

$$Arc\ Length = \frac{88}{5}\,cm$$

$$Arc\ Length = 17.6\,cm$$

Then the correct option is A)

10. Correct option: (B)

If the diameter of a circle and the side of a square are equal, it means the radius of the circle (r) is equal to half the side length of the square (s).

Let A_{circle} be the area of the circle and A_{square} be the area of the square.

$$A_{circle} = \pi r^2$$

$$A_{square} = s^2$$

Since $r = \frac{s}{2}$, we can express the area of the circle in terms of the area of the square:

$$A_{circle} = \pi(r^2) = \pi(\left(\frac{s}{2}\right)^2) = \pi\left(\frac{s^2}{4}\right)$$

Now, we can find the ratio of the areas:

$$\frac{A_{circle}}{A_{square}} == \frac{\pi\left(\frac{s^2}{4}\right)}{s^2} = \left(\frac{\pi}{4}\right)$$

Then the correct option is B)

11. Correct option: (A)

To find the radius of the road roller, we can use the following formula:

$$Distance = Circumference\ of\ the\ roller \times Number\ of\ rotations$$

In this case, the distance is 2200 meters, and the number of rotations is 200.

Let r be the radius of the roller.

$$Circumference\ of\ the\ roller = 2\pi r$$

Now, we can set up the equation:

$$2200\ meters = (2\pi r) \times 200\ rotations$$

Let's solve for r:

$$r = \frac{2200\ meters}{2\pi \times 200}$$

$$r = \frac{22}{4\pi}\ meters = \frac{22}{4 \times \frac{22}{7}} = \frac{7}{4}\ m$$

Then the correct option is A)

12. Correct option: (D)

To find the perimeter of a semicircle, we'll need to first calculate its radius using the given area and then use that radius to find the semicircle's circumference (which is half of the circumference of a full circle).

Let A be the area of the semicircle, and r be its radius.

The formula for the area of a semicircle is:

$$A = \left(\frac{1}{2}\right) \times \pi \times r^2$$

Given that the area is 1925 sq. cm:

$$1925 = \left(\frac{1}{2}\right) \times \pi \times r^2$$

Now, solve for r:

$$r^2 = \frac{2 \times 1925}{\pi} = \frac{2 \times 1925}{\frac{22}{7}} = \frac{2 \times 1925}{\pi} = \frac{3850 \times 7}{22} = 175 \times 7 = 1225$$

Now, take the square root of both sides to find r:

$$r = \sqrt{1225} = 35\ cm$$

Now that we have the radius, we can calculate the semicircle's circumference using the formula:

$$Circumference = \pi \times r = \frac{22}{7} \times 35 = 22 \times 5 = 110\ cm$$

Perimeter of the semicircle is

$$P = Circumference\ of\ semicircle + diameter$$

$$P = 110\ cm + 2 \times 35\ cm = 180\ cm$$

So, the perimeter of the semicircle is 180 cm.

Then the correct option is D)

13. Correct option: (C)

The area of a ring, also known as an annulus, can be calculated using the formula:

$$Area = \pi \times (R^2 - r^2)$$

Where:
- R is the outer radius (25 m in this case)
- r is the inner radius (18 m in this case)
Now, plug in the values:

$$Area = \pi \times (25^2 - 18^2)$$
$$Area = \pi \times (625 - 324)$$
$$Area = \pi \times 301$$
$$Area = \frac{22}{7} \times 301\ sq.m = 22 \times 43$$

Then the correct option is C)

14. Correct option: (D)

To find the area of the ring (annulus), we can use the formula:

$$Area = \pi \times (R^2 - r^2)$$

Where:
- R is the outer radius (4 cm in this case).
- r is the inner radius (3 cm in this case).
Now, plug in the values:

$$Area = \pi \times (4^2 - 3^2)$$
$$Area = \pi \times (16 - 9)$$
$$Area = \pi \times 7$$
$$Area = 22\ sq.cm$$

Then the correct option is D)

15. Correct option: (A)

To find the length of the bounding arc of a sector, we can use the formula:

$$Arc\ Length = \left(\frac{\theta}{360}\right) \times 2\pi r$$

Where:
- θ is the angle of the sector in degrees.
Given that the radius is 21 m, we still need to find the angle θ. We can use the formula for the area of a sector to find θ:

$$Area\ of\ Sector = \left(\frac{\theta}{360}\right) \times \pi \times r^2$$

Given that the area of the sector is 231 sq. cm, we need to convert this to square meters (since the radius is in meters):

$$Area\ of\ Sector = \left(\frac{\theta}{360}\right) \times \pi \times (21\ m)^2$$
$$231\ sq.cm = \left(\frac{\theta}{360}\right) \times \pi \times (21\ m)^2$$

Now, solve for θ:

$$\frac{\theta}{360} = \frac{231}{\pi \times 21^2}$$
$$\frac{\theta}{360} = \left(\frac{231}{\pi \times 21^2}\right) = \frac{11}{\frac{22}{7} \times 21} = \frac{11}{66} = \frac{1}{6}$$

Now that we have the angle $\frac{\theta}{360}$, we can calculate the arc length:

$$Arc\ Length = \frac{1}{6} \times 2\pi \times 21\ m = \frac{1}{6} \times 2 \times \frac{22}{7} \times 21\ m = 22\ m$$
$$Arc\ Length = 22\ m$$

Then the correct option is A)

16. Correct option: (C)

To find the perimeter of the sector, we need to calculate the sum of the arc length and the two radii.

Calculate the perimeter of the sector, which is the sum of the arc length and the two radii:

$$Perimeter = 2r + Arc\ Length$$

$$Perimeter = 2 \times 8\,cm + 12\,cm = 16\,cm + 12\,cm = 28\,cm$$

Then the correct option is C)

17. Correct option: (A)

To find the number of times a wheel with a radius of 35 cm must rotate to travel 660 m, we can use the formula for the circumference of a circle:

$$Circumference = 2\pi r$$

Where:

- Circumference is the distance traveled in one full rotation (in this case, the distance traveled in one rotation of the wheel).

First, calculate the circumference of the wheel:

$$Circumference = 2 \times \left(\frac{22}{7}\right) \times 35\,cm = 220\,cm$$

Now, divide the total distance by the circumference of one rotation to find the number of rotations:

$$Number\ of\ Rotations = \frac{Total\ Distance}{Circumference\ of\ One\ Rotation}$$

$$Number\ of\ Rotations = \frac{660\,cm}{220\,cm} = 3$$

Then the correct option is A)

18. Correct option: (B)

The formula for the circumference of a circle is:

$$Circumference = 2\pi r$$

Where C is the circumference, π is approximately $22/7$, and r is the radius.

Given the circumference is 264 cm, we can solve for the radius (r):

$$264\,cm = 2 \times \left(\frac{22}{7}\right) \times r$$

Now, isolate r:

$$r = \frac{264\,cm}{2 \times \left(\frac{22}{7}\right)}$$

$$r = \frac{264\,cm}{\left(\frac{44}{7}\right)}$$

$$r = (264\,cm) \times \left(\frac{7}{44}\right)$$

$$r = 42\,cm$$

So, the radius of the circle is 42 cm.

Then the correct option is B)

19. Correct option: (D)

To find the perimeter of a semicircle, we need to calculate the sum of the semicircle's curved boundary (half of its circumference) and the straight boundary (diameter).

1. Calculate the curved boundary (half of the circumference): $Circumference = 2\pi r$

Where r is the radius, which is 28 cm in this case.

$$Curved\ Boundary = \left(\frac{1}{2}\right) \times (2\pi r) = \pi r = \pi \times 28\,cm = 88\,cm$$

2. Calculate the straight boundary (diameter):

$$Diameter = 2r = 2 \times 28\,cm = 56\,cm$$

$$Perimeter = Curved\,Boundary + Diameter = 88\,cm + 56\,cm = 144\,cm$$

So, the perimeter of the semicircle is 144 cm.
Then the correct option is D)

20. Correct option: (B)

The length of the wire is equal to the circumference of the circle formed.
Given that the length of the wire is 9 units, the circumference of the circle is also 9 units.
Then the correct option is B)

LEVEL – II

1. Correct option: (A)

To find the length of the arc of a sector, we can use the following formula:

$$Arc\,Length = (\theta/360) \times 2\pi r$$

Where:
- θ is the angle of the sector in degrees.

First, we need to find the angle $\left(\frac{\theta}{360}\right)$. To do that, use the formula for the area of a sector:

$$Area = \left(\frac{\theta}{360}\right) \times \pi \times r^2$$

Given that the area is 88 sq. cm, we can solve for $\frac{\theta}{360}$:

$$88\,sq.cm = \left(\frac{\theta}{360}\right) \times \pi \times (8\,cm)^2$$

$$\frac{\theta}{360} = \frac{88}{(\pi \times 64)} = \frac{7}{16}$$

$$Arc\,Length = \left(\frac{\theta}{360}\right) \times 2\pi \times 8\,cm$$

$$Arc\,Length = \left(\frac{7}{16}\right) \times 2 \times \left(\frac{22}{7}\right) \times 8\,cm$$

$$Arc\,Length = 22\,cm$$

Then the correct option is A)

2. Correct option: (B)

To find the area of a semicircle, we'll first need to calculate its radius using the given perimeter, and then we can use the radius to find the area.
The perimeter of a semicircle consists of the curved boundary (half of the circumference) and the straight boundary (diameter).
Let P be the perimeter.

$$P = Curved\,Boundary + Diameter$$

Given that the perimeter is 144 cm, you can set up the equation:

$$144\,cm = \left(\frac{1}{2}\right) \times (2\pi r) + d$$

Now, simplify this equation: $144\,cm = \pi r + d$

Since the diameter (d) is twice the radius (r), we can rewrite the equation as:

$$144\,cm = \pi r + 2r$$

Now, factor out r: $144\,cm = r(\pi + 2)$

Now, solve for r:

$$r = \frac{144\,cm}{\pi + 2} = \frac{144 \times 7}{36}cm = 28\,cm$$

Now, we have the radius (r), and we can use it to find the area (A) of the semicircle:

$$A = \frac{\pi r^2}{2}$$

Plug in the value of r:

$$A = \frac{\pi(28^2)}{2}$$

Now, calculate the area:

$$A = \frac{22}{7} \times \frac{28}{2} \times 28 = 1232\ cm^2$$

Then the correct option is B)

3. Correct option: (D)

The area of a ring (annulus) can be calculated using the formula:

$$Area = \pi \times (R^2 - r^2)$$

Where:

- R is the outer radius (55 m in this case).
- r is the inner radius (41 m in this case).

Now, plug in the values:

$$Area = \pi \times (55^2 - 41^2)$$
$$Area = \pi \times (3025 - 1681)$$
$$Area = \pi \times 1344$$
$$Area = 4224\ sq.m$$

Then the correct option is D)

4. Correct option: (C)

The ratio of the area of a sector to the area of the whole circle is given by:

$$\frac{Area\ of\ Sector}{Area\ of\ Circle} = \frac{\left(\frac{\theta}{360}\right) \times \pi r^2}{\pi r^2} = \left(\frac{\theta}{360}\right)$$

Where:

- θ is the angle of the sector in degrees (90° in this case).

Now, let's calculate this ratio:

$$\frac{Area\ of\ Sector}{Area\ of\ Circle} = \left(\frac{90}{360}\right)$$
$$\frac{Area\ of\ Sector}{Area\ of\ Circle} = \left(\frac{1}{4}\right)$$

The ratio of the area of the sector to the area of the circle is 1:4.

Then the correct option is C)

5. Correct option: (B)

To find the perimeter of a sector, we can use the formula:

The perimeter of the sector consists of the curved boundary (arc length) and two radii.

$$Arc\ Length\ (curved\ boundary) = \left(\frac{\theta}{360}\right) \times 2\pi r$$
$$Arc\ Length = \left(\frac{72}{360}\right) \times 2\pi \times 105\ m$$
$$Arc\ Length = \left(\frac{1}{5}\right) \times 2\pi \times 105\ m$$
$$Arc\ Length = 42\pi\ m$$

Now, let's calculate the perimeter of the sector:

$$Perimeter = Arc\ Length + 2r$$
$$Perimeter = 42\pi\ m + 2 \times 105\ m$$
$$Perimeter = 42\pi\ m + 210\ m = 42 \times \frac{22}{7} + 210\ m = (132 + 210)m = 342$$

So, the perimeter of the sector is 342 m.

Then the correct option is B)

6. Correct option: (D)

The tip of the minute hand moves in a circular path as the clock's minute hand moves around the clock face. To find how far it moves in 60 minutes, we can use the formula for the circumference of a circle:

$$Circumference = 2\pi r$$

Where:

- r is the length of the minute hand, which is 14 cm in this case.

$$Circumference = 2\pi \times 14\,cm$$

Now, calculate the circumference:

$$Circumference = 28\pi\,cm$$

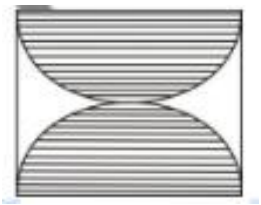
$$Circumference = 88\,cm$$

So, the tip of the minute hand moves approximately 88 cm in 60 minutes.

Then the correct option is D)

7. Correct option: (D)

Given figure



Looking at figure we can conclude that the area of shaded region is same as sum of areas of the two semicircles with diameter as 14 cm

Let's calculate the sum of the areas of the two semicircles.

Each semicircle has a diameter equal to the side length of the square, which is 14 cm. The radius (r) of each semicircle is half of the diameter, so:

$$r = \frac{14\,cm}{2} = 7\,cm$$

Now, we can find the area of one semicircle:

$$Area\ of\ one\ semicircle = \left(\frac{1}{2}\right) \times \pi \times r^2$$

$$Area\ of\ one\ semicircle = \left(\frac{1}{2}\right) \times \frac{22}{7} \times (7\,cm)^2$$

$$Area\ of\ one\ semicircle = 77\,cm^2$$

Since there are two semicircles, we can sum their areas:

$$Total\ area\ of\ two\ semicircles = 2 \times 77\,cm^2 = 154\,cm^2$$

The sum of the areas of the two semicircles is 154 square centimeters.

Then the correct option is D)

8. Correct option: (B)

To find the area of the path around the circular lawn, we can first calculate the area of the outer circle (the lawn plus the path) and then subtract the area of the circular lawn itself.

Let A_1 be the area of the circular lawn, and A_2 be the area of the outer circle (lawn + path). Given that A_1 is 2464 sq. m, we need to find A_2.

$$A_1 = \pi \times r^2\ (for\ the\ circular\ lawn)$$

$$2464 = \pi \times r^2$$

Now, solve for r:

$$r^2 = \frac{2464}{\pi}$$

$$r^2 = 784$$

$$r = \sqrt{784}$$

$$r = 28\,m\ (radius\ of\ the\ circular\ lawn)$$

Now, calculate the area of the outer circle (lawn + path):

$$A_2 = \pi \times (r + 7)^2$$
$$A_2 = \pi \times (28 + 7)^2$$
$$A_2 = \pi \times (35^2)$$
$$A_2 = \pi \times 1225\,sq.m = 3850\,sq.m$$

Now, subtract the area of the circular lawn from the area of the outer circle to find the area of the path:

$$Area\ of\ the\ path = A_2 - A_1$$
$$Area\ of\ the\ path = 3850\,sq.m - 2464\,sq.m$$
$$Area\ of\ the\ path = 1386\,sq.m$$

Then the correct option is B)

9. Correct option: (C)

Let the radius of the inner circle be "r" meters, and the width of the track (the difference between the radii of the two circles) is 14 meters.

The radius of the outer circle is then (r + 14) meters.

The area of the track (the area between the two concentric circles) is given as 2,144 square meters. We can calculate it using the formula for the area of a ring (annulus):

$$Area\ of\ Track = \pi \times (R^2 - r^2)$$

Where R is the radius of the outer circle, and r is the radius of the inner circle.

Substituting the values:

$$1{,}144 = \pi \times ((r + 14)^2 - r^2)$$

Now, let's solve for r:

$$1{,}144 = \pi \times \big((2r + 14) \times 14\big) = 28\pi r + 196\pi \Rightarrow 28 \times \frac{22}{7} \times r + 196 \times \frac{22}{7}$$
$$1144 = 88r + 616$$
$$1144 - 616 = 88r \Rightarrow 528 = 88r \Rightarrow r = 6$$

So, the radius of the inner circle is approximately 6 meters.

The radius of the outer circle is (r + 14) meters:

Radius of the outer circle = 14 + 6 = 20 meters

Then the correct option is C)

10. Correct option: (A)

Let the radii of the two circles be $5x$ and $8x$, where x is a positive constant representing the ratio between the radii.

The area of a circle is given by the formula: $Area = \pi \times r^2$

So, the area of the first (smaller) circle is $\pi \times (5x)^2 = 25\pi x^2$, and the area of the second (bigger) circle is $\pi \times (8x)^2 = 64\pi x^2$.

Now, find the difference in their areas: $64\pi x^2 - 25\pi x^2 = 39\pi x^2$

It is given that the difference in their areas is 351π sq. cm, so:

$$39\pi x^2 = 351\pi$$

Now, divide both sides by 39π to solve for x^2:

$$x^2 = \frac{351\pi}{39\pi}$$
$$x^2 = 9 \Rightarrow x = 3$$

Now that we have found the value of x, we can find the area of the bigger circle (second circle):

$$Area\ of\ the\ bigger\ circle = \pi \times (8x)^2$$
$$Area\ of\ the\ bigger\ circle = \pi \times (8 \times 3)^2 = 576\pi$$

So, the area of the bigger circle is 576π sq. cm.

Then the correct option is A)

11. Correct option: (A, C)

To find the area of a circle when the circumference is given, we can use the following formulas: $1. Circumference\ (C) = 2\pi r$

$2. Area\ (A) = \pi r^2$

Given that the circumference is 176 m, we can first find the radius (r) using the circumference formula: $176 = 2\pi r$

Now, solve for r: $r = \frac{176}{2\pi} = 28\,m$

Now that we have the radius, we can find the area: $A = \pi r^2 = \pi \times (28\,m)^2 = 2464\,m^2$

So, the area of the circle is 2464 m².

Then the correct option is both A) and C)

12. Correct option: (B, D)

The relationship between the area (A) and circumference (C) of a circle is as follows:

$$C = 2\sqrt{\pi A} = \sqrt{4\pi A}$$

Then the correct option is both B) and D)

13. Correct option: (C)

To find the area of the ring-shaped region enclosed between two circles with radii of 20 cm and 15 cm, we can use the formula for the area of a ring (annulus):

$$Area = \pi \times (R^2 - r^2)$$

Where:

- R is the radius of the larger circle (20 cm).
- r is the radius of the smaller circle (15 cm).

Now, plug in the values: $Area = \pi \times (20^2 - 15^2)$

$$Area = \pi \times (400 - 225)$$

$$Area = \pi \times 175\,cm^2 = 175 \times \frac{22}{7}\,cm^2 = 25 \times 22\,cm^2 = 550 cm^2$$

Then the correct option is C)

14. Correct option: (B)

To find the length of an arc of a sector, we can use the formula:

$$Arc\ Length = \left(\frac{\theta}{360}\right) \times 2\pi r$$

Where:

- θ is the angle of the sector in degrees.

In this case, the angle θ is given as 135 degrees, and the radius r is 8.4 cm. Plug these values into the formula:

$$Arc\ Length = \left(\frac{135}{360}\right) \times 2\pi \times 8.4\,cm$$

Now, calculate the arc length:

$$Arc\ Length = \left(\frac{3}{8}\right) \times 2 \times \left(\frac{22}{7}\right) \times 8.4\,cm = \frac{3}{8} \times 22 \times 1.2\,cm = 19.8\,cm$$

So, the length of the arc of the sector is 19.8 cm.

Then the correct option is B)

15. Correct option: (C)

To find the area of a sector, we can use the formula: $Area = \left(\frac{\theta}{360}\right) \times \pi \times r^2$

Where:

- θ is the angle of the sector in degrees.

Now, plug in the values: $Area = \left(\frac{135}{360}\right) \times \left(\frac{22}{7}\right) \times (8.4\,cm)^2$

$$Area = \left(\frac{3}{8}\right) \times \left(\frac{22}{7}\right) \times (8.4\,cm)^2$$

Now, calculate the area:

$$Area = \left(\frac{3}{8}\right) \times \left(\frac{22}{7}\right) \times (8.4\,cm)^2 = \frac{3}{8} \times \frac{22}{7} \times 8.4 \times 8.4\,cm^2 = 3 \times 11 \times 0.3 \times 8.4\,cm^2 = 83.16\,cm^2$$

Then the correct option is C)

16. Correct option: (A)

Let's calculate the circumference and diameter of the circle.

$$Circumference\ of\ the\ circle = 2\pi r$$

$$Diameter\ of\ the\ circle = 2r$$

Given the radius of the sector is $8.4\,cm$, we can calculate the circumference and diameter:

$$Circumference = 2\pi \times 8.4\,cm = 16.8\pi\,cm$$

$$Diameter = 2 \times 8.4\,cm = 16.8\,cm$$

So, the circumference and diameter of the circle are 16.8π cm and 16.8 cm, respectively.

$$Circumference = 2\pi \times 8.4\,cm = 16.8\pi\,cm = 16.8 \times \frac{22}{7}\,cm = 2.4 \times 22\,cm = 52.8\,cm$$

Then the correct option is A)

17. Correct Matching is: a-P, b-T, c-S, d-Q

a) Circumference of a semi-circle **(P)** = $2r + \pi r$

Explanation: The formula for the circumference of a semicircle is half the circumference of a full circle, and added with diameter of circle.

b) Area of a circle **(T)** = πr^2

Explanation: The area of a full circle is given by πr^2.

c) Diameter of a circle **(S)** = 2r

Explanation: The diameter of a circle is twice the radius.

d) Area of a semi-circle **(Q)** = $\frac{\pi r^2}{2}$

Explanation: The formula for the area of a semicircle is half the area of a full circle, which is πr^2 divided by 2.

18. Correct Answer is: (5)

To find the width of the ring, subtract the smaller radius (2 cm) from the larger radius (7 cm):

$$Width = Larger\ Radius - Smaller\ Radius$$

$$Width = 7\,cm - 2\,cm$$

$$Width = 5\,cm$$

So, the width of the ring is 5 cm.

19. Correct Answer is: (8)

To find $\frac{1}{44}$ times the circumference of a circle with a radius of 56 cm, simply multiply the circumference by $\frac{1}{44}$:

$$Circumference = 2\pi r, where\ r\ is\ the\ radius.$$

$$Circumference = 2\pi \times 56\,cm$$

$$=> \left(\frac{1}{44}\right) \times (2\pi \times 56\,cm) = \left(\frac{2}{44}\right) \times (\frac{22}{7} \times 56\,cm)$$

$$=> \frac{56}{7} = 8$$

So, $\frac{1}{44}$ times the circumference of a circle with a radius of 56 cm is 8.

20. Correct Answer is: (9)

When a wire of length 9 units is bent to form a circle, the length of the wire becomes equal to the circumference of the circle. So, the circumference of the circle formed from the wire is 9 units.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

Given that the circumference is 44 cm, we can rearrange the formula to solve for the radius:

$$r = \frac{C}{2\pi}$$

Substitute the given circumference:

$$r = \frac{44\ cm}{2\pi} = \frac{44}{2 \times \frac{22}{7}}$$

$$r = 7\ cm$$

Now that we have the radius, we can calculate the area of the circle using the formula $A = \pi r^2$:

$$A = \pi \times (7\ cm)^2$$

$$A = 154\ sq\ cm$$

Then the correct option is B)

2. Correct option: (B)

The formula for the circumference of a circle is $C = 2\pi r$, where C is the circumference and r is the radius of the circle.

Given the radius is 15.4 cm, we can plug it into the formula:

$$C = 2\pi \times 15.4\ cm$$

$$C = 2 \times \frac{22}{7} \times 15.4 = 44 \times 2.2 = 96.8$$

Then the correct option is B)

3. Correct option: (B)

The formula for the circumference of a circle is C = 2πr, where C is the circumference and r is the radius of the circle.

Given that the circumference is 176 m, we can rearrange the formula to solve for the radius:

$$r = \frac{C}{2\pi}$$

Substitute the given circumference: $r = \frac{176\ m}{2 \times \frac{22}{7}} = 16 \times \frac{7}{4} = 28\ m$

Now that we have the radius, we can calculate the area of the circle using the formula

$$A = \pi r^2$$

$$A = \pi \times (28\ m)^2$$

$$A = 2464\ sq.m$$

Then the correct option is B)

4. Correct option: (A)

The formula for the area of a semicircle is:

$$A = \frac{\pi \times r^2}{2}$$

Given that we're using π as $\frac{22}{7}$ and the radius is 7 cm, we can calculate the area as follows:

$$A = \frac{\frac{22}{7} \times 7\ cm \times 7\ cm}{2}$$

$$A = (11) \times 7cm^2 = 77\ cm^2$$

Then the correct option is A)

5. Correct option: (C)

Let C be the circumference of the circle, and P be the perimeter of the square.

The formula for the circumference of a circle is C = 2πr, where r is the radius of the circle.

The formula for the perimeter of a square is P = 4s, where s is the length of one side of the square.

Given that C = P, we can set up an equation: $2\pi r = 4s$

Now, let's solve for r in terms of s: $r = \frac{4s}{2\pi}$

$$r = \frac{2s}{\pi}$$

Now, we can calculate the ratio of the areas of the circle and the square.

The area of a circle is given by $A_{circle} = \pi r^2$, and the area of a square is given by $A_{square} = s^2$.

So, the ratio of their areas is:

$$\frac{A_{circle}}{A_{square}} = \frac{\pi r^2}{s^2} = \frac{\pi\left(\frac{(2s)}{\pi}\right)^2}{s^2} = \frac{\left(\frac{4s^2\pi}{\pi^2}\right)}{s^2} = \frac{4}{\pi}$$

Then the correct option is C)

6. Correct option: (D)

To find the length of the arc of a sector, we can use the following formula:

$$Arc\ Length = \left(\frac{\theta}{360}\right) \times 2\pi r$$

Where:

- θ is the angle of the sector in degrees (75° in this case).

Let's calculate it:

$$Arc\ Length = \left(\frac{75}{360}\right) \times 2 \times \left(\frac{22}{7}\right) \times 42\ cm$$

First, simplify the fraction:

$$Arc\ Length = \left(\frac{5}{24}\right) \times 2 \times \left(\frac{22}{7}\right) \times 42\ cm$$

Now, perform the calculations step by step:

$$Arc\ Length = \frac{5}{12} \times 134\ cm$$

$$Arc\ Length = 55\ cm$$

Then the correct option is D)

7. Correct option: (C)

To find the perimeter of a sector, we can use the following steps:

1. Calculate the angle (θ) of the sector using the formula for the area of a sector:

$$Area = \left(\frac{\theta}{360}\right) \times \pi \times r^2$$

Given that the area is 70 sq. cm and the radius is 7 cm:

$$70 = \left(\frac{\theta}{360}\right) \times \pi \times (7\ cm)^2$$

Solve for $\frac{\theta}{360}$:

$$\frac{\theta}{360} = \frac{70}{\pi \times 49} = \frac{70}{\frac{22 \times 49}{7}} = \frac{10}{22}$$

2. Once we have the angle, we can find the perimeter of the sector:

$$Perimeter = \left(\frac{\theta}{360}\right) \times 2\pi r + 2r$$

Plug in the values:

$$Perimeter = \left(\frac{10}{22}\right) \times 2\pi \times 7\,cm + 2 \times 7\,cm$$
$$Perimeter = \frac{10}{22} \times 2 \times \frac{22}{7} \times 7 + 14\,cm$$
$$Perimeter = 20\,cm + 14\,cm$$
$$Perimeter = 34\,cm$$

Then the correct option is C)

8. Correct option: (D)

The area of a ring (annulus) can be calculated using the formula:

$$Area = \pi \times (R^2 - r^2)$$

Where:

- R is the outer radius
- r is the inner radius

In this case, the outer radius is "a + b" and the inner radius is "a - b."

So, we can plug these values into the formula:

$$Area = \pi \times ((a + b)^2 - (a - b)^2)$$

Now, let's simplify the expression inside the parentheses:

$$(a + b)^2 - (a - b)^2 = (a + b + a - b) \times \big(a + b - (a - b)\big) = 2a \times 2b = 4ab$$

Now, subtract the second expression from the first:

$$Area = \pi \times 4ab$$

Then the correct option is D)

9. Correct option: (C)

To find the length of the bounding arc of a sector, we can use the formula:

$$Arc\ Length = \left(\frac{\theta}{360}\right) \times 2\pi r$$

Where:

- θ is the angle of the sector in degrees.

Given that the radius is 42 m, we still need to find the angle θ. We can use the formula for the area of a sector to find θ:

$$Area\ of\ Sector = \left(\frac{\theta}{360}\right) \times \pi \times r^2$$

Given that the area of the sector is 924 sq. cm, we need to convert this to square meters (since the radius is in meters):

$$Area\ of\ Sector = \left(\frac{\theta}{360}\right) \times \pi \times (42\,m)^2$$
$$924\,sq.cm = (\theta/360) \times \pi \times (42\,m)^2$$

Now, solve for θ:

$$\frac{\theta}{360} = \frac{924}{\pi \times 42^2}$$

$$\frac{\theta}{360} = \left(\frac{924}{\pi \times 42^2}\right) = \frac{924}{\frac{22}{7} \times 42 \times 42} = \frac{22 \times 42}{22 \times 6 \times 42} = \frac{1}{6}$$

Now that we have the angle $\frac{\theta}{360}$, we can calculate the arc length:

$$Arc\ Length = \frac{1}{6} \times 2\pi \times 42\,m = \frac{1}{6} \times 2 \times \frac{22}{7} \times 42\,m = 44\,m$$
$$Arc\ Length = 44\,m$$

Then the perimeter of the sector

$$P = Arc\ Length + 2 \times radius = 44 + 84 = 128\,cm$$

Then the correct option is C)

10. Correct option: (C)

The formula for the circumference of a circle is:

$$Circumference = 2\pi r$$

Where C is the circumference, π (pi) is approximately $22/7$, and r is the radius.

Given the radius is 56 cm, we can calculate the circumference as follows:

$$Circumference = 2 \times \left(\frac{22}{7}\right) \times 56\,cm$$

$$Circumference = 2 \times \left(\frac{22}{7}\right) \times 56\,cm$$

$$Circumference = \left(\frac{44}{7}\right) \times 56\,cm$$

$$Circumference = 352\,cm$$

So, the circumference of the circle is 352 cm.

Then the correct option is C)

11. Correct option: (B, D)

To find the perimeter of a semicircle, we can calculate the sum of the curved boundary (arc length) and the diameter of the semicircle.

Let's first calculate the diameter (D) of the semicircle. The area of the semicircle is given as 1925 sq. cm, and we can use the formula for the area of a semicircle to find the diameter:

$$Area\ of\ Semicircle = \left(\frac{\pi}{2}\right) \times r^2$$

Where r is the radius of the semicircle

$$1925\,sq.cm = (\pi/2) \times r^2$$

$$r^2 = \frac{1925\,sq.cm \times 2}{\pi}$$

$$r^2 = 1225\,cm^2$$

$$r = \sqrt{1225}cm$$

$$r = 35\,cm$$

Now that we have the radius, we can calculate the perimeter of the semicircle:

$$Perimeter = Arc\ Length + Diameter$$

$$Arc\ Length = \left(\frac{1}{2}\right) \times 2\pi r$$

$$Arc\ Length = \pi r$$

$$Arc\ Length = \pi \times 35\,cm$$

Now, calculate the perimeter:

$$Perimeter = \pi \times 35\,cm + 2 \times 35\,cm$$

$$Perimeter = 35 \times \frac{22}{7}\,cm + 70\,cm = 180\,cm$$

So, the perimeter of the semicircle is 180 cm.

Then the correct option is both B) and D)

12. Correct option: (A, B)

To find the length of the bounding arc of a sector, we can use the formula:

$$Arc\ Length = \left(\frac{\theta}{360}\right) \times 2\pi r$$

Where:

- θ is the angle of the sector in degrees.

In this case, the radius is 7 cm, and the angle of the sector is 240°. Plug these values into the formula:

$$Arc\ Length = \left(\frac{240}{360}\right) \times 2\pi \times 7\,cm$$

$$Arc\ Length\ = \left(\frac{2}{3}\right) \times 2 \times \left(\frac{22}{7}\right) \times 7\ cm$$

Now, simplify:

$$Arc\ Length\ = \left(\frac{2}{3}\right) \times 44\ cm$$

$$Arc\ Length = \frac{88}{3}\ cm$$

$$Arc\ Length = 29\frac{1}{3}\ cm$$

Then the correct option is both A) and B)

13. Correct option: (B)

To find the cost of fencing a semicircular garden of radius 7 meters at Rs. 10 per meter, we can follow these steps:

1. Calculate the perimeter of the semicircular garden.
2. Multiply the perimeter by the cost per meter to find the total cost.

1. Perimeter of the semicircular garden:

The perimeter consists of the curved boundary (arc length) and the straight boundary (diameter).

$$Arc\ Length\ = \left(\frac{\theta}{360}\right) \times 2\pi r$$

$$Arc\ Length\ = \left(\frac{180}{360}\right) \times 2\pi \times 7\ m = \frac{22}{7} \times 7\ cm = 22\ cm$$

$$Diameter\ = 2r\ = 2 \times 7\ m\ = 14\ m$$

$$Total\ Perimeter\ = Arc\ Length\ + Diameter$$

$$Total\ Perimeter\ = 22\ m\ + 14\ m = 36\ m$$

2. Calculate the total cost:

$$Total\ Cost\ = Total\ Perimeter\ \times Cost\ per\ Meter$$

$$Total\ Cost\ = 36\ meters \times Rs.\frac{10}{meter} = Rs.360$$

So, the cost of fencing the semicircular garden is Rs. 360.

Then the correct option is B)

14. Correct option: (A)

To find the ratio of the circumference of the circle to the length of the arc of the sector, we can use the formulas for both:

1. Circumference of the Circle:

$$Circumference\ = 2\pi r$$

$$Circumference\ = 2\pi\ \times 10\ m\ = 20\pi\ m$$

2. Length of the Arc of the Sector:

$$Length\ of\ Arc\ = (\theta/360)\ *\ 2\pi r$$

$$Length\ of\ Arc\ = \left(\frac{72}{360}\right) \times 2\pi \times 10\ m\ = \left(\frac{1}{5}\right) \times 20\pi\ m\ = 4\pi\ m$$

Now, let's find the ratio:

$$Ratio\ = \frac{20\pi\ m}{4\pi\ m} = 5:1$$

So, the ratio of the circumference of the circle to the length of the arc of the sector is: 5 : 1

Then the correct option is A)

15. Correct Matching is: a-R, b-S, c-P, d-Q

a) Area of sector **(R)** = $\left(\frac{x}{360}\right) \times \pi r^2$

Explanation: The area of a sector is calculated as a fraction of the total circle's area based on the central angle 'x' out of 360 degrees.

b) Length of an arc of a sector **(S)** = $\left(\frac{x}{360}\right) \times 2\pi r$

Explanation: The length of an arc of a sector is a fraction of the total circumference of the circle based on the central angle 'x' out of 360 degrees.

c) Perimeter of a sector **(P)** = $l + 2r$

Explanation: The perimeter of a sector is the sum of the length of the arc (l) and twice the radius (2r).

d) If 'l' is the length of arc, then area is **(Q)** = $\left(\frac{lr}{2}\right)$

Explanation: The area of a sector can be found by multiplying the length of the arc (l) by half the radius (r) of the sector.

16. Correct Answer is: (2)

To find $\frac{1}{72}$ times the perimeter of a semicircle with a radius of 28 cm, first calculate the perimeter of the semicircle and then multiply it by $\frac{1}{72}$.

$$Perimeter\ of\ a\ semicircle\ = (\pi + 2)r$$

$$Perimeter\ of\ the\ semicircle\ = (\frac{22}{7} + 2) \times 28\ cm$$

Now, calculate the perimeter:

$$Perimeter\ = (88 + 56) = 144\ cm$$

Now, multiply it by $\frac{1}{72}$:

$$\frac{1}{72} \times 144 = 2$$

So, $\frac{1}{72}$ times the perimeter of a semicircle with a radius of 28 cm is 2.

17. Correct Answer is: (3)

To find the number of times a wheel of radius 35 cm will be rotated to travel 660 cm, we can use the formula:

$$Number\ of\ rotations\ = \frac{Distance\ traveled}{Circumference\ of\ the\ wheel}$$

First, calculate the circumference of the wheel:

$$Circumference\ = 2\pi r$$

$$Circumference\ = 2\pi \times 35\ cm = 220\ cm$$

Now, calculate the number of rotations:

$$Number\ of\ rotations\ = \frac{660\ cm}{220\ cm} = 3$$

So, the wheel will be rotated 3 times to travel 660 cm.

18. Correct Answer is: (6)

To find the value of "k," which represents the area of the path, we can start by subtracting the area of the circular lawn from the total area of the circular lawn and the path combined.

$$Area\ of\ the\ circular\ lawn\ = 2464\ sq.m$$

$$Width\ of\ the\ path\ = 7\ m$$

$$Total\ area\ (lawn + path)\ = Area\ of\ the\ circular\ lawn + Area\ of\ the\ path$$

$$Total\ area\ = 2464\ sq.m + Area\ of\ the\ path$$

We are given that the area of circular lawn is 2464 sq. m.

$$\pi r^2 = 2464 => r^2 = 2464 \times \frac{7}{22} = 784 => r = 28m$$

We are given that width of new area is 7m.

So total radius of circular path $7m + 28m = 35m$

Then the area of the path can be calculated as

$$=> \pi(R^2 - r^2) = \pi(35^2 - 28^2) => \pi(7)(63) = \frac{22}{7} \times 7 \times 63 = 22 \times 63 = 1386\ sq.m$$

Comparing the values

$$138k = 1386 => k = 6$$

19. Correct Answer is: (8)

To find the perimeter of a sector, we need to consider both the arc length (the curved part) and the two radii (the straight parts).

Given:

$$Area\ of\ the\ sector\ =\ 924\ sq.cm$$
$$Radius\ of\ the\ sector\ =\ 42\ cm$$

First, calculate the angle of the sector using the formula for the area of a sector:

$$Area\ of\ Sector\ =\ \left(\frac{\theta}{360}\right)\times\ \pi r^2$$
$$924\ =\ \left(\frac{\theta}{360}\right)\ \times\ \pi(42^2) => \left(\frac{\theta}{360}\right) = \frac{924}{\frac{22\times 42\times 42}{7}} = \frac{924}{22\times 6\times 42} = \frac{7}{42} = \frac{1}{6}$$

Now, calculate the arc length (the curved part of the perimeter) using the angle θ:

$$Arc\ Length\ =\ \left(\frac{\theta}{360}\right)\times\ 2\pi r$$
$$Arc\ Length = \frac{1}{6}\times 2\pi \times 42 = 14\pi = 44\ cm$$

Now, calculate the perimeter of the sector:

$$Perimeter\ =\ 44\ cm\ +\ 2\ \times 42\ cm = 128\ cm$$

Comparing the result

$$120 + k = 128 => k = 8$$

20. Correct Answer is: (0.5)

To find the radius of a circle when given its circumference, we can use the formula for the circumference of a circle:

$$Circumference\ =\ 2\pi r$$

Given that the circumference is 264 cm:

$$264\ =\ 2\pi r$$

Now, isolate the radius (r):

$$r\ = \frac{264}{2\pi} = \frac{264\times 7}{2\times 22} = 42$$

Comparing the results

$$\frac{21}{k} = 42 => k = \frac{1}{2} = 0.5$$

SURFACE AREAS AND VOLUMES
LEVEL-1

1. Correct option: (C)

To calculate the volume of a cuboid, we can use the formula:

$$Volume\ =\ Length\ \times\ Breadth\ \times\ Height$$

In this case, the length is 15 meters, the breadth is 12 meters, and the height is 10 meters.

$$Volume\ =\ 15\ m\ \times\ 12\ m\ \times\ 10\ m$$
$$Volume\ =\ 1{,}800\ cubic\ meters$$

So, the volume of the cuboid is 1,800 cubic meters.

Then the correct option is C)

2. Correct option: (C)

$$Volume\ =\ Length\ \times\ Breadth\ \times\ Height$$

In this case:

$$Length\ =\ (1\ +\ x)$$

$$Breadth = (1 - x)$$
$$Height = (1 + x^2)$$

Now, plug these values into the formula:

$$Volume = (1 + x) \times (1 - x) \times (1 + x^2)$$

Simplify this expression further:

$$Volume = (1 + x) \times (1 - x) \times (1 + x^2)$$
$$Volume = (1 - x\text{^}2) \times (1 + x^2)$$

Now, apply the difference of squares formula: $a^2 - b^2 = (a + b) \times (a - b)$

$$Volume = (1^2 - x^2) \times (1 + x^2)$$

$$Volume = (1 - x^2) \times (1 + x^2)$$

Now, apply the difference of squares formula again:

$$Volume = (1 - x^4)$$

Then the correct option is C)

3. Correct option: (D)

To calculate the total surface area of a cuboid, we can use the formula:

$$Total\ Surface\ Area = 2(lw + lh + bh)$$

Given the dimensions:

Length (l) = 32 cm

Breadth (b) = 25 cm

Height (h) = 15 cm

Plug these values into the formula:

$$Total\ Surface\ Area = 2(32\ cm \times 25\ cm + 32\ cm \times 15\ cm + 25\ cm \times 15\ cm)$$
$$Total\ Surface\ Area = 2(800\ cm^2 + 480\ cm^2 + 375\ cm^2)$$
$$Total\ Surface\ Area = 2(1655\ cm^2)$$
$$Total\ Surface\ Area = 3310\ cm^2$$

So, the total surface area of the cuboid is 3310 square centimeters.

Then the correct option is D)

4. Correct option: (B)

To find the height of the cuboid when we know its volume, length, and breadth, we can use the formula for the volume of a cuboid:

$$Volume = Length \times Breadth \times Height$$

Given that the volume is 720 cubic centimeters, the length is 12 cm, and the breadth is 10 cm. We can plug these values into the formula and solve for the height:

$$720\ cu.cm = 12\ cm \times 10\ cm \times Height$$

Now, solve for Height:

$$Height = \frac{720\ cu.cm}{12\ cm \times 10\ cm}$$
$$Height = \frac{720\ cu.cm}{120\ cm^2}$$
$$Height = 6\ cm$$

So, the height of the cuboid is 6 cm.

Then the correct option is B)

5. Correct option: (D)

Given the lateral surface area is 520 square cm, the length is 14 cm, and the breadth is 12 cm, we can calculate the height as follows:

$$Lateral\ Surface\ Area = 2 \times (Length \times Height + Breadth \times Height)$$
$$520\ sq.cm = 2 \times (14\ cm \times Height + 12\ cm \times Height)$$
$$520\ sq.cm = 2 \times (26\ cm \times Height)$$

$$260\ sq.cm\ =\ 26\ cm\ \times\ Height$$
$$Height\ =\frac{260\ sq.cm}{26\ cm}$$
$$Height\ =\ 10\ cm$$

Now that we know the height is 10 cm, we can calculate the volume of the cuboid:

$$Volume\ =\ 14\ cm\ \times\ 12\ cm\ \times\ 10\ cm$$
$$Volume\ =\ 1680\ cubic\ cm$$

So, the volume of the cuboid is 1680 cubic centimeters.
Then the correct option is D)

6. Correct option: (A)

Let's call the lengths of the two cuboids L and l, their breadths B and b, and their heights H and h.
Given that the volumes of the two cuboids are the same, we can write the following equation: $Volume_1 = Volume_2$
Now, the volume of a cuboid is given by the formula:

$$Volume\ =\ Length\ \times\ Breadth\ \times\ Height$$
$$L\ \times\ B\ \times\ H\ =\ l\times b\times h$$

Now, we know that the lengths are in the ratio 5:4, which means: $\frac{L}{l}=\frac{5}{4}$
And we also know that the breadths are in the ratio 2:3, which means: $\frac{B}{b}=\frac{2}{3}$
We can write L and l in terms of a common variable, say x:

$$L\ =\ 5x$$
$$l\ =\ 4x$$

And we can write B and b in terms of a common variable, say y:

$$B\ =\ 2y$$
$$b=\ 3y$$

Now, we can substitute these expressions into the equation for volume:

$$(5x)\ \times\ (2y)\ \times\ H\ =\ (4x)\ \times\ (3y)\ \times h$$
$$10xyH=\ 12xyh$$
$$10H\ =\ 12h$$
$$\frac{H}{h}=\frac{12}{10}=\frac{6}{5}$$

So, the ratio of the heights of the two cuboids is 6:5.
Then the correct option is A)

7. Correct option: (B)

To find the height of the cuboid, we can use the formula for the volume of a cuboid:

$$Volume\ =\ Area\ of\ Base\ \times\ Height$$

Given that the volume is 440 cm³ and the area of its base is 88 sq.cm, we can plug these values into the formula:

$$440\ cm^3\ =\ 88\ sq.cm\ \times\ Height$$
$$Height\ =\frac{440\ cm^3}{88\ sq.cm}$$
$$Height\ =\ 5\ cm$$

So, the height of the cuboid is 5 cm.
Then the correct option is B)

8. Correct option: (A)

The lateral surface area of a cube can be calculated using the formula:

$$Lateral\ Surface\ Area\ =\ 4\ \times\ (Side\ Length)^2$$

In this case, the side length of the cube is 5 cm.

$$Lateral\ Surface\ Area\ =\ 4\ \times\ (5\ cm)^2$$

$$Lateral\ Surface\ Area\ =\ 4\ \times\ 25\ square\ cm$$
$$Lateral\ Surface\ Area\ =\ 100\ square\ cm$$

So, the lateral surface area of the cube is 100 square centimeters.
Then the correct option is A)

9. Correct option: (B)

For a cube, the total surface area includes the area of all six faces, and the lateral surface area includes the area of only the four side faces (excluding the top and bottom).
The formulae are as follows:

$$1.\ Total\ Surface\ Area\ (TSA)\ =\ 6\ \times\ (Side\ Length)^2$$
$$2.\ Lateral\ Surface\ Area\ (LSA)\ =\ 4\ \times\ (Side\ Length)^2$$

Now, let's calculate the ratio:

$$\frac{TSA}{LSA} = \frac{6\ \times\ (Side\ Length)^2}{4\ \times\ (Side\ Length)^2}$$
$$TSA/LSA\ =\ \left(\frac{6}{4}\right)\ \times\ \left(\frac{(Side\ Length)^2}{(Side\ Length)^2}\right)$$
$$\frac{TSA}{LSA}\ =\ \left(\frac{3}{2}\right)$$

So, the ratio of the total surface area to the lateral surface area of a cube is 3:2.
Then the correct option is B)

10. Correct option: (C)

1. Calculate the length of one of the cube's edges using the diagonal length.
2. Use the edge length to find the surface area.

Given that the length of the diagonal of the cube is $8\sqrt{3}$ cm, we can find the edge length (e) using the Pythagorean Theorem:

$$e^2\ +\ e^2\ +\ e^2\ =\ \left(8\sqrt{3}cm\right)^2$$
$$3e^2\ =\ 192\ cm^2$$
$$e^2\ =\ 64\ cm^2$$
$$e\ =\ \sqrt{64\ cm^2}$$
$$e\ =\ 8\ cm$$

Now that we have found the edge length (e), we can calculate the surface area of the cube using the formula: $Surface\ Area\ =\ 6\ \times\ (Edge\ Length)^2 = 6e^2$

$$Surface\ Area\ =\ 6\ \times\ (8\ cm)^2$$
$$Surface\ Area\ =\ 6\ \times\ 64\ cm^2$$
$$Surface\ Area\ =\ 384\ cm^2$$

So, the surface area of the cube is 384 square centimeters.
Then the correct option is C)

11. Correct option: (D)

1. Length of Diagonal (a):

The length of a diagonal of a cube can be found using the Pythagorean Theorem, which relates the diagonal length to the edge length (e) of the cube. The formula is:

$$a\ =\ \sqrt{e^2\ +\ e^2\ +\ e^2}\ =\ \sqrt{3e^2}\ =\ \sqrt{3}e$$

2. Volume of Cube (V):

The volume of a cube is calculated as:

$$V\ =\ e^3$$

Then

$$a\ =\ \sqrt{3}e\ =>\ e = \frac{a}{\sqrt{3}}$$

So volume of the cube

$$V = e^3 = \left(\frac{a}{\sqrt{3}}\right)^3 = \frac{a^3}{3\sqrt{3}} => 3\sqrt{3}V = a^3$$

Then the correct option is D)

12. Correct option: (A)

Given that the volume of the cube is 1,000 cm³, we can find the edge length (e) using the formula for the volume of a cube:

$$Volume = Edge\ Length^3$$
$$1{,}000\ cm^3 = e^3$$
$$e = \sqrt[3]{1{,}000\ cm^3}$$
$$e = 10\ cm$$

Now that we have found the edge length (e), we can calculate the total surface area of the cube using the formula:

$$Total\ Surface\ Area = 6 \times (Edge\ Length)^2$$
$$Total\ Surface\ Area = 6 \times (10\ cm)^2$$
$$Total\ Surface\ Area = 6 \times 100\ cm^2$$
$$Total\ Surface\ Area = 600\ cm^2$$

So, the total surface area of the cube is 600 square centimeters.

Then the correct option is A)

13. Correct option: (A)

Given that the area of one end face of the triangular prism is $64\sqrt{3}$ sq. cm and its length (height) is 60 cm, we can plug these values into the formula:

$$Volume = \left(64\sqrt{3} sq.cm\right) \times (60\ cm)$$
$$Volume = 3840\sqrt{3}\ cubic\ cm$$

So, the volume of the triangular prism is 3840√3 cubic centimeters.

Then the correct option is A)

14. Correct option: (A)

The formula for the area of the base of a regular hexagonal prism with side length "a" is:

$$Area\ of\ Base = \left(\frac{3\sqrt{3}}{2}\right) \times (a^2) = \left(\frac{6\sqrt{3}}{4}\right) \times (a^2)$$

Here, "a" is the length of one side of the hexagon.

Then the correct option is A)

15. Correct option: (A)

1. Area of Hexagonal Base:

The area of a regular hexagon can be calculated using the formula:

$$Area\ of\ Hexagon = \left(\frac{3\sqrt{3}}{2}\right) \times (Side\ Length)^2$$

In this case, the side length is 'a'.

2. Area of Rectangular Side:

Each rectangular side consists of a rectangle with a length equal to the side length of the hexagon ('a') and a height equal to the height of the prism ('h').

Now, we can calculate the surface area:

$$Surface\ Area = 2 \times \left[\left(\frac{3\sqrt{3}}{2}\right) \times (a^2)\right] + 6 \times (a \times h)$$

So, the surface area of the hexagonal prism is given by this formula.

Then the correct option is A)

16. Correct option: (C)

The formula for the lateral surface area (LSA) of a regular prism with an equilateral triangle as its base is:

$$LSA = Perimeter\ of\ Base \times Height$$

Given that the base is an equilateral triangle with a side length of 12 cm and the height of the prism is 25 cm, we can calculate the LSA as follows:

$$Perimeter\ of\ Base = 3 \times Side\ Length$$

$$Perimeter\ of\ Base = 3 \times 12\ cm = 36\ cm$$

Now, use the LSA formula: $LSA = 36\ cm \times 25\ cm = 900\ square\ cm$

So, the lateral surface area of the prism is 900 square centimeters.

Then the correct option is C)

17. Correct option: (A)

To calculate the lateral surface area (LSA) of a triangular prism, you can use the formula:

$$LSA = Perimeter\ of\ the\ Base \times Height$$

In this case, the triangular prism has sides of 15 cm, 20 cm, and 25 cm. To find the perimeter of the base, which is a triangle, we simply add the lengths of the three sides:

$$Perimeter\ of\ the\ Base = 15\ cm + 20\ cm + 25\ cm = 60\ cm$$

Given that the height of the triangular prism is 40 cm, we can now calculate the LSA:

$$LSA = Perimeter\ of\ the\ Base \times Height = 60\ cm \times 40\ cm = 2400\ square\ cm$$

So, the lateral surface area of the triangular prism is 2400 square centimeters.

Then the correct option is A)

18. Correct option: (B)

To calculate the volume of a prism, we can use the formula:

$$Volume = (Area\ of\ Base) \times Height$$

In this case, the base of the prism is an equilateral triangle with a side length of 8 cm and a height of 12 cm. To find the area of the equilateral triangle base, we can use the formula for the area of an equilateral triangle:

$$Area\ of\ Base = \left(\frac{\sqrt{3}}{4}\right) \times (Side\ Length)^2 = \left(\frac{\sqrt{3}}{4}\right) \times (8\ cm)^2$$

$$Area\ of\ Base = 16\sqrt{3}\ cm^2$$

Now, we can calculate the volume of the prism:

$$Volume = (Area\ of\ Base) \times Height$$

$$Volume = \left(16\sqrt{3} cm^2\right) \times 12\ cm$$

$$Volume = 192\sqrt{3}\ cubic\ cm$$

So, the volume of the prism is 192√3 cubic centimeters.

Then the correct option is B)

19. Correct option: (A)

To calculate the volume of a regular hexagonal prism, you can use the formula:

$$Volume = (Area\ of\ Base) \times Height$$

In this case, the base of the prism is a regular hexagon with side length 30 cm, and the height of the prism is 6 cm.

To find the area of the regular hexagonal base, we can use the formula for the area of a regular hexagon:

$$Area\ of\ Base = \left(\frac{3\sqrt{3}}{2}\right) \times (Side\ Length)^2$$

$$Area\ of\ Base = \left(\frac{3\sqrt{3}}{2}\right) \times (30\ cm)^2$$

$$Area\ of\ Base = \left(\frac{3\sqrt{3}}{2}\right) \times 900\ cm^2$$

$$Area\ of\ Base = 1350\sqrt{3} cm^2$$

Now, we can calculate the volume of the prism:

$$Volume = (Area\ of\ Base) \times Height$$
$$Volume = \left(1350\sqrt{3}cm^2\right) \times 6\ cm$$
$$Volume = 8100\sqrt{3}cubic\ cm \approx 14029.2\ cm^3$$

So, the volume of the regular hexagonal prism is $14029.2\ cm^3$
Then the correct option is A)

20. Correct option: (C)

To calculate the volume of the prism, we can use the formula:

$$Volume = (Area\ of\ Base) \times Height$$

In this case, the base of the prism is a rhombus with diagonals of 28 cm and 96 cm, and the height of the prism is 32 cm.
To find the area of the rhombus base, we can use the formula for the area of a rhombus:

$$Area\ of\ Base = \left(\frac{1}{2}\right) \times (Product\ of\ Diagonals)$$
$$Area\ of\ Base = \left(\frac{1}{2}\right) \times (28\ cm \times 96\ cm)$$
$$Area\ of\ Base = \left(\frac{1}{2}\right) \times 2688\ cm^2$$
$$Area\ of\ Base = 1344\ cm^2$$

Now, we can calculate the volume of the prism:

$$Volume = (Area\ of\ Base) \times Height$$
$$Volume = 1344\ cm^2 \times 32\ cm$$
$$Volume = 43{,}008\ cubic\ cm$$

So, the volume of the prism is 43,008 cubic centimeters.
Then the correct option is C)

LEVEL - II

1. Correct option: (A, D)

The formula for the lateral surface area (LSA) of a rectangular cuboid is given by:

$$LSA = 2 \times (Length \times Height + Width \times Height)$$

In this case, the dimensions of the cuboid are:

$$Length = 12\ cm$$
$$Width = 8\ cm$$
$$Height = 5\ cm$$

Now, plug these values into the formula:

$$LSA = 2 \times (12\ cm \times 5\ cm + 8\ cm \times 5\ cm)$$
$$LSA = 2 \times (60\ cm^2 + 40\ cm^2)$$
$$LSA = 2 \times 100\ cm^2$$
$$LSA = 200\ cm^2 = 2(10^2)sq.cm$$

So, the lateral surface area of the cuboid is 200 square centimeters.
Then the correct option is A, D)

2. Correct option: (D)

We can find the diagonal of a cuboid using the Pythagorean theorem. The diagonal (d) can be calculated using the formula:

$$d = \sqrt{Length^2 + Breadth^2 + Height^2}$$

In this case, the dimensions of the cuboid are:

$$Length = 16\ cm$$
$$Breadth = 8\ cm$$
$$Height = 6\ cm$$

Now, plug these values into the formula:

$$d = \sqrt{[(16\,cm)^2 + (8\,cm)^2 + (6\,cm)^2]}$$
$$d = \sqrt{256\,cm^2 + 64\,cm^2 + 36\,cm^2}$$
$$d = \sqrt{356\,cm^2}$$
$$d = \sqrt{356}cm$$

Then the correct option is D)

3. Correct option: (A, D)

When we double each side of a cube, we are essentially scaling it up uniformly.
To find how the volume changes, we can use the scaling factor of 2.
The volume of a cube is given by the formula:

$$Volume = (Side\ Length)^3$$

If we double the side length, the new side length becomes 2 times the original side length.
Therefore, the new volume will be:

$$New\ Volume = (2 \times Side\ Length)^3 = 2^3 \times (Side\ Length)^3 = 8 \times (Side\ Length)^3$$

So, when each side of a cube is doubled, its volume becomes 8 times the original volume.
Then the correct option is A, D)

4. Correct option: (A)

To calculate the lateral surface area (LSA) of a triangular prism, we can use the formula:

$$LSA = Perimeter\ of\ the\ Base \times Height$$

In this case, the triangular prism has sides of 3 cm, 4 cm, and 5 cm, and its height is 4 cm.
First, calculate the perimeter of the triangular base:

$$Perimeter\ of\ the\ Base = 3\,cm + 4\,cm + 5\,cm = 12\,cm$$

Now, use the LSA formula:

$$LSA = Perimeter\ of\ the\ Base \times Height = 12\,cm \times 4\,cm = 48\,square\ cm$$

So, the lateral surface area of the triangular prism is 48 square centimeters.
Then the correct option is A)

5. Correct option: (B)

To find the total surface area (TSA) of a triangular prism, we need to calculate the lateral surface area (LSA) and add it to the area of the two triangular bases.
1. Calculate the Lateral Surface Area (LSA):

$$LSA = Perimeter\ of\ the\ Base \times Height$$
$$Perimeter\ of\ the\ Base = 3\,cm + 4\,cm + 5\,cm = 12\,cm$$
$$LSA = 12\,cm \times 4\,cm = 48\,square\ cm$$

2. Calculate the Area of One Triangular Base:
The triangular base is a right-angled triangle with legs of 3 cm and 4 cm. You can use the formula for the area of a right-angled triangle:

$$Area\ of\ One\ Base = \left(\frac{1}{2}\right) \times Base \times Height = \left(\frac{1}{2}\right) \times 3\,cm \times 4\,cm = 6\,square\ cm$$

3. Calculate the Total Surface Area (TSA):

$$TSA = 2 \times Area\ of\ One\ Base + LSA$$
$$TSA = 2 \times 6\,square\ cm + 48\,square\ cm$$
$$= 12\,square\ cm + 48\,square\ cm = 60\,square\ cm$$

So, the total surface area of the triangular prism is 60 square centimeters.
Then the correct option is B)

6. Correct option: (A)

To find the volume of a triangular prism, we can use the formula:

$$Volume = (Area\ of\ Base) \times Height$$

In this case, the triangular prism has a triangular base with sides of 3 cm, 4 cm, and 5 cm, and its height is 4 cm.
First, calculate the area of the triangular base. We can use Heron's formula for the area of a triangle when we know the side lengths:

$$Semi_perimeter\ (s) = \frac{3\ cm\ +\ 4\ cm\ +\ 5\ cm}{2} = 6\ cm$$

$$Area\ of\ Base\ = \sqrt{[s\ \times\ (s\ -\ side1) \times\ (s\ -\ side2) \times\ (s\ -\ side3)]}$$

$$Area\ of\ Base\ = \sqrt{[6\ cm\ \times\ (6\ cm\ -\ 3\ cm) \times\ (6\ cm\ -\ 4\ cm) \times\ (6\ cm\ -\ 5\ cm)]}$$

$$Area\ of\ Base\ = \sqrt{[6\ cm\ \times\ 3\ cm\ \times\ 2\ cm\ \times\ 1\ cm]} = \sqrt{36\ cm^4} = 6\ cm^2$$

Now, we can calculate the volume of the triangular prism:

$$Volume\ = (Area\ of\ Base)\ \times\ Height\ = 6\ cm^2\ \times\ 4\ cm\ = 24\ cubic\ cm$$

So, the volume of the triangular prism is 24 cubic centimeters.
Then the correct option is A)

7. Correct option: (A)

To find the side length of a cube when you know its lateral surface area, we can use the formula for the lateral surface area (LSA) of a cube:

$$LSA\ = 4\ \times\ (Side\ Length)^2$$

Given that the LSA is 576 cm², we can set up the equation:

$$576\ cm^2\ = 4\ \times\ (Side\ Length)^2$$

Now, solve for the side length (S):

$$(Side\ Length)^2 = \frac{576\ cm^2}{4}$$

$$(Side\ Length)^2 = 144\ cm^2$$

$$Side\ Length\ = \sqrt{144\ cm^2}$$

$$Side\ Length\ = 12\ cm$$

So, the side length of the cube is 12 centimeters.
Then the correct option is A)

8. Correct option: (D)

To find the side length of a cube when you know its surface area, we can use the formula for the surface area (SA) of a cube:

$$SA\ = 6\ \times\ (Side\ Length)^2$$

Given that the surface area is 600 cm², we can set up the equation:

$$600\ cm^2\ = 6\ \times\ (Side\ Length)^2$$

Now, solve for the side length (S):

$$(Side\ Length)^2 = \frac{600\ cm^2}{6}$$

$$(Side\ Length)^2 = 100\ cm^2$$

$$Side\ Length\ = \sqrt{100\ cm^2}$$

$$Side\ Length\ = 10\ cm$$

So, the side length of the cube is 10 centimeters.
Then the correct option is D)

9. Correct option: (C)

To find the total surface area (TSA) of the cuboid formed by a rectangle with dimensions length (l), breadth (b), and height (h), given that the area of the rectangle is 96 cm², we can use the formula for TSA:

$$TSA\ = 2(lb\ +\ bh\ +\ lh)$$

First, we have $lb\ = 96\ cm^2$.
Then

$$TSA = 2(lb + bh + lh) = 2(lb + 5b + 5l)$$

Since $lb = 96\,cm^2$:

$$TSA = 2(96 + 5b + 5l)$$
$$TSA = 2(96 + 5b + 5l)$$
$$TSA = 2(96 + 5(b + l))$$
$$TSA = 2\left(96 + 5\left(l + \frac{l}{96}\right)\right)$$

10. Correct option: (B)

To find the volume of a cuboid, we can use the formula:

$$Volume = Length \times Breadth \times Height$$

In this case, the dimensions of the cuboid are:

$$Length = 22\,cm$$
$$Breadth = 12\,cm$$
$$Height = 7.5\,cm$$

Now, plug these values into the formula:

$$Volume = 22\,cm \times 12\,cm \times 7.5\,cm$$
$$Volume = 1980\,cubic\,cm$$

So, the volume of the cuboid is 1980 cubic centimeters.

Then the correct option is B)

11. Correct option: (C, D)

Given that the length, width, and height of a rectangular solid are in the ratio 3:2:1, let's represent these dimensions as follows:

$$Length = 3x$$
$$Width = 2x$$
$$Height = x$$

The volume of the box is given as 48 cm³, and we can find it using the formula:

$$Volume = Length \times Width \times Height$$
$$48\,cm^3 = (3x) \times (2x) \times (x)$$
$$48\,cm^3 = 6x^3$$
$$x^3 = \frac{48\,cm^3}{6}$$
$$x^3 = 8\,cm^3 => x = 2\,cm$$

Then

$$Length = 3x = 3 \times 2\,cm = 6\,cm$$
$$Width = 2x = 2 \times 2\,cm = 4\,cm$$
$$Height = x = 2\,cm$$

Now, calculate the total surface area (TSA) using the formula for a rectangular solid:

$$TSA = 2 \times (Length \times Width + Width \times Height + Height \times Length)$$
$$TSA = 2 \times (6\,cm \times 4\,cm + 4\,cm \times 2\,cm + 2\,cm \times 6\,cm)$$
$$TSA = 2 \times (24\,cm^2 + 8\,cm^2 + 12\,cm^2)$$
$$TSA = 2 \times 44\,cm^2$$
$$TSA = 88\,cm^2$$

So, the total surface area of the box is 88 square centimeters.

Then the correct option is C, D)

12. Correct option: (A, D)

The ratio of the surface area (A) to the volume (V) of a cuboid can be expressed as A/V. In general, for a cuboid with dimensions x, y, and z, the surface area (A) and volume (V) are given by:

$$Surface\ Area\ (A) = 2(xy + yz + xz)$$

$$Volume\,(V) = xyz$$

So, A/V can be calculated as:

$$\frac{A}{V} = \frac{[2(xy + yz + xz)]}{xyz} = 2\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$$

This ratio represents how much surface area there is relative to the volume of the cuboid. It is a measure of how "spread out" or "compact" the shape is.

Then the correct option is A, D)

13. Correct option: (A, C)

If the volumes of two cubes are in the ratio of 8:1, then the ratio of their edges (side lengths) can be found by taking the cube root of the volume ratio.

The volume of a cube is proportional to the cube of its edge length ($V = side^3$).

So, if the volume ratio is 8:1, it means the cube of the edge ratio is also 8:1.

Taking the cube root of 8 gives:

$$\sqrt[3]{8}: 1 = 2: 1$$

So, the ratio of their edges is 2:1.

Then the correct option is A, C)

14. Correct option: (B)

To find the diagonal of a cuboid with length (L), breadth (B), and height (H), we can use the following formula: $Diagonal = \sqrt{L^2 + B^2 + H^2}$

In this case, the length is 5 meters, the breadth is 4 meters, and the height is 3 meters. Plug these values into the formula:

$$Diagonal = \sqrt{5^2 + 4^2 + 3^2}$$
$$Diagonal = \sqrt{25 + 16 + 9}$$
$$Diagonal = \sqrt{50}$$

So, the diagonal of the cuboid is $\sqrt{50}$ meters.

Then the correct option is B)

15. Correct option: (C)

The lateral surface area (LSA) of a cuboid can be calculated using the formula:

$$LSA = 2 \times (Length \times Height + Breadth \times Height)$$

In this case, the length (L) is 5 meters, the breadth (B) is 4 meters, and the height (H) is 3 meters. Plug these values into the formula:

$$LSA = 2 \times (5\,meters \times 3\,meters + 4\,meters \times 3\,meters)$$
$$LSA = 2 \times (15\,square\,meters + 12\,square\,meters)$$
$$LSA = 2 \times 27\,square\,meters$$
$$LSA = 54\,square\,meters$$

So, the lateral surface area of the cuboid is 54 square meters.

Then the correct option is C)

16. Correct option: (D)

A cuboid has 12 edges, and each edge consists of one length, one breadth, and one height. Given the dimensions of the cuboid:

$$Length\,(L) = 5\,meters$$
$$Breadth\,(B) = 4\,meters$$
$$Height\,(H) = 3\,meters$$

To find the lengths of all 12 edges, we can calculate the following:

1. Length of 4 vertical edges:

$$4 \times L = 4 \times 5\,meters = 20\,meters$$

2. Length of 4 horizontal edges along the width:

$$4 \times B = 4 \times 4\,meters = 16\,meters$$

3. Length of 4 horizontal edges along the height:

$$4 \times H = 4 \times 3\,meters = 12\,meters$$

Then 20 meters + 16 meters + 12 meters = 48 meters

So, the total length of all 12 edges of the cuboid is 48 meters.

Then the correct option is D)

17. Correct Matching is: a-S, b-T, c-Q, d-P

a) Diagonal of the cuboid with dimensions length (l), breadth (b), and height (h) is given by:-→(S)

$$Diagonal = \sqrt{l^2 + b^2 + h^2}$$

b) The total surface area (TSA) of the cuboid with dimensions length (l), breadth (b), and height (h) is given by:-→(T)

$$TSA = 2(lb + bh + hl)$$

c) Diagonal of a cube with edge length (a) units is given by:-→(Q)

$$Diagonal = \sqrt{(a^2 + a^2 + a^2)} = \sqrt{(3a^2)} = a\sqrt{3}$$

d) The length of all 12 edges of a cube with edge length (a) units is calculated as follows:-→(P)

$$Length\ of\ 4\ vertical\ edges: 4 \times a = 4a$$
$$Length\ of\ 4\ horizontal\ edges\ along\ the\ width: 4 \times a = 4a$$
$$Length\ of\ 4\ horizontal\ edges\ along\ the\ height: 4 \times a = 4a$$
$$Total\ length\ of\ all\ 12\ edges = 4a + 4a + 4a = 12a$$

18. Correct Answer is: (5)

To find the height (h) of the cuboid when we know its base area (B) and volume (V), we can use the formula: $Volume\ (V) = Base\ Area\ (B) \times Height\ (h)$

Given: Base Area (B) = 180 cm²

Volume (V) = 900 cm³

Now, plug in the values and solve for the height (h):

$$900\ cm^3 = 180\ cm^2 \times h$$
$$h = \frac{900\ cm^3}{180\ cm^2}$$
$$h = 5\ cm$$

So, the height of the cuboid is 5 centimeters.

19. Correct Answer is: (4)

To find the side length of a cube when you know its total surface area (TSA), we can use the formula: TSA = 6 × (Side Length)^2

Given that the total surface area is 96 cm², we can set up the equation:

$$96\ cm^2 = 6 \times (Side\ Length)^2$$

Now, solve for the side length (S):

$$(Side\ Length)^2 = \frac{96\ cm^2}{6}$$
$$(Side\ Length)^2 = 16\ cm^2$$
$$Side\ Length = \sqrt{16\ cm^2}$$
$$Side\ Length = 4\ cm$$

So, the side length of the cube is 4 centimeters.

20. Correct Answer is: (5)

To find the height (h) of a triangular prism when you know its volume (V) and base area (A), we can use the formula:

$$Volume\ (V) = Base\ Area\ (A) \times Height\ (h)$$

Given:

$$Volume\ (V) = 40\ cm^3$$
$$Base\ Area\ (A) = 8\ cm^2$$

Plug in the values and solve for the height (h):

$$40\ cm^3 = 8\ cm^2 \times h$$

$$h = \frac{40}{8} = 5\ cm$$

So, the height of the triangular prism is 5 centimeters.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

To calculate the lateral surface area of a cuboid, we can use the formula:

$$Lateral\ Surface\ Area = 2 \times (Length \times Height + Breadth \times Height)$$

In this case, the length is 0.8 cm, the breadth is 0.6 cm, and the height is 0.05 cm. Plug these values into the formula:

$$Lateral\ Surface\ Area = 2 \times (0.8\ cm \times 0.05\ cm + 0.6\ cm \times 0.05\ cm)$$

$$Lateral\ Surface\ Area = 2 \times (0.04\ cm^2 + 0.03\ cm^2)$$

$$Lateral\ Surface\ Area = 2 \times 0.07\ cm^2$$

$$Lateral\ Surface\ Area = 0.14\ cm^2$$

So, the lateral surface area of the cuboid is 0.14 square centimeters.

Then the correct option is B)

2. Correct option: (D)

To find the ratio of the volumes of the two cuboids, we can simply calculate the volumes of both cuboids and then take the ratio.

$Cuboid_1$:

$$Length\ (L) = 10\ cm$$

$$Breadth\ (B) = 8\ cm$$

$$Height\ (H) = 6\ cm$$

$$Volume\ of\ Cuboid_1 = L \times B \times H = 10\ cm \times 8\ cm \times 6\ cm = 480\ cubic\ cm$$

$Cuboid_2$:

$$Length\ (l) = 15\ cm$$

$$Breadth\ (b) = 12\ cm$$

$$Height\ (h) = 9\ cm$$

$$Volume\ of\ Cuboid_2 = l \times b \times h = 15\ cm \times 12\ cm \times 9\ cm = 1620\ cubic\ cm$$

$$Ratio = \frac{Volume\ of\ Cuboid_1}{Volume\ of\ Cuboid_2} = \frac{480\ cubic\ cm}{1620\ cubic\ cm}$$

Simplify the ratio:

$$Ratio = \frac{8}{27}$$

So, the ratio of the volumes of the two cuboids is 8:27.

Then the correct option is D)

3. Correct option: (A)

$$Total\ Surface\ Area = 2 \times (Area\ of\ Base) + Lateral\ Surface\ Area$$

Given:

$$Total\ Surface\ Area = 40\ m^2$$

$$Lateral\ Surface\ Area = 26\ m^2$$

$$40\ m^2 = 2 \times (Area\ of\ Base) + 26\ m^2$$

$$40\ m^2 - 26\ m^2 = 2 \times (Area\ of\ Base)$$

$$14\ m^2 = 2 \times (Area\ of\ Base)$$

$$Area\ of\ Base = \frac{14\ m^2}{2}$$

$$Area\ of\ Base = 7\ m^2$$

So, the area of the base of the cuboid is 7 square meters.
Then the correct option is A)

4. Correct option: (A)

To find the volume of a cube, we can use the formula:

$$Volume = Edge\ Length^3$$

In this case, the edge length of the cube is given as 15 cm.

$$Volume = (15\ cm)^3$$
$$Volume = 15\ cm \times 15\ cm \times 15\ cm$$
$$Volume = 3375\ cubic\ cm$$

So, the volume of the cube is 3375 cubic centimeters.
Then the correct option is A)

5. Correct option: (D)

Let V_1 be the volume of the original cube, and V_2 be the volume of the new cube after doubling each edge length. The scaling factor is 2 because we doubled each side, so:

$$V_2 = (2^3) \times V_1$$
$$V_2 = 8 \times V_1$$

So, the volume of the new cube is 8 times the volume of the original cube.
Then the correct option is D)

6. Correct option: (B)

1. Calculate the length of one of the cube's edges using the total surface area.
2. Use the edge length to find the volume.

Given that the total surface area of the cube is 3456 sq. cm, we can find the edge length (e) using the formula for the total surface area of a cube:

$$Total\ Surface\ Area = 6 \times (Edge\ Length)^2$$
$$3456\ sq.cm = 6 \times (Edge\ Length)^2$$
$$(Edge\ Length)^2 = \frac{3456\ sq.cm}{6}$$
$$(Edge\ Length)^2 = 576\ sq.cm$$
$$Edge\ Length = \sqrt{576\ sq.cm}$$
$$Edge\ Length = 24\ cm$$

Now that we have found the edge length (e), we can calculate the volume of the cube using the formula:

$$Volume = (Edge\ Length)^3$$
$$Volume = (24\ cm)^3$$
$$Volume = 13{,}824\ cubic\ cm$$

So, the volume of the cube is 13,824 cubic centimeters.
Then the correct option is B)

7. Correct option: (B)

Given that the total surface area of the cube is 600 cm², we can find the edge length (e) using the formula for the total surface area of a cube:

$$Total\ Surface\ Area = 6 \times (Edge\ Length)^2$$
$$600\ cm^2 = 6 \times (Edge\ Length)^2$$
$$600\ cm^2 = 6 \times (Edge\ Length)^2$$
$$100\ cm^2 = (Edge\ Length)^2$$
$$Edge\ Length = \sqrt{100\ cm^2}$$
$$Edge\ Length = 10\ cm$$

Now that we have found the edge length (e), we can calculate the length of the diagonal (d) of the cube using the Pythagorean theorem:

$$d^2 = e^2 + e^2 + e^2$$
$$d^2 = 10^2 + 10^2 + 10^2$$

$$d^2 = 100 + 100 + 100$$
$$d^2 = 300$$
$$d = \sqrt{300}$$
$$d = 10\sqrt{3}\ cm$$

So, the length of the diagonal of the cube is $10\sqrt{3}$ centimeters.
Then the correct option is B)

8. Correct option: (C)

Let's call the edge length of the first cube "a" and the edge length of the second cube "b."
The volume ratio is given as 8:1, which means:

$$\frac{Volume\ of\ first\ cube}{Volume\ of\ second\ cube} = \frac{8}{1}$$

Now, we know that the volume of a cube is proportional to the cube of its edge length. So:

$$\frac{a^3}{b^3} = \frac{8}{1}$$
$$a^3 = 8b^3$$
$$\sqrt[3]{a^3} = \sqrt[3]{8b^3}$$
$$a = 2b$$

So, the ratio of their edges is 2:1.
Then the correct option is C)

9. Correct option: (D)

The volume of a triangular prism can be calculated using the formula:

$$Volume = (Area\ of\ Base) \times Height$$

In this case, the base of the triangular prism is a right-angled isosceles triangle with equal sides of 15 cm and a height of 12 cm. To find the area of the base, we can use the formula for the area of a right-angled triangle:

$$Area\ of\ Base = \left(\frac{1}{2}\right) \times Base \times Height$$
$$Area\ of\ Base = \left(\frac{1}{2}\right) \times 15\ cm \times 15\ cm = (1/2) \times 225\ cm^2 = 112.5\ cm^2$$

Now, we can calculate the volume of the triangular prism:

$$Volume = Area\ of\ Base \times Height = 112.5\ cm^2 \times 12\ cm = 1350\ cubic\ cm$$

So, the volume of the triangular prism is 1350 cubic centimeters.
Then the correct option is D)

10. Correct option: (C)

To find the total surface area (TSA) of the triangular prism:
1. Calculate the lateral surface area (LSA) using

$$LSA = Perimeter\ of\ the\ Base \times Height$$
$$Perimeter\ of\ the\ Base = 12\ cm + 16\ cm + 20\ cm = 48\ cm$$
$$Height = 15\ cm$$
$$LSA = 48\ cm \times 15\ cm = 720\ square\ cm.$$

2. Calculate the area of the triangular base using Heron's formula:

$$Semi_perimeter\ (s) = \frac{12\ cm + 16\ cm + 20\ cm}{2} = 24\ cm$$
$$Area\ (A) = \sqrt{[s \times (s - 12\ cm) \times (s - 16\ cm) \times (s - 20\ cm)]} = 96\ cm^2.$$

3. Calculate TSA:

$$TSA = LSA + 2 \times Base\ Area = 720\ square\ cm + 2 \times 96\ square\ cm = 912\ square\ cm.$$

So, the total surface area of the triangular prism is 912 square centimeters.
Then the correct option is C)

11. Correct option: (B, C)

The length of the diagonal of a cube can be related to its edge length using the Pythagorean Theorem.
Let "d" be the length of the diagonal, and "s" be the edge length.
According to the Pythagorean theorem, for a cube:

$$d^2 = s^2 + s^2 + s^2$$
$$d^2 = 3s^2$$

Given that the length of the diagonal is $14\sqrt{3}$ cm, we can set up the equation:

$$\left(14\sqrt{3}\right)^2 = 3s^2$$
$$3s^2 = 588$$
$$s^2 = \frac{588}{3}$$
$$s^2 = 196$$
$$s = \sqrt{196}$$
$$s = 14\,cm$$

Now that we have found the edge length "s" of the cube (14 cm), we can calculate the surface area (SA) of the cube. The surface area of a cube is given by:

$$SA = 6s^2$$
$$SA = 6(14\,cm)^2$$
$$SA = 6(196\,cm^2)$$
$$SA = 1176\,cm^2$$

So, the surface area of the cube is 1176 square centimeters.
Then the correct option is B, C)

12. Correct option: (C, D)

The volume of a cube is given by the formula: $Volume = Side\ Length^3$
In this case, the side length (edge) of the cube is given as 5 cm, so we can calculate the volume as:

$$Volume = (5\,cm)^3 = 5\,cm \times 5\,cm \times 5\,cm = 125\,cubic\ centimeters\ (cm^3)$$

So, the volume of the cube is 125 cubic centimeters.
Then the correct option is C, D)

13. Correct option: (C)

The diagonal (D) of a cube can be calculated using the Pythagorean theorem. If "a" is the edge length of the cube, then the diagonal is given by:

$$D = \sqrt{a^2 + a^2 + a^2}$$

In this case, the edge length (a) is 5 cm. Plug this value into the formula:

$$D = \sqrt{[(5\,cm)^2 + (5\,cm)^2 + (5\,cm)^2]}$$
$$D = \sqrt{25\,cm^2 + 25\,cm^2 + 25\,cm^2}$$
$$D = \sqrt{75\,cm^2}$$
$$D = \sqrt{25\,cm^2 \times 3}$$
$$D = 5\,cm \times \sqrt{3}$$

So, the diagonal of the cube is $5\sqrt{3}$ cm.
Then the correct option is C)

14. Correct option: (A)

The total surface area (TSA) of a cube can be calculated using the formula:

$$TSA = 6 \times (Edge\ Length)^2$$

In this case, the edge length (a) of the cube is given as 5 cm. Plug this value into the formula:

$$TSA = 6 \times (5\,cm)^2$$

$$TSA = 6 \times 25\,cm^2$$
$$TSA = 150\,cm^2$$

So, the total surface area of the cube is 150 square centimeters.
Then the correct option is A)

15. Correct option: (D)

The volume of a cube can be calculated using the formula:

$$Volume = (Edge\ Length)^3$$

In this case, the edge length (a) of the cube is given as 5 cm. Plug this value into the formula:

$$Volume = (5\,cm)^3$$
$$Volume = 125\,cubic\ centimeters\ (cm^3)$$

So, the volume of the cube is 125 cubic centimeters.
Then the correct option is D)

16. Correct Matching is: a-Q, b-P, c-S, d-R

a) If the edge of a cube is 3 times more, then the new volume will be $(3^3) = 27$ times the original volume.-→(Q)

b) If each dimension of a cuboid is doubled, then the volume becomes $2^3 = 8$ times the original volume.-→(P)

c) The value of $\frac{V}{A}$ for a cuboid with dimensions a, b, and c, where the total surface area is A and the volume is V, is:-→(S)

$$\frac{V}{A} = \frac{abc}{2ab + 2bc + 2ca} = \frac{1}{2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)}$$

d) The value of $\frac{A}{V}$ for a cuboid with dimensions a, b, and c, where the total surface area is A and the volume is V, is:-→(R)

$$\frac{A}{V} = \frac{2ab + 2bc + 2ca}{abc} = \mathbf{2}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

17. Correct Answer is: (4)

To determine the number of soaps of size 10cm x 5cm x 5cm that can be put in a box of a cuboid size 20cm x 10cm x 5cm, we need to calculate the volume of each soap and the volume of the box.

The volume of each soap is given as:

$$Volume\ of\ one\ soap = Length \times Breadth \times Height = 10cm \times 5cm \times 5cm = 250\,cm^3$$

The volume of the box is given as:

$$Volume\ of\ the\ box = Length \times Breadth \times Height = 20cm \times 10cm \times 5cm$$
$$= 1000\ cubic\ centimeters\ (cm^3)$$

Now, divide the volume of the box by the volume of one soap to find the number of soaps that can fit in the box:

$$Number\ of\ soaps = \frac{Volume\ of\ the\ box}{Volume\ of\ one\ soap} = \frac{1000\,cm^3}{250\,cm^3} = 4$$

So, we can fit 4 soaps of given size into a box of a cuboid size
$20cm \times 10cm \times 5cm$.

7. Algebra, Exponents and Powers
Solutions

LEVEL – I

1. Correct option: (C)
The expression for the statement "x multiplied by 6 and added to 13" is:

$$6x + 13$$

Then the correct option is C)

2. Correct option: (A)
The statement "One-fifth of the product of 'x' and 'y' is subtracted from P" can be expressed as:

$$P - \left(\frac{1}{5}\right)xy$$

In this expression:
- 'x' and 'y' represent two variables.
- The product of 'x' and 'y' is calculated as xy.
- One-fifth of that product is $\left(\frac{1}{5}\right)xy$.
- Finally, that value is subtracted from P to obtain the expression.

Then the correct option is A)

3. Correct option: (B)
In the expression $5x^2y + 2x + 7$ the numerical term is the term that does not have any variable (x or y) attached to it.
So, the numerical term in the given expression is: B) 7

4. Correct option: (C)
The false statement is:
C) A literal coefficient is a constant.
A literal coefficient is not always a constant; it always a variable.

5. Correct option: (B)
Given expression:

$$12x + 25y - 25 + 25xy + 7xy^2 - 5x^2y - 5xy + 10x^2y^2 + 6xy$$

Expression after grouping the like terms

$$10x^2y^2 - 5x^2y + 7xy^2 + 26xy + 12x + 25y - 25$$

Then the numerical coefficient of maximum number of like terms in the expression is 26.
Therefore the correct option is B)

6. Correct option: (C)
An expression that contains two unlike terms is called a Binomial.
So, the correct answer is: C) Binomial

7. Correct option: (C)
A Binomial expression consists of two unlike terms. Let's analyze the given options:
A) $x^2 + (2x)x \Rightarrow 3x^2$ - This is a Monomial because it has only one term.
B) $4xy + (2x)(2y) \Rightarrow 8xy$ - This is a Monomial because it has only one term.
C) $2x^2 - 4$ - This is a Binomial because it has two terms ($2x^2$ and -4).
D) $(3x)y - x^2 - 3yx \Rightarrow -x^2$ - This is a Monomial because it has only one term.
So, the correct answer is: C) $2x^2 - 4$

8. Correct option: (B)
A Trinomial expression consists of three unlike terms. Let's analyze the given options:
A) $(2x^2)y - 4x + (2x)(2xy) \Rightarrow -4x$ - This is a Monomial because it has only one term.

B) $x^2y - 3x^2 - (2x)(2y) =>$ - This is a Trinomial because it has three unlike terms.
C) $-2xy + xy + y^2 => y^2 - xy$ - This is a Binomial because it has two unlike terms only.
D) $(2x)y^2 - y^2(2xy) + (-y)(2xy) => -y^2(2xy)$ - This is a Monomial because it has only one term.
So, the correct answer is: B) $x^2y - 3x^2 - (2x)(2y)$

9. Correct option: (A)

To find the degree of the polynomial, add up the exponents of each term and select the highest sum.
The degree of the algebraic expression $4x^2 + 3xyz + y^2$ is indeed 3 because the highest power of variable's add up (in this case, the variable 3xyz) is 3.
Then the correct option is A)

10. Correct option: (C)

To find the degree of the polynomial, add up the exponents of each term and select the highest sum.
For the given expression $pq - 6p^2q^2 + q^2p + 9$
$pq: 1 + 1 = 2$
$-6p^2q^2: 2 + 2 = 4$
$q^2p: 2 + 1 = 3$
$9: 0 + 0 = 0$
Therefore the highest degree of the expression is 4.
Then the correct option is C)

11. Correct option: (A)

To find the sum of the expressions
$4xy^2 - 7x^2y, 12x^2y - 6xy^2, -3x^2y + 6xy^2$

$$4xy^2 - 7x^2y + 12x^2y - 6xy^2, +(-3x^2y) + 6xy^2$$

Now, combine like terms by adding the coefficients (the numbers in front of the variables) that have the same variables:

$$(4 - 6 + 6)xy^2 + (12 - 3 - 7)x^2y = 4xy^2 + 2x^2y$$

So, the sum of the expressions is $4xy^2 + 2x^2y$.
Then the correct option is A)

12. Correct option: (B)

To find the sum of the expressions
$6x^2 - 4x, 2x - 3x^2 + 4, 4x^2 - 5x + 3$

$$6x^2 - 4x + 2x - 3x^2 + 4 + 4x^2 - 5x + 3$$

Now, combine like terms by adding the coefficients (the numbers in front of the variables) that have the same variables:

$$(6 - 3 + 4)x^2 + (-4 + 2 - 5)x + (4 + 3) = 7x^2 - 7x + 7$$

So, the sum of the expressions is $7x^2 - 7x + 7$.
Then the correct option is B)

13. Correct option: (B)

To subtract $-5x^2y$ from $6xy - x^2y$, we can follow these steps:
1. Start with the expression: $6xy - x^2y$
2. Subtract $-5x^2y$ from it: $6xy - x^2y - (-5x^2y)$
3. To subtract a negative number, we can change it to addition:

$$6xy - x^2y + 5x^2y$$

4. Now, we can combine like terms by adding the coefficients that have the same variables: $(6xy + 5x^2y) - x^2y => 6xy + 4x^2y$
Then the correct option is B)

14. Correct option: (C)

To find the sum of the expressions we can simply add them together term by term:

$$\left(\frac{7}{2}\right)x^3 - \left(\frac{3}{2}\right)x^2 + 5 + \left(\frac{7}{2}\right)x^2 - \left(\frac{3}{2}\right)x^3 + \frac{1}{3} + \left(\frac{3}{2}\right)x^3 - \left(\frac{1}{2}\right)x^2 + \frac{2}{3}$$

Now, let's combine like terms:

$$\left[\left(\frac{7}{2}\right)x^3 - \left(\frac{3}{2}\right)x^3 + \left(\frac{3}{2}\right)x^3\right] + \left[-\left(\frac{3}{2}\right)x^2 + \left(\frac{7}{2}\right)x^2 - \left(\frac{1}{2}\right)x^2\right] + \left[5 + \frac{1}{3} + \frac{2}{3}\right]$$

Combine the coefficients for each power of x and the constants:

$$\frac{7}{2}x^3 + \left(-\left(\frac{3}{2}\right) + \frac{7}{2} - \frac{1}{2}\right)x^2 + \left(5 + \frac{1}{3} + \frac{2}{3}\right)$$

$$=> \left(\frac{7}{2}\right)x^3 + \left(\frac{3}{2}\right)x^2 + 6$$

Then the correct option is C)

15. Correct option: (A)

From the question we can write as:

$$\left(\frac{1}{3} - \frac{5}{3}y^2\right) - \left(-\frac{7}{3}y^2 + \frac{1}{2}y\right)$$

To subtract a negative expression, change it to addition:

$$\left(\frac{1}{3} - \frac{5}{3}y^2\right) + \left(\frac{7}{3}y^2 - \frac{1}{2}y\right)$$

Now, combine like terms by adding or subtracting the coefficients that have the same variable (y) or constant terms:

$$\left(\frac{7}{3}y^2 - \frac{5}{3}y^2\right) - \frac{1}{2}y + \frac{1}{3} = \frac{2}{3}y^2 - \frac{1}{2}y + \frac{1}{3}$$

Then the correct option is A)

16. Correct option: (B)

To find the perimeter of a triangle with side lengths of $3x, 4x, \text{and } 6x$, simply add up the lengths of all three sides:

$$\text{Perimeter} = 3x + 4x + 6x$$

Now, combine like terms by adding the coefficients (the numbers in front of x):

$$\text{Perimeter} = (3 + 4 + 6)x$$

$$\text{Perimeter} = 13x$$

So, the perimeter of the triangle is 13x.

Then the correct option is B)

17. Correct option: (A)

The product of xy and x^2y is calculated by multiplying the two expressions together:

$$xy \times x^2y = (xy) \times (x^2y)$$

Now, use the properties of exponents to simplify:

$$(xy) \times (x^2y) = x^1 \times y^1 \times x^2 \times y^1$$

When multiplying variables with the same base (in this case, x, y), we add their exponents. So, combining the exponents:

$$x^{1+2} \times y^{1+1} = x^3 \times y^2$$

Therefore, the product of xy and x^2y is x^3y^2.

Then the correct option is A)

18. Correct option: (B)

To find the product of $4x, -2xy, 4x^3, -5x^2y^2, \text{and } -2xy^2$, we simply multiply all of these terms together:

$$(4x) \times (-2xy) \times (4x^3) \times (-5x^2y^2) \times (-2xy^2)$$

Now, let's multiply the coefficients and the variables separately:

Coefficients: $(4) \times (-2) \times (4) \times (-5) \times (-2) = -320$
Variables:

$$x \times xy \times x^3 \times x^2y^2 \times xy^2 = x^8y^5$$

Putting it all together:

$$=> -320x^8y^5$$

Then the correct option is B)

19. Correct option: (A)

To find the area of a rectangle, we simply multiply its length and breadth.
In this case, the length of the rectangle is $3x - 2y^2$, and the breadth is $4x^2 - 5y$. So, the area (A) of the rectangle is:

$$A = (\text{Length}) \times (\text{Breadth})$$
$$A = (3x - 2y^2) \times (4x^2 - 5y)$$

Now, let's multiply these two expressions together using the distributive property:

$$A = (3x \times 4x^2) + \big(3x \times (-5y)\big) + (-2y^2 \times 4x^2) + (-2y^2 \times (-5y))$$

Now, simplify each term:

$$A = 12x^3 - 15xy - 8x^2y^2 + 10y^3$$

Then the area of rectangle is $A = 12x^3 - 15xy - 8x^2y^2 + 10y^3$
Therefore the correct option is A)

20. Correct option: (C)

Let's denote the length of the cuboid as L, the breadth as B, and the height as H.
According to the given information:
1. The length (L) is 2 times its breadth (B), so $L = 2B$.
2. The height (H) is one-third of the length (L), so $H = \left(\frac{1}{3}\right)L$.
Now, we mentioned that the breadth is x cm, so B = x.
We need to find the volume (V) of the cuboid, which is given by the formula:

$$V = \text{Length} \times \text{Breadth} \times \text{Height}$$

Substitute the values we found for L, B, and H:

$$V = (2x) \times (x) \times \left(\frac{1}{3}\right)(2x)$$

Now, simplify the expression: $V = (2x) \times (x) \times \left(\frac{2}{3x}\right) = \frac{4}{3}x^3$
So, the volume of the cuboid is $\left(\frac{4}{3}\right)x^3$ cubic centimeters.
Therefore the correct option is C)

LEVEL-II

1. Correct option: (A)

Firstly the price for 1 dozen of apple is x, and y dozens of apples is needed.
After discount the price for 1 dozen is x-4 and (y-6) dozens is needed now.
The total amount to pay is the product of the price per dozen of apples and the number of dozens of apples.

$$\text{Amount to pay} = (x - 4) \times (y - 6)$$

Therefore the correct option is A)

2. Correct option: (B)

The area of a rectangle is given by the formula: $\text{Area} = \text{Length} \times \text{Breadth}$
If the length is increased by 6 units and the breadth is decreased by 9 units, the new length (l') and new breadth (b') will be:

$$\text{New Length } (l') = l + 6$$
$$\text{New Breadth } (b') = b - 9$$

Now, we can calculate the area of the new rectangle: $\text{New Area} = l' \times b'$

$$\text{New Area} = (l + 6) \times (b - 9) = lb + 6b - 9l - 54$$

This is the area of the rectangle after the changes in length and breadth have been made.
Therefore the correct option is B)

3. Correct option: (D)

To subtract the product of $3x^2(2x - 3y)$ from the product of $6x(x^2 + 4y)$, follow these steps:

1. Calculate the product of $3x^2(2x - 3y)$:
   $3x^2(2x - 3y) = 6x^3 - 9x^2y$
2. Calculate the product of $6x(x^2 + 4y)$:
   $6x(x^2 + 4y) = 6x^3 + 24xy$
3. Now, subtract the result from step 1 from the result from step 2:
   $$(6x^3 + 24xy) - (6x^3 - 9x^2y)$$
4. Distribute the negative sign into the second expression inside the parentheses:
   $$6x^3 + 24xy - 6x^3 + 9x^2y$$
5. Now, combine like terms:
   $(6x^3 - 6x^3) + (24xy + 9x^2y)$

So, the result of subtracting the product of $3x^2(2x - 3y)$ from the product of $6x(x^2 + 4y)$ is $24xy + 9x^2y$.
Therefore, the correct option is D)

4. Correct option: (C)

$$\text{Total Sum: } 6y + 13x - 7x^2$$

Given Expressions: $(3x^2 - 4x + 5y) + (3x - 10x^2 - 8y)$
Now, subtract the sum of the two given expressions from the total sum:

$$\text{Third Expression} = \text{Total Sum} - \text{Sum of Given Expressions}$$

$$\text{Third Expression} = (6y + 13x - 7x^2) - [(3x^2 - 4x + 5y) + (3x - 10x^2 - 8y)]$$

Now, distribute the negative sign inside the brackets:

$$\text{Third Expression} = 6y + 13x - 7x^2 - 3x^2 + 4x - 5y - 3x + 10x^2 + 8y$$

$$\text{Third Expression} = (6y - 5y + 8y) + (13x - 3x + 4x) + (-7x^2 - 3x^2 + 10x^2)$$

$$\text{Third Expression} = 9y + 14x$$

So, the third expression is $9y + 14x$
Therefore, the correct option is C)

5. Correct option: (A)

The perimeter of a rectangle is given by the formula:

$$\text{Perimeter} = 2(\text{Length} + \text{Breadth})$$

In this case, the length of the rectangle is $6x^2 - 4y^2$, and the breadth is $2x^2 - 8xy$.
So, the perimeter is:

$$\text{Perimeter} = 2(6x^2 - 4y^2 + 2x^2 - 8xy)$$

Now, let's simplify the expression inside the parentheses:

$$\text{Perimeter} = 2(8x^2 - 4y^2 - 8xy)$$

$$\text{Perimeter} = 16x^2 - 8y^2 - 16xy$$

So, the perimeter of the rectangle is $16x^2 - 8y^2 - 16xy$.
Then the correct option is A)

6. Correct option: (C)

If the population of a certain species of insects is currently $\frac{x^2}{2}$, and it becomes 3y times itself after one week, we can calculate the population after one week:

$$\text{Population after one week} = (3y) \times \left(\frac{x^2}{2}\right)$$

Now, to find the population after 5 weeks, we need to keep multiplying it by 3y for each week:

$$\text{Population after 5 weeks} = (3y)^5 \times \left(\frac{x^2}{2}\right)$$

To simplify, raise 3y to the 5th power:

$$\text{Population after 5 weeks} = 3^5 \times y^5 \times \left(\frac{x^2}{2}\right)$$

$$\text{Population after 5 weeks} = 243 \times y^5 \times \left(\frac{x^2}{2}\right)$$

Then the correct option is C)

7. Correct option: (C)

To subtract $y^2 - 7 - 2y$ from $8 - y + 2y^2$, we need to perform the subtraction operation term by term:

$(8 - y + 2y^2) - (y^2 - 7 - 2y)$

Now, distribute the negative sign into the second set of parentheses:

$$8 - y + 2y^2 - y^2 + 7 + 2y$$

Next, combine like terms:

$$(2y^2 - y^2) + (-y + 2y) + (8 + 7)$$

Simplify each set of parentheses:

$$y^2 + y + 15$$

So, the result of subtracting $y^2 - 7 - 2y$ from $8 - y + 2y^2$ is $y^2 + y + 15$.

Then the correct option is C)

8. Correct option: (C)

To find the value of the quadratic expression $ax^2 + bx + c$ at $x = -\frac{b}{a}$, we can simply substitute $-\frac{b}{a}$ for x in the expression and calculate:

$$ax^2 + bx + c \text{ at } x = -\frac{b}{a}:$$

$$= a\left(-\frac{b}{a}\right)^2 + b\left(-\frac{b}{a}\right) + c$$

$$= a\left(\frac{b^2}{a^2}\right) - \left(\frac{b^2}{a}\right) + c$$

Now, simplify:

$$= \frac{b^2}{a} - \frac{b^2}{a} + c$$

$$= \left(\frac{b^2}{a} - \frac{b^2}{a}\right) + c$$

$$= 0 + c$$

So, the value of the quadratic expression $ax^2 + bx + c$ at $x = -\frac{b}{a}$ is simply c.

Then the correct option is C)

9. Correct option: (C)

Given $a = 0, b = 1, \text{and } c = 2$, we can calculate the expression $a^2 + b + c$ as follows:

$$a^2 + b + c = (0)^2 + 1 + 2$$

Now, simplify:

$$a^2 + b + c = 0 + 1 + 2$$

$$a^2 + b + c = 3$$

So, when a = 0, b = 1, and c = 2, the value of $a^2 + b + c$ is 3.

Then the correct option is C)

10. Correct option: (D)

The sum of $x + 1$ and the reciprocal of $(x - 1)$ can be expressed as:

$$(x + 1) + \frac{1}{x - 1}$$

To simplify this expression further, we can find a common denominator:

$$\frac{[(x + 1)(x - 1) + 1]}{x - 1}$$

Now, expand the numerator:

$$\frac{x^2 - 1 + 1}{x - 1}$$

Simplify the numerator:

$$\frac{x^2}{x - 1}$$

So, the sum of x + 1 and the reciprocal of (x - 1) is $\frac{x^2}{x-1}$.

None of the option matches with this.

Then the correct option is D)

11. Correct option: (B, C)

The statement can be represented as an algebraic expression as follows:

1. The sum of p and q is: p + q.
2. The difference between p and q is: p - q.
3. Taking away 5 from their difference: (p - q) - 5.

Now, combine the two parts:

The sum of p and q is multiplied by taking away 5 from their difference: $(p + q)((p - q) - 5)$

$$(p + q)\big((p - q) - 5\big) = (p + q)(p - q) - 5(p + q) = p^2 - q^2 - 5(p + q)$$

Then the correct option is B), C).

12. Correct option: (A, B)

To find the total number of persons in the large hall, we can multiply the number of rows by the number of benches in each row and then multiply that by the number of persons each bench can accommodate.

Given:

$$\text{Number of rows of benches} = 4a^2b$$

$$\text{Number of benches in each row} = 16a^3b^3$$

$$\text{Number of persons each bench can accommodate} = 3ab^2$$

Total number of persons

$$= (\text{Number of rows}) \times (\text{Number of benches per row}) \times (\text{Number of persons per bench})$$

$$\text{Total number of persons} = (4a^2b) \times (16a^3b^3) \times (3ab^2)$$

Now, multiply the coefficients and combine the variables with the same base:

$$\text{Total number of persons} = (4 \times 16 \times 3) \times (a^2 \times a^3 \times a) \times (b \times b^3 \times b^2)$$

$$\text{Total number of persons} = 192a^6b^6 = 2^6 \times 3a^6b^6$$

Then the correct option is A), B).

13. Correct option: (B, D)

To find the value of 2A - B, first, we'll find the expressions for 2A and then subtract B from it.

Given:

$$A = x^2 + 2x + 2$$

$$B = 5x^2 - x + 4$$

Now, let's calculate 2A:

$$2A = 2(x^2 + 2x + 2)$$

Now, distribute the 2 into each term inside the parentheses:

$$2A = 2x^2 + 4x + 4$$

Now, subtract B from 2A:

$$2A - B = (2x^2 + 4x + 4) - (5x^2 - x + 4)$$

Now, distribute the negative sign into the second expression inside the parentheses:

$$2A - B = 2x^2 + 4x + 4 - 5x^2 + x - 4$$

$$2A - B = (2x^2 - 5x^2) + (4x + x) + (4 - 4)$$

Simplify each set of parentheses:

$$2A - B = -3x^2 + 5x + 0$$

So, the value of 2A - B is $-3x^2 + 5x$.

Then the correct option is B), D).

14. Correct option: (C)

To find the perimeter of a triangle with sides A, B, and C, we simply need to add the lengths of these sides together.

Given:

$$A = 5x^2y + 2xy^2 - 4xy$$

$$B = 3xy^2 + 4x^2y - 2$$

$$C = 7x^2y - 5$$

Now, add the lengths of these sides to find the perimeter:

$$\text{Perimeter} = A + B + C$$

$$P = (5x^2y + 2xy^2 - 4xy) + (3xy^2 + 4x^2y - 2) + (7x^2y - 5)$$

$$P = 5x^2y + 2xy^2 - 4xy + 3xy^2 + 4x^2y - 2 + 7x^2y - 5$$

$$P = (5x^2y + 4x^2y + 7x^2y) + (2xy^2 + 3xy^2 - 4xy) - 7$$

$$P = 16x^2y + 5xy^2 - 4xy - 7$$

So, the perimeter of the triangle with sides A, B, and C is $16x^2y + 5xy^2 - 4xy - 7$.

Then the correct option is C)

15. Correct option: (B)

To find the value of 2B + C - A, simply substitute the given expressions for A, B, and C into the expression and simplify: $2B + C - A$

$$= 2(3xy^2 + 4x^2y - 2) + (7x^2y - 5) - (5x^2y + 2xy^2 - 4xy)$$

Now, distribute the coefficients:

$$= 6xy^2 + 8x^2y - 4 + 7x^2y - 5 - 5x^2y - 2xy^2 + 4xy$$

$$= (6xy^2 - 2xy^2) + (8x^2y + 7x^2y - 5x^2y) + (-4 - 5 + 4xy)$$

Now, simplify each set of parentheses:

$$2B + C - A = 4xy^2 + 10x^2y - 9 + 4xy$$

So, the value of $2B + C - A$ is $4xy^2 + 10x^2y + 4xy - 9$.

Then the correct option is B)

16. Correct option: (D)

The additive inverse of a number or expression is what we add to it to get zero.

$$A + B - C$$

The additive inverse would be:

$$-(A + B - C)$$

Now, let's substitute the expressions for A, B, and C:

$$-(5x^2y + 2xy^2 - 4xy + 3xy^2 + 4x^2y - 2 - 7x^2y + 5)$$

Now, distribute the negative sign to each term inside the parentheses:

$$-5x^2y - 2xy^2 + 4xy - 3xy^2 - 4x^2y + 2 + 7x^2y - 5$$

Now, combine like terms:

$$(-5x^2y - 4x^2y + 7x\text{^}2y) + (-2xy^2 - 3xy^2) + (4xy + 2 - 5)$$

Combine the like terms further:

$$-2x^2y - 5xy^2 + 4xy - 3$$

So, the additive inverse of A + B - C is

$$-2x^2y - 5xy^2 + 4xy - 3.$$

Then the correct option is D)

17. Correct Matching is: a-T, b-S,c-P,d-R

a) $(x^2 + 2)^2$

$(x^2 + 2)^2 = x^4 + 4 + 4x^2$ Highest power of x = 4

Then the degree of the above expression is 4.-→(T)

b) $(x^2 + 2)(x + 3)$

$(x^2 + 2)(x + 3) = x^3 + 2x + 3x^2 + 6$ Highest power of x = 3

Then the degree of the above expression is 3.-→(S)

c) 2

Degree of a constant term is always 0.

Then the degree of the above expression is 0.-→P)

d) $(x + 3)(x - 3)$

$(x + 3)(x - 3) = x^2 - 9$ Highest power of x = 2

Then the degree of the above expression is 2.-→(R)

18. Correct Answer is: (2)

The degree of a polynomial expression is determined by the highest power of the variable in the expression.

In the expression $(x + 2)(x^n + 1)$, we have two factors: (x+2) and $(x^n + 1)$.

The degree of the first factor (x+2) is 1 because the highest power of x is 1.

The degree of the second factor $(x^n + 1)$ is n because it contains the term x^n , which has the highest power of x.

To find the degree of the entire expression, we add the degrees of the factors:

$$\text{Degree of expression} = \text{Degree of } (x + 2) + \text{Degree of } (x^n + 1) = n + 1$$

We are given that degree of the expression is 3.

$$n + 1 = 3 \Rightarrow n = 2$$

Then the value of n is 2.

19. Correct Answer is: (5)

Let's substitute the values of a, b, c, and d into the expression and calculate:

$$(cd - ab) - (bd - ac)$$

Substitute the values: $(4 \times 3 - 1 \times 2) - (2 \times 4 - 1 \times 3)$

Now, perform the multiplications and subtractions inside the parentheses:

$$(12 - 2) - (8 - 3)$$

Now, continue simplifying: $(10) - (5) = 5$

So, the value of $(cd - ab) - (bd - ac)$ when $a = 1, b = 2, c = 3,$ and $d = 4$ is 5.

20. Correct Answer is: (2)

To find the value of x in the equation $7x - 5 = 3x + 3$, we need to isolate x on one side of the equation.

$$7x - 3x - 5 = 3x - 3x + 3$$
$$4x - 5 = 3$$
$$4x - 5 + 5 = 3 + 5$$
$$4x = 8$$
$$\frac{4x}{4} = \frac{8}{4}$$
$$x = 2$$

So, the value of x is 2.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

The correct statement for the expression "4x + 9" is:

A) 4 is multiplied by x and added to 9

2. Correct option: (D)

To find the number of terms in the expression:

$$6x^2y + 2xy + 5xy^2 + 9x + 6y + 20x^2y^2 + (-2x^3) - 7$$

Count all the individual terms:

1. $6x^2y$
2. $2xy$
3. $5xy^2$
4. $9x$
5. $6y$
6. $20x^2y^2$
7. $-2x^3$
8. -7

There are a total of 8 terms in the given expression.

3. Correct option: (A)

An expression that contains three like terms is called a Trinomial.

So, the correct answer is: A) Trinomial

4. Correct option: (D)

A Monomial expression consists of three unlike terms. Let's analyze the given options:

D) $(2x)y^2 - y^2(2xy) + (-y)(2xy) => -y^2(2xy)$ - This is a Monomial because it has only one term.

So, the correct answer is: (D)

5. Correct option: (C)

To find the sum of the expressions $3a^2b, -4a^2b,$ and $9a^2b$

$$3a^2b + (-4a^2b) + 9a^2b$$

Now, combine like terms by adding the coefficients (the numbers in front of the variables) that have the same variables (a^2b):

$$(3 - 4 + 9)a^2b$$
$$= (8)a^2b$$

So, the sum of the expressions is $8a^2b$.

Then the correct option is C)

6. Correct option: (D)

To subtract $3a - 2b$ from $5b - 6a$, we can directly do subtraction:

$$5b - 6a - (3a - 2b) = 5b - 6a - 3a + 2b = 7b - 9a$$

Then the correct option is D)

7. Correct option: (B)

To find out what should be taken away from $3x^2 - 4y^2 + 5xy + 20$ to get $-x^2 - y^2 + 6xy + 20$, we need to calculate the difference between the two expressions:

$$(3x^2 - 4y^2 + 5xy + 20) - (-x^2 - y^2 + 6xy + 20)$$

Now, distribute the negative sign across the terms inside the second set of parentheses:

$$3x^2 - 4y^2 + 5xy + 20 + x^2 + y^2 - 6xy - 20$$

Next, combine like terms by adding or subtracting the coefficients that have the same variables:

$$(3x^2 + x^2) + (-4y^2 + y^2) + (5xy - 6xy) + (20 - 20)$$

Now, perform the arithmetic operations within each set of parentheses:

$$4x^2 - 3y^2 - xy$$

So, to get from $3x^2 - 4y^2 + 5xy + 20$ to $-x^2 - y^2 + 6xy + 20$, you need to subtract $4x^2 - 3y^2 - xy$.

Then the correct option is B)

8. Correct option: (C)

To find the product of 4xy and −6xyz, we simply multiply the two expressions together:

$$(4xy) \times (-6xyz)$$

Now, apply the rules of multiplication:

$$4 \times (-6) \times x \times x \times y \times z \times y$$

Multiply the coefficients:

$$-24 \times x \times x \times y \times z \times y$$

$$-24 \times x^2 \times y^2 \times z = -24x^2y^2z$$

Then the correct option is C)

9. Correct option: (D)

To find the perimeter of a rectangle, we add up the lengths of all four sides.

In this case, the length of the rectangle is 6x and the breadth (width) is $4y + 3$. A rectangle has two pairs of equal sides, so we have:

1. Two sides of length 6x.
2. Two sides of length 4y + 3.

Now, calculate the perimeter:

$$\text{Perimeter} = 2(\text{Length} + \text{Breadth})$$

$$\text{Perimeter} = 2(6x + 4y + 3)$$

$$\text{Perimeter} = 12x + 8y + 6$$

So, the perimeter of the rectangle is 12x + 8y + 6.

Then the correct option is D)

10. Correct option: (B)

To find the total distance Rahul moves in 6y steps, we can simply multiply the distance he covers in one step (8x) by the number of steps (6y).

$$\text{Distance moved} = (\text{Distance per step}) \times (\text{Number of steps})$$

$$\text{Distance moved} = (8x) \times (6y)$$

Now, multiply the terms together:

Distance moved = 48xy

So, Rahul covers a distance of 48xy centimeters in 6y steps.

Therefore the correct option is B)

11. Correct option: (B, D)

To determine what should be subtracted from $a^2 + b^2 + 2ab$ to get $-4ab + 2b^2$, we can set up the subtraction as follows:

$$(a^2 + b^2 + 2ab) - (\text{What should be subtracted}) = -4ab + 2b^2$$

Now, let's find out what should be subtracted:

$$\text{What should be subtracted} = (a^2 + b^2 + 2ab) - (-4ab + 2b^2)$$

Now, subtract the expressions inside the parentheses:

$$= a^2 + b^2 + 2ab + 4ab - 2b^2$$

Combine like terms: $= (a^2 + 6ab - b^2) \Rightarrow a^2 + b(-b + 6a)$

Then the correct option is B), D).

12. Correct option: (A, B)

Like terms are terms that have the same variables raised to the same powers. Let's identify the like terms in each set:

$$\text{A) } 7xy \text{ and } \frac{8}{7}xy$$

$$\text{Like terms: } 7xy \text{ and } \frac{8}{7}xy$$

$$\text{B) } 9x^3y^2 \text{ and } -7x^3y^2$$

$$\text{Like terms: } 9x^3y^2 \text{ and } -7x^3y^2$$

$$\text{C) } 71\,xy^2 \text{ and } \frac{8}{3}x$$

- These terms do not have the same variables raised to the same powers, so they are not like terms.

$$\text{D) } 3x^2y \text{ and } 8xy$$

- These terms do not have the same variables raised to the same powers, so they are not like terms.

So, the like terms in each set are as follows:

$$\text{A) } 7xy \text{ and } \frac{8}{7}xy$$

$$\text{B) } 9x^3y^2 \text{ and } -7x^3y^2$$

Then the correct options are A), B).

13. Correct option: (A)

A) $2x^2y$ is a monomial because it contains only one term.

B) x + y is a binomial because it contains two terms.

C) $x + y^2$ is also a binomial because it contains two terms.

Then the correct option is A)

14. Correct option: (D)

A binomial is an algebraic expression containing exactly two terms. Let's identify the binomial from the options:

A) 8abc - This is a monomial because it contains only one term.

B) 2 + x + y - This is not a binomial; it's a trinomial because it contains three terms (2, x, and y).

C) x^2y - This is a monomial because it contains only one term.

D) $x + y^2$ - This is a binomial because it contains two terms.

So, the binomial among the given options is D) $x + y^2$.

15. Correct Matching is: a-S, b-R, c-Q, d-P

a) $(x+y)^2-(x-y)^2 =$

Solving the expression

$$x^2+y^2+2xy-(x^2+y^2-2xy)=4xy$$

Then we can say that a) -→(S)

b) $(ax+b)(cx+d) =$

Solving the expression

$$acx^2+cbx+adx+bd=cax^2+(ad+bc)x+bd$$

Then we can say that b) -→(R)

c) $\left(x-\frac{1}{xy}\right)\left(xy+\frac{1}{x}\right)$

Solving the expression

$$\left(x^2y-1+1-\frac{1}{x^2y}\right)=x^2y-\frac{1}{x^2y}$$

Then we can say that c) -→(Q)

d) $(ax+d)(cx+b) =$

Solving the expression

$$acx^2+cdx+abx+bd=cax^2+(ab+cd)x+bd$$

Then we can say that d) -→(P)

16. Correct Answer is: (0)

Given the values of A, B, and C as follows:

$$A = -2x + 3y - 5z$$
$$B = -4x - 9y - 3z$$
$$C = 6x + 6y + 8z$$

We want to find the value of A + B + C, which is the sum of these three expressions:

$$A + B + C = (-2x + 3y - 5z) + (-4x - 9y - 3z) + (6x + 6y + 8z)$$

Now, simply combine like terms by adding the coefficients of the same variables:

$$= (-2x - 4x + 6x) + (3y - 9y + 6y) + (-5z - 3z + 8z)$$

Now, perform the additions:

$$A + B + C = 0 + 0 + 0 = 0$$

So, the value of A + B + C is 0.

17. Correct Answer is: (2)

To find the coefficient of x^2 in the expression $(3y + x^2)(xy + 2)$, we need to perform polynomial multiplication. The term x^2 can be obtained from multiplying x^2 in the first factor and the constant 2 in the second factor.

So, multiply these two terms: $x^2 \times 2 = 2x^2$

The coefficient of x^2 in the expression is 2.

18. Correct Answer is: (6)

To find the number of terms in the expression $(x + xy - 2)(x + y)$, we can use the distributive property to multiply the two polynomial:

$$(x + xy - 2)(x + y)$$

When you multiply each term in the first binomial by each term in the second binomial, we will generate several terms:

$$x^2 + x^2y - 2x + xy + xy^2 - 2y$$

Now, simplify each term:

$$x^2 + xy + x^2y + xy^2 - 2x - 2y$$

Now, count the number of terms:

There are 6 terms in the expression $(x + xy - 2)(x + y)$.

19. Correct Answer is: (8)

To solve for x in the equation 3x + 9 = 33, we'll need to isolate x.

First, subtract 9 from both sides of the equation:

$$3x + 9 - 9 = 33 - 9$$

This simplifies to: $3x = 24$

Now, divide both sides by 3 to solve for x:

$$\frac{3x}{3} = \frac{24}{3}$$
$$x = 8$$

So, the solution to the equation is x = 8.

20. Correct Answer is: (8)

To find the length of one side of a square when we know its area, we can use the formula for the area of a square:

$$\text{Area} = \text{side}^2$$
$$64 = \text{side}^2$$

Now, to solve for the side length, we can take the square root of both sides:

$$\sqrt{64} = \sqrt{\text{side}^2}$$
$$8 = \text{side}$$

So, the length of one side of the square is 8 cm.

Multiplications, Divisions, Lcm & Hcf

LEVEL-I

1. Correct option: (B)

Given expression

$$(x^6 - 7x^5 + 4x^4) \div x^2$$

To divide the expression we can apply the rules of exponents. When we divide two terms with the same base we subtract the exponents.

Divide each term separately:

$$\left(\frac{x^6}{x^2} - \frac{7x^5}{x^2} + \frac{4x^4}{x^2}\right) = x^4 - 7x^3 + 4x^2$$

So, the result of $(x^6 - 7x^5 + 4x\text{^}4) \div x^2$ is:

$$x^4 - 7x^3 + 4x^2$$

Then the correct option is B)

2. Correct option: (A)

To divide the expression abx^2 by $(-abx)$, we can simplify as follows:

$$abx^2 \div (-abx) = \frac{abx^2}{-abx}$$

Now, divide both the numerator and the denominator by $-abx$

$$\frac{abx^2}{-abx} = -\frac{x \times abx}{abx} = -x$$

So, the result of $abx^2 \div (-abx)$ is $-x$.

Then the correct option is A)

3. Correct option: (B)

To divide the expression $16p^6q^7r^8$ by $-4p^2q^3r^4$ we can simplify as follows:

$$\frac{16p^6q^7r^8}{-4p^2q^3r^4} = -\frac{(4p^4q^4r^4)(4p^2q^3r^4)}{4p^2q^3r^4} = -4p^4q^4r^4$$

So, the result of $16p^6q^7r^8 \div (-4p^2q^3r^4)$ is

$$-4p^4q^4r^4$$

Then the correct option is B)

4. Correct option: (B)

Given expression

$$\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \div \left(\frac{5}{6}x\right)$$

Find the reciprocal of $\left(\frac{5}{6}x\right)$, which is $\left(\frac{6}{5x}\right)$.

Now, multiply the expression by the reciprocal:

$$\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \times \left(\frac{6}{5x}\right)$$

Now, perform the multiplication for each term:

$$1.\left(-\frac{5}{2}x^2\right) \times \left(\frac{6}{5x}\right) = \left(-\frac{5}{2}\right) \times \left(\frac{6}{5}\right)x = -3x$$

$$2.\left(\frac{5}{3}xy\right) \times \left(\frac{6}{5x}\right) = 2y$$

$$3.\left(\frac{10}{3}x\right) \times \left(\frac{6}{5x}\right) = 4$$

So, the result of $\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \div \left(\frac{5}{6}x\right)$ is:

$$-3x + 2y + 4$$

Then the correct option is B)

5. Correct option: (D)

Given expression $\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \div \left(\frac{3}{8}ax\right)$

Find the reciprocal of $\left(\frac{3}{8}ax\right)$, which is $\left(\frac{8}{3ax}\right)$.

Now, multiply the expression by the reciprocal: $\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \times \left(\frac{8}{3ax}\right)$

Now, perform the multiplication for each term:

$$\frac{1}{4}a^2x \times \frac{8}{3ax} - \frac{1}{10}abx \times \frac{8}{3ax} - \frac{3}{8}acx \times \frac{8}{3ax} = \frac{2}{3}a - \frac{4}{15}b - c$$

So, the result of

$$\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \div \left(\frac{3}{8}ax\right)$$

is: $\frac{2}{3}a - \frac{4}{15}b - c$

Then the correct option is D)

6. Correct option: (C)

To simplify the expression $(-48x^6 - 6x^3) \div (-12x^2)$, we can divide each term in the numerator by the denominator.

1. Divide $-48x^6$ by $-12x^2$:

$$\frac{-48x^6}{-12x^2} = 4x^{6-2} = 4x^4$$

2. Divide $-6x^3$ by $-12x^2$:

$$\frac{-6x^3}{-12x^2} = \left(\frac{1}{2}\right)x^{3-2} = \left(\frac{1}{2}\right)x$$

So, the simplified expression is: $4x^4 + \left(\frac{1}{2}\right)x$

Then the correct option is C)

7. Correct option: (A)

To simplify the expression $(x^2 - 11x + 30) \div (x - 6)$, we can use factorization.

We can also simplify the expression by factoring the numerator:

$x^2 - 11x + 30$ can be factored as $(x - 5)(x - 6)$.

Now, we can cancel out the common factor $(x - 6)$ in the numerator and denominator:

$$\frac{[(x - 5)(x - 6)]}{x - 6} = x - 5$$

So, the simplified expression is $x - 5$.

Then the correct option is A)

8. Correct option: (C)

To simplify the expression $(12a^2 + 10a - 2) \div (6a - 1)$, We can use polynomial long division

Polynomial Long Division Perform polynomial long division by dividing each term of the numerator by the denominator:

$$\begin{array}{r} 2a + 2 \\ \hline 6a - 1 \;|\; 12a^2 + 10a - 2 \\ -(12a^2 - 2a) \\ \hline 12a - 2 \\ -(12a - 2) \\ \hline 0 \end{array}$$

So, the result of the division is 2a + 2.

Then the correct option is C)

9. Correct option: (A)

In the context of multinomials, the term "prime" is not commonly used as it is for integers. The concept of primality primarily applies to integers.

In the context of integers, a number is considered prime if it is divisible by only 1 and itself (option A). However, this concept doesn't directly apply to multinomials, which are algebraic expressions with multiple terms involving variables and coefficients. Multinomials are not typically classified as prime or non-prime.

10. Correct option: (A)

The product of two polynomials, $p(x)$ and $q(x)$, can be expressed in terms of their (HCF) and least common multiple (LCM) as follows:

$$p(x) \times q(x) = [HCF(p(x), q(x))] \times [LCM(p(x), q(x))]$$

This equation expresses the product of two polynomials in terms of their HCF and LCM. This is a useful relation when dealing with polynomial factorization and simplification.

Then the correct option is A)

11. Correct option: (A)

To find the highest common factor (HCF) of two polynomials, we need to determine the highest power of the variable that is common to both polynomials.

In this case, we have the polynomials x^4 and $8x^3$.

The highest power of x that is common to both polynomials is x^3.

Therefore, the HCF of x^4 and $8x^3$ is x^3.

Then the correct option is A)

12. Correct option: (C)

1. Factorize each expression: $22h^2t = 2 \times 11 \times h \times h \times t$

$$33ht^2 = 3 \times 11 \times h \times t \times t$$

2. Now, identify the common factors:

The common factors among these expressions are 11, h, and t.

3. Calculate the HCF by multiplying these common factors:

$$HCF = 11 \times h \times t = 11ht$$

Then the correct option is C)

13. Correct option: (D)

To find the three numbers in the ratio 2:3:5 with a highest common factor (HCF) of 6, we can follow these steps:

1. Write the numbers in the form of 2x, 3x, and 5x, where x is a common multiplier.

2. Find the HCF of these numbers $HCF(2x, 3x, 5x) = x$

Equate it with the given HCF as $x = 6$

Then the numbers are $2x = 12, 3x = 18, 5x = 30$

So, the three numbers are 12, 18, and 30, which corresponds to option D) 12, 18, and 30.

14. Correct option: (B)

If three monomials are in the ratio x:y:z and their highest common factor (HCF) is xyz, then we can express the monomials as follows:

$$\text{First monomial: HCF (xyz)} \times x = x^2yz$$

$$\text{Second monomial: HCF (xyz)} \times y = xy^2z$$

$$\text{Third monomial: HCF (xyz)} \times z = xyz^2$$

So, the three monomials are $x^2yz, xy^2z,$ and xyz^2.

Then the correct option is B)

15. Correct option: (C)

To find the least common multiple (LCM) of the two expressions abc and cd, we'll need to factorize each expression and then determine the LCM based on their prime factors.

1. Factorize the expressions: $abc = a \times b \times c, cd = c \times d$

2. Determine the LCM by taking the highest power of each prime factor:

$$\text{LCM} = a \times b \times c \times d$$

So, the LCM of abc and cd is abcd.

Then the correct option is C)

16. Correct option: (D)

1. Factorize each expression:

$3x^2y^2z^2$: Prime factors include 3, x, y, and z.

$4xy^2z^4$: Prime factors include 2, x, y, and z.

2. Now, identify the prime factors and their respective highest powers:

– LCM will have the prime factors 2, 3, x, y, and z.

– The highest power of 2 is $2^2 = 4$.

– The highest power of 3 is $3^1 = 3$.

– The highest power of x is x^2.

– The highest power of y is y^2.

– The highest power of z is z^4.

3. Calculate the LCM by multiplying these highest powers:

$$\text{LCM} = 4 \times 3 \times x^2 \times y^2 \times z^4$$

Then the correct option is D)

17. Correct option: (C)

1. Factorize each expression:

$7p^3q^4$: Prime factors include 7, p^3, and q^4.

$8pq^5$: Prime factors include 2^3, p, and q^5.

$2p^6q^2$: Prime factors include 2, p^6, and q^2.

2. Now, identify the prime factors and their respective highest powers:

– LCM will have the prime factors 2^3, 7, p^6, and q^5.

– The highest power of 2 is 2^3.

– The highest power of 7 is $7^1 = 7$.

– The highest power ofp is p^6.

– The highest power of q is q^5.

3. Calculate the LCM by multiplying these highest powers:

$$\text{LCM} = 2^3 \times 7 \times p^6 \times q^5 = 56p^6q^5$$

Then the correct option is C)

18. Correct option: (B)

1. Expand each expression: $(x + y)(x - y)$

$(x + y)^2 = (x + y)(x + y)$

$(x - y)^2 = (x - y)(x - y)$

2. Identify the prime factors and their respective highest powers:

highest power of the $(x + y)$is 2

highest power of the $(x - y)$is 2

3. Calculate the LCM by multiplying these highest powers: $\text{LCM} = (x - y)^2(x + y)^2$

Then the correct option is B)

19. Correct option: (B)

To find the least common multiple (LCM) and highest common factor (HCF) of the two expressions $A = (x + y)x$ and $B = (x + y)y$, we can start by analyzing their factors and expressions:

1. $A = x + y$

Factors: $x, (x + y)$

2. $B = (x + y)y$

Factors: $(x + y), y$

Now, let's find the LCM and HCF:
To find the LCM, we need to take the highest power of each factor that appears in both expressions.

$$\text{LCM} = (x + y)xy$$

To find the HCF, we need to take the lowest power of each common factor. In this case, the common factors are $x + y$.

$$\text{HCF} = x + y$$

So, the LCM of A and B is $(x + y)xy$, and the HCF of A and B is $x + y$.
Then the correct option is B)

20. Correct option: (A)

We can use the relationship between LCM, HCF, and the product of two numbers to find the unknown monomial: $\text{LCM} * \text{HCF} = \text{Product of the two numbers}$
Plugging in the given values: $(12a^2b^4c) \times (a^2b^2) = (12a^2b^2c) \times \text{Monomial}_2$
Now, simplify:

$$12a^2b^4c \times \frac{a^2b^2}{12a^2b^2c} = \text{Monomial}_2 = a^2b^4$$

Then the correct option is A)

LEVEL-II

1. Correct option: (C)

To find the highest common factor (HCF) of the two expressions we can compare their prime factors and take the lowest power of each common factor.
First, let's factorize both expressions:

$$A = 2^7 \times 3 \times x^4 \times y^5 \times z^3$$
$$B = 2^3 \times 3^3 \times x^2 \times y^3 \times z^3$$

Now, identify the common factors and their lowest powers:

$$\text{Common factors: } 2, 3, x, y, z$$
$$\text{HCF} = 2^3 \times 3 \times x^2 \times y^3 \times z^3 = 24x^2y^3z^3$$

So, the highest common factor (HCF) of $A = 384x^4y^5z^3$ and $B = 216x^2y^3z^3$ is $24x^2y^3z^3$.
Then the correct option is C)

2. Correct option: (B)

To factor the expression $2x(x + y) + 3(x + y)$, we can factor out the common binomial factor $(x + y)$:

$$2x(x + y) + 3(x + y) = (x + y)(2x + 3)$$

So, the binomial factor of the expression is $(x + y)$.
Then the correct option is B)

3. Correct option: (A)

To find the monomials B and C when the highest common factor (HCF) is given as $3^4x^2a^3b^7$, we need to express B and C in a way that their HCF matches the given value.
Let's break down the HCF into its prime factors:

$$\text{HCF} = 3^4 \times x^2 \times a^3 \times b^7$$

Now, let's look at the answer choices:

$$\text{A) } 324x^2a^5b^9, 405x^4a^3b^7$$
$$\text{B) } 324x^2a^5b^9, 405x^4ab^4$$
$$\text{C) } 324x^2a^2b^3, 405x^4ab^4$$
$$\text{D) } 324x^2a^2b^3, 405x^4ab^4$$

We can see that option A has the correct prime factors to match the HCF:

$$B = 324x^2a^5b^9$$
$$C = 405x\text{^}4a^3b^7$$

Then the correct option is A)

4. Correct option: (B)

To find the least common multiple (LCM) and highest common factor (HCF) of two numbers or expressions, we can compare their prime factors.

Let's start by factoring both A and B:

$$A = 2^3 \times 3^2 \times x^2 \times y^2 \times z^2$$
$$B = 2^4 \times 3 \times x \times y \times z$$

Now, let's calculate the LCM and HCF:

LCM (Least Common Multiple):

To find the LCM, we need to take the highest power of each prime factor that appears in both A and B:

$$\text{For 2, the highest power is } 2^4.$$
$$\text{For 3, the highest power is } 3^2.$$
$$\text{For x, the highest power is } x^2.$$
$$\text{For y, the highest power is } y^2.$$
$$\text{For z, the highest power is } z^2.$$

So, the LCM is: $LCM = 2^4 3^2 x^2 y^2 z^2$

HCF (Highest Common Factor):

To find the HCF, we need to take the lowest power of each common prime factor:

$$\text{For 2, the lowest power is } 2^3.$$
$$\text{For 3, the lowest power is 3.}$$
$$\text{For x, the lowest power is x.}$$
$$\text{For y, the lowest power is y.}$$
$$\text{For z, the lowest power is z.}$$

So, the HCF is: $HCF = 2^3 \times 3 \times x \times y \times z$

Now, you have the LCM and HCF:

$$LCM = 2^4 \times 3^2 \times x^2 \times y^2 \times z^2$$
$$HCF = 2^3 \times 3 \times x \times y \times z$$

Then the correct option is B)

5. Correct option: (C)

To find the product of two polynomials when their least common multiple (LCM) and highest common factor (HCF) are given, we can use the following relationship:

$$\text{Product of polynomials} = LCM \times HCF$$

In this case, the LCM is given as $6x^2y^2$, and the HCF is given as $2xy^2$.

Now, multiply the LCM and HCF to find the product of the polynomials:

$$\text{Product} = (6x^2y^2) \times (2xy^2)$$

Multiply the coefficients and combine the like terms:

$$\text{Product} = (6 \times 2) \times (x^2 \times x) \times (y^2 \times y^2)$$
$$\text{Product} = 12x^3y^4$$

So, the product of the two polynomials is $12x^3y^4$.

Then the correct option is C)

6. Correct option: (C)

To find the product of the expressions $(m + 2)(m + 5)(m + 3)$, we can use the distributive property to multiply them step by step:

1. First, multiply the first two expressions:

$(m + 2)(m + 5) = m(m) + m(5) + 2(m) + 2(5) = m^2 + 5m + 2m + 10 = m^2 + 7m + 10$

2. Now, multiply the result from step 1 by the third expression (m+3):

$$(m^2 + 7m + 10)(m + 3) = m^2(m) + m^2(3) + 7m(m) + 7m(3) + 10(m) + 10(3)$$
$$= m^3 + 3m^2 + 7m^2 + 21m + 10m + 30$$
$$= m^3 + 10m^2 + 31m + 30$$

So, the product of the expressions $(m + 2)(m + 5)(m + 3)$ is equal to:

$$m^3 + 10m^2 + 31m + 30$$

Then the correct option is C)

7. Correct option: (A)

To find the product of the expressions $(2x^2 + 3x + 4)((2x^2)^2 + (3x)^2 + (4)^2 - 6x^3 - 12x - 8x^2)$, we can simplify the inner expression first:

After solving the inner expression

$$4x^4 + 9x^2 + 16 - 6x^3 - 12x - 8x^2$$

Next, let's combine like terms in the inner expression:

$$4x^4 - 6x^3 + x^2 - 12x + 16$$

Now, let's multiply this simplified inner expression by $(2x^2 + 3x + 4)$:

$$(2x^2 + 3x + 4)(4x^4 - 6x^3 + x^2 - 12x + 16)$$

To find the product, we can use the distributive property to multiply each term in the first expression by each term in the second expression and then combine like terms:

First, distribute $(2x^2 + 3x + 4)$ to each term in the second expression:

$$(2x^2)(4x^4) + (2x^2)(-6x^3) + (2x^2)(x^2) + (2x^2)(-12x) + (2x^2)(16) +$$
$$(3x)(4x^4) + (3x)(-6x^3) + (3x)(x^2) + (3x)(-12x) + (3x)(16) +$$
$$(4)(4x^4) + (4)(-6x^3) + (4)(x^2) + (4)(-12x) + (4)(16)$$

Now, calculate each of these products:

$$8x^6 - 12x^5 + 2x^4 - 24x^3 + 32x^2 +$$
$$12x^5 - 18x^4 + 3x^3 - 36x^2 + 48x +$$
$$16x^4 - 24x^3 + 4x^2 - 48x + 64$$

Now, combine like terms:

$$8x^6 - 45x^3 + 64$$

Now, simplify further:

So, the product of the expressions is $8x^6 - 45x^3 + 64$

Then the correct option is A)

8. Correct option: (A)

To simplify the expression $\frac{a^9 - b^9}{a^3 - b^3}$, we can use the formula for the difference of cubes, which states that: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

So, the expression becomes:

$$\frac{a^9 - b^9}{a^3 - b^3} = \frac{[(a^3)^3 - (b^3)^3]}{a^3 - b^3}$$

Now, notice that we have a difference of cubes in the numerator, which we can factor:

$$(a^3)^3 - (b^3)^3 = (a^3 - b^3)(a^6 + a^3b^3 + b^6).$$

So, the expression becomes:

$$\frac{[(a^3 - b^3)(a^6 + a^3b^3 + b^6)]}{[a^3 - b^3]}$$

Now, we can cancel the common factor $(a^3 - b^3)$ from the numerator and the denominator:

$$\frac{[(a^3 - b^3)(a^6 + a^3b^3 + b^6)]}{[a^3 - b^3]} = (a^6 + a^3b^3 + c^6)$$

So, the simplified expression is: $a^6 + a^3b^3 + b^6$.

Then the correct option is A)

9. Correct option: (C)

To find the highest common factor (HCF) of two numbers or expressions, we can compare their prime factorizations and take the lowest power of each common prime factor.

Let's factorize both A and B:

$$A = 384x^4y^5z^3$$
$$B = 256x^2y^3z^3$$

Now, let's find the HCF:

- For the numbers 384 and 256, we can find their common factors:

$$384 = 2^7 \times 3$$
$$256 = 2^8$$

So, the common factor for both is 2^7.

For x, the common factor is x^2 (the lowest power).

For y, the common factor is y^3 (the lowest power).

For z, the common factor is z^3 (the lowest power).

Now, multiply all these common factors together: $HCF = 2^7 \times x^2 \times y^3 \times z^3$

So, the HCF of A and B is $2^7 \times x^2 \times y^3 \times z^3$.

Then the correct option is C)

10. Correct option: (C)

1. Factorize each expression: $22h^2t = 2 \times 11 \times h \times h \times t$

$$33ht^2 = 3 \times 11 \times h \times t \times t$$

2. Now, identify the common factors:

The common factors among these expressions are 11, h, and t.

3. Calculate the HCF by multiplying these common factors:

$$HCF = 11 \times h \times t = 11ht$$

Then the correct option is C)

11. Correct option: (A, C)

Let's analyze the given expressions: $1.\ 14x^2y^2$

$2.\ 21x^2y^2$

$3.\ 35x^4y^5z$

Numerical Coefficients:

- The numerical coefficients are 14, 21, and 35.

Literals (variables and their exponents):

- The literals for the first two expressions are x^2y^2.
- The literal for the third expression is x^4y^5z.

Now, let's consider each statement:

A) The HCF of numerical coefficients is 7.

- This statement is true because the numerical coefficients 14, 21, and 35 have a common factor of 7.

B) The HCF of numerical coefficients is 35.

- This statement is not true. The highest common factor (HCF) of the numerical coefficients is 7, not 35.

C) The HCF of literals is x^2y^2.

- This statement is true because the first two expressions have the common literal factor x^2y^2.

D) The HCF of literals is x^4y^5z.

- This statement is not true. The HCF of literals for the given expressions is x^2y^2, not x^4y^5z.

So, the correct statements are A) The HCF of numerical coefficients is 7 and C) The HCF of literals is x^2y^2.

12. Correct option: (C, D)

Let's analyze the given expressions:

$$1.\ 12x^2y^3z$$
$$2.\ 6xy^2z^4$$
$$3.\ 8x^4yz^3$$

Numerical Coefficients: - The numerical coefficients are 12, 6, and 8.

Literals (variables and their exponents):

- The literals for the first expression are x^2y^3z.
- The literals for the second expression are xy^2z^4.
- The literals for the third expression are x^4yz^3.

Now, let's consider each statement:

A) The HCF of numerical coefficients is 6.

- This statement is false because the highest common factor (HCF) of 12, 6, and 8 is indeed 2.

B) The LCM of numerical coefficients is 36.

- This statement is not true. The least common multiple (LCM) of the numerical coefficients 12, 6, and 8 is 24, not 36.

C) The HCF of literals is xyz.

- This statement is true because the HCF of the literals $x^2y^3z, xy^2z^4,$ and x^4yz^3 is xyz.

D) The LCM of literals is $x^4y^3z^4$.

- This statement is true because the LCM of the literals $x^2y^3z, xy^2z^4,$ and x^4yz^3 is $x^4y^3z^4$.

So, the correct statements are C) The HCF of literals is xyz, and D) The LCM of literals is $x^4y^3z^4$.

13. Correct option: (A, C)

To determine what expression divides evenly into the given expression $a^2 + b^2 + 2b$, we can factor the expression further to find its divisors.

$a^2 + b^2 + 2b$ Can be factored as follows: $a^2 + b^2 + 2b = (a+b)^2$

Now $a^2 + b^2 + 2b = (a+b)^2$

We can say that

$$\frac{(a+b)^2}{a+b} = (a+b), \frac{(a+b)^2}{(a+b)^2} = 1$$

Therefore, the expression $a^2 + b^2 + 2ab$ is divide by $(a+b)^2$, $(a+b)$.

14. Correct option: (A)

To find the highest common factor (HCF) of the two polynomials $g(x) = 15x^3y^3z$ and $f(x) = 25xy^3z^2$, we can use the relationship between the LCM (Least Common Multiple), HCF, and the product of the two polynomials:

$$\text{LCM} \times \text{HCF} = \text{Product of the two polynomials}$$

Given that the LCM is $75x^3y^3z^2$, we can rearrange the equation to find the HCF:

$$\text{HCF} = \frac{\text{Product of the two polynomials}}{\text{LCM}}$$

First, calculate the product of the two polynomials:

$$\text{Product} = g(x) \times f(x) = (15x^3y^3z) \times (25xy^3z^2) = 375x^4y^6z^3$$

Now, divide the product by the LCM to find the HCF:

$$\text{HCF} = (375x^4y^6z^3)/(75x^3y^3z^2)$$

Now, simplify:

$$\text{HCF} = 5x^{4-3} \times y^{6-3} \times z^{3-2} = 5xy^3z$$

So, the highest common factor (HCF) of $g(x) = 15x^3y^3z$ and $f(x) = 25xy^3z^\wedge2$ is $5xy^3z$.

Then the correct option is A)

15. Correct option: (B)

We know that $\text{LCM} \times \text{HCF} = \text{Product of the two monomials}$

In this case, the HCF is given as $5p^2q^3$. The two monomials are:

$$1.\ 15p^3q^4$$
$$2.\ 20m^2p^2q^3$$

Now, we can use the formula to find the LCM:

$$\text{LCM} \times \text{HCF} = (15p^3q^4) \times (20m^2p^2q^3)$$

Now, divide both sides by the HCF to find the LCM:

$$\text{LCM} = ((15p^3q^4) \times (20m^2p^2q^3))/(5p^2q^3)$$

Now, simplify:

$$\text{LCM} = \frac{300m^2p^5q^7}{5p^2q^3} = 60m^2p^3q^4$$

So, the least common multiple (LCM) of $15p^3q^4$ and $20m^2p^2q^3$, given their HCF is $5p^2q^3$, is $60m^2p^3q^4$.

Then the correct option is B)

16. Correct option: (A)

We know that $\text{LCM} \times \text{HCF} = \text{Product of the two monomials}$

In this case, we have: $\text{LCM} = 36k^3m^2n^5$

$$\text{HCF} = 216k^3m^3n^5$$
$$\text{One of the monomials} = 72k^2m^2n^4$$

Now, we can use the formula to find the other monomial:

$$\text{Other monomial} = \frac{36k^3m^2n^5 \times 216k^3m^3n^5}{72k^2m^2n^4}$$

Cancel common factors: $= 108k^4m^3n^6$

So, the other monomial is $= 108k^4m^3n^6$

Then the correct option is A)

Given $A = 2x^2 - 3x + 5, B = 5x + 2, C = x^2 + 3x - 7$

17. Correct Matching is: a-Q,b-T,c-S,d-P

a)

Find the LCM by taking the highest power of each factor that appears in both expressions:

- The LCM of $x^3 \times y^2$ and $-5 \times y^2$ is: $\text{LCM} = x^3 \times y^2 \times (5)$

So, the LCM of x^3y^2 and $-5y^2$ is $5x^3y^2$.

a) -→(Q)

b)

The least common multiple (LCM) of $4x^3, 6y^2,$ and $8z^2$ is:

$$\text{LCM}(4x^3, 6y^2, 8z^2) = \text{LCM}(2^2x^3, 2 \times 3y^2, 2^3z^2) = 2^3 \times 3x^3y^2z^2$$

So, the LCM is $24x^3y^2z^2$.

b) -→(T)

c)

The least common multiple (LCM) of $2x^3y^3, 10y^2x^3,$ and $18xy$ is:

$$\text{LCM}(2x^3y^3, 10y^2x^3, 18xy) = \text{LCM}(2x^3y^3, 2 \times 5y^2x^3, 2 \times 3^2xy) = 2 \times 5 \times 3^2x^3y^3$$

So, the LCM is $90x^3y^3$.

c) -→(S)

d)

The least common multiple (LCM) of $25x^3, 15x^2,$ and $45x^3$ is:

$$\text{LCM}(5^2x^3, 3 \times 5x^2, 3^2 \times 5x^3) = 225x^3$$

So, the LCM is $225x^3$.

d) -→(P)

18. Correct Answer is: (6)

To find the least common multiple (LCM) of $2x^2, 3x^4,$ and $6x^8$ when x = 1, simply substitute x = 1 into each term and calculate:

$$\text{LCM}(2x^2, 3x^4, 6x^8) = \text{LCM}(2(1)^2, 3(1)^4, 6(1)^8) = \text{LCM}(2, 3, 6)$$

Now, find the LCM of the numbers 2, 3, and 6:

$$\text{LCM}(2, 3, 6) = 6$$

So, the LCM of $2x^2, 3x^4,$ and $6x^8$ when x = 1 is 6.

19. Correct Answer is: (3)

To find the highest common factor (HCF) of $3x^2, 9y^3,$ and $3z^4$, we can identify the common factors for each term and take the lowest power of those common factors:

only common factor is 3

$$\text{HCF}(3x^2, 9y^3, 3z^4) = 3$$

So, the HCF is 3.

20. Correct Answer is: (7)

If we want to find the quotient of $(3x^2 - 2x - 1)$ when divided by (x - 1) and x = 2, we can substitute the value of x into the expression: $\frac{3x^2 - 2x - 1}{x - 1}$

Now, substitute x = 2:

$$\frac{3(2)^2 - 2(2) - 1}{2 - 1}$$

$$\frac{3(4) - 2(2) - 1}{1}$$

Simplify the numerator: $\frac{12 - 4 - 1}{1} = 7$

The quotient is 7.

STUDENT PRACTICE QUESTIONS

1. Correct option: (A)

Given expression $(10x^7 - 8x^6 + 3x^4) \div x^3$

Divide each term separately:

$$\frac{10x^7}{x^3} - \frac{8x^6}{x^3} + \frac{3x^4}{x^3} = 10x^4 - 8x^3 + 3x$$

So, the result of $(10x^7 - 8x^6 + 3x^4) \div x^3$ is:

$$10x^4 - 8x^3 + 3x$$

Then the correct option is A)

2. Correct option: (A)

To simplify the expression $(3x^2 - x - 14) \div (x + 2)$ using factorization, we can first factor the numerator and then cancel out common factors:

1. Factor the numerator $(3x^2 - x - 14)$:
   The quadratic expression $3x^2 - x - 14$ factors as $(3x - 7)(x + 2)$.
2. Now, rewrite the expression with the factored numerator:
   $[(3x - 7)(x + 2)] \div (x + 2)$
3. Cancel out the common factor (x + 2) in both the numerator and denominator:

$$\frac{(3x - 7)(x + 2)}{x + 2} = 3x - 7$$

So, the simplified expression is $3x - 7$

Then the correct option is A)

3. Correct option: (A)

To simplify the expression $(15a^2 + 17ax - 4x^2) \div (3a + 4x)$ using factorization, we can first factor the numerator and then cancel out common factors:

1. Factor the numerator $(15a^2 + 17ax - 4x^2)$:
   The quadratic expression $15a^2 + 17ax - 4x^2$ factors as $(3a + 4x)(5a - x)$.
2. Now, rewrite the expression with the factored numerator:
$$[(3a + 4x)(5a - x)] \div (3a + 4x)$$
3. Cancel out the common factor $(3a + 4x)$ in both the numerator and denominator:
$$\frac{(3a + 4x)(5a - x)}{3a + 4x} = 5a - x$$

So, the simplified expression is $5a - x$.

Then the correct option is A)

4. Correct option: (D)

To find the highest common factor (HCF) of the two polynomials pq and qrs, we can identify the common factors in each term and then determine the HCF:

1. Break down each polynomial into its prime factors:
$$pq = p \times q$$
$$qrs = q \times r \times s$$
2. Now, identify the common factors:
   The only common factor between pq and qrs is q.

Then the correct option is D)

5. Correct option: (B)

1. Factorize each expression:
$$4x^2y = 2 \times 2 \times x \times x \times y$$
$$6xy = 2 \times 3 \times x \times y$$
$$8x^2y^2 = 2 \times 2 \times 2 \times x \times x \times y \times y$$
2. Now, identify the common factors:
   The common factors among these expressions are 2, x, and y.
3. Calculate the HCF by multiplying these common factors:
$$HCF = 2 \times x \times y = 2xy$$

Then the correct option is B)

6. Correct option: (D)

If three monomials are in the ratio $xy : yz : zx$ and their highest common factor (HCF) is xyz, we can express the monomials as follows:

1. First monomial: $HCF\ (xyz) \times (xy) = xyz \times xy = x^2y^2z$
2. Second monomial: $HCF\ (xyz) \times (yz) = xyz \times yz = xy^2z^2$
3. Third monomial: $HCF\ (xyz) \times (zx) = xyz \times zx = x^2yz^2$

Then the correct option is D)

7. Correct option: (C)

To find the least common multiple (LCM) of the three expressions $xy, yz,$ and xz, we can determine the LCM based on their prime factors.

1. Factorize each expression:
   xy: Prime factors include x and y.
   yz: Prime factors include y and z.
   xz: Prime factors include x and z.
2. Calculate the LCM by multiplying these prime factors:
$$LCM = x \times y \times z$$

Then the correct option is C)

8. Correct option: (A)

1. Factorize each expression:

$120x^2y^3$: Prime factors include $2, 3, 5, x^2$, and y^3.

$36x^6y^8$: Prime factors include $2, 3, x^6$, and y^8.

2. Now, identify the prime factors and their respective highest powers:

– LCM will have the prime factors $2, 3, 5, x^6$, and y^8.

– The highest power of 2 is $2^3 = 8$.

– The highest power of 3 is $3^1 = 3$.

– The highest power of 5 is $5^1 = 5$.

– The highest power of x is x^6.

– The highest power of y is y^8.

3. Calculate the LCM by multiplying these highest powers:

$\text{LCM} = 8 \times 3 \times 5 \times x^6 \times y^8 = 120x^6y^8$

Then the correct option is A)

9. Correct option: (A)

1. Factorize each expression:

a^2bc: Prime factors include a^2, b, c.

ab^2c: Prime factors include a, b^2, c.

abc^2: Prime factors include a, b, c^2.

2. Now, identify the prime factors and their respective highest powers:

– LCM will have the prime factors a^2, b^2, and c^2.

– The highest power of a is a^2.

– The highest power of b is b^2.

– The highest power of c is c^2.

3. Calculate the LCM by multiplying these highest powers:

$$\text{LCM} = a^2 \times b^2 \times c^2$$

Then the correct option is A)

10. Correct option: (C)

Given that the least common multiple (LCM) of two monomials is $8x^2y^2$ and the highest common factor (HCF) is $2xy$, and we already have one of the monomials as $4xy^2$, let's find the second monomial.

The LCM can be expressed as the product of the two monomials divided by their HCF:

$$\text{LCM} = \frac{\text{Monomial}_1 \times \text{Monomial}_2}{\text{HCF}}$$

Plugging in the values:

$$8x^2y^2 = \frac{4xy^2 \times \text{Monomial}_2}{2xy}$$

$$\text{Monomial}_2 = 8x^2y^2 \times \frac{2xy}{4xy^2} = 4x^2y$$

So, the second monomial is $4x^2y$.

Then the correct option is C)

11. Correct option: (A, B, C)

To determine what expression divides evenly into the given expression $a^2 - b^2$, we can factor the expression further to find its divisors.

$a^2 - b^2$ Can be factored as follows:

$$a^2 - b^2 = (a + b)(a - b)$$

Now we can say that

$$=> \frac{(a + b)(a - b)}{a + b} = (a - b), \frac{(a + b)(a - b)}{a - b} = (a + b)$$

$$\frac{(a+b)(a-b)}{(a+b)(a-b)} = 1$$

Therefore, the expression $a^2 - b^2$ is divide by
$(a+b), (a-b), (a+b)(a-b)$.

12. Correct option: (A, B, C ,D)

To determine what expression divides evenly into the given expression $a^4 - b^4$, we can factor the expression further to find its divisors.
$a^4 - b^4$ Can be factored as follows:

$$a^4 - b^4 = (a^2 + b^2)(a^2 - b^2)$$

Further we can factor this as

$$(a^2 + b^2)(a^2 - b^2) = (a^2 + b^2)(a - b)(a + b)$$

Now we can say that

$$=> \frac{(a^2 + b^2)(a+b)(a-b)}{a+b} = (a^2 + b^2)(a-b)$$

$$\frac{(a^2 + b^2)(a+b)(a-b)}{a-b} = (a^2 + b^2)(a+b)$$

$$\frac{(a^2 + b^2)(a+b)(a-b)}{(a+b)(a-b)} = (a^2 + b^2)$$

$$\frac{(a^2 + b^2)(a+b)(a-b)}{(a^2 + b^2)} = (a+b)(a-b)$$

Therefore, the expression $a^4 - b^4$ is divide by
$(a+b), (a-b), (a+b)(a-b), (a^2 + b^2)$

13. Correct option: (B)

To find the product of A and B, we need to multiply the expressions A and B:

$$A \times B = (2x^2 - 3x + 5) \times (5x + 2)$$

To calculate the product, we can use the distributive property and multiply each term in A by each term in B: $A \times B = (2x^2 \times 5x) + (2x^2 \times 2) + (-3x \times 5x) + (-3x \times 2) + 25x + 10$
Now, simplify each term: $A \times B = 10x^3 + 4x^2 - 15x^2 - 6x + 25x + 10$
Combine like terms: $A \times B = 10x^3 - 11x^2 + 19x + 10$
So, the product of A and B is $10x^3 - 11x^2 + 19x + 10$
Then the correct option is B)

14. Correct option: (C)

To find the product of B and C, we can simply multiply the two polynomials together:

$$B \times C = (5x + 2)(x^2 + 3x - 7)$$

Use the distributive property to perform the multiplication:

$$= 5x \times (x^2 + 3x - 7) + 2 \times (x^2 + 3x - 7)$$
$$= 5x^3 + 15x^2 - 35x + 2x^2 + 6x - 14$$

Simplify each term: $= 5x^3 + (15x^2 + 2x^2) + (-35x + 6x) - 14$

$$= 5x^3 + 17x^2 - 29x - 14$$

So, the product of B and C is $5x^3 + 17x^2 - 29x - 14$.
Then the correct option is C)

15. Correct Matching is: a-S,b-T,c-Q,d-P

a)

To find the highest common factor (HCF) of x^3y^2 and $-5y^2$, follow these steps:
1. Identify the common factors in both expressions.

$$\text{common factor}(y)$$

2. Determine the HCF by taking the lowest power of each common factor.

$$\text{HCF}(x^3y^2, -5y^2) = y^2$$

The HCF of x^3y^2 and $-5y^2$ is y^2.

a) -→(S)

b)

The highest common factor (HCF) of $4x^3, 6y^2,$ and $8z$ is: $\text{HCF}(4x^3, 6y^2, 8z) = 2$

So, the HCF is 2.

b) -→(T)

c)

The highest common factor (HCF) of $2x^3y^3$, $10x^3y^2,$ and $14xy$ is:

$$\text{HCF}(2x^3y^3, 10x^3y^2, 14xy) = 2xy$$

So, the HCF is 2xy.

c) -→(Q)

d)

The highest common factor (HCF) of $5x^3, -15x^2,$ and $45x$ is: $\text{HCF}(5x^3, -15x^2, 45x) = 5x$

So, the HCF is 5x.

d) -→(P)

16. Correct Answer is: (4)

To find the least common multiple (LCM) of $abc, 2a^2b,$ and $4a^3b^2c$ when a = 1, b = 1, and c = 1, simply substitute these values into the expressions and calculate the LCM:

$$\text{LCM}(abc, 2a^2b, 4a^3b^2c) = \text{LCM}(1, 2 \times 1, 4 \times 1)$$

Now, calculate the LCM:

$$\text{LCM}(1, 2, 4) = 4$$

So, the LCM of $abc, 2a^2b,$ and $4a^3b^2c$ when a = 1, b = 1, and c = 1 is 4.

17. Correct Answer is: (5)

To find the highest common factor (HCF) of $35x^2y, 5xy^3,$ and $25x^2y^2$ when x = 1 and y = 1, simply substitute these values into the expressions and calculate the HCF:

$$\text{HCF}(35x^2y, 5xy^3, 25x^2y^2) = \text{HCF}(35,5,25)$$

Now, calculate the HCF: $\text{HCF}(35, 5, 25) = 5$

So, the HCF of $35x^2y, 5xy^3,$ and $25x^2y^2$ when x = 1 and y = 1 is 5.

18. Correct Answer is: (9)

To divide $153x^3y^2$ by $17x^3y^2$, we can simply cancel out the common factors in the numerator and denominator:

$$\frac{153x^3y^2}{17x^3y^2}$$

Now, cancel out the common factors:

$$\frac{153x^3y^2}{17x^3y^2} = \frac{153}{17} = 9$$

So, the result of dividing $153x^3y^2$ by $17x^3y^2$ is 9.

19. Correct Answer is: (0)

First, find the product of the expressions $(1 + x)(1 - x)(1 + x^2)$:

Now, use the difference of squares formula to simplify the first two factors:

$$(1 + x)(1 - x) = 1^2 - x^2 = 1 - x^2$$

So, the expression becomes:

$$(1 - x^2)(1 + x^2)$$

Now, apply the difference of squares formula again:

$$(1 - x^2)(1 + x^2) = 1^2 - x^4 = 1 - x^4$$

Now, we have the simplified expression:

$$1 - x^4$$

Now, put x = 1 into the simplified expression:

$$1 - (1)^4 = 1 - 1 = 0$$

So, when we put x = 1 into the expression $(1 + x)(1 - x)(1 + x^2)$, we get the result 0.

20. Correct Answer is: (3)

To divide $x(3x^2 - 108)$ by $(x^3 - 36x)$, follow these steps:

1. Factor both the numerator and denominator.

Numerator: $x(3x^2 - 108) = 3x^3 - 108x$

Denominator: $x^3 - 36x = x(x^2 - 36) = x(x - 6)(x + 6)$

2. Rewrite the division problem with the factored expressions:

$$\frac{3x^3 - 108x}{[x(x - 6)(x + 6)]}$$

3. Now, divide the numerator by the denominator. We can cancel common factors:

$$\frac{3x^3 - 108x}{[x(x - 6)(x + 6)]} = \frac{3x(x-6)(x+6)}{x(x-6)(x+6)} = 3$$

So, $\frac{x(3x^2 - 108)}{x^3 - 36x}$ simplifies to 3.

Factorization And Algebraic Identities

LEVEL-I

1. Correct option: (C)

To find the value of $a^2 + b^2$ when ab = 6 and a + b = 5, we can use the identity:

$$(a + b)^2 = a^2 + b^2 + 2ab$$

We already know that a + b = 5 and ab = 6, so we can substitute these values into the identity:

$$(5)^2 = a^2 + b^2 + 2(6)$$

Simplify:

$$25 = a^2 + b^2 + 12$$
$$25 - 12 = a^2 + b^2$$
$$13 = a^2 + b^2$$

So, the value of $a^2 + b^2$ is 13.

Then the correct option is C)

2. Correct option: (C)

To find the value of $x^4 + \frac{1}{x^4}$ given that $x + \frac{1}{x} = 5$, we can use the relationship between the sum of squares and powers of x:

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}$$

We know that $x + \frac{1}{x} = 5$, so we can substitute this value into the equation:

$$(5)^2 = x^2 + 2 + \frac{1}{x^2}$$
$$25 = x^2 + 2 + \frac{1}{x^2}$$
$$25 - 2 = x^2 + \frac{1}{x^2}$$
$$23 = x^2 + \frac{1}{x^2}$$

Now, square both sides of the equation:

$$(23)^2 = \left(x^2 + \frac{1}{x^2}\right)^2$$

$$529 = x^4 + 2 + \frac{1}{x^4}$$

$$529 - 2 = x^4 + \frac{1}{x^4}$$

$$527 = x^4 + \frac{1}{x^4}$$

So, the value of $x^4 + \frac{1}{x^4}$ is 527.

Then the correct option is C)

3. Correct option: (A)

Start with the equation $x^2 + \frac{1}{x^2} = 18$.

$$x^2 + \frac{1}{x^2} + 2 = 18 + 2$$

$$x^2 + \frac{1}{x^2} + 2 = 20$$

Now, take the square root of both sides:

$$\sqrt{x^2 + \frac{1}{x^2} + 2} = \sqrt{20}$$

$$\sqrt{\left(x + \frac{1}{x}\right)^2} = \sqrt{20} = x + \frac{1}{x}$$

So, the value of $x + \frac{1}{x}$ is$\sqrt{20}$.

Then the correct option is A)

4. Correct option: (B)

To calculate $(980)^2 - (20)^2$, we can use the difference of squares formula:

$$a^2 - b^2 = (a + b)(a - b)$$

In this case, a = 980 and b = 20:

$$(980)^2 - (20)^2 = (980 + 20)(980 - 20)$$

Now, calculate the values inside the parentheses:

$$(1000)(960) = 960000$$

So, $(980)^2 - (20)^2$ is equal to 960,000.

Then the correct option is B)

5. Correct option: (C)

To calculate 995 × 1004 using the distributive property:

$$995 \times 1004 = 995 \times (1000 + 4)$$

Now, apply the distributive property:

$$995 \times (1000 + 4) = 995 \times 1000 + 995 \times 4$$

Now, calculate each part of the expression:

$$995 \times 1000 = 995{,}000$$

$$995 \times 4 = 3{,}980$$

Now, add these two results together:

$$995{,}000 + 3{,}980 = 998{,}980$$

So, 995 ×1004 is equal to 998,980.

Then the correct option is C)

6. Correct option: (A)

To calculate 81^2 using the method:

$$(80 + 1)^2 = 80^2 + 1 + 2 \times 80$$

Now, calculate each part of the expression:

$$80^2 = 6{,}400$$
$$1 = 1$$
$$2 \times 80 = 160$$

Now, apply these values to the expression:

$$6{,}400 + 1 + 160 = 6{,}561$$

So, 81^2 is equal to 6,561.

Then the correct option is A)

7. Correct option: (C)

To factor the expression $1 - 100a^6b^4c^2$, we can use the difference of squares formula:

$$a^2 - b^2 = (a + b)(a - b)$$

In this case, we have:

$$a = 1$$
$$b = 10a^3b^2c$$

Now, apply the difference of squares formula:

$$1 - 100a^6b^4c^2 = (1 - 10a^3b^2c)(1 + 10a^3b^2c)$$

So, the factored form of $1 - 100a^6b^4c^2\ is\ (1 - 10a^3b^2c)(1 + 10a^3b^2c)$.

Then the correct option is C)

8. Correct option: (C)

In our expression, we have a similar pattern:

$$(2a + b + m + n)^2 - (a - b + c)^2$$

Now, apply the difference of squares formula:

$$[(2a + b + m + n) + (a - b + c)][(2a + b + m + n) - (a - b + c)]$$

Now, simplify each factor:

$$1.\,(2a + b + m + n) + (a - b + c) = 3a + m + n + c$$
$$2.\,(2a + b + m + n) - (a - b + c) = a + 2b + m + n - c$$

So, the simplified expression is:

$$(3a + m + n + c)(a + 2b + m + n - c)$$

Then the correct option is C)

9. Correct option: (A)

We can use the difference of squares identity to simplify the expression

$$\frac{(147.75)^2 - (117.75)^2}{265.5}$$
$$a^2 - b^2 = (a + b)(a - b)$$

In this case, let a = 147.75 and b = 117.75. So, we have:

$$\frac{(147.75)^2 - (117.75)^2}{265.5} = \frac{[(147.75 + 117.75)(147.75 - 117.75)]}{265.5}$$

Now, simplify inside the parentheses:

$$\frac{(265.5)(30)}{265.5} = 30$$

So, $\frac{(147.75)^2 - (117.75)^2}{265.5}$ is equal to 30.

Then the correct option is A)

10. Correct option: (D)

We can rewrite the middle term $(7xy)$ using sum of two terms as:

$$12x^2 + 15xy - 8xy - 10y^2$$

Now, group the terms and factor by grouping:

$$(12x^2 + 15xy) - (8xy + 10y^2)$$

Factor the GCF (Greatest Common Factor) from each group:

$$3x(4x + 5y) - 2y(4x + 5y)$$

Now, notice that both terms have a common factor of (4x + 5y). Factor it out:

$$(4x + 5y)(3x - 2y)$$

So, the factored form of $12x^2 + 7xy - 10y^2$ is (4x + 5y)(3x - 2y).

Then the correct option is D)

11. Correct option: (A)

To find a common factor of the expression $3x(x-4) - 5(x-4)$, we can notice that both terms have the factor (x-4). We can factor this out:

$$3x(x - 4) - 5(x - 4)$$

Now, factor out the common factor (x - 4):

$$(x - 4)(3x - 5)$$

So, the factors of the expression $3x(x-4) - 5(x-4) is\ (x - 4), (3x - 5)$.

Then the correct option is A)

12. Correct option: (A)

To factor the quadratic expression $3x^2 + 11x + 6$, we can look for two binomials that multiply to give you the original expression.

First, find two numbers that multiply to the product of the leading coefficient (3) and the constant term (6), which is $3 \times 6 = 18$, and add up to the coefficient of the middle term (11).

The two numbers that meet these criteria are 9 and 2, because $9 \times 2 = 18$ and 9 + 2 = 11.

Now, rewrite the middle term ($11x$) using these two numbers:

$$3x^2 + 9x + 2x + 6$$

Now, group the terms and factor by grouping:

$$(3x^2 + 9x) + (2x + 6)$$

Factor the GCF (Greatest Common Factor) from each group:

$$3x(x + 3) + 2(x + 3)$$

Now, notice that both terms have a common factor of (x + 3). Factor it out:

(x + 3)(3x + 2)

So, the factored form of $3x^2 + 11x + 6$ is $(x + 3)(3x + 2)$.

Then the correct option is A)

13. Correct option: (B)

We can rewrite the middle term (-x) using these two numbers:

$$2x^2 - 6x + 5x - 15$$

Now, we can group the terms and factor by grouping:

$$(2x^2 - 6x) + (5x - 15)$$

Now, factor the GCF (Greatest Common Factor) from each group:

$$2x(x - 3) + 5(x - 3)$$

Now, notice that both terms have a common factor of (x - 3). Factor it out:

$$(x - 3)(2x + 5)$$

So, the factored form of $2x^2 - x - 15$ is $(x - 3)(2x + 5)$.

Then the correct option is B)

14. Correct option: (B)

To factor the expression $a^2 - b + ab - a$, we can group the terms and factor by grouping:

$$a^2 - b + ab - a$$

Now, group the terms:

$$(a^2 + ab) - (a + b)$$

Now, factor the GCF (Greatest Common Factor) from each group:

$$a(a + b) - 1(a + b)$$

Now, notice that both terms have a common factor of (a + b). Factor it out:

$$(a + b)(a - 1)$$

So, the factored form of $a^2 - b + ab - a$ is $(a + b)(a - 1)$.

Then the correct option is B)

15. Correct option: (B)

To factor the expression $a^2 - b^2 - a - b$, we can use the difference of squares and factor by grouping:

$$a^2 - b^2 - a - b$$

First, notice that $a^2 - b^2$ is a difference of squares, which can be factored as $(a + b)(a - b)$:

$$= (a + b)(a - b) - (a + b)$$

Now, factor out the common factor of (a + b):

$$= (a + b)(a - b - 1)$$

So, the factored form of $a^2 - b^2 - a - b$ is $(a + b)(a - b - 1)$.

Then the correct option is B)

16. Correct option: (A)

To find the factor of given expression we follow as:

$$n^4 + 4 = n^4 + 4n^2 + 4 - 4n^2 = (n^2 + 2)^2 - (2n)^2$$

Now, we have a difference of squares, which can be factored using the difference of squares formula:

$$(a^2 - b^2) = (a + b)(a - b)$$

In this case:

$$a = n^2 + 2$$
$$b = 2n$$

So, applying the difference of squares formula:

$$(n^4 + 4) = ((n^2 + 2) + 2n)((n^2 + 2) - 2n)$$
$$= (n^2 + 2n + 2)(n^2 - 2n + 2)$$

So, the factored form of $n^4 + 4$ is $(n^2 + 2n + 2)(n^2 - 2n + 2)$.

Then the correct option is A)

17. Correct option: (D)

The expression $x^3 + 3x^2 + 3x - 7$ can be factored using the guess method.

Let's first find the roots of the polynomial:

1. By trying potential roots, we find that x = 1 is a root.

Now that we have found one root, which is x = 1

$$\begin{array}{r} x^2 + 4x + 7 \\ \hline x - 1 \,|\, x^3 + 3x^2 + 3x - 7 \\ -(x^3 - x^2) \\ \hline 4x^2 + 3x \\ -(4x^2 - 4x) \\ \hline 7x - 7 \\ -(7x - 7) \\ \hline 0 \end{array}$$

The quotient is $x^2 + 4x + 7$. We've factored the polynomial as:

$$x^3 + 3x^2 + 3x - 7 = (x - 1)(x^2 + 4x + 7)$$

The quadratic factor $x^2 + 4x + 7$ cannot be factored further using real numbers, so this is the final factorization of the polynomial as

$$(x - 1)(x^2 + 4x + 7)$$

Then the correct option is D)

18. Correct option: (B)

The value of $(a-b)^3 + (b-c)^3 + (c-a)^3$ is:

$$= (a-b)^3 + (b-c)^3 + (c-a)^3$$

Now, we can use the identity for the sum of cubes:

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - xz)$$

In this case: $x = a - b$

$$y = b - c$$

$$z = c - a$$

So, the identity becomes:

$$(a - b)^3 + (b - c)^3 + (c - a)^3 - 3(a - b)(b - c)(c - a)$$

$$= (a-b+b-c+c-a)((a-b)^2 + (b-c)^2 + (c-a)^2 - (a - b)(b - c) - (b - c)(c - a) - (a - b)(c - a)$$

$$= (0)[other\ terms] = 0$$

$$(a - b)^3 + (b - c)^3 + (c - a)^3 - 3(a - b)(b - c)(c - a) = 0$$

$$(a - b)^3 + (b - c)^3 + (c - a)^3 = 3(a - b)(b - c)(c - a)$$

So, the value of $(a-b)^3 + (b-c)^3 + (c-a)^3$ is $3(a-b)(b-c)(a-c)$

Then the correct option is B)

19. Correct option: (B)

Given the equations: $x^2 + y^2 = 29$

$$xy = 2$$

We want to find the value of $x^4 + y^4$.

First, square both sides of equation (1): $(x^2 + y^2)^2 = 29^2$

Expand the left side: $x^4 + 2x^2y^2 + y^4 = 841$

From equation (2), we know that $xy = 2$, so $(xy)^2 = 2^2 = 4$.

Now, substitute this into the equation we derived: $x^4 + 8 + y^4 = 841$

$$x^4 + y^4 = 841 - 8$$

$$x^4 + y^4 = 833$$

So, the value of $x^4 + y^4$ is 833.

Then the correct option is B)

20. Correct option: (B)

Given the equations: $3x + 5y = 11$

$$xy = 2$$

We want to find the value of $9x^2 + 25y^2$.

First, let's square equation (1): $(3x + 5y)^2 = 11^2$

Expand the left side: $9x^2 + 30xy + 25y^2 = 121$

Now, we know $xy = 2$ from equation (2). Substitute this into the equation:

$$9x^2 + 30(2) + 25y^2 = 121$$

$$9x^2 + 60 + 25y^2 = 121$$

Now, subtract 60 from both sides: $9x^2 + 25y^2 = 121 - 60$

$$9x^2 + 25y^2 = 61$$

So, the value of $9x^2 + 25y^2$ is 61.

Then the correct option is B)

LEVEL-II

1. Correct option: (B)

Given the equations:

$$x^2 + y^2 = 25$$
$$xy = 2$$

We want to find the value of $3(x+y)^2 - 2(x-y)^2$.

First, let's find expressions for $(x+y)^2$ and $(x-y)^2$:

$$(x+y)^2 = x^2 + 2xy + y^2$$
$$(x-y)^2 = x^2 - 2xy + y^2$$

Now, substitute the value of xy from equation (2):

$$(x+y)^2 = x^2 + 2(2) + y^2 = x^2 + 4 + y^2$$
$$(x-y)^2 = x^2 - 2(2) + y^2 = x^2 - 4 + y^2$$

Now, we can calculate $3(x+y)^2 - 2(x-y)^2$:

$$3(x+y)^2 - 2(x-y)^2 = 3(x^2 + 4 + y^2) - 2(x^2 - 4 + y^2)$$

Now, distribute the constants:

$$3x^2 + 12 + 3y^2 - 2x^2 + 8 - 2y^2 = x^2 + y^2 + 20$$

Now, we know that $x^2 + y^2 = 25$ from equation (1), so:

$$x^2 + y^2 + 20 = 25 + 20 = 45$$

So, the value of $3(x+y)^2 - 2(x-y)^2$ is 45.

Then the correct option is B)

2. Correct option: (A)

Given the equation:

$$(m-5)^2 + (b-4)^2 + (a-1)^2 = 0$$

We want to find the value of $m \times b \times a - 10$.

Let's work with the equation first:

$$(m-5)^2 + (b-4)^2 + (a-1)^2 = 0$$

Since all the terms on the left side of the equation are squared and non-negative, the only way for the sum to equal 0 is if each term is equal to 0 individually. Therefore:

$$(m-5)^2 = 0 \Rightarrow m = 5$$
$$(b-4)^2 = 0 \Rightarrow b = 4$$
$$(a-1)^2 = 0 \Rightarrow a = 1$$
$$m \times b \times a - 10 = 5 \times 4 \times 1 - 10 = 20 - 10 = 10$$

So, $m \times b \times a - 10$ is equal to 10.

Then the correct option is A)

3. Correct option: (B)

To simplify the expression $(2ab - bc)^2 + 4ab^2c$, first expand and then combine like terms:

$$(2ab - bc)^2 + 4ab^2c$$

Expand the square of the binomial:

$$(4a^2b^2 - 4ab^2c + b^2c^2) + 4ab^2c$$

Now, combine like terms:

$$4a^2b^2 - 4ab^2c + b^2c^2 + 4ab^2c = 4a^2b^2 + b^2c^2$$

Now, factor out the common term b^2:

$$b^2(4a^2 + c^2)$$

So, the simplified form of $(2ab - bc)^2 + 4ab^2c$ is $b^2(4a^2 + c^2)$.

Then the correct option is B)

4. Correct option: (C)

Let the two numbers be x and y. We are given two pieces of information:

$$x - y = 6$$

$$x + y = 18$$

Now, we want to find the difference of their squares, which is $(x^2 - y^2)$.

We can use the identity $a^2 - b^2 = (a + b)(a - b)$ to find the difference of their squares:

$$(x^2 - y^2) = (x + y)(x - y)$$

We already know that $x + y = 18$ from the second equation, and $x - y = 6$ from the first equation.

Now, we can calculate their difference of squares:

$$(x^2 - y^2) = (18)(6) = 108$$

So, the difference of their squares is 108.

Then the correct option is C)

5. Correct option: (A)

Let the first number be x and the second number be y. We are given two pieces of information:

1. The cube of a number is 8 times the cube of another number: $x^3 = 8y^3$

2. The sum of the cubes of the numbers is 243: $x^3 + y^3 = 243$

We can use the first piece of information to express one of the variables in terms of the other. Taking the cube root of both sides of $x^3 = 8y^3$, we get:

$$x = 2y$$

Now, substitute this expression for x into the second equation:

$$(2y)^3 + y^3 = 243$$

$$8y^3 + y^3 = 243$$

Combine like terms:

$$9y^3 = 243$$

Now, divide both sides by 9 to solve for y^3:

$$y^3 = \frac{243}{9}$$

$$y^3 = 27$$

$$y = 3$$

Now that we have found the value of y, we can find the value of x using x = 2y:

$$x = 2 \times 3$$

$$x = 6$$

So, the two numbers are $x = 6$ and $y = 3$, and the difference between them is:

$$x - y = 6 - 3 = 3$$

Then the correct option is A)

6. Correct option: (D)

The equation given is:

$$(a - 1)^2 + (b - 2)^2 + (c - 1)^2 = 0$$

In order for the sum of squares to be equal to zero, each square term must be equal to zero because the square of any real number is non-negative. Therefore:

$$(a - 1)^2 = 0 => a = 1$$

$$(b - 2)^2 = 0 => b = 2$$

$$(c - 1)^2 = 0 => c = 1$$

Now that we have found the values of a, b, and c, we can find their product:

$$a \times b \times c = 1 \times 2 \times 1 = 2$$

So, the value of $a \times b \times c$ is 2.

Then the correct option is D)

7. Correct option: (C)

Let the two numbers be x and y. We are given two pieces of information:

1. The sum of the numbers is 9: $x + y = 9$

2. The sum of their squares is 41: $x^2 + y^2 = 41$
We want to find the product of the numbers, which is $x \times y$.
We can use the identity $(x + y)^2 = x^2 + y^2 + 2xy$ to relate the sum of squares to the product of the numbers:

$$(x + y)^2 = x^2 + y^2 + 2xy$$

Plug in these values:

$$(9)^2 = 41 + 2xy$$
$$81 = 41 + 2xy$$
$$81 - 41 = 2xy$$
$$40 = 2xy$$
$$\frac{40}{2} = xy$$
$$20 = xy$$

So, the product of the numbers x and y is 20.
Then the correct option is C)

8. Correct option: (B)

We can express the middle term $(-4m)$ using these two numbers:

$$m^2 - 4m - 21 = m^2 - 7m + 3m - 21$$

Next, group the terms and factor by grouping:

$$(m^2 - 7m) + (3m - 21)$$

Now, factor out the greatest common factor from each group:

$$m(m - 7) + 3(m - 7)$$

Notice that we now have a common factor of (m - 7) in both terms. Factor it out:

$$(m - 7)(m + 3)$$

So, the factored form of $m^2 - 4m - 21$ *is* $(m - 7)(m + 3)$.
Then the correct option is B)

9. Correct option: (A)

To factor the expression $x^2 + y^2 - 4z^2 + 2xy$, we can look for groups of terms that can be factored together.
Notice that $x^2 + 2xy + y^2$ is a perfect square trinomial, which can be factored as $(x + y)^2$.
So, the expression can be rewritten as:

$$(x + y)^2 - 4z^2$$

Now, we have a difference of squares, which can be factored as the product of the sum and difference of the square roots:

$$[(x + y) + 2z][(x + y) - 2z]$$

So, the factored form of $x^2 + y^2 - 4z^2 + 2xy$ is
$[(x + y) + 2z][(x + y) - 2z]$.
Then the correct option is A)

10. Correct option: (C)

Given expression

$$39y^3(50y^2 - 98) \div 26y^2(5y + 7)$$

Solving left side of expression

$$39y^3(50y^2 - 98) = 39y^3 \times 2(25y^2 - 49) = 78y^3((5y)^2 - (7)^2)$$
$$= 78y^3(5y + 7)(5y - 7)$$

Putting back into expression

$$\frac{78y^3(5y + 7)(5y - 7)}{26y^2(5y + 7)} = 3y(5y - 7)$$

Therefore solution of the expression is $3y(5y - 7)$
Then the correct option is C)

11. Correct option: (B, C)

The expression $36p^2 - 49q^2$ can be factored using the difference of squares formula:

$$a^2 - b^2 = (a + b)(a - b)$$

In this case, a = 6p and b = 7q. So, we can factor the expression as follows:

$$36p^2 - 49q^2 = (6p + 7q)(6p - 7q)$$

So, the factored form of $36p^2 - 49q^2 is\ (6p + 7q)(6p - 7q)$.

Then the correct option is B, C)

12. Correct option: (B, C, D)

To factor the expression $x^2 + \frac{x}{6} - \frac{1}{6}$, we can look for two binomials whose product equals the given expression.

The expression can be rewritten as:

$$x^2 + \frac{x}{6} - \frac{1}{6} = \frac{6x^2 + x - 1}{6}$$

Now, let's try to factor the numerator, $6x^2 + x - 1$.

$$6x^2 + x - 1 = 6x^2 + 3x - 2x - 1$$

Now, group the terms and factor by grouping:

$$(6x^2 + 3x) - (2x + 1)$$

Factor out the greatest common factor from each group:

$$3x(2x + 1) - 1(2x + 1)$$

$$(2x + 1)(3x - 1)$$

So, the factored form of $x^2 + x/6 - 1/6$ is $\frac{(2x+1)(3x-1)}{6}$.

Then the correct option is B, C, and D)

13. Correct option: (C, D)

To calculate the value of the expression we can first find the difference of squares in the numerator:

$$\frac{(734.783)^2 - (644.783)^2}{1379.566}$$

Now, simplify the terms within the brackets:

$$\frac{(734.783 - 644.783)(734.783 + 644.783)}{1379.566}$$

Notice that the term 1379.566 cancels out in the numerator and denominator: 90

So, the value of expression is 90.

Then the correct option is C, and D)

14. Correct option: (A)

To factor the expression $A = 6x^2 - 12xy + 18y^2$, we can look for common factors among the terms.

First, notice that each term contains a factor of 6, so we can factor 6 out:

$$A = 6(x^2 - 2xy + 3y^2)$$

Therefore, the factored form of $6x^2 - 12xy + 18y^2$ is $6(x^2 - 2xy + 3y^2)$

Then the correct option is A)

15. Correct option: (B)

To factor the expression $B = 0.8ab - 0.2a^2b^2$, we can look for common factors among the terms.

First, notice that each term contains a factor of 0.2ab, so we can factor 6 out:

$$B = 0.2ab(4 - ab)$$

Therefore, the factored form of $B = 0.8ab - 0.2a^2b^2$ is

$$B = 0.2ab(4 - ab)$$

Then the correct option is B)

16. Correct option: (B)

To factor the expression $C = \frac{2}{3}x^2 + \frac{4}{9}xy - \frac{8}{9}xz$, we can look for common factors among the terms.

First, notice that each term contains a factor of $\frac{2}{3}x$, so we can factor 6 out:

$$C = \frac{2}{3}x^2 + \frac{4}{9}xy - \frac{8}{9}xz = \frac{2}{3}x\left(x + \frac{2}{3}y - \frac{4}{3}z\right)$$

Therefore, the factored form of $C = \frac{2}{3}x^2 + \frac{4}{9}xy - \frac{8}{9}xz$ is

$$\frac{2}{3}x\left(x + \frac{2}{3}y - \frac{4}{3}z\right)$$

Then the correct option is B)

17. Correct Matching is: a-S, b-R, c-Q, d-P

Let's solve and match the expressions:

a) $lm^2 - mn^2 - lm + n^2$

We can factor this expression by grouping:

$$lm^2 - mn^2 - lm + n^2 = (lm^2 - lm) - (mn^2 - n^2) = lm(m - 1) - n^2(m - 1)$$

Now, factor out the common factor of (m - 1):

$$lm(m - 1) - n^2(m - 1) = (m - 1)(lm - n^2)$$

So, a) matches with S) m - 1.

b) $x^3 - 2x^2y + 3xy^2 - 6y^3$

This expression can be factored as follows:

$$x^3 - 2x^2y + 3xy^2 - 6y^3 = x^2(x - 2y) + 3y^2(x - 2y)$$

Now, factor out the common factor of (x - 2y):

$$x^2(x - 2y) + 3y^2(x - 2y) = (x - 2y)(x^2 + 3y^2)$$

So, b) matches with R) x - 2y.

c) $a(a - 2b - c) + 2bc$

Expand the expression:

$$a(a - 2b - c) + 2bc = a^2 - 2ab - ac + 2bc$$

Now, let's factor by grouping:

$$a^2 - 2ab - ac + 2bc = (a^2 - 2ab) - (ac - 2bc) = a(a - 2b) - c(a - 2b)$$

Now, factor out the common factor of (a - 2b):

$$a(a - 2b) - c(a - 2b) = (a - 2b)(a - c)$$

So, c) matches with Q) a - c.

d) $abx^2 + (ay - b)x - y$

Factor out the common factor of x in the second term:

$$abx^2 + (ay - b)x - y = abx^2 + axy - xb - y$$

Now, let's factor by grouping:

$$abx^2 + axy - xb - y = xb(ax - 1) + y(ax - 1) = (ax - 1)(xb + y)$$

So, d) matches with P) bx + y.

18. Correct Answer is: (2 or -2)

Let's translate the given statement into an equation:

"If x is multiplied by itself and subtracted from 9, we get the result as 5."

This can be written as an equation:

$$9 - x^2 = 5$$

Now, let's solve for x:

$$9 - x^2 = 5$$
$$9 - 5 - x^2 = 0$$
$$4 - x^2 = 0$$

$$4 = x^2$$
$$x = \pm\sqrt{4}$$

x can have two values:

$$x = 2 \; or \; x = -2$$

So, there are two possible solutions: $x = 2$ and $x = -2$.

19. Correct Answer is: (9)

Given expression

$$a + b + c = 10, a^2 + b^2 + c^2 = 42$$

We have to find the value of $ab + bc + ca$

To find the required value we will use the given information as

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Putting the known values to find the value of unknown as

$$10^2 = 42 + 2(ab + bc + ca) => \frac{100 - 42}{2} = ab + bc + ca = 29$$

Then the unit digit of the expression is 9.

20. Correct Answer is: (7)

We have the following equations: $(a + b + c + d) + \frac{1}{a+b+c+d} = 3$

We want to find the value of $(a + b + c + d)^2 + \frac{1}{(a+b+c+d)^2}$

To find the value of required expression we can use the given equation as

$$\left((a + b + c + d) + \frac{1}{a + b + c + d}\right)^2 = 3^2 = 9$$
$$(a + b + c + d)^2 + \frac{1}{(a + b + c + d)^2} + 2 = 9$$
$$(a + b + c + d)^2 + \frac{1}{(a + b + c + d)^2} = 7$$

Then the value of required expression is 7.

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

First, square the equation $x - \frac{1}{x} = 3$:

$$\left(x - \frac{1}{x}\right)^2 = 3^2$$
$$x^2 - 2 + \frac{1}{x^2} = 9$$
$$x^2 + \frac{1}{x^2} = 9 + 2$$
$$x^2 + \frac{1}{x^2} = 11$$

Now, square this equation again: $\left(x^2 + \frac{1}{x^2}\right)^2 = 11^2$

$$x^4 + 2 + \frac{1}{x^4} = 121$$
$$x^4 + \frac{1}{x^4} = 121 - 2$$
$$x^4 + \frac{1}{x^4} = 119$$

So, the value of $x^4 + \frac{1}{x^4}$ is 119.

Then the correct option is B)

2. Correct option: (C)

To calculate $(121)^2 - (21)^2$, we can use the difference of squares formula:

$$a^2 - b^2 = (a + b)(a - b)$$

In this case, a = 121 and b = 21:

$$(121)^2 - (21)^2 = (121 + 21)(121 - 21)$$

Now, calculate the values inside the parentheses:

$$(142)(100) = 14200$$

Then the correct option is C)

3. Correct option: (C)

To calculate 103×96 using the method

$$(100 + 3)(100 - 3 - 1) = (100^2 - 3^2) - (100 + 3)$$

Now, calculate each part of the expression:

$$100^2 = 10{,}000$$
$$3^2 = 9$$
$$100 + 3 = 103$$

Now, apply these values to the expression:

$$(10{,}000 - 9) - 103$$
$$10{,}000 - 9 = 9{,}991$$
$$9{,}991 - 103 = 9{,}888$$

So, 103×96 is equal to 9,888.

Then the correct option is C)

4. Correct option: (B)

To find 399^2 using the method:

$$399^2 = (400 - 1)^2$$

Now, apply the formula for squaring a binomial:

$$(400 - 1)^2 = 400^2 - 2 \times 400 * 1 + 1^2$$

Now, calculate each part of the expression:

$$400^2 = 160{,}000$$
$$2 \times 400 \times 1 = 800$$
$$1^2 = 1$$

Now, apply these values to the expression:

$$160{,}000 - 800 + 1 = 159{,}201$$

So, 399^2 is equal to 159,201.

Then the correct option is B)

5. Correct option: (D)

To factor the expression $a^2b^4c^6 - x^{16}$, we can use the difference of squares formula:

$$a^2 - b^2 = (a + b)(a - b)$$

In this case, we can consider $a^2b^4c^6$ as a single term and x^{16} as another single term. Now, apply the difference of squares formula:

$$a^2b^4c^6 - x^{16} = (a^2b^4c^6 + x^8)(a^2b^4c^6 - x^8)$$

Now, we have factored the expression into two terms:

$$(a^2b^4c^6 + x^8) \; and \; (a^2b^4c^6 - x^8)$$

Then the correct option is D)

6. Correct option: (A)

Factorization of $x^2 + 6x + 8$ using a perfect square identity as:

$$x^2 + 6x + 8 = x^2 + 6x + 9 - 1$$

Now, we used the perfect square identity for the first three terms:

$$= (x + 3)^2 - 1$$

Now, we can use the difference of squares formula:

$$= [(x + 3) + 1][(x + 3) - 1]$$
$$= (x + 4)(x + 2)$$

So, the factors of $x^2 + 6x + 8$ are (x + 4) and (x + 2).
Then the correct option is A)

7. Correct option: (C)

We can rewrite the middle term $(11x)$ using these two numbers:

$$4x^2 + 12x - 1x - 3$$

Now, we can group the terms and factor by grouping:

$$(4x^2 + 12x) - (1x + 3)$$

Now, factor the GCF (Greatest Common Factor) from each group:

$$4x(x + 3) - 1(x + 3)$$

Now, notice that both terms have a common factor of (x + 3). Factor it out:

$$(x + 3)(4x - 1)$$

So, the factored form of $4x^2 + 11x - 3$ is $(x + 3)(4x - 1)$.
Then the correct option is C)

8. Correct option: (A)

To factor the expression $a^2 - b^2 - c^2 - 2bc$, we can use the difference of squares and factor by grouping:

$$a^2 - b^2 - c^2 - 2bc$$

First, notice that $-b^2 - c^2 - 2bc$ is a expansion of square as

$$= a^2 - (b^2 + c^2 + 2bc) = a^2 - (b + c)^2$$

Using the difference of squares property as

$$a^2 - (b + c)^2 = (a + b + c)(a - b - c)$$

So, the factored form of $a^2 - b^2 - c^2 - 2bc$ is $(a + b + c)(a - b - c)$.
Then the correct option is A)

9. Correct option: (A)

To find the factor of given expression we follow as:

$$n^4 + 4 = n^4 + 4n^2 + 4 - 4n^2 = (n^2 + 2)^2 - (2n)^2$$

Now, we have a difference of squares, which can be factored using the difference of squares formula:

$$(a^2 - b^2) = (a + b)(a - b)$$

In this case:

$$a = n^2 + 2$$
$$b = 2n$$

So, applying the difference of squares formula:

$$(n^4 + 4) = ((n^2 + 2) + 2n)((n^2 + 2) - 2n)$$
$$= (n^2 + 2n + 2)(n^2 - 2n + 2)$$

So, the factored form of $n^4 + 4$ is $(n^2 + 2n + 2)(n^2 - 2n + 2)$.
Then the correct option is A)

10. Correct option: (A)

We can express it as a sum of two squares:

$$x^4 + x^2 + 1 = (x^4 + 2x^2 + 1) - x^2 = (x^2 + 1)^2 - x^2$$

Now, we have a difference of squares:

$$(a^2 - b^2) = (a + b)(a - b)$$

In this case:

$$a = x^2 + 1$$
$$b = x$$

So, applying the difference of squares formula:

$$x^4 + x^2 + 1 = ((x^2 + 1) + x)((x^2 + 1) - x)$$
$$= (x^2 + x + 1)(x^2 - x + 1)$$

So, the factored form of $x^4 + x^2 + 1$ is $(x^2 + x + 1)(x^2 - x + 1)$.
Then the correct option is A)

11. Correct option: (A, C)

To calculate the value of expression, we can first find the difference of squares in the numerator:

$$\frac{(67.542)^2 - (32.458)^2}{75.458 - 40.374}$$

Now, simplify the terms within the brackets:

$$\frac{(67.542 + 32.458)(67.542 - 32.458)}{35.084}$$

Notice that the term 35.084 cancels out in the numerator and denominator: 100
So, the value of expression is 100.

12. Correct option: (A, B, C)

To factor the expression $(a^4 - 13a^3 + 36a^2)$, we can look for common factors among the terms.
First, notice that each term contains a factor of a^2 so we can factor out:

$$\left(a^4 - 13a^3 + 36a^2 = a^2(a^2 - 13a + 36)\right)$$

Now, we need to factor the quadratic expression $a^2 - 13a + 36$.
So, we can further factor the expression as:

$$(a^2(a - 9)(a - 4))$$

Therefore, the factored form of expression is $(a^2(a - 9)(a - 4))$.
Then the correct option is A, B and C)

13. Correct option: (D)

To find the values of a, b, and c in the equation expression, we can start by multiplying both sides of the equation by the denominators on the right-hand side to eliminate the fractions:

$$ax^2 + bx + c = (4x - 3)(3x + 2)$$

Now, we'll expand the right side of the equation:

$$[ax^2 + bx + c = 12x^2 - x - 6]$$

Now, we can equate the coefficients of the corresponding terms on both sides of the equation:

$$1. For\ x^2\ term: (a = 12)$$
$$2. For\ x\ term: (b = -1)$$
$$3. For\ the\ constant\ term: (c = -6)$$

So, the values of a, b, and c are:

$$a = 12$$
$$b = -1$$
$$c = -6$$

Then the correct option is D)

14. Correct option: (D)

To find the expression, we can start by multiplying both sides of the equation by the denominators on the right-hand side to eliminate the fractions:

$$ax^2 + bx + c = (4x - 3)(3x + 2)$$

Now, we'll expand the right side of the equation:

$$[ax^2 + bx + c = 12x^2 - x - 6]$$

Therefore the expression is $12x^2 - x - 6$
Then the correct option is D)

15. Correct Matching is: a-Q, b-P, c-S, d-R

Let's solve and match the expressions:

a) The value of $3x(a+b) - (a+b)x + 2x(a+b)$

First, simplify take common (a+b):

$$(a+b)(3x - x + 2x) = (a+b)(4x)$$

So, a) matches with Q) $4(a+b)x$.

b) The value of $6x^2(a-2b) + 3x(2b-a)$

First, simplify each term:

$$6x^2(a-2b) = 6ax^2 - 12bx^2$$
$$3x(2b-a) = 6bx - 3ax$$
$$(6ax^2 - 12bx^2) + (6bx - 3ax)$$
$$6ax^2 - 12bx^2 + 6bx - 3ax$$
$$(6ax^2 - 3ax) + (6bx - 12bx^2)$$
$$3ax(2x-1) - 6bx(2x-1)$$
$$(3ax - 6bx)(2x-1)$$

So, b) matches with P) $(a-2b)3x(2x-1)$.

c) The value of $(x+y)^2 - (x-y)(x+y)$

First, expand the terms:

$$(x+y)^2 = x^2 + 2xy + y^2$$
$$(x-y)(x+y) = x^2 - y^2$$

Now, subtract the second expression from the first:

$$(x^2 + 2xy + y^2) - (x^2 - y^2)$$
$$2y^2 + 2xy = 2y(x+y)$$

So, c) matches with S) $2y(x+y)$.

d) The value of $a(x-y) + b(y-x)$

First, distribute the terms:

$$a(x-y) = ax - ay$$
$$b(y-x) = by - bx$$

Now, combine the terms:

$$(ax - ay) + (by - bx)$$
$$x(a-b) - y(a-b)$$

Now, factor out the common factor of (a - b):

$$(a-b)(x-y)$$

So, d) matches with R) $(a-b)(x-y)$.

16. Correct Answer is: (3 or -3)

Given expression

$$(1+a)(1-a+a^2)(1-a)(1+a+a^2) = -728$$

Solving further as

$$(1+a)(1-a)\big((1+a^2)-a\big)\big((1+a^2)+a\big) = -728$$

Using the formula of difference of squares we can write as

$$(1-a^2)((1+a^2)^2 - a^2) = (1-a^2)(1+a^4+a^2)$$

Further simplifying the terms

$$(1-a^4) + a^4(1-a^2) = 1 - a^6 = -728 \Rightarrow a^6 = 729$$
$$a^6 = 729 \Rightarrow a = -3 \text{ or } 3$$

Then the solution of the given expression is 3 or -3.

17. Correct Answer is: (4)

To find the value of $\left(\frac{A+B+C}{3}\right)$ with given expressions A, B, and C, first, we'll calculate the sum A + B + C, and then we'll divide it by 3.

Given:

$$A = 3a - 2b$$
$$B = 13a - 4b$$
$$C = -2a + 5b$$

Now, let's calculate A + B + C:

$$A + B + C = (3a - 2b) + (13a - 4b) + (-2a + 5b)$$

Now, combine like terms:

$$(3a + 13a - 2a) + (-2b - 4b + 5b)$$
$$(14a - b)$$
$$\frac{A + B + C}{3} = \frac{14a - b}{3}$$

Now, substitute the values a = 1 and b = 2:

$$\frac{A + B + C}{3} = \frac{14(1) - 2}{3}$$

Simplify:

$$\frac{14 - 2}{3} = \frac{12}{3} = 4$$

So, when a = 1 and b = 2, the value of expression is 4.

18. Correct Answer is: (1)

We have two equations: $1.\ 3x + 4y = 13$

$$2.\ xy = 7$$

We want to find the value of $9x^2 + 16y^2$.

First, let's square equation 1 to get rid of the square roots:

$$(3x + 4y)^2 = 13^2$$
$$9x^2 + 24xy + 16y^2 = 169$$

Now, let's substitute the value of xy from equation 2:

$$9x^2 + 24(7) + 16y^2 = 169$$
$$9x^2 + 168 + 16y^2 = 169$$

Now, let's rearrange the terms and simplify: $9x^2 + 16y^2 = 169 - 168$

$$9x^2 + 16y^2 = 1$$

So, the value of $9x^2 + 16y^2$ is 1.

19. Correct Answer is: (1)

We have the following equations: $1.\ a + b + c = 1$

$$2.\ a^2 + b^2 + c^2 = 9$$
$$3.\ a^3 + b^3 + c^3 = 1$$

We want to find the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$.

Let's start by squaring equation 1: $(a + b + c)^2 = 1^2$

$$a^2 + b^2 + c^2 + 2ab + 2ac + 2bc = 1$$

Now, we have equation 2, which is $a^2 + b^2 + c^2 = 9$. We can substitute this into the above equation: $9 + 2ab + 2ac + 2bc = 1$

Now, simplify: $2ab + 2ac + 2bc = 1 - 9$

$$2ab + 2ac + 2bc = -8$$
$$ab + ac + bc = -4$$

Let's take equation 3: $a^3 + b^3 + c^3 = 1$

We can write as $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - (ab + bc + ac))$

Putting the values from the above equations as $1 - 3abc = 9 - (-4) = 13 \Rightarrow abc = -4$

Now we have to find the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab+bc+ca}{abc} = \frac{-4}{-4} = 1$

Then the value of the expression with given conditions is 1.

20. Correct Answer is: (2)

We have the following equations: $x^2 + y^2 + z^2 = x + y + z = xyz = 1$

We want to find the value of $x^3 + y^3 + z^3$.

Sum of cubes can be written as

$$x^3 + y^3 + z^3 = (x + y + z)\big(x^2 + y^2 + z^2 - (xy + yz + zx)\big) + 3xyz$$

Putting the known values as $x^3 + y^3 + z^3 = \big(1 - (xy + yz + zx)\big) + 3 = 2 + xy + yz + zx$

To find the value of $xy + yz + zx$ we take equation 1 as

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx) => (xy + yz + zx) = 0$$

Then the value of $x^3 + y^3 + z^3 = 2 + xy + yz + zx = 2$

Exponents and Powers

LEVEL – I

1. Correct option: (B)

3^2 is equal to 3 multiplied by itself, which is 9.

Now, multiply 9 by 5: $5 \times 9 = 45$

Then the correct option is B)

2. Correct option: (A)

$$4^3 = 4 \times 4 \times 4 = 64$$

$$5^2 = 5 \times 5 = 25$$

Then $4^3 \times 5^2 = 1600$

Then the correct option is A)

3. Correct option: (D)

$$(-3)^2 = -3 \times -3 = 9$$

$$6^4 = (2 \times 3)^4 = 2^4 \times 3^4 = 16 \times 81 = 1296$$

Then $(-3)^2 \times 6^4 = 11664$

Then the correct option is D)

4. Correct option: (C)

$$(-1)^{2015} = -1\ (odd\ power)$$

$$(-1)^{2016} = 1\ (even\ power)$$

Then $(-1)^{2015} \times (-1)^{2016} = -1 \times 1 = -1$

Then the correct option is C)

5. Correct option: (B)

1. Cube $\left(-\frac{2}{3}\right)$:

$$\left(-\frac{2}{3}\right)^3 = \left(-\frac{2}{3}\right) \times \left(-\frac{2}{3}\right) \times \left(-\frac{2}{3}\right) = \left(-\frac{8}{27}\right)$$

2. Square $\left(-\frac{4}{5}\right)$:

$$(-4/5)\text{^}2 = (-4/5) \times (-4/5) = \left(\frac{16}{25}\right)$$

3. Multiply the results from steps 1 and 2 together:

$$\left(-\frac{8}{27}\right) \times \left(\frac{16}{25}\right) = \frac{-8 \times 16}{27 \times 25} = \frac{-128}{675}$$

Then the correct option is B)

6. Correct option: (B)

$$(-3)^2 = 9$$

$$\left(-\frac{5}{12}\right)^2 = \frac{25}{144}$$

Then $(-3)^2 \times \left(\frac{5}{12}\right)^2 = 9 \times \frac{25}{144} = \frac{25}{16}$

Then the correct option is B)

7. Correct option: (B)

The exponential form of the expression "$2 \times 2 \times 2 \times 5 \times 5$" is written as: $= (2^3) \times (5^2)$

In this form, we use the exponents to represent the repeated multiplication of numbers. So, "$2 \times 2 \times 2$" is written as "2^3," and "5×5" is written as "5^2."

Then the correct option is B)

8. Correct option: (B)

The exponential form of 3600 in prime factorization form is: $3600 = 2^3 \times 3^2 \times 5^2$

(Use Prime factorization)

In this form, we break down 3600 into its prime factors, where 2, 3, and 5 are the prime numbers and their respective exponents indicate how many times each prime number is multiplied to obtain 3600. So, 2 is raised to the power of 3, 3 is raised to the power of 2, and 5 is raised to the power of 2.

Then the correct option is B)

9. Correct option: (D)

The exponential form of $-\frac{1}{243}$ is:

$$243 = 3 \times 3 \times 3 \times 3 \times 3 = 3^5$$

$$-1 = (-1)^5$$

Then $$-\frac{1}{243} = \frac{(-1)^5}{3^5} = \left(-\frac{1}{3}\right)^5$$

Then the correct option is D)

10. Correct option: (B)

Given expression $3^2 \times 3^4 \times 3^8$

For same base we can use the following formula

$$a^b \times a^c = a^{b+c}$$

Then $$3^2 \times 3^4 \times 3^8 = 3^{2+4} \times 3^8 = 3^{2+4+8} = 3^{14}$$

Then the correct option is B)

11. Correct option: (C)

Given expression $\{(5^2)^3 \times 5^4\} \div (5^7)$

For same base we can use the following formula

$$(a^b)^c = a^{bc}, a^b \times a^c = a^{b+c}, a^b \div a^c = a^{b-c}$$

Then $$\{(5^2)^3 \times 5^4\} \div (5^7) = \{5^6 \times 5^4\} \div 5^7$$

$$= 5^{10} \div 5^7 = 5^{10-7} = 5^3$$

Then the correct option is C)

12. Correct option: (D)

Given expression

$$\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5}$$

To solve the given expression we use prime factorization of each number as

$$\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5} = \frac{3^5 \times (2 \times 5)^5 \times (5^2)}{5^7 \times (2 \times 3)^5}$$

$$= \frac{3^5 \times 2^5 \times 5^5 \times 5^2}{5^7 \times 2^5 \times 3^5}$$

For same base we can use the following formula

$$(a^b)^c = a^{bc}, a^b \times a^c = a^{b+c}, a^b \div a^c = a^{b-c}$$

Then

$$\frac{3^5 \times 10^5 \times 25}{5^7 \times 6^5} = \frac{3^5 \times 2^5 \times 5^5 \times 5^2}{5^7 \times 2^5 \times 3^5}$$

$$= \left(\frac{3^5}{3^5}\right) \times \left(\frac{2^5}{2^5}\right) \times \left(\frac{5^{5+2}}{5^7}\right)$$
$$= 5^{5+2-7} = 5^0 = 1$$

Then the correct option is D)

13. Correct option: (B)

To express 729 × 64 as the product of prime factors, we can do the following:

1. Prime factorization of 729: $729 = 3^6$
2. Prime factorization of 64: $64 = 2^6$
3. Multiply the prime factorizations: $(3^6) \times (2^6)$

So, the product 729 × 64 is equal to $(3^6) \times (2^6)$ in prime factorization.

Then the correct option is B)

14. Correct option: (C)

1. Prime factorization of 135: $135 = 5 \times 3^3$

Then given equation $5^{n-2} \times 3^{2n-3} = 135 = 5 \times 3^3$

Equating the exponents of the same bases $5^{n-2} = 5^1 => n - 2 = 1 => n = 3$

Or $3^{2n-3} = 3^3 => 2n - 3 = 3 => 2n = 6 => n = 3$

So value of n is 3

Then the correct option is C)

15. Correct option: (C)

Given expression $\frac{p}{q} = \left(\frac{2}{3}\right)^2 \div \left(\frac{6}{7}\right)^0$

We can write $\left(\frac{6}{7}\right)^0 = 1$

Then $\frac{p}{q} = \left(\frac{2}{3}\right)^2 \div \left(\frac{6}{7}\right)^0 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$

So $\frac{q}{p} = \frac{9}{4}$

Therefore $\left(\frac{q}{p}\right)^3 = \left(\frac{9}{4}\right)^3 = \frac{9\times9\times9}{4\times4\times4} = \frac{729}{64}$

Then the correct option is C)

16. Correct option: (A)

Given expression $7^{2n+1} \div 49 = 7^3$

We can write $7^{2n+1} \div 7^2 = 7^{2n+1-2} = 7^{2n-1}$

Then $7^{2n-1} = 7^3$

So $2n - 1 = 3 => 2n = 4 => n = 2$

Therefore value of n is 2

Then the correct option is A)

17. Correct option: (D)

Given expression

$$\frac{10 \times 5^{n+1} + 25 \times 5^n}{3 \times 5^{n+2} + 10 \times 5^{n+1}}$$

We can write

$$\frac{2 \times 5 \times 5^{n+1} + 5^2 \times 5^n}{3 \times 5^{n+2} + 2 \times 5 \times 5^{n+1}} = \frac{2 \times 5^{n+2} + 5^{n+2}}{3 \times 5^{n+2} + 2 \times 5^{n+2}} = \frac{5^{n+2}(2+1)}{5^{n+2}(3+2)}$$

Then

$$5^{n+2-(n+2)}\left(\frac{3}{5}\right) = \frac{3}{5}$$

Therefore

$$\frac{10 \times 5^{n+1} + 25 \times 5^n}{3 \times 5^{n+2} + 10 \times 5^{n+1}} = \frac{3}{5}$$

Then the correct option is D)

18. Correct option: (D)

The standard form of the number 3,80,12,000 in scientific notation is:

$$= 3.8012 \times 10^7$$

In scientific notation, the number is expressed as a decimal (between 1 and 10) multiplied by a power of 10 (with an appropriate exponent). In this case, we've moved the decimal point to the right, and the exponent indicates the number of places the decimal point was moved. So, 3.8012×10^7 represents 38,012,000 in standard form.

Then the correct option is D)

19. Correct option: (C)

The diameter of the sun, 1,400,000,000 meters, in standard form is:

$$1.4 \times 10^9 \, meters$$

In standard form (scientific notation), a number is expressed as a decimal (between 1 and 10) multiplied by a power of 10 (with an appropriate exponent). In this case, we've moved the decimal point to the right, and the exponent indicates the number of places the decimal point was moved. So, 1.4×10^9 meters represents 1,400,000,000 meters in standard form.

Then the correct option is C)

20. Correct option: (C)

The standard form of 0.0003628 is: $= 3.628 \times 10^{-4}$

In scientific notation, numbers less than 1 are expressed as a decimal (between 1 and 10) multiplied by a negative power of 10. In this case, we've moved the decimal point four places to the right, so the exponent is -4. So, 0.0003628 is written as 3.628×10^{-4} in standard form.

Then the correct option is C)

LEVEL-II

1. Correct option: (D)

Given expression $(1)^{2011}$

We know that $1^n = 1\{n \in R\}$

So the value of given expression is also 1

Then the correct option is D)

2. Correct option: (D)

The base of the expression $-4x^5$ is "x."

In this expression, "x" is the variable raised to the fifth power ($-4x^5$). The base represents the variable or number that is being raised to a certain power. In this case, "x" is the base, and it is raised to the fifth power.

Then the correct option is D)

3. Correct option: (A)

The 2nd power of a number is called the "Square" of the number.

So, the correct answer is: A) Square

4. Correct option: (A)

The expression $3 \times 9 \times 27 \times 81$ can be written in exponent form as follows:

$$3 \times 9 \times 27 \times 81 = 3^1 \times 3^2 \times 3^3 \times 3^4$$

Now, using the properties of exponents (when we multiply numbers with the same base, we add their exponents), we can simplify this:

$$3^1 \times 3^2 \times 3^3 \times 3^4 = 3^{1+2+3+4} = 3^{10}$$

So, $3 \times 9 \times 27 \times 81$ in exponent form is 3^{10}.

Then the correct option is A)

5. Correct option: (B)

Given expression $\left(\frac{1}{5}\right)^x = (125)^4$

We can write $5^{-x} = (5^3)^4 = 5^{12}$

Comparing the exponents of the same base $-x = 12 => x = -12$

Then the correct option is B)

6. Correct option: (C)

We can raise power $\frac{1}{a}$ to both of side as

$$(x^a)^{\frac{1}{a}} = ((a+2)^a)^{\frac{1}{a}}$$

$$x = ((a+2)^a)^{\frac{1}{a}}$$

$$x = ((a+2)^a)^{\frac{1}{a}} = (a+2)^{\frac{a}{a}} = (a+2)^1$$

So, $x = a + 2 => x - a = 2$

Then the correct option is C)

7. Correct option: (C)

Given expression $[1 + 2^2 + 3^3 + 4^4 + \cdots + 100^{100}]^0$

We know that

$$(x)^0 = 1, \{x \neq 0\}$$

So value of the given expression is

$$[1 + 2^2 + 3^3 + 4^4 + \cdots + 100^{100}]^0 = 1$$

Then the correct option is C)

8. Correct option: (C)

Given expression $7^n = 1$

We know that $(x)^0 = 1, \{x \neq 0\}$

So we can write

$$n = 0$$

Then the correct option is C)

9. Correct option: (D)

Given expressions $a^{\frac{1}{3}} + b^{\frac{1}{3}} + c^{\frac{1}{3}} = 0$

Then we have to find the value of $(a + b + c)^3$

To find the value of $a + b + c$ we can use the identity

$$x + y + z = 0 => x^3 + y^3 + z^3 = 3xyz$$

So

$$a^{\frac{1}{3}} + b^{\frac{1}{3}} + c^{\frac{1}{3}} = 0 => \left(a^{\frac{1}{3}}\right)^3 + \left(b^{\frac{1}{3}}\right)^3 + \left(c^{\frac{1}{3}}\right)^3 = 3a^{\frac{1}{3}}b^{\frac{1}{3}}c^{\frac{1}{3}}$$

$$a + b + c = 3a^{\frac{1}{3}}b^{\frac{1}{3}}c^{\frac{1}{3}}$$

Therefore

$$(a + b + c)^3 = \left(3a^{\frac{1}{3}}b^{\frac{1}{3}}c^{\frac{1}{3}}\right)^3 = 27abc$$

Then the correct option is D)

10. Correct option: (D)

Given expressions $3^n = 243$

Then we have to find the value of n

For 243 we can write $243 = 3 \times 3 \times 3 \times 3 \times 3 = 3^5$

Then

$$3^n = 3^5 => n = 5$$

Then the correct option is D)

11. Correct option: (A)

To write the number 0.0000021 in scientific notation, we need to express it as a number between 1 and 10 multiplied by a power of 10. Here's how we can do it:

0.0000021 can be expressed as 2.1×10^{-6}

Then the correct option is A)

12. Correct option: (C)

Given equation $5^x = 1000$

We have to find the value of 5^{x+2}

$$5^{x+2} = 5^x \times 5^2 = 5^x \times 25$$

Substitute the value of 5^x

$$5^{x+2} = 1000 \times 25 = 25000$$

Then the correct option is C)

13. Correct option: (D)

Given that $\frac{a+b+c}{3} = 3; (mean\ of\ a, b, c = 3)$

We have to find the value of $\sqrt[3]{(7^{a+b-c})(7^{b+c-a})(7^{c+a-b}}$

We can write $\sqrt[3]{(7^{a+b-c})(7^{b+c-a})(7^{c+a-b}} = \sqrt[3]{7^{a+b-c+b+c-a+c+a-b}}$

$$= \sqrt[3]{7^{a+b+c}}$$

Substitute the value of $(a + b + c) = 9$

$$\sqrt[3]{7^{a+b+c}} = \sqrt[3]{7^9} = 7^{9\times\frac{1}{3}} = 7^3$$

Therefore, the value of $\sqrt[3]{(7^{a+b-c})(7^{b+c-a})(7^{c+a-b}} = 7^3$

Then the correct option is D)

14. Correct option: (A)

Given expression

$$\frac{5^{2000} + 5^{1999}}{5^{1999} - 5^{1997}}$$

So we can write

$$\frac{5000^{1997}(5^3 + 5^2)}{5000^{1997}(5^2 - 1)} = \frac{125 + 25}{24} = \frac{150}{24} = \frac{75}{12} = \frac{25}{4}$$

Therefore

$$\frac{5^{2000} + 5^{1999}}{5^{1999} - 5^{1997}} = \frac{25}{4}$$

Then the correct option is A)

15. Correct option: (A)

When the number "-1" is raised to an odd positive integral power, it gives the result of -1.

For example:

$$-1^1 = -1$$
$$-1^3 = -1$$
$$-1^5 = -1$$

and so on...

In general, when we raise -1 to any odd positive integral power (an odd positive integer), the result will be -1.

Then the correct option is A)

16. Correct option: (B)

Given expression $(1024)^{-\frac{1}{5}}$

So we can write $(1024)^{-\frac{1}{5}} = (2^{10})^{-\frac{1}{5}} = 2^{-\frac{10}{5}} = 2^{-2} = \frac{1}{4}$

Then the correct option is B)

17. Correct option: (B)

Given expression $\frac{x^{m+n}\times x^{n+l}\times x^{l+m}}{\left(x^m\times x^n\times x^l\right)^2}$

So we can write $(x^{m+n+n+l+l+m} = x^{2(m+n+l)}$

$$\left(x^m \times x^n \times x^l\right)^2 = x^{2m+2n+2l} = x^{(2(m+n+l))}$$

Therefore

$$\frac{x^{m+n} \times x^{n+l} \times x^{l+m}}{(x^m \times x^n \times x^l)^2} = \frac{x^{2(m+n+l)}}{x^{(2(m+n+l))}} = 1$$

Then the correct option is B)

18. Correct option: (A)

Given expression

$$2^n = 4096$$

We have to find the value of

$$2^n = 4096 = 2^{12}$$
$$n = 12$$

We can write

$$2^{n-5} = 2^7 = 128$$

So value

$$2^{n-5} = 128$$

Then the correct option is A)

19. Correct option: (D)

Given expression

$$abc = 0$$

We have to find the value of

$$\frac{\{(x^a)^b\}^c}{\{(x^b)^c\}^a}$$

We can write

$$\{(x^a)^b\}^c = x^{abc}$$
$$\left\{\left(x^b\right)^c\right\}^a = x^{abc}$$

So value

$$\frac{\{(x^a)^b\}^c}{\{(x^b)^c\}^a} = x^{abc-abc} = x^0 = 1$$

Then the correct option is D)

20. Correct option: (B)

Given expression

$$6^{-2} \times y = 216$$

Where number y is the number we require

So we can write $\frac{y}{6^2} = 216 => y = 6^2 \times 216 = 6^2 \times 6^3 = 6^5$

Therefore the value of y is 6^5

Then the correct option is B)

STUDENT PRACTICE QUESTIONS

1. Correct option: (C)

$$3^2 = 3 \times 3 = 9$$
$$2^3 = 2 \times 2 \times 2 = 8$$

Then $3^2 \times 2^3 = 9 \times 8 = 72$

Then the correct option is C)

2. Correct option: (B)

$$(-2)^5 = -2 \times -2 \times -2 \times -2 \times -2 = -32$$

$$3^2 = 9$$

Then $(-2)^5 \times 9 = -288$

Then the correct option is B)

3. Correct option: (D)

$$(-1)^{2001} + (-1)^{2002} + .. + (-1)^{2009} + (-1)^{2010}$$

The given expression represents the sum of power of -1 with odd and even power.

So $-1 + 1 \ldots . + (-1) + 1 = 0$

Then $(-1)^{2001} + (-1)^{2002} + .. + (-1)^{2009} + (-1)^{2010} = 0$

Then the correct option is D)

4. Correct option: (C)

Given expression

$$\left[(3^2 - 2^2) \div \left(\frac{1}{5}\right)^2\right]$$

So $3^2 - 2^2 = 9 - 4 = 5$

$$\left(\frac{1}{5}\right)^2 = \frac{1}{25}$$

Therefore $\left[(3^2 - 2^2) \div \left(\frac{1}{5}\right)^2\right] = 5 \div \left(\frac{1}{25}\right) = 5 \times \frac{25}{1} = 125$

Then the correct option is C)

5. Correct option: (B)

The exponential form of the number 3125 is written as: $3125 = 5^5$

In this form, we express 3125 as the fifth power of 5, where 5 is raised to the exponent 5.

Then the correct option is B)

6. Correct option: (C)

Given expression $25^{n-1} + 100 = 5^{2n-1}$

We can write $25^{n-1} + 4 \times 25 = 5^{2n-1}$

$$5^{2n-2} + 4 \times 5^2 = 5^{2n-1}$$

$$4 \times 5^2 = 5^{2n-1} - 5^{2n-2} = 5^{2n-2}(5 - 1) = 5^{2n-2} \times 4$$

So $4 \times 5^2 = 4 \times 5^{2n-2} => 5^2 = 5^{2n-2}$

We have same base so we can equate the exponents of the numbers

$$2 = 2n - 2 => 2n = 4 => n = 2$$

So value of n is 2

Then the correct option is C)

7. Correct option: (C)

Given expression

$$\frac{9^n \times 3^2 \times 3^n - 27^n}{(3^3)^5 \times 2^3} = \frac{1}{27}$$

We can write

$$\frac{9^n \times 3^2 \times 3^n - 27^n}{(3^3)^5 \times 2^3} = \frac{3^{2n} \times 3^2 \times 3^n - 3^{3n}}{3^{15} \times 2^3} = \frac{3^{2n+2+n} - 3^{3n}}{3^{15} \times 2^3} = \frac{3^{3n}(3^2 - 1)}{3^{15} \times 2^3}$$

$$\frac{3^{3n}}{3^{15}} = 3^{3n-15}$$

Then $3^{3n-15} = \frac{1}{27} = 3^{-3}$

Therefore $3n - 15 = -3 => 3n = -3 + 15 = 12 => 3n = 12 => n = 4$

So value of n is 4

Then the correct option is C)

8. Correct option: (A)

The standard form of the number 86,000,000 is: 8.6×10^7

In scientific notation, the number is expressed as a decimal (between 1 and 10) multiplied by a power of 10 (with an appropriate exponent). In this case, we've moved the decimal point to the right, and the exponent indicates the number of places the decimal point was moved. So, 8.6×10^7 represents 86,000,000 in standard form.

Then the correct option is A)

9. Correct option: (A)

The standard form of 0.0000387×10^3 is: $= 3.87 \times 10^{-5} \times 10^3 = 3.87 \times 10^{-2}$

Then the correct option is A)

10. Correct option: (B)

The standard form of 0.00832 is: $= 8.32 \times 10^{-3}$

In scientific notation, numbers less than 1 are expressed as a decimal (between 1 and 10) multiplied by a negative power of 10. In this case, we've moved the decimal point three places to the right, so the exponent is -3. So, 0.00832 is written as 8.32×10^{-3} in standard form.

Then the correct option is B)

11. Correct option: (B)

Given equation $\left(\frac{-4}{5}\right) \times \left(-\frac{4}{5}\right) \times \left(-\frac{4}{5}\right)$

We can write $(-1) \times (-1) \times (-1) \times \left(\frac{4}{5}\right)^3 = (-1)^3 \times \left(\frac{4}{5}\right)^3 = -\frac{64}{125}$

Then the correct option is B)

12. Correct option: (B)

Given expression $(a^{p+q})^{p-q} \times (a^{q+r})^{q-r} \times (a^{r+p})^{r-p}$

We can write (using difference of squares)

$$a^{p^2-q^2} \times a^{q^2-r^2} \times a^{r^2-p^2}$$

Using the property of the exponents $a^{p^2-q^2+q^2-r^2+r^2-p^2} = a^0 = 1$

So $(a^{p+q})^{p-q} \times (a^{q+r})^{q-r} \times (a^{r+p})^{r-p} = 1$

Then the correct option is B)

13. Correct option: (C)

Given expressions $a^x = b, b^y = c, c^z = a$

Then we have to find the value of xyz

To find the value of xyz we can substitute the values as

$$a^x = b \Rightarrow (b^y) = (a^{xy}) = c \Rightarrow c^z = (a^{xyz}) = a$$

Then $a^{xyz} = a \Rightarrow xyz = 1$

Then the correct option is C)

14. Correct option: (C)

Given expressions $\frac{a^m}{b^n}, m = n$

Then $\frac{a^m}{b^n} = \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m = \left(\frac{a}{b}\right)^n$

15. Correct option: (A)

To write the number 35,000,000 in scientific notation, we need to express it as a number between 1 and 10 multiplied by a power of 10. Here's how we can do it:

35,000,000 can be expressed as 3.5×10^7

In this form, the decimal point is moved seven places to the right to make the number between 1 and 10, and the exponent 7 indicates the number of places the decimal point was moved. So, 35,000,000 in scientific notation is 3.5×10^7.

Then the correct option is A)

16. Correct option: (B)

Given expression $\left[\left(\frac{a}{b}\right)^{\sqrt{99}-\sqrt{97}}\right]^{\sqrt{99}+\sqrt{97}}$

To solve above expression we can use difference of square

$$\left[\left(\frac{a}{b}\right)^{(\sqrt{99}-\sqrt{97})(\sqrt{99}+\sqrt{97})}\right] = \left(\frac{a}{b}\right)^{99-97} = \left(\frac{a}{b}\right)^{2}$$

Then the correct option is B)

17. Correct option: ($\frac{x+y}{xy}$)

Given expression $(x^{-1}+y^{-1})$

Solving further $x^{-1}=\frac{1}{x}, y^{-1}=\frac{1}{y} => (x^{-1}+y^{-1}) = \left(\frac{1}{x}+\frac{1}{y}\right)$

$$\left(\frac{1}{x}+\frac{1}{y}\right) = \frac{x+y}{xy}$$

18. Correct option: (C)

Given that $\left(\frac{3}{5}\right)^4 \times \left(\frac{3}{5}\right)^3 = \left(\frac{3}{5}\right)^{2n-1}$

We have to find the value of n

We can write $\left(\frac{3}{5}\right)^4 \times \left(\frac{3}{5}\right)^3 = \left(\frac{3}{5}\right)^7$

Equating the exponents of same base $2n-1=7 => 2n=8 => n=\frac{8}{2}=4$

So, value of n is 4.

Then the correct option is C)

19. Correct option: (D)

Given expression $27^{x+1} = 9^{x+3} = 3^y$

So we can write

$$27^{x+1} = 9^{x+3}$$

$$3^{3(x+1)} = 3^{2(x+3)} => 3(x+1) = 2(x+3)$$

$$3x+3 = 2x+6 => 3x-2x = 6-3 => x = 3$$

Also

$$9^{x+3} = 3^y = 9^{3+3} = 9^6 = 3^{12}$$

So

$$y = 12$$

Therefore the value of (x, y) = (3, 12)

Then the correct option is D)

20. Correct option: (B)

Given expression

$$2^a.3^b.7^c = 1176$$

We have to find the value of

$$a^a b^b 7^{-c}$$

$$1176 = 2^3 \times 3 \times 7^2$$

We can write

$$2^a.3^b.7^c = 2^3.3^1.7^2 => a=3, b=1, c=2$$

So value

$$a^a b^b 7^{-c} = 3^3.1^1.7^{-2} = \frac{27}{49}$$

Then the correct option is B)

8. Square Roots and Cube Roots

Solutions

Squares and Square Roots

LEVEL – I

1. Correct option: (B)

Squares of even numbers are always even. When you square an even number, the result is a multiple of 2, making it an even number.

2. Correct option: (D)

Given equation $2(x-3)^2 = 288$

We have to solve above equation to find the value of x

1. Dividing by 2 and taking square root of both side

$$(x-3)^2 = \frac{288}{2} = 144 => \sqrt{(x-3)^2} = \sqrt{144} = \pm 12$$

2. Further solving the equation $x - 3 = \pm 12 => x = 15\ or\ -9$

Therefore, the values of x that satisfy the given equation are $x = 15$ and $x = -9$

So the Correct option: will be D)

3. Correct option: (C)

Given value $x^2 = 2401$

We have to solve above equation to find the value of x

Step 1: Prime factorization of 2401: $[2401 = 7 * 7 * 7 * 7 = 7^4]$

Step 2: Take the square root of the prime factorization: $[\sqrt{2401} = \sqrt{7^4} = 7^2 = 49]$

So, the square root of 2401 is 49.

4. Correct option: (D)

The area of a square is calculated using the formula:

$$Area = side\ length * side\ length$$

In this case, you have the area as $2\frac{1054}{625} = \frac{2304}{625}\ cm^2$

Let's call the side length of the square "x".

So, we can set up the equation: $(x) * (x) = 2304/625$

Solving for x: $x^2 = 2304/625$

To isolate x, take the square root of both sides:

$$x = \sqrt{\frac{2304}{625}}$$

$$x = \frac{\sqrt{2304}}{\sqrt{625}}$$

$$x = \frac{48}{25}$$

$$x = 1.92$$

So, the length of each side of the square field is $1.92\ cm$.

5. Correct option: (B)

To find the number that must be subtracted from 13245 to get a perfect square, we need to look for a perfect square that is closest to 13245. We can start by finding the square root of 13245.

$$\sqrt{13245} = \sqrt{3 * 5 * 883} \approx 115.11$$

The perfect squares closest to 13245 are $115^2 = 13225$ and $116^2 = 13456$.

If we subtract 13225 from 13245, we get 20. If we subtract 13456 from 13245, we get -211.

Since the problem asks for a positive number, the answer must be 20.

So, the Correct option: is: B) 20

6. Correct option: (C)

If we define $\sqrt{-1}$ as "i," then the square of "i" would be: $i^2 = \left(\sqrt{-1}\right)^2 = (-1)$

So, if we define $\sqrt{-1}$ as "i," then the value of i^2 would be -1.

7. Correct option: (B)

To find the square root of 256 we first do prime factorization of 256.

To find the prime factorization of 256, we can start by dividing it by the smallest prime number, which is 2: Prime factorization of $256 = 2^8$

Now we have to find $\sqrt{256} = \sqrt{2^8} = 2^4$

So 256 is square of $2^4 = 16$

8. Correct option: (B)

The given number, 2215, is indeed odd.

Consequently, the square of 2215 is an odd number. This aligns with the general rule that the square of an odd number results in an odd number, whereas the square of an even number yields an even number. Therefore, the Correct option: is B) Odd Number.

9. Correct option: (D)

To find the unit digit of the square of 109, you can follow this approach:

1. Calculate the square of the last digit of the number. In this case, the last digit of 109 is 9. So, $9^2 = 81$.

2. The unit digit of the square of the entire number is the same as the unit digit of the result obtained in step 1. In this case, the unit digit of 81 is 1.

So, the unit digit of the square of 109 is option D) 1.

10. Correct option: (A)

The square of 2158 is an even number.

If you square an even number, the result is always an even number. In this case, since 2158 is an even number, its square will also be even.

So, the Correct option: is: A) Even

11. Correct option: (A)

To make a non-square number into a perfect square, you need to multiply it by the smallest number that will result in the perfect square.

Let's find the prime factorization of 75: $75 = 3 * 5 * 5$

To make 75 a perfect square, you need to multiply it by 3, which will result in:

$75 * 3 = 225$

Now, 225 is a perfect square (15^2).

So, the Correct option: is: A) 3

12. Correct option: (C)

To find the square root of 576 we first do prime factorization of 576.

To find the prime factorization of 576, we can start by dividing it by the smallest prime number, which are 2:

Prime factorization of $576 = 2^6 * 3^2$

Now we have to find $\sqrt{576} = \sqrt{2^6 * 3^2} = 2^3 * 3$

So $\sqrt{576} = 24$

13. Correct option: (A)

We have to find the value of $\sqrt{4\frac{73}{324}} = \sqrt{\frac{1369}{324}} = \frac{\sqrt{1369}}{\sqrt{324}}$

As $1369 = 37 * 37 = 37^2 => \sqrt{1369} = 37$ *and* $324 = 18 * 18 => \sqrt{324} = 18$

$$\sqrt{4\frac{73}{324}} = \frac{37}{18} = 2\frac{1}{18}$$

14. Correct option: (C)

To find the square root of the least number that must be added to 780 to obtain a perfect square, follow these steps:

1. Find the next perfect square after 780. The next perfect square after 780 is 784 ($28^2 = 784$).

2. Subtract 780 from 784: $784 - 780 = 4$.

So, the square root of the least number that must be added to 780 to obtain a perfect square is the square root of 4, which is 2.

The Correct option: is: C) 2

15. Correct option: (C)

Let's work through this step by step:

1. Calculate the perimeter of the rectangle: $Perimeter = 2 * (Length + Breadth)$

$$Perimeter = 2 * (14.2 + 3.8) = 2 * 18 = 36\, meters$$

2. Since the area of the square is given to be equal to the perimeter of the rectangle, we equate the two: $Side\ of\ Square^{\,2} = area\ of\ the\ square$

$$side\ of\ the\ square = \sqrt{area\ of\ the\ square} = \sqrt{36} = 6\, meters$$

So, the Correct option: is: C) 6

16. Correct option: (B)

To find the perfect square which is the least among the four-digit numbers, we need to find the smallest perfect square greater than or equal to 1000.

The smallest perfect square greater than or equal to 1000 is ($32^2 = 1024$).

So, the least perfect square among the four-digit numbers is 1024, and its square root is 32.

The Correct option: is: B) 32

17. Correct option: (C)

Let's denote the three numbers in the ratio of $1:2:3$ $as\ x, 2x, and\ 3x$. According to the given information, we have the sum of their cubes:

$$x^3 + (2x)^3 + (3x)^3 = 4500$$

Simplify the equation:

$$x^3 + 8x^3 + 27x^3 = 4500$$

$$36x^3 = 4500$$

Now, solve for x:

$$x^3 = \frac{4500}{36}$$

$$x^3 = 125$$

Take the cube root of both sides: $x = 5$

So, the three numbers are 5, 10, and 15.

Among these numbers, the highest number is 15.

Therefore, the correct answer is: C) 15

18. Correct option: (A)

We have to find the value of $\sqrt{98 * 169 * 450} = V$

Prime factors of the given numbers $98 = 7 * 7 * 2, 169 = 13 * 13, 450 = 3 * 3 * 2 * 5 * 5$

Combining the numbers

$$98 * 169 * 450 = 7 * 7 * 13 * 13 * 2 * 2 * 3 * 3 * 5 * 5 = 7^2 * 13^2 * 2^2 * 3^2 * 5^2$$

$$\sqrt{98 * 169 * 450} = \sqrt{7^2 * 13^2 * 2^2 * 3^2 * 5^2} = 7 * 13 * 2 * 3 * 5 = 2730$$

So the Correct option: is A)

19. Correct option: (D)

The given information states that the product of a number and itself is 3969. This means that the number is the square root of 3969.

Let's find the square root of 3969: $\sqrt{3969} = \sqrt{63 * 63} = 63$

So, the correct answer is: D) 63

20. Correct option: (A)

Given equation

$$\sqrt{2-\frac{17}{121}}=1+\frac{x}{11}$$

We have to solve above equation to find the value of the x

$$\sqrt{2-\frac{17}{121}}=\sqrt{\frac{225}{121}}=\frac{15}{11}=1+\frac{x}{11}$$

$$\frac{15}{11}=\frac{4+11}{11}=\frac{4}{11}+1=1+\frac{x}{11}=>\frac{x}{11}=\frac{4}{11}=>x=4$$

So the Correct option: is A)

LEVEL – II

1. Correct option: (C)

Given $\sqrt{-2}*\sqrt{2}=V$

We have to find the value of V

Using the properties of square roots, we can simplify the expression:

$$\sqrt{-2}*\sqrt{2}=\sqrt{-2*2}=\sqrt{-4}$$

Now, the square root of -4 is not a real number, but it can be expressed as a complex number: $\sqrt{-4}=2i$

So, the value of V is: C) +2i

2. Correct option: (B)

A perfect square is a number that can be expressed as the square of an integer. To find the least two-digit perfect square, we need to look for the smallest perfect square greater than or equal to 10 (the smallest two-digit number).

Let's calculate the squares of integers starting from 1:

$$1^2=1$$
$$2^2=4$$
$$3^2=9$$
$$4^2=16$$

The least two-digit perfect square is 16.

So, the correct answer is: B) 16

3. Correct option: (D)

To determine the smallest perfect square number that is divisible by 8, 12, 15, and 20, we proceed by calculating their least common multiple (LCM). This entails identifying the lowest common multiple of these numbers. Subsequently, our solution will entail selecting the number that is both a multiple of the LCM and a perfect square**.**

$$8=2*2*2,12=3*2*2,15=3*5,20=5*2*2$$

Then LCM will be $2*2*2*3*5=120$

From our options the least square number which is divisible by each of 8, 12, 15 and 20 is 3600.

4. Correct option: (B)

To determine which of the given numbers not a perfect square is, we can look at the last digit of each number.

A perfect square has a unique pattern of digits in its last position, depending on the square root of the number.

Let's analyze each option:

A) 12544 - The last digit is 4, which is a valid last digit for a perfect square.

B) 23457 - The last digit is 7, which is not a valid last digit for a perfect square.
C) 100489 - The last digit is 9, which is a valid last digit for a perfect square.
D) 5329 - The last digit is 9, which is a valid last digit for a perfect square.
From the options provided, option B (23457) is not a perfect square since its last digit is 7, which does not fit the pattern of perfect squares.
So, the correct answer is: B) 23457

5. Correct option: (D)

Let's denote the original number as "x". According to the given information, the process is as follows:
1. Multiply the number by half of itself: $0.5 * x * x = 0.5x^2$
2. Add 10 to the product: $0.5x^2 + 10$
According to the problem, the final result is 82: $0.5x^2 + 10 = 82$
Now, we can solve for x:

$$0.5x^2 = 82 - 10$$
$$0.5x^2 = 72$$
$$x^2 = 144$$

Take the square root of both sides: $x = \pm 12$
So, the correct answer is: D) 12

6. Correct option: (B)

Let's solve for the length of each side of the square field using the given area.
Given area = $\frac{1440}{49}\ sq\ cm$
Let's assume the length of each side of the square field is "x" cm.
The formula for the area of a square is: $Area = side * side$
Given that the area is $\frac{1440}{49}\ sq\ cm$ and the side length is x cm: $x * x = \frac{1440}{49}$
Solving for x:

$$x^2 = \frac{1440}{49}$$
$$x = \sqrt{\frac{1440}{49}}$$
$$x = \frac{\sqrt{1440}}{\sqrt{49}}$$
$$x = \frac{12\sqrt{10}}{7}$$

Now, let's compare the value of x with the options provided:
A) 2.50: This is not close to $(12\sqrt{10})/7$.
B) 5.42: This is close to $(12\sqrt{10})/7$.
C) 2.30: This is not close to $(12\sqrt{10})/7$.
D) 1.92: This is close to $(12\sqrt{10})/7$.
So, the correct answer is: B) 5.42

7. Correct option: (A)

Given the expression as $\frac{0.144}{2.5} \times \frac{25}{0.36} \times \frac{2.89}{0.04}$
Solving the decimal points

$$\frac{144}{25} \times \frac{1000}{10} \times \frac{25}{36} \times \frac{1}{100} \times \frac{289}{4} \times \frac{100}{100} = \frac{144}{25} \times \frac{25}{36} \times \frac{289}{4}$$

Putting the prime factors in place of numbers as

$$\frac{2 * 2 * 2 * 2 * 3 * 3 * 5 * 5 * 17 * 17}{5 * 5 * 6 * 6 * 2 * 2} = 17 * 17$$

$$\frac{0.144}{2.5}\times\frac{25}{0.36}\times\frac{2.89}{0.04}=17*17$$

We have to find the square root of the expression

$$\sqrt{\frac{0.144}{2.5}\times\frac{25}{0.36}\times\frac{2.89}{0.04}}=\sqrt{17^2}=17$$

So the Correct option: is A)

8. Correct option: (B)

To find the greatest two-digit number that is a perfect square, we can start by finding the largest perfect square less than 100.

The largest perfect square less than 100 is 81 (9^2).

So, the correct answer is: B) 81

9. Correct option: (B)

Given the expression as $\sqrt{313+\sqrt{100+7*3}}$

Solving the inner square root

$$\sqrt{100+7*3}=\sqrt{100+21}=\sqrt{121}=11$$

Putting the inner square root in expression

$$\sqrt{313+11}=\sqrt{324}=18$$

So the Correct option: is B)

10. Correct option: (C)

Given equation

$$\sqrt{1+\frac{64}{225}}=\frac{x}{15}$$

We have to solve above equation to find the value of the x

$$\sqrt{1+\frac{64}{225}}=\sqrt{\frac{289}{225}}=\frac{17}{15}=\frac{x}{15}=>x=17$$

So the Correct option is: C)

11. Correct option: (A)

To find the greatest three-digit number that is a perfect square, we need to identify the largest perfect square that is less than 1000.

The largest perfect square less than 1000 is 961 (31^2).

So, the correct answer is: A) 961

12. Correct option: (D)

We are given the equation: $3(x-5)^2=675$

Let's solve for x: $(x-5)^2=225$

$$x-5=\pm\sqrt{225}$$
$$x-5=\pm15$$

Now, solve for x in both cases:

$$1.\ x-5=15$$
$$x=15+5$$
$$x=20$$
$$2.\ x-5=-15$$
$$x=-15+5$$
$$x=-10$$

Out of the options given, only one solution matches: D) 20

So, x can be 20.

13. Correct option: (D)

Given equation

$$\sqrt{0.01 + \sqrt{0.0009}} = x$$

We have to solve above equation to find the value of the x

$$\sqrt{0.0009} = \sqrt{\frac{9}{10000}} = \frac{3}{100} = 0.03$$

Putting back in the equation to find the value of x

$$\sqrt{0.01 + 0.03} = \sqrt{0.04} = \sqrt{\frac{4}{100}} = \frac{2}{10} = 0.2$$

So the Correct option is: D)

14. Correct option: (C)

Given equation $\sqrt{243 + \sqrt{12 + \sqrt{169}}} = x$

We have to solve above equation to find the value of the x

$$\sqrt{169} = 13 => \sqrt{12 + 13} = \sqrt{25} = 5$$

Putting back in the equation to find the value of x

$$\sqrt{243 + 5} = \sqrt{248}$$

$$225 < 248 => \sqrt{225} < \sqrt{248} => 15 < \sqrt{248}$$

So the only option higher than the 15 is option C)

15. Correct option: (C)

A perfect square is a number that can be expressed as the product of an integer with itself. Let's check the options:

A) 1777 - This is not a perfect square, as it cannot be expressed as the product of an integer with itself. (7 cannot be the last digit of any perfect square)

B) 1638 - This is not a perfect square. (8 cannot be the last digit of any perfect square)

C) 2500 - This is a perfect square, as it can be expressed as 50 * 50.

D) 23457 - This is not a perfect square. (7 cannot be the last digit of any perfect square)

So, the correct answer is: C) 2500

16. Correct option: (A)

Square root of 1471369 using long division method

```
1213
                 01 47 13 69
1     |  01
      |  047
22    |  044
      |    0313
241   |    0241
      |      07269
2423  |           7269
      |              0
```

Therefore, 1471369 are square of 1213.

17. Correct option: (C)

We are given the equation: $5(x - 1)^2 = 180$

Let's solve for x: $(x - 1)^2 = 36$

$$x - 1 = \pm\sqrt{36}$$

$$x - 1 = \pm 6$$

Now, solve for x in both cases: $1. x - 1 = 6$

$$x = 7$$

$$2. x - 1 = -6$$

$$x = -5$$

Out of the options given, only one solution matches: C) 7

So, x can be 7.

18. Correct option: (A)

Given that the area of the square field is $4225/2500\ sq.cm$, let's find the length of each side of the field.

The area of a square is given by the formula:

$$Area = side * side$$

Given that the area is $4225/2500\ sq.cm$ and the side length is "x" cm: $x * x = \frac{4225}{2500}$

Solving for x:

$$x^2 = \frac{4225}{2500}$$

$$x = \sqrt{\frac{4225}{2500}}$$

$$x = \sqrt{\frac{169}{100}}$$

$$x = \frac{13}{10}$$

$$x = 1.3$$

Among the provided options, the correct answer is: A) 1.30

19. Correct option: (D)

Given equation

$$\sqrt{274 + \sqrt{211 + \sqrt{196}}} = x$$

We have to solve above equation to find the value of the x

$$\sqrt{196} = 14 => \sqrt{211 + 14} = \sqrt{225} = 15$$

Putting back in the equation to find the value of x

$$\sqrt{274 + 15} = \sqrt{289} = 17$$

So the Correct option: is D)

20. Correct option: (B)

Given equation

$$\sqrt{0.04 + \sqrt{0.0025}} = x$$

We have to solve above equation to find the value of the x

$$\sqrt{0.0025} = \sqrt{\left(\frac{25}{10000}\right)} => \frac{5}{100} = 0.05$$

Putting back in the equation to find the value of x

$$\sqrt{0.04 + 0.05} = \sqrt{0.09} = \sqrt{\left(\frac{9}{100}\right)} = \frac{3}{10} = 0.3$$

So the Correct option: is B)

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

1. The number is multiplied by $2\frac{1}{3}$ of itself: Let's call the original number "x." The result of this step is $2\frac{1}{3}x * x = \frac{7}{3}x * x$.

2. 61 are subtracted from the product: After the first step, we have $\frac{7}{3}x * x$. When we subtract 61 from this, we get $\frac{7}{3}x * x - 61$.

3. The final result is 9200: According to the problem, the final result after all these operations is 9200. So, we can set up an equation:

$$\frac{7}{3}x * x - 61 = 9200$$

Now, we can solve this equation to find the original number "x."

$$\frac{7}{3}x * x = 9200 + 61$$

$$\frac{7}{3}x^2 = 9261$$

Now, to solve for x, divide both sides of the equation by $(\frac{7}{3})$: $x^2 = \frac{9261}{\frac{7}{3}} = 9261 * \frac{3}{7}$

$$x^2 = 3969$$

4. Prime factor of 3969 is $3 * 3 * 3 * 3 * 7 * 7 = 3^4 * 7^2$

$$x^2 = 3969 = 3^4 * 7^2 => x = \sqrt{3^4 * 7^2} = 3^2 * 7$$

So, the original number is $x = 63$.

2. Correct option: (C)

Long division method for square root

1. Divide the digits of the number into pairs of segments starting with the digit in the unit's place.

Let's identify each pair and the remaining final digit (in case there is an odd count of digits in the number) as a segment.

For example: 841 is divided as (08 41)

2. After dividing the digits into segments, start from the leftmost segment. The largest number whose square is equal to or just less than the first segment is taken as the **divisor** and also as the quotient (so that the product is the square).

For example: 4 is the closest perfect square to 8, the first segment 8

3. Subtract the square of the divisor from the first segment and bring down the next segment to the right of the remainder to get the new dividend.

For example: 8 - 4 = 4 is concatenated with next segment 41.

New dividend = 441

4. Now, the new divisor is obtained by taking two times the previous quotient (which was 2 in the above example as 2^2 = 9) and concatenating it with a suitable digit which is also taken as the next digit of the quotient, chosen in such a way that the product of the new divisor and this digit is equal to, or just less than the new dividend.

For example: Two times quotient 2 is 4.

49 times 9 are 441 which are closest to the new dividend.

5. Repeat steps (2), (3) and (4) till all the segments have been taken up. Now, the quotient so obtained is the required square root of the given number.

Square root of 841 using long division method

29

	08 41
2	04
49	441
	441
	0

Therefore, 841 are square of 29.

3. Correct option: (C)

To find the unit digit of the square of 1234, you can follow this approach:

1. Calculate the unit digit of the square of the last digit of the number. In this case, the last digit of 1234 is 4. $So, 4^2 = 16$.

2. The unit digit of the square of the entire number is the same as the unit digit of the result obtained in step 1. In this case, the unit digit of 16 is 6.

So, the unit digit of the square of 1234 is 6.

Therefore, the Correct option: is: C) 6

4. Correct option: (C)

To make a non-square number into a perfect square, you need to divide it by the smallest number that will result in the perfect square.

Let's find the prime factorization of 2352:

$$2352 = 2 * 2 * 2 * 2 * 3 * 7 * 7$$

For the number to become a perfect square, you need to divide it by the smallest number that eliminates the odd powers of its prime factors. In this case, you need to divide by 3, as that will cancel out one of the odd powers of 3.

So, when you divide 2352 by 3, you get: $2352/3 = 774$

Now, 774 is a perfect square (28^2).

So, the Correct option: is: C) 3

5. Correct option: (C)

We have to find the value of $\sqrt{0.9} * \sqrt{1.6} = \sqrt{\frac{9}{10}} * \sqrt{\frac{16}{10}} = \frac{\sqrt{144}}{\sqrt{100}}$

As $144 = 12 * 12 = 12^2 => \sqrt{144} = 12\ and\ 100 = 10 * 10 => \sqrt{100} = 10$

$$\sqrt{0.9} * \sqrt{1.6} = \frac{12}{10} = 1.2$$

6. Correct option: (B)

We have to find the value of $\sqrt{98} * \sqrt{162} =$

As $98 = 7 * 7 * 2\ and\ 162 = 2 * 9 * 9$

$$\sqrt{98} * \sqrt{162} = \sqrt{98 * 162} = \sqrt{7 * 7 * 2 * 2 * 9 * 9} = \sqrt{7^2 * 2^2 * 9^2} = 7 * 9 * 2$$

Then the value of $\sqrt{98} * \sqrt{162}$ is 126

Option B)

7. Correct option: (C)

To determine the smallest perfect square number that is divisible by 8, 9, and 10, we proceed by calculating their least common multiple (LCM). This entails identifying the lowest common multiple of these numbers. Subsequently, our solution will entail selecting the number that is both a multiple of the LCM and a perfect square**.**

$$8 = 2 * 2 * 2, 9 = 3 * 3, 10 = 2 * 5$$

Then LCM will be $2 * 2 * 2 * 3 * 3 * 5 = 360$

From our options the least square number which is divisible by each of 8, 9, 10 is 3600.

8. Correct option: (D)

This is an arithmetic series where the first term (a_1) is 1, the common difference d is 2, and the last term a_n is 125.

The formula for the sum of an arithmetic series is given by: $[S = \frac{n}{2}(a_1 + a_n)]$

First, calculate n using the formula for the nth term of an arithmetic series:

$$[a_n = a_1 + (n - 1)\,d]$$
$$[125 = 1 + (n - 1)\,2]$$
$$[n = 63]$$

Now, use the formula for the sum of an arithmetic series:

$$[S = \frac{n}{2}(a_1 + a_n)]$$
$$[S = \frac{63}{2}(1 + 125)]$$
$$[S = 63 * 63 = 3969]$$

So, the sum of the series given is 3969.

9. Correct option: (C)

Square root of 2916 using long division method

	54
	$\overline{29}\ \overline{16}$
5	25
104	416
	416
	0

Therefore, 2916 are square of 54.

10. Correct option: (B)

We have to find the value of $\sqrt{0.000256} = V$

Rewrite the number in ration format

$$\sqrt{\frac{256}{1000000}} = \sqrt{\frac{16 * 16}{1000 * 1000}} = \frac{16}{1000} = 0.016$$

So the Correct option: is B)

Cubes and Cube Roots

LEVEL – I

1. Correct option: (D)

Cube of any number represents as $ab => (ab)^3 = a^3b^3$

Then, $(2a)^3 = 8a^3$

So, the Correct option: is D)

2. Correct option: (A)

We are given the number as 2197

When the unit digit of a number is 7, its cube root will have a unit digit of 3.

In other words, if the ones digit of a number is 3 then the ones digit of its cube will be 7.

This is a pattern that holds true for numbers ending in 7 and their cube roots.

So, the Correct option: is A)

3. Correct option: (A)

We are given the expression as $1^3 + 2^3 + 3^3 + 4^3 + 5^3$

Solving the cubes of the number we get $1 + 8 + 27 + 64 + 125 = 225$

So, the Correct option: will be A)

We can calculate the sum of cubes of consecutive natural number starting from 1 using the formula.

$$\sum_{1}^{n} n^3 = 1^3 + 2^3 + 3^3 + 4^3 + \cdots + n^3 = \frac{n^2 * (n+1)^2}{4} => \frac{5^2 * (5+1)^2}{4} = \frac{25 * 36}{4} = 25 * 9 = 225$$

4. Correct option: (D)

A perfect cube is a number that can be expressed as the cube of an integer. Let's check the options:

$$A)\ 1000 = 10^3. This\ is\ a\ perfect\ cube.$$
$$B)\ 1728 = 12^3. This\ is\ a\ perfect\ cube.$$
$$C)\ 27000 = 30^3. This\ is\ a\ perfect\ cube.$$
$$D)\ 6400 = 2^6 * 10^2. This\ is\ not\ a\ perfect\ cube.$$

So, the 6400 is not a perfect cube.

D) 6400

5. Correct option: (A)

Given expression as $\sqrt[3]{5-\frac{10}{27}}$

Solving further

$$\sqrt[3]{\frac{5*27-10}{27}} = \sqrt[3]{\frac{5*27-5*2}{27}} = \sqrt[3]{\frac{5*(27-2)}{27}} = \sqrt[3]{\frac{5*(25)}{27}} = \sqrt[3]{\frac{5*5*5}{27}} = \sqrt[3]{\frac{5^3}{3^3}} = \frac{5}{3}$$

So, the Correct option: will be A) $5/3$

6. Correct option: (A)

To determine the smallest number by which 8640 must be divided so that the quotient becomes a perfect cube, we need to factorize 8640 and then identify the factors required to transform the quotient into a perfect cube.

The prime factorization of 8640 is: $8640 = 2^6 * 3^3 * 5$

To ensure that the quotient becomes a perfect cube, all prime factors must have powers that are multiples of 3.

So, the smallest number by which 8640 must be divided to get a perfect cube quotient is:

$$\frac{8640}{5} = 2^6 * 3^3 * \frac{5}{5} = 2^6 * 3^3$$

Among the given options, the correct answer is: A) 5

7. Correct option: (D)

The concept of a "Hardy-Ramanujan number" is related to a story involving the famous mathematician Srinivasa Ramanujan and the British mathematician G.H. Hardy.

In the story, Hardy visited Ramanujan in the hospital and mentioned that he arrived in a taxi with the rather uninteresting number 1729. Ramanujan immediately responded that 1729 is actually an interesting number because it is the smallest number which can be expressed as the sum of two cubes in two different ways:

$$1.\ 1729 = 1^3 + 12^3$$
$$2.\ 1729 = 9^3 + 10^3$$

This property of 1729 being expressible as the sum of two cubes in multiple ways is what led to it being called a "Hardy-Ramanujan number."

So, the correct answer is: D) 1729

8. Correct option: (A)

The cube of 24 is: $24^3 = 13824$

So, the correct answer is: A) 13824

9. Correct option: (B)

Cube of a negative number will also be negative

The cube of -21 is: $(-21)^3 = (21)^3 * (-1)^3 = -(21)^3 = -9261$

So, the correct answer is: B) -9261

10. Correct option: (D)

If the digit in the one's place of a number is 2, then let's consider the cubes of numbers that end with 2:

$$2^3 = 8$$
$$12^3 = 1728$$
$$22^3 = 10648$$

As you can see, in all these cases, the last digit of the cube is 8.
So, the correct answer is option D) 8.

11. Correct option: (A)

Let's calculate the cubes of the given options:

$$A)\ (-1)^3 = -1$$
$$B)\ (-2)^3 = -8$$
$$C)\ (-3)^3 = -27$$
$$D)\ (-4)^3 = -64$$

Among the given options, only option A) -1 is equal to its cube.
Therefore, the correct answer is A) -1.

12. Correct option: (B)

The correct answer is: B) Cube of a negative number is always negative.
Let's take the example of the number -2.

$$-2^3 = -2 * -2 * -2 = -8$$

As you can see, the cube of -2, (-8) is negative. This demonstrates that the cube of a negative number is indeed negative.
This rule applies to all negative numbers. When you raise a negative number to an odd power like 3, the result will be negative. If you raise a negative number to an even power like 2, the result will be positive.

13. Correct option: (B)

The correct answer is: B) Negative
The cube of a negative number is always negative. When you raise a negative number to an odd power like 3 (cube it), the result will be negative. If you raise a negative number to an even power like 2, the result will be positive.

14. Correct option: (C)

If a number ends with one zero, it means the number is a multiple of 10. Let's consider the cube of such a number: $(10)^3 = 1000$
The cube of 10, which has one zero at the end, results in a number with three zeroes at the end.
Therefore, the correct answer is: C) Three

15. Correct option: (C)

A perfect cube is a number that can be expressed as the cube of an integer. Let's examine the options:

$$A)\ 216 = 6^3$$
$$B)\ 1000 = 10^3$$
$$C)\ 243 = 3^5$$
$$D)\ 1331 = 11^3$$

Among the given options, option C) 243 is not a perfect cube because it can be expressed as 3^5, which is not the cube of an integer.
Therefore, the correct answer is C) 243.

16. Correct option: (C)

Each prime factor appears 3 times in its cube.
So, the correct answer is C) 3.

17. Correct option: (C)

The unit digit of the cube of a number ending with 6 can be determined by looking at the pattern of the units digits of the cubes of numbers ending with 6:

$$6^3 = 216$$
$$16^3 = 4096$$
$$26\text{^}3 = 17576$$
$$\ldots$$

And so on.
From this pattern, you can observe that the unit digit of the cube of a number ending with 6 is always 6.
So, the correct answer is C) 6.

18. Correct option: (B)
The cube of an even number is always **even**.
Let's take an even number as an example, such as 4.

$$4^3 = 4 * 4 * 4 = 64$$

As you can see, the cube of 4 (an even number) is 64, which is also an even number.
This pattern holds true for all even numbers. When you cube an even number, the result will always be an even number.
Therefore, the correct answer is B) Even.

19. Correct option: (C)
The given number is $-\frac{a^{-9}}{b^{-6}}$.
Let's simplify the given expression as $-\frac{b^6}{a^9}$
To find the cube root of the expression

$$\sqrt[3]{-\frac{b^6}{a^9}} = \sqrt[3]{\frac{b^6}{a^9}} * \sqrt[3]{-1} = -\frac{b^2}{a^3} => -\frac{a^{-3}}{b^{-2}}$$

Which is equal to the option (C).

20. Correct option: (B)
The given expression $\sqrt[3]{2744} = a + 2$
Prime factorization of the number 2744 is $7^3 \times 2^3$
Let's simplify the given expression as $\sqrt[3]{7^3 \times 2^3} = 7 * 2 = 14$
Further simplification $14 = a + 2 => a = 12$
So, the Correct option: is B)

LEVEL – II

1. Correct option: (A)
Given expression $\sqrt[3]{5} \times \sqrt[3]{5} \times \sqrt[3]{5}$
We can collect all term in one cube root (all has same power) $\sqrt[3]{5 \times 5 \times 5} = \sqrt[3]{5^3} = 5$

2. Correct option: (A)
Given expression $\sqrt[3]{a^3 + 3a^2b + 3ab^2 + b^3}$
We know cube of any binomial term $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
So, expression in cube root can be written as $\sqrt[3]{a^3 + 3a^2b + 3ab^2 + b^3} = \sqrt[3]{(a + b)^3} = a + b$

3. Correct option: (A)
Let's start by finding the side length of the cube, given its volume.
The formula for the volume of a cube is: $Volume = side\ length^3$
Given that the volume is 2197 cm^3, we can solve for the side length: $side\ length^3 = 2197$

$$side\ length = \sqrt[3]{2197} = 13\ cm$$

Now that we know the side length is 13 cm, we can calculate the surface area of the cube.
The formula for the surface area of a cube is: $Surface\ Area = 6 * (side\ length)^2$
Substitute the value of the side length:

$$Surface\ Area\ =\ 6\ *\ (13\ cm)^2$$
$$Surface\ Area\ =\ 6\ *\ 169\ cm^2$$
$$Surface\ Area\ =\ 1014\ cm^2$$

So, the correct answer is A) $1014\ cm^2$.

4. Correct option: (C)

We are given the expression as $1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3 + 7^3 + 8^3 + 9^3 + 10^3$

We can calculate the sum of cubes of consecutive natural number starting from 1 using the formula.

$$\sum_{1}^{n} n^3 = 1^3 + 2^3 + 3^3 + 4^3 + \cdots + n^3 = \frac{n^2 * (n+1)^2}{4} => (n = 10)$$

$$=> \frac{10^2 * (10+1)^2}{4} = \frac{100 * 121}{4} = 25 * 121 = 3025$$

So, the Correct option: is C)

5. Correct option: (A)

Given equation $25a^4 = a^6$

We have to find the value of a^3.

Solving for a $25 = \frac{a^6}{a^4} => a^2 = 25 => a = \pm 5$

So, the value of $a^3$is $a^3 = -125$ or 125

So, the Correct option: is A)

6. Correct option: (B)

To make 361 a perfect cube, we need to find the smallest number by which it must be multiplied to achieve this. Let's analyze the prime factorization of 361:

$$361\ =\ 19^2$$

To make 361 a perfect cube, we need each prime factor's exponent to be a multiple of 3. In this case, the prime factorization of 361 already has a prime factor (19) with an exponent of 2, which is not a multiple of 3.

To make it a perfect cube, we need to multiply it by 19 again:

$$361\ *\ 19\ =\ 6859$$

Now, the prime factorization of 6859 is: $6859\ =\ 19^3$

So, the smallest number by which 361 must be multiplied to make the product a perfect cube is 19.

The correct answer is B) 19.

7. Correct option: (A)

Given expression $\sqrt[3]{5\frac{10023}{125}} = \sqrt[3]{\frac{10648}{125}}$

Solving further

$$\sqrt[3]{\frac{10648}{125}} => \sqrt[3]{\frac{2*2*2*11*11*11}{5*5*5}} = \sqrt[3]{\frac{2^3 \times 11^3}{5^3}} = \frac{22}{5}$$

So, the Correct option: is A)

8. Correct option: (C)

The volume of a cube is given by the formula: $Volume\ =\ side\ length^3$

Given that the side length of the cubical box is 1.5 m, we can calculate its volume in cubic meters: $Volume\ =\ (1.5\ m)^3\ =\ 3.375\ m^3$

Now, to convert the volume from cubic meters to cubic centimeters, we know that:

$$1\ m\ =\ 100\ cm$$
$$So, 1\ m^3\ =\ (100\ cm)^3\ =\ 1{,}000{,}000\ cm^3$$

Therefore, the volume of the cubical box in cubic centimeters is:

$$Volume = 3.375\,m^3 * 1{,}000{,}000\frac{cm^3}{m^3} = 3{,}375{,}000\,cm^3 = 3.375 \times 10^6 cm^3$$

So, the volume of the cubical box is 3,375,000 cm^3 with Correct option: as C)

9. Correct option: (B)

Let's translate the given information into an equation:

The cube of a number x is 25 times x: $x^3 = 25x$

Now, we need to solve for x:

Divide both sides of the equation by x: $x^2 = 25$

Take the square root of both sides: $x = \pm\sqrt{25}$

$$x = \pm 5$$

However, the given condition specifies that x cannot be equal to 0 or -5. So, the possible value of x satisfying the condition is 5.

The correct answer is B) 5.

10. Correct option: (B)

Given expression

$$\sqrt[6]{(\frac{1728}{1000})^2} = (\frac{1728}{1000})^{2\times\frac{1}{6}} = (\frac{1728}{1000})^{\frac{1}{3}}$$

Solving further

$$\sqrt[3]{\frac{1728}{1000}} => \sqrt[3]{\frac{2^6 \times 3^3}{10^3}} = \frac{2^2 \times 3}{10} = \frac{12}{10} = 1.2$$

So, the Correct option: is B)

11. Correct option: (D)

To determine the least number by which 648 must be multiplied to make it a perfect cube, let's analyze its prime factorization: $648 = 2^3 * 3^4$

In order for 648 to become a perfect cube, each prime factor's exponent must be a multiple of 3.

The prime factorization already has the necessary powers of 2.

For 648 to become a perfect cube, we need to increase the exponent of 3 to 3 and leave the exponent of 2 as it is (since it's already a multiple of 3):

$$2^3 * 3^4 * 3\text{^}2 = 2^3 * 3^6$$

So, we need to multiply 648 by 3^2 to make it a perfect cube.

Therefore, the least number by which 648 must be multiplied to make it a perfect cube is 9.

The correct answer is D) 9.

12. Correct option: (B)

Let's denote the side length of the cube as "s". The formula for the surface area of a cube is given by: $Surface\ Area = 6 * (side\ length)^2$

Given that the surface area is 96 m², we can set up the equation: $96 = 6 * s^2$

Now, let's solve for "s" (side length):

$$s^2 = \frac{96}{6}$$
$$s^2 = 16$$
$$s = \sqrt{16}$$
$$s = 4$$

The side length of the cube is 4 meters. Now, let's calculate the volume of the cube using the formula: $Volume = (side\ length)^3$

$$Volume = 4^3 = 64\,m^3$$

So, the correct answer is: B) 64 m³

13. Correct option: (B)

The side length of the cubical box is given as 2.1 meters, but the volume is mentioned to be in the form of cubic centimeters (cm^3).

To proceed, we need to ensure that the units are consistent. Let's convert the side length to centimeters: $1\ meter = 100\ centimeters$

So, if the side length is 2.1 meters, then it's equivalent to:

$$2.1\ meters * 100\ centimeters/meter = 210\ centimeters$$

Now that we have the side length in centimeters, we can calculate the volume of the cubical box using the formula for the volume of a cube:

$$Volume = (side\ length)^3$$

$$Volume = (210\ cm)^3 = 9{,}261{,}000\ cm^3$$

So, the volume of the cubical box is $9{,}261{,}000\ cm^3$ which is equal to the option B) $9.261 \times 10^6 cm^3$

14. Correct option: (B)

To find the largest two-digit perfect cube, let's list the cubes of two-digit numbers and see which one is the largest: $2\text{^}3 = 8$

$$3\text{^}3 = 27$$

$$4\text{^}3 = 64$$

$$5\text{^}3 = 125\ (This\ is\ too\ large\ for\ a\ two-digit\ number)$$

So, among the options given, the largest two-digit perfect cube is $4^3 = 64$.

Therefore, the correct answer is: 64

15. Correct option: (C)

Let's denote the volumes of the two cubes as V_1 and V_2, and the side lengths of the cubes as a_1 and a_2, respectively.

Given that the volumes are in the ratio 1:64, we can write: $\frac{V_1}{V_2} = \frac{1}{64}$

Substitute the given volume ratio into the equation: $\frac{1}{64} = \frac{(a_1{}^3)}{(a_2{}^3)}$

Take the cube root of both sides: $\left(\frac{a_1}{a_2}\right) = \frac{1}{4}$

This tells us that the side length of the smaller cube is one-fourth of the side length of the larger cube.

The ratio of the area of the faces of the smaller cube to that of the larger cube is the square of the ratio of their side lengths: $\left(\frac{a_1}{a_2}\right)^2 = (\frac{1}{4})^2$

So, the ratio of the area of the face of one cube to that of the other cube is 1:16.

The Correct option: is C)

16. Correct option: (D)

Given expression $P = \sqrt[3]{\sqrt[3]{a^9}}$

Solving further $\sqrt[3]{a^9} => a^3 => \sqrt[3]{a^3} = a$

So, the value of given expression is $\sqrt[3]{\sqrt[3]{a^9}} = a$

So, the Correct option: is D)

17. Correct option: (A)

Let's denote the unknown number as "x". According to the information given: $\frac{\sqrt[3]{x}}{7} = 49$

To solve for "x", we can start by isolating the cube root of "x" by multiplying both sides of the equation by 7: $\sqrt[3]{x} = 7 * 49$

$$x = 343^3$$

So, the number is 343^3.

18. Correct option: (C)

Let's find the largest four-digit perfect cube.

To do this, we can find the cube root of the largest four-digit number and then check if its cube is still within the four-digit range.

The largest four-digit number is 9999.

The cube root of 9999 is approximately 21.54 (rounded to two decimal places).

Now, let's check the cubes of 21 and 22:

$$21^3 = 9261$$
$$22^3 = 10648$$

The largest four-digit perfect cube is 9261, which is 21^3.

So, the correct answer is: C) 9261

19. Correct option: (B)

If the dimensions of the pond are double those of the tank, then the volume of the pond will be eight times that of the tank. This is because volume is proportional to the cube of the dimensions ($V \propto l^3$).

Given that the tank holds 27000 liters of water, let's denote its volume as $V_tank = 27000\ liters$.

The volume of the pond (V_pond) would be: $V_pond = 8 * V_tank$

$$V_pond = 8 * 27000 = 216000\ liters$$

So, the correct answer is: B) 216000 $liters$

20. Correct option: (C)

Let's denote the natural number as "n". According to the information given: $n^3 - n^2 = 48$

We want to solve for "n". Let's factor out n^2 from both terms on the left side of the equation: $n^2 * (n - 1) = 48$

Now we have a product of two numbers equal to 48. We need to find a combination of factors that satisfy this equation and also consider that "n" is a natural number.

Possible factor pairs of 48 are: (1, 48), (2, 24), (3, 16), (4, 12), and (6, 8).

However, we are looking for a pair $(n^2, n - 1)$. Among the factor pairs, only (3, 16) would work:

$$n^2 = 16$$
$$n = \sqrt{16}$$
$$n = 4$$

So, the correct answer is: C) 4

STUDENT PRACTICE QUESTIONS

1. Correct option: (B)

We are given the number as 17576

Prime factorization of the number $17576 = 2^3 \times 13^3$

So, the cube root of the given number will be $\sqrt[3]{17576} = \sqrt[3]{2^3 \times 13^3} = 2 * 13 => 26$

So, the Correct option: is B)

2. Correct option: (D)

The cube of 0.8 is: $0.8^3 = 0.512$

So, the correct answer is: D) 0.512

3. Correct option: (B)

The cube root of 0.000216 is 0.06.

$$\sqrt[3]{0.000216} = \sqrt[3]{\frac{216}{1000000}} = \sqrt[3]{\frac{6^3}{100^3}} = \frac{6}{100}$$

So, the correct answer is: B) 0.06

4. Correct option: (A)

To determine which number is a perfect cube, we need to find the cube root of each number and see if it's an integer.

A) $1331 = 11^3$

B) 1441 is not a perfect cube.

C) 1661 is not a perfect cube.

D) 1301 is not a perfect cube.

So, the correct answer is: A) 1331

5. Correct option: (C)

The cube of (a^6) can be calculated by rising (a^6) to the power of 3:

$$[(a^6)^3 = a^{\{6 \times 3\}} = a^{\{18\}}]$$

So, the cube of (a^6) is $(a^{\{18\}})$.

6. Correct option: (B)

We can determine the last digit of the cube of 293 without actually computing the cube. The last digit of the cube is determined by the last digit of the base number (in this case, 293, last digit is 3).

The last digit of 293 is 3. When you multiply any number ending in 3 by itself three times, the last digit of the result will be 7.

So, the correct answer is still B) 7.

7. Correct option: (A)

The cube root of 2197 is 13. Therefore, the cube form of 2197 can be expressed as:

$$[\,2197 = 13^3\,]$$

8. Correct option: (C)

A perfect cube is a number that can be expressed as the cube of an integer. Among the options provided:

$$A)\ 10000 = 10^4 = 2^4 * 5^4$$

$$B)\ 243 = 3^5$$

$$C)\ 343 = 7^3$$

$$D)\ 270000 = 30^3 * 10$$

The only number that is a perfect cube from the given options is: $C)\ 343 = 7^3$

So, the correct answer is option C) 343.

9. Correct option: (A)

To find the unit digit of the cube of a number without actually calculating the cube, you can use the fact that the unit digit of the cube is determined solely by the unit digit of the original number.

In this case, the unit digit of 3331 is 1. When you multiply any number that ends in 1 by itself three times (cube it), the unit digit of the result will always be 1.

Therefore, the correct answer is: A) 1

10. Correct option: (B)

A perfect cube is a number that can be expressed as the cube of an integer. The cube root of a number represents the value that, when cubed, gives you the original number.

In this case, the cube root of 243 is approximately 6.079. Since 6^3 is equal to 216 and 7^3 is equal to 343, it's clear that 243 is not a perfect cube, as it's not the cube of any integer.

Therefore, the correct answer is: B) No

11. Correct option: (A)

To find the smallest natural number by which 53240 should be divided to get a perfect cube quotient, we need to factorize 53240 and then divide it by the appropriate factors.

The prime factorization of 53240 is: $53240 = 2^3 * 5 * 11^3$

A perfect cube quotient would require that each prime factor is raised to a multiple of 3 in the divisor. The prime factors are 2, 5, and 11.

For the factor 2, it's already raised to the power of 3, so it's fine.
For the factor 11, it's already raised to the power of 3, so it's fine.
For the factor 5, it needs to be lowered to the power of 0.
This is to ensure that the quotient becomes a perfect cube.
So, the divisor should be 5.
The smallest natural number that should divide 53240 to give a perfect cube quotient is 5.
Therefore, the correct answer is: A) 5

12. Correct option: (A)

To determine the smallest number by which 675 must be multiplied to obtain a perfect cube, let's analyze the prime factorization of 675: $675 = 3^3 * 5^2$
For the product to become a perfect cube, we need each prime factor's exponent to be a multiple of 3.
For the factor 3 it's already raised to the power of 3, so it's fine.
For the factor 5, it needs to be raised to the power of 1 more (3 - 2 = 1) to make it a perfect cube.
So, the number by which 675 must be multiplied to obtain a perfect cube is: 5
Therefore, the correct answer is: 5

13. Correct option: (B)

The cube of $(\frac{a^3}{b^6})$ can be calculated by cubing both the numerator and the denominator separately:

$$\left(\frac{a^3}{b^6}\right)^3 = \frac{(a^3)^3}{(b^6)^3}$$
$$= \frac{a^{3*3}}{b^{6*3}}$$
$$= \frac{a^9}{b^{18}}$$

So, the cube of $(\frac{a^3}{b^6})$ is$(\frac{a^9}{b^{18}})$.
So, the Correct option: will be B)

14. Correct option: (C)

To find the cube root of 15625 using prime factorization, let's factorize 15625:

$$15625 = 5\text{^}6$$

Since the prime factorization of 15625 contains only one prime factor raised to a power of 6, the cube root can be obtained by taking the cube root of that prime factor's power:

$$Cube\ root\ of\ 15625 = Cube\ root\ of\ 5^6 = 5^{\frac{6}{3}} = 5^2 = 25$$

So, the correct answer is C) 25.

15. Correct option: (A)

The volume of a cube is given by the formula: $Volume = side\ length^3$
Given that the volume of the cube is 125 cm³, we can set up an equation:

$$125 = side\ length^3$$

To find the side length, we need to take the cube root of both sides:

$$side\ length = cube\ root\ of\ 125$$
$$side\ length = 5$$

So, the correct answer is A) 5 cm.

16. Correct option: (B)

Given number is 46656
To find the number of digits in its cube root, let's find the cube root using prime factorization.

To find the prime factorization of 46656, we can break down the number into its prime factors:

The prime factorization of 46656 is: $46656 = 2^6 * 3^6$

$$\sqrt[3]{46656} = \sqrt[3]{2^6 * 3^6} = 2^2 * 3^2 = 36$$

Number of digits in the cube root of 46656 is 2

17. Correct option: (A)

The given number is 8000.

The given number has 3 trailing zeroes, and when its cube root is taken, it will have one trailing zero. As we know, if a number has one trailing zero, then its cube will have 3 trailing zeroes.

18. Correct option: (B)

The given number is $\sqrt[3]{343} \times \sqrt[3]{-64}$

Let's simplify the given expression as $\sqrt[3]{7 * 7 * 7} \times \sqrt[3]{4 * 4 * 4} \times \sqrt[3]{-1}$

Further simplification $-1 \times \sqrt[3]{7^3} \times \sqrt[3]{4^3} = -1 * 7 * 4 = -28$

So, the Correct option: is B)

19. Correct option: (B)

Given expression $\sqrt[3]{x + 20} = 10$

Cubing both side $(\sqrt[3]{x + 20})^3 = 10^3$

Simplify further $x + 20 = 10^3 = 1000 => x = 980$

20. Correct option: (B)

We are given the expression as $31 + 33 + 35 + 37 + 39 + 41$

We can calculate the sum as 216

216 can be written as 6^3

So, the Correct option: is B)

21. Correct option: (B)

Given expression $\sqrt[3]{81} \times \sqrt[3]{192} \times \sqrt[3]{24}$

We can collect all term in one cube root (all has same power)

$$\sqrt[3]{81 \times 192 \times 24} = \sqrt[3]{3 \times 3 \times 3 \times 3 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 2 \times 2 \times 2}$$

$$\sqrt[3]{3^6 \times 2^9} = 3^2 \times 2^3 => 72$$

So, the Correct option: is B)

22. Correct option: (B)

Given the expression as $22^3 - 33^3 + 11^3$

Solving the cubes and finding the value of the given expression

$$10648 - 35937 + 1331 = -23958$$

So, the Correct option: is B)

23. Correct option: (B)

Given expression $\sqrt[3]{30} \times \sqrt[3]{12} \times \sqrt[3]{75}$

We can collect all term in one cube root (all has same power)

$$\sqrt[3]{30 \times 12 \times 75} = \sqrt[3]{3 \times 5 \times 2 \times 3 \times 2 \times 2 \times 3 \times 5 \times 5}$$

$$\sqrt[3]{3^3 \times 2^5 \times 5^3} = 3 \times 2 \times 5 => 30$$

So, the Correct option: is B)

24. Correct option: (C)

Given expression $\sqrt[3]{5} \times \sqrt[3]{50} \times \sqrt[3]{500}$

We can collect all term in one cube root (all has same power)

$$\sqrt[3]{5 \times 50 \times 500} = \sqrt[3]{5 \times 2 \times 5 \times 5 \times 2 \times 2 \times 5 \times 5 \times 5}$$

$$\sqrt[3]{2^3 \times 5^6} = 2 \times 5^2 => 50$$

So, the Correct option: is C)

25. Correct option: (B)

Given expression $\sqrt[3]{\frac{35937}{1331}} + \sqrt[3]{\frac{125}{8}}$

Solving further $\sqrt[3]{\frac{3^3 \times 11^3}{11^3}} + \sqrt[3]{\frac{5^3}{2^3}} = \sqrt[3]{3^3} + \sqrt[3]{\frac{5^3}{2^3}} = 3 + \frac{5}{2} => \frac{11}{2}$

So, the Correct option: is B)

26. Correct option: (D)

Given expression $\sqrt[3]{30} \times \sqrt[3]{12} \times \sqrt[3]{75}$

We can collect all term in one cube root (all has same power)

$$\sqrt[3]{30 \times 12 \times 75} = \sqrt[3]{3 \times 5 \times 2 \times 3 \times 2 \times 2 \times 3 \times 5 \times 5}$$

$$\sqrt[3]{3^3 \times 2^5 \times 5^3} = 3 \times 2 \times 5 => 30$$

So, the Correct option: is D)

27. Correct option: (A)

Given expression $(a+b+c)(a^2+b^2+c^2+2ab+2ac+2bc)$

As we know that $(a+b+c)^2 = a^2+b^2+c^2+2ab+2ac+2bc$

Further simplify $(a+b+c)(a^2+b^2+c^2+2ab+2ac+2bc) = (a+b+c)(a+b+c)^2$

$$(a+b+c)(a+b+c)^2 = (a+b+c)^3$$

We have to find the cube root of the given expression

$$\sqrt[3]{(a+b+c)(a^2+b^2+c^2+2ab+2ac+2bc)} = \sqrt[3]{(a+b+c)^3}$$

$$= (a+b+c)$$

So, the Correct option: is A)

28. Correct option: (A)

Given expression $2\sqrt[3]{x+3} = 3$

Simplify the equation to find the value of x $\sqrt[3]{x+3} = \frac{3}{2}$

Cubing both of the side $(\sqrt[3]{x+3})^3 = (\frac{3}{2})^3 => x+3 = \frac{27}{8}$

$$x = \frac{27}{8} - 3 => \frac{3}{8}$$

So, the Correct option: is A)

29. Correct option: (B)

The formula for the surface area of a cube is given by:

$$Surface\ Area\ =\ 6\ *\ (side\ length)^2$$

Given that the surface area is 54 m², we can set up the equation:

$$54\ =\ 6\ *\ (side\ length)^2$$

Now, let's solve for the side length of the cube:

$$(side\ length)^2\ = \frac{54}{6}$$

$$(side\ length)^2\ =\ 9$$

$$side\ length\ =\ \sqrt{9}$$

$$side\ length\ =\ 3$$

The side length of the cube is 3 meters. Now, let's calculate the volume of the cube using the formula:

$$Volume\ =\ (side\ length)^3$$

$$Volume\ =\ 3^3\ =\ 27\ m^3$$

So, the volume of the cube with a surface area of 54 m² is 27 m^3.

30. Correct option: (D)

Given expression $\sqrt[3]{-4913} + \sqrt[3]{0.064} = x$

To find the value of x we have to calculate the cube roots of the given values

$$\sqrt[3]{-4913} = \sqrt[3]{-4913} = -17$$

$$\sqrt[3]{0.064} = \frac{4}{10} = 0.4$$

Putting the values in expression $-17 + 0.4 => -16.6$

So, the Correct option: is D)

31. Correct option: (B)

Given expression

$$\sqrt[3]{-\frac{1728}{2744}} = -(\frac{1728}{2744})^{\frac{1}{3}}$$

Solving further

$$\sqrt[3]{\frac{1728}{2744}} => \sqrt[3]{\frac{2^6 \times 3^3}{2^3 \times 7^3}} = \frac{2^2 \times 3}{2 \times 7} = \frac{6}{7}$$

So, the value of given expression is

$$\sqrt[3]{-\frac{1728}{2744}} = -\frac{6}{7}$$

So, the Correct option: is B)

32. Correct option: (D)

Given expression

$$P = \sqrt[3]{\frac{343 \times 125}{0.064}}$$

Solving further

$$P = \sqrt[3]{\frac{7^3 \times 5^3}{\frac{4^3}{10^3}}} = \sqrt[3]{\frac{7^3 \times 5^3 \times 10^3}{4^3}} = \frac{7 \times 5 \times 10}{4} = \frac{350}{4} = 87.5$$

So, the value of given expression is

$$P = \sqrt[3]{\frac{343 \times 125}{0.064}} = 87.5$$

So, the Correct option: is D)

33. Correct option: (D)

Given expression $P = \sqrt[3]{-3\frac{3}{8}}$

Solving further $P = \sqrt[3]{-\frac{27}{8}} = -\sqrt[3]{\frac{3^3}{2^3}} = -\frac{3}{2}$

So, the value of given expression is $P = \sqrt[3]{-3\frac{3}{8}} = -\frac{3}{2}$

So, the Correct option: is D)

34. Correct option: (A)

Given $512 = 8^3 \; and \; 3.375 = 1.5^3$

Given expression $P = \sqrt[3]{512} \times \sqrt[3]{3.375}$

Putting the given values in the equation

Solving further $P = \sqrt[3]{512} \times \sqrt[3]{3.375} = \sqrt[3]{8^3} \times \sqrt[3]{1.5^3} = 8 \times 1.5$

So, the value of given expression is $P = \sqrt[3]{512} \times \sqrt[3]{3.375} = 12$

So, the Correct option: is A)